



CHEMISTRY

Senior Three



Student's Book

Experimental version

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FOREWORD

Dear Student,

The Rwanda Basic Education Board (REB) is honored to present the Chemistry student book for Senior three. This textbook is designed to support the effective implementation of the Competence Based Curriculum (CBC), ensuring student-centered teaching practices that align with Rwanda's vision of empowering students to reach their full potential, actively participate in society, and pursue meaningful careers.

As Rwanda advances toward building a knowledge-based economy, education plays a pivotal role. Teachers are central to this transformation, guiding students to connect classroom knowledge with real-life experiences, thus fostering individual growth and national development.

This student's book serves as a comprehensive resource for teaching Chemistry at Senior three. The subject is an integral part of the CBC, aimed at equipping students with essential knowledge, skills, values, and attitudes necessary for becoming responsible and competent citizens. Through this subject, students are encouraged to develop problem-solving abilities and contribute positively to their communities.

In pursuit of quality education, the Government of Rwanda emphasizes the importance of using teaching and learning materials that are aligned with the curriculum.

I sincerely thank everyone who contributed to the refinement of this textbook. Special recognition goes to the REB team who coordinated the entire process. We thank all partners, teachers, and individuals who were involved in the process from start to the end. Every feedback was highly valued and helped to improve learning outcomes of our students.



Dr. MBARUSHIMANA Nelson

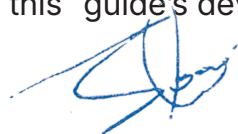
Director General, REB

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I would like to extend my sincere appreciation to everyone who contributed to the refinement of the Chemistry student's book for Senior three. This achievement would not have been possible without the active involvement of various stakeholders.

Special thanks go to Rwanda Basic Education Board (REB) staff, subject experts, University lecturers and teachers for their valuable work in the adaptation of this textbook.

I am also grateful to all our education development partners whose support was instrumental throughout the process. I acknowledge University lectures, Teachers, Partners and all stakeholders for their contributions in editing of this guide's development.



MURUNGI Joan

Head of Curriculum, Teaching and Learning Resources Department/ REB

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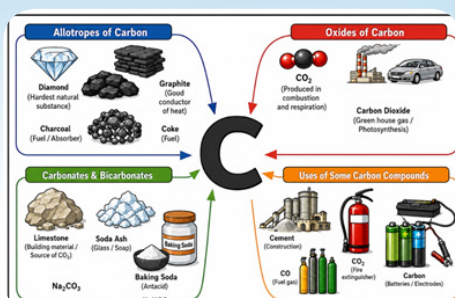
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UNIT 1

CARBON AND ITS INORGANIC COMPOUNDS



Key Unit Competence:

Relate the physical and chemical properties of carbon and its inorganic compounds to their preparation, uses and effects on the environment.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Describe the allotropic forms of carbon based on their properties.
- Explain the properties of carbon and its compounds.
- Explain the impact of carbon compounds on the environment.

Skills

2

- Prepare the inorganic compounds of carbon.
- Test the presence of carbon dioxide.
- Differentiate hard water from soft water based on their properties.
- Interpret the carbon cycle.
- Analyse the role of carbon dioxide in the process of photosynthesis and combustion.

Attitudes and values

3

- Appreciate the importance of natural resources.
- Develop self- confidence during presentations.
- Develop a culture of working together in groups.
- Show concern for environmental conservation by understanding the effects of carbon dioxide emissions on climate change.

1.0. introduction

Carbon is one of the most abundant elements found in the nature. It is found as free element or in many compounds, natural or man-made. This unit deals with the different forms of carbon. It explains its main physical and chemical properties and analyses the behaviour of its most important inorganic derivatives. Common uses of inorganic carbon containing compounds are also addressed. Impact of excess carbon dioxide in the environment is also highlighted.

Introductory activity



Observe and analyse the picture below.



Identify the things in the pictures. Which element the item or part of it, is made of?

The items shown in the figure above are all made from or contain carbon element. Carbon is one of the most important elements. Carbon exists as well in the elemental form as diamond and graphite and in combined forms in many compounds. Coal, oil, natural gas, limestone and other metal carbonates are all compounds of carbon. Charcoal and the positive electrode in dry cells are all composed of carbon.

Allotropic forms of carbon, their physical properties and uses.

1.1. Allotropic forms of carbon, their physical properties and uses.

1.1.1 Definition of allotropy

Learning activity 1.1.1



Use textbooks and/or internet to find the meaning of allotropy and answer the following questions:

1. What is allotropy? Get the meaning from the dictionary, textbooks or the internet.
2. Name two allotropes of carbon that you have identified from this activity.

Atoms in certain elements often may arrange themselves in different patterns. When this occurs, it results in different forms of that element called allotropes. Allotropy is therefore the existence of an element in two or more forms. Elements that exhibit allotropy include carbon, sulphur and phosphorus among others. Allotropes have same chemical properties but different physical properties. The allotropic forms of carbon are diamond and graphite. These are crystalline forms of carbon. But also, carbon exists in non-crystalline forms called amorphous forms among which we find charcoal, soot, black carbon.

1.1.2. Physical properties of allotropic forms of carbon

Learning activity 1.1.2



Using textbooks and from the internet research on Physical properties of allotropic forms of carbon. Use for example this link, <https://flexbooks.ck12.org/cbook/cbse-chemistry-class-10/section/4.2/primary/lesson/allotropic-forms-of-carbon-10378583/>

Describe and explain the physical properties of crystalline and non-crystalline allotropic forms of carbon.

Carbon exists in two main crystalline forms, namely diamond and graphite. The less common crystalline forms of carbon are fullerenes and graphene. Non-crystalline forms of carbon also exist and are called amorphous carbon. These include charcoal, coal, black carbon, and soot.

A. Diamond

Diamond occurs naturally in many countries. Thus, diamond is a natural resource.

i. Structure of diamond

In diamond, each carbon atom is covalently bonded to four other carbon atoms giving a regular tetrahedron shape as shown in figure 1.1 (a) below. This results in a rigid closely interlocked three dimensional structure. The atoms form a giant atomic structure. All the four valence electrons in diamond are used in the bonding, which greatly contributes to the physical properties of diamond.

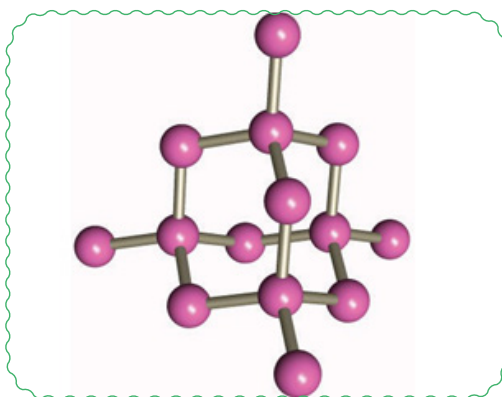


Figure 1.1 (a): Structure of diamond

ii. Physical properties of diamond

- Diamond stands as the hardest naturally occurring substance, a consequence of the exceptionally strong covalent bonds forming a three-dimensional tetrahedral network throughout its carbon atom structure.
- When expertly cut and polished, diamond exhibits a colourless, crystalline transparency with a characteristic brilliant lustre. This property is due to its remarkably high refractive index, which enables superior light dispersion and reflection.
- With a density of approximately 3.5 g/cm^3 , diamond is substantially denser than its allotropic counterpart, graphite. This dense structure contributes to its extraordinary thermal stability, with a melting point of approximately 4200°C . Such extreme thermal resistance stems from the substantial energy required to rupture the extensive network of strong covalent bonds permeating its lattice structure.
- Unlike graphite, diamond functions as an electrical insulator due to the absence of delocalized electrons; all four valence electrons of each carbon atom are localized within covalent bonds, leaving none available for electrical conduction.

B. Graphite

Graphite is a dark grey shiny crystalline solid. It is the most abundant allotrope of carbon. It occurs naturally but a superior form of it is made by heating anthracite (a variety of coal) in an electric furnace.

i. Structure of graphite

In graphite, each carbon atom is covalently bonded to three other atoms while the fourth electron is delocalised: it is not attached to any particular atom but belongs to the entire structure. These delocalised electrons are free to move through the graphite structure making graphite a good conductor of electricity. Graphite exists in hexagonal layers of carbon atoms. The layers are held together by weak Van der Waal's forces of attraction as shown in figure 1.1 (b). This makes them slide easily over each other.

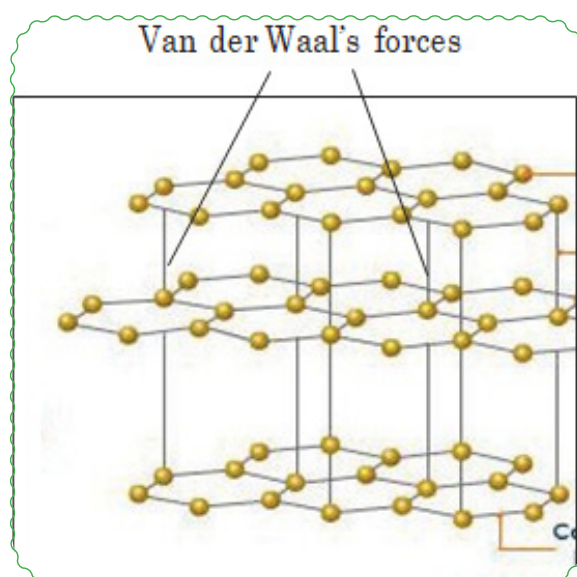


Figure 1.1(b): Structure of graphite

ii. Physical properties of graphite

- Graphite is a soft, black and shiny material with a greasy feel.
- Graphite easily flakes off, i.e detaches and falls away from a surface in small, thin, brittle pieces.
- Has relatively high melting and boiling points due to the strong covalent bonds holding its atoms together.
- It has a density of 2.25g/cm^3 .
- Graphite is a good conductor of electricity due to the presence of delocalised electrons in its structure.
- It is opaque since-parallel layers arranged on top of one another blocks light from penetrating through.

1.1.3. Uses of the allotropic forms of carbon

Learning activity 1.1.3



Using textbooks and from the internet, research the uses of diamond and graphite.

a) Uses of diamond

- Diamond's exceptional physical properties make it useful in many industrial, technological, and decorative applications.
- Diamond is the hardest known material. This makes it essential for manufacturing cutting tools including drill bits for cutting metals and ceramics, glass cutters, and industrial grinding wheels. Diamond can cut materials that would quickly wear down ordinary tools.
- Diamond sparkles brilliantly due to its high refractive index. This property splits light into colours. These qualities make diamond the most prized gemstone for jewellery.

b) Uses of graphite

The properties of graphite determine its uses. Some uses of graphite include the following:

- It is used as electrodes in dry cells and fuel cells. This is because it is a good conductor of electricity.
- Graphite is used as a carbon raiser in the production of steel. It gives steel its strengthening characteristics.
- It is used in advanced high-friction applications such as car brakes and clutches thanks to its high thermal and electrical conductivity.
- When mixed with clay, it is used to make pencil 'leads'. Its layered structure allows it to slide easily on paper when writing.

- At high temperatures, graphite can be used as a lubricant in place of grease or oil due to its high boiling point.
- At high temperatures, graphite can be used as a lubricant in place of grease or oil. Unlike conventional lubricants, graphite remains stable due to its high boiling point. It is used to make furnace linings, brushes for electric motors and generators.

Application activity 1.1



1. What is meant by allotropy?
2. What are the crystalline forms of carbon?
3. Outline three physical properties of each of diamond and graphite.
4. Explain why graphite conducts electricity whereas diamond does not.
5. What is the role of the rod in a dry cell?
6. Explain why diamond is used for drill bits while graphite is used for pencil leads.

<https://unacademy.com/content/jeemain/study-material/chemistry/uses-of-allotropes-and-oxides-of-carbon/>

1.2. Chemical properties of carbon

1.2.1 Reaction of carbon with oxygen

Learning activity 1.2.1



Investigating the reaction of carbon with oxygen.

Materials

Deflagrating spoon, Bunsen burner, gas jar of oxygen, calcium hydroxide solution

Procedure

1. Place a piece of charcoal in a deflagrating spoon and heat it strongly using a Bunsen burner flame until the charcoal glows red-hot.
2. Lower the red-hot charcoal into a gas jar of oxygen. What do you observe?

3. After sometime, pass the product in the jar through a solution of calcium hydroxide and observe what happens.

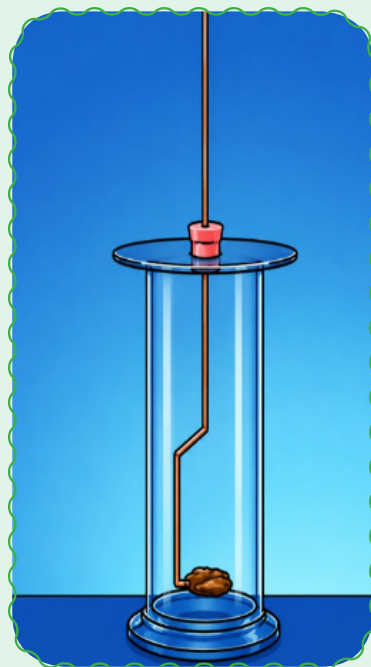
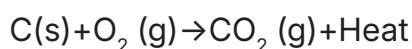


Figure 1.2.1: Burning charcoal in oxygen.

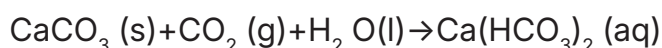
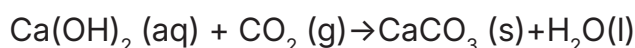
Guiding questions

1. Why did burning stop after sometime?
2. What caused the white precipitate when the gas formed in the jar was passed through the calcium hydroxide solution?

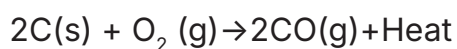
Carbon burns in enough supply of oxygen to form carbon dioxide gas. The reaction is exothermic and hence burning continues even after heating has been stopped.



When carbon dioxide is passed through calcium hydroxide, it reacts to form calcium carbonate and a white precipitate is observed. The white precipitate would disappear if carbon dioxide gas is bubbled in excess due to formation of calcium hydrogen carbonate.



Carbon reacts with limited oxygen supply to form carbon monoxide.



1.2.2 Reaction of carbon with carbon dioxide

Learning activity 1.2.2



Investigating the reaction of carbon and carbon dioxide

Materials: Charcoal, Source of carbon dioxide, Source of heat, Combustion tube

Procedure:

1. Set up the apparatus as shown below:

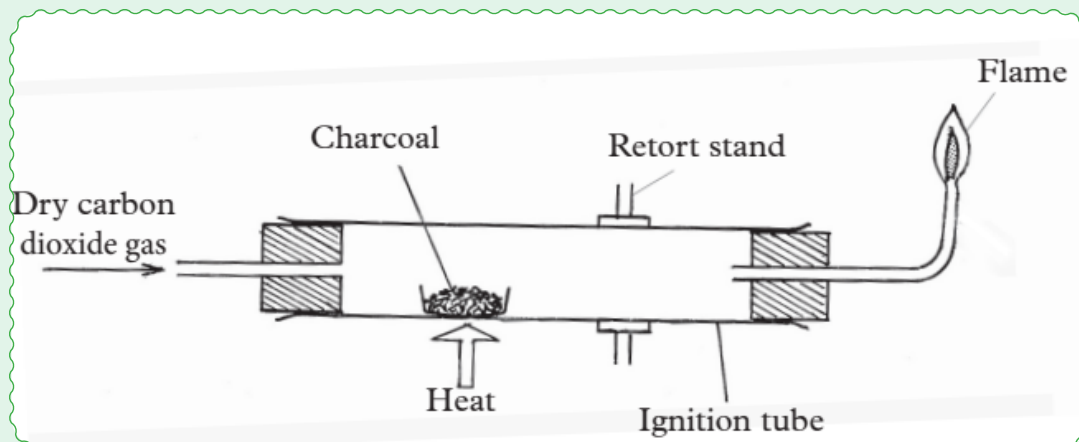


Fig 1.2.2: Experimental set-up for the reaction between charcoal and carbon dioxide

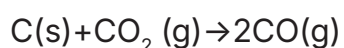
2. Heat the charcoal until it is red-hot.
3. Pass dry carbon dioxide gas through the heated charcoal.
4. Ignite the gas coming out through the other side of the combustion tube as shown above.

Precaution: Avoid inhaling the gas produced. It is poisonous!

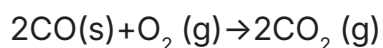
Guiding questions

1. What observations did you make in the ignition tube?
2. Write a balanced chemical equation for the reaction that takes place in the combustion tube.

When carbon dioxide is passed through heated charcoal, it reacts to form carbon monoxide:



The carbon monoxide produced then burns with a blue flame:



The reaction is used to produce carbon monoxide that is used in the extraction of some metals.

1.2.3 Reaction of carbon with iron (III) oxide

Learning activity 1.2.3



Experiment on the reaction between carbon and iron (III) oxide

Materials: Crucible, Iron (III) oxide, Source of heat, Powdered charcoal, Tripod stand, Wire gauze

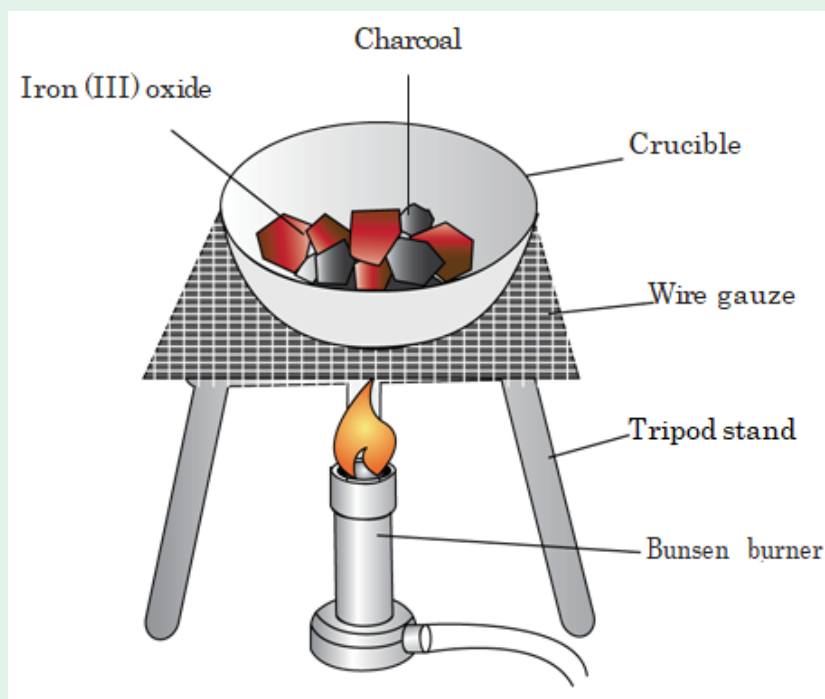


Figure: 1.2.3: Set up for the reaction between carbon and iron (III) oxide

Procedure

1. Mix charcoal powder with iron (III) oxide in a crucible. Note the initial colour of the mixture.
2. Apply heat on the crucible until it is red-hot. Note the colour of the final product.

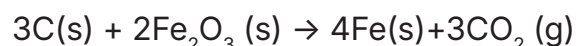
Colour of the mixture at the beginning	Colour of the mixture after heating

Guiding questions

1. Describe the colour change from that of the inside mixture and that of the final product. Give reasons for these changes.
2. Write a chemical equation for the reaction that took place.

When black carbon in charcoal is heated with red-brown iron (III) oxide, grey iron metal and carbon dioxide are produced. This exothermic redox reaction involves carbon reducing iron by removing oxygen atoms while being oxidized to carbon dioxide. The

reaction continues even after heating stops and is commonly used for metal extraction from oxides.



Other examples of metals that be obtained from their oxides in this way include zinc, lead, copper and tin.

1.2.4 Reaction of carbon with concentrated sulphuric acid and nitric acid.

Learning activity 1.2.4



Investigating the reaction of carbon with concentrated sulphuric acid and nitric acid

Materials: Two test tubes, Concentrated sulphuric acid, Cork, delivery tube, Retort stand, Concentrated nitric acid, Powdered charcoal, Spatula, Bunsen burner, Tripod stand, Calcium hydroxide solution (lime water)

Procedure

1. Put a spatula full of charcoal powder in a test tube.
2. Add about 10 cm³ of concentrated sulphuric acid, warm and observe.
3. Repeat the experiment using concentrated nitric acid instead of sulphuric acid.

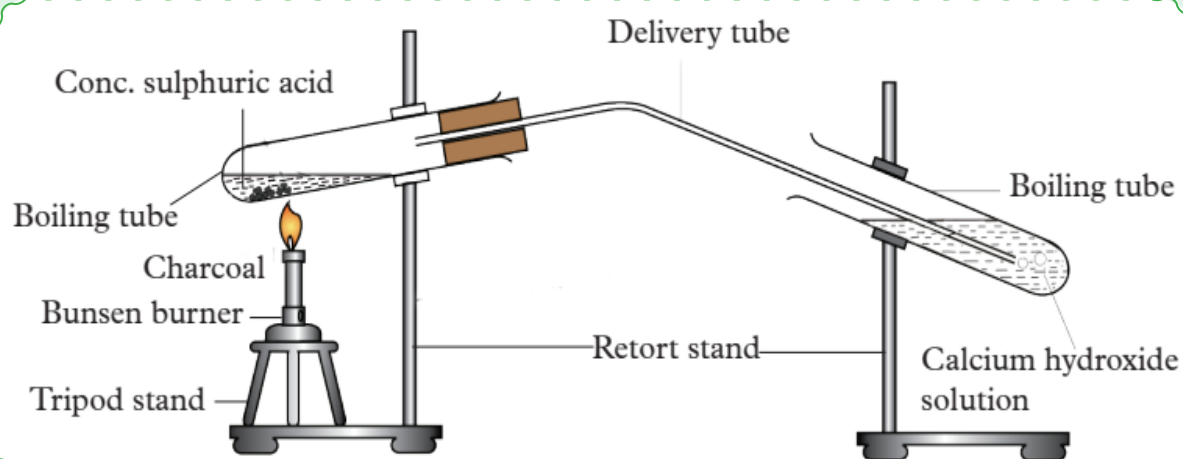


Figure 1.2.4: Reaction of carbon with concentrated acids

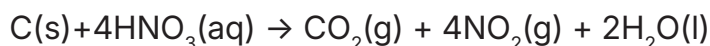
Guiding questions

1. What observations did you make on reacting each acid with carbon?
2. Write balanced chemical equations for the reactions that took place.

Carbon reacts with concentrated sulphuric acid to form carbon dioxide, sulphur dioxide and water.



With concentrated nitric acid, it forms carbon dioxide, nitrogen dioxide and water as shown below.



Note: These acids oxidise carbon to carbon dioxide. It is the carbon dioxide gas that turns lime water milky.

Application activity 1.2



- Carbon reacts with oxygen according to the equations below:
 - $\text{C(s)} + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{Heat}$
 - $2\text{C(s)} + \text{O}_2(\text{g}) \rightarrow 2\text{CO(g)} + \text{Heat}$
 - What condition is required in (a) and (b) above?
 - What does heat in the equations indicate about the two reactions?
- Carbon reacts with carbon dioxide according to the equation below.
$$\text{CO}_2(\text{g}) + \text{C(s)} \rightarrow 2\text{CO(g)}$$
 - Identify the substance that loses oxygen and the one that gains oxygen from the reaction above.
 - Explain the importance of this reaction in daily life.
- Write a balanced chemical equation of the reaction between carbon and iron (III) oxide.
- Name other metal oxides reduced by carbon to their respective metals.
- Write balanced chemical equations for the reactions that take place when:
 - Carbon reacts with concentrated nitric acid.
 - Carbon reacts with concentrated sulphuric acid.

1.3 Inorganic compounds of carbon

Commonly known inorganic compounds of carbon include: oxides of carbon, carbonates and hydrogen carbonates.

1.3.1. Carbon oxides

The two oxides of carbon are carbon dioxide and carbon monoxide.

A. Carbon dioxide

A.1. Preparation methods of carbon dioxide

Learning activity 1.3.1 (a)



Laboratory preparation of carbon dioxide

Materials: Flat-bottomed flask, Dropping funnel, Two wash bottles, Delivery tubes, Gas jars, Cardboard covers, Calcium carbonate (marble chips or limestone), Dilute hydrochloric acid, Distilled water, Concentrated sulphuric acid.

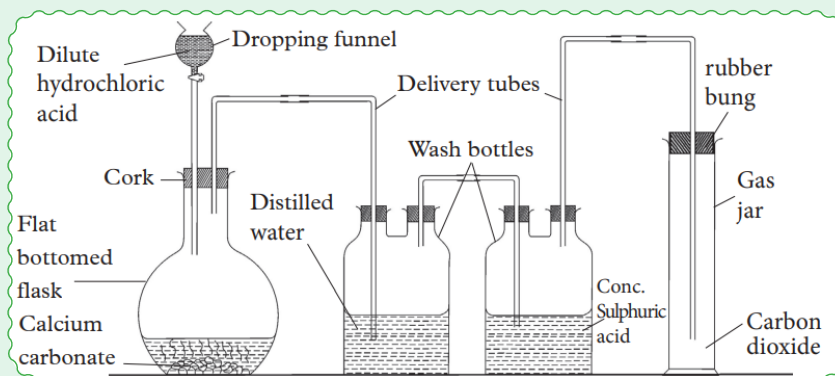


Figure 1.3.1(a): Laboratory preparation of carbon dioxide

Procedure

1. Arrange the apparatus as shown above.
2. To the limestone or marble chips, add dilute hydrochloric acid dropwise.
3. Record your observation.
4. Collect the gas produced by downward delivery method (upward displacement of air) or over warm water.
5. Collect a few jars of the gas and preserve them for the next experiment.

Guiding questions

1. Why is the gas collected by downward delivery method or over warm water?
2. What is the role of water and concentrated sulphuric acid in the wash bottles?
3. What would happen if dilute sulphuric acid is used instead of hydrochloric acid in the above experiment?

Dilute hydrochloric acid reacts with marble chips (calcium carbonate) to produce carbon dioxide gas, calcium chloride, and water.



Carbon dioxide is collected by downward delivery because it is denser than air. Alternatively, it can be collected over warm water since it has low solubility in warm water. The solubility of gases decreases as temperature increases.

The gas first passes through water to remove any spray of hydrochloric acid. It then passes through concentrated sulfuric acid to remove moisture. Anhydrous calcium chloride in a U-tube can also be used as a drying agent.

Although all mineral acids react with metal carbonates to produce carbon dioxide, dilute sulfuric acid cannot be used with calcium carbonate. This is because the reaction produces insoluble calcium sulfate, which forms a coating on the marble chips. This coating prevents further reaction, so the reaction stops before completion.

A.2. Properties of carbon dioxide

Learning activity 1.3.1 (b)



b) Investigating the properties of carbon dioxide

Materials: Nitric acid, Gas-jars full of carbon dioxide, Test-tubes, Magnesium ribbon, A pair of tongs, Candle, Blue litmus paper, Two water troughs, Beehive shelves, Calcium hydroxide solution, Sodium hydroxide solution, Dilute nitric acid, Deflagrating spoon.

Procedure

1. Examine the gas jar full of carbon dioxide. Note down the colour of the gas and its smell.
2. Pass carbon dioxide through a test-tube containing calcium hydroxide solution for a short time as shown below and record your observations.

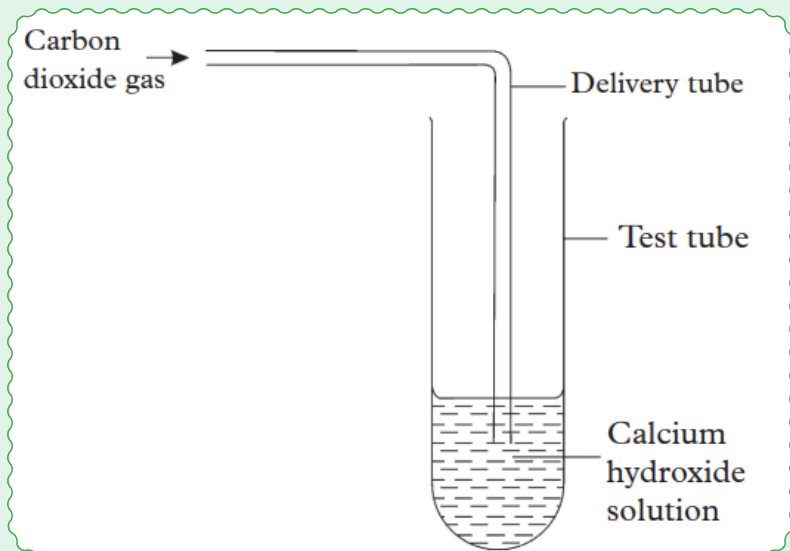


Figure 1.3.1 (b): Testing for carbon dioxide

- Continue passing more of the gas into the same test-tube until there is no further change and record your observations.
- Pass carbon dioxide into a test-tube containing distilled water and then dip a blue litmus paper into the resulting solution. Record your observations.
- Invert a gas-jar full of carbon dioxide over a burning candle as shown below. Note down what happens.

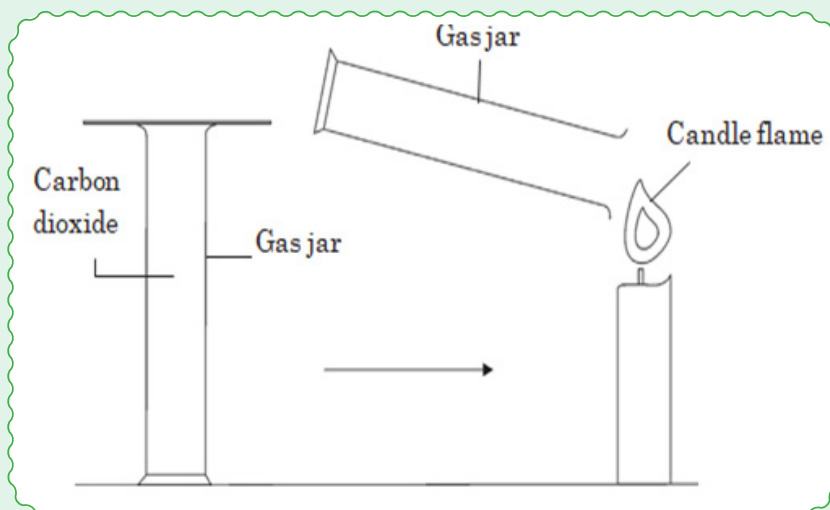


Fig 1.3.1 (c): Testing whether carbon dioxide supports combustion.

- Plunge a piece of burning magnesium ribbon quickly in a gas-jar full of carbon dioxide as shown below and record your observations.

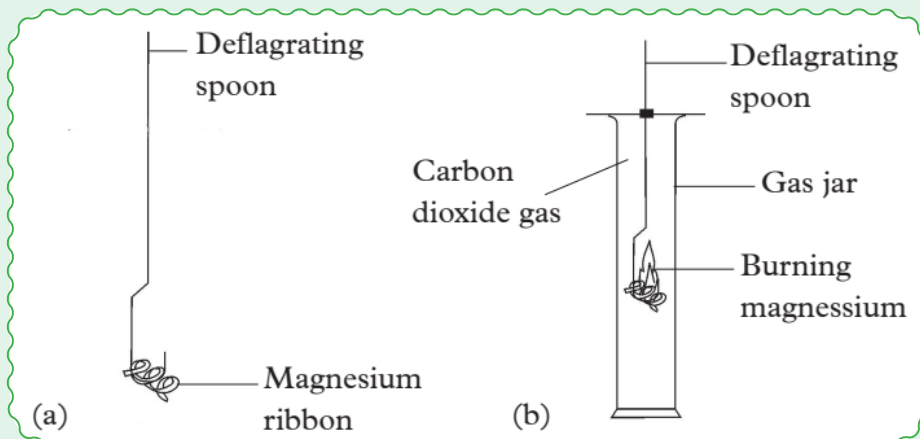


Fig.1.3.1 (d): Testing the effect of carbon dioxide on burning magnesium

- Add about 50 cm³ of dilute nitric acid to the mixture in the gas jar.
 - Note down your observations.
 - Write a chemical equation for the reaction.

8. Invert a gas jar full of carbon dioxide in a trough containing sodium hydroxide solution as shown below. Shake the gas jar and wait for some time.

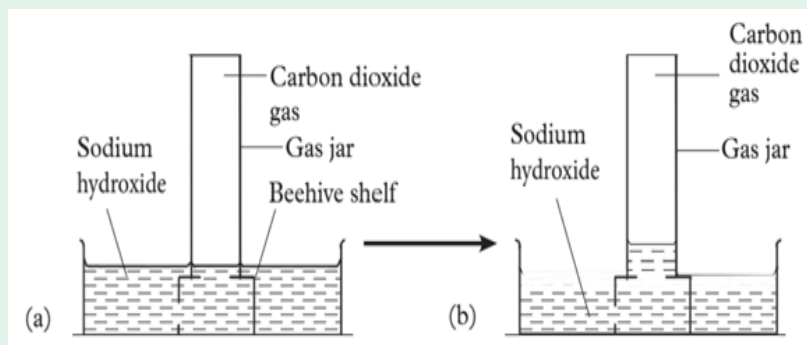


Fig 1.3.1(e): Dissolving carbon dioxide in sodium hydroxide solution

9. Repeat procedure (8) above using water instead of sodium hydroxide.
- Note any water level changes inside the gas jar.
 - What happens to the level of sodium hydroxide solution?
 - Write chemical equations for the reactions that occur.

Guiding questions

1. Write balanced chemical equations for the following:
 - a) Dissolving little amount of carbon dioxide into calcium hydroxide solution.
 - b) Dissolving excess carbon dioxide gas into calcium hydroxide solution.
2. Explain why a solution of carbon dioxide in water turns blue litmus paper red.
3. Write a balanced chemical equation for the reaction between carbon dioxide and water.
4. Carbon dioxide does not support burning but magnesium continues to burn in its presence. Explain.

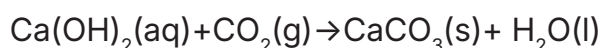
The following are some of the properties of carbon dioxide:

- Physical properties

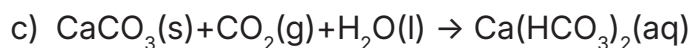
- a) Carbon dioxide is a colourless and odourless gas.
- b) It is denser than air. That is why it can be collected by downward delivery.
- c) It is slightly soluble in water.

- Chemical properties

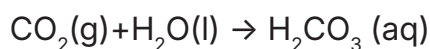
- a) Carbon dioxide reacts with little amounts of calcium hydroxide to form calcium carbonate in form of a white precipitate.



- b) When excess carbon dioxide is passed through the white precipitate, the precipitate dissolves. This is due to the formation of soluble calcium hydrogen carbonate.

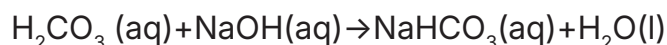


Carbon dioxide moderately reacts with water to form carbonic acid which is characterised by slightly acidic properties.

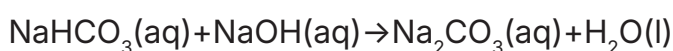


- d) Since carbonic acid is dibasic, i.e. has two replaceable hydrogen atoms, it can form two types of salts:

❖ The first hydrogen replacement produces sodium hydrogen carbonate which is an acidic salt since it has one replaceable hydrogen atom:



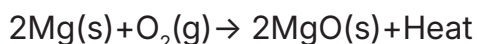
❖ The second hydrogen replacement produces sodium carbonate which is a normal salt since it has no replaceable hydrogen atom:



Note: Potassium hydroxide solution reacts in a similar way but its hydrogen carbonate is more soluble and does not precipitate. That is why potassium hydroxide solution is preferred as an absorbent for carbon dioxide.

- e) Carbon dioxide extinguishes a burning candle showing that it does not support combustion.

- f) Burning magnesium continues to burn in carbon dioxide because magnesium is more reactive than carbon, allowing it to reduce CO_2 to form white magnesium oxide (MgO) and black carbon (C). The intense heat from the burning magnesium splits the CO_2 molecules, providing the oxygen necessary to sustain combustion. The ribbon burns with a bright white light, leaving black, powdery spots (carbon) and white ash (MgO) on the container.



This reaction highlights that CO_2 cannot extinguish magnesium fires, as magnesium steals the oxygen from the CO_2 .

- g) The addition of dilute nitric acid enables the black carbon particles to be seen more clearly because the acid reacts with magnesium oxide forming magnesium nitrate solution and water. Although carbon dioxide does not support combustion, magnesium continues to burn in it. The heat produced by the intense burning of magnesium is sufficient to decompose the carbon dioxide into constituent elements: carbon and oxygen. This oxygen produced supports the continued combustion to magnesium oxide.

A.3. Uses of carbon dioxide

Learning activity 1.3.1 (c)



Using textbooks and from the internet, research on the uses of carbon dioxide then record your findings.

used in baking of cakes and bread among others. It is also an ingredient in some health salts to relieve constipation.



Figure 1.3.1(f) Bakery

Carbon dioxide is used as preservative in the production of mineral water and carbonated drinks.



Figure 1.3.1(g): A girl opening a soda

Solid carbon dioxide commonly known as dry ice is a preferred refrigerant to ordinary ice (solid water). This is because it sublimates at room temperature forming gaseous carbon dioxide and therefore leaves no residue unlike ordinary ice.



Figure 1.3.1(h): Dry ice

Some fire extinguishers contain sodium hydrogen carbonate or sodium carbonate solution and sulphuric acid. When mixed by inversion or pressing the plunger, carbon dioxide is produced due to the reaction between carbonate with sulphuric acid.



A high pressure is set which forces a froth of carbon dioxide through the jet. It is directed onto the burning substances.



Figure 1.3.1 (i): Using a fire extinguisher to put out a fire.

In making rain, dry ice (solid carbon dioxide) is spread in the clouds to accelerate the condensation process. Small aircrafts are sometimes used to spread dry ice in the sky.

Application activity 1.3.1

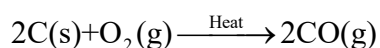


1. Explain why dilute sulphuric acid is not used for the laboratory preparation of carbon dioxide.
2. Write a balanced chemical equation of the reaction between calcium carbonate and dilute sulphuric acid.
3. Explain why carbon dioxide is collected by downward delivery.
4. State two physical properties of carbon dioxide gas.
5. Write equation to show how sodium hydroxide reacts with carbon dioxide.
6. Describe what happens when carbon dioxide is passed through a solution of calcium hydroxide (lime water). Use equations in your explanations.
7. Carbon dioxide was passed through burning magnesium. State what was likely to be observed. Write equation for the reaction.
8. The product in 7 was dissolved in water and a red litmus paper dipped into the solution formed.
 - a) State what was observed.
 - b) Write balanced equations for the reaction.
 - c) Explain how dry ice is a better refrigerant than ordinary ice.
 - d) Use equation to explain how carbon dioxide is used as a fire extinguisher.

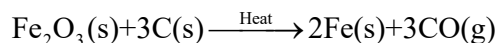
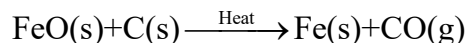
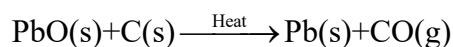
B. Carbon monoxide

B.1.Preparation methods of carbon monoxide

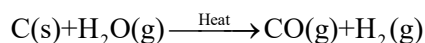
1. This is a colourless and odourless gas. It is slightly denser than air. Carbon monoxide is produced by incomplete combustion of various fuels. These include coal, wood, charcoal, fractions of oil and natural gas.



2. It can also be prepared by reduction of metallic oxide with carbon (coal or coke).



3. By passing steam over red-hot coke.



4. It is a poisonous and very toxic gas. This is because when inhaled, it binds to haemoglobin blocking off oxygen. This can lead to death over prolonged exposure to the gas.

5. Carbon monoxide gas can be prepared in lab by heating formic acid or oxalic acid with concentrated sulfuric acid.

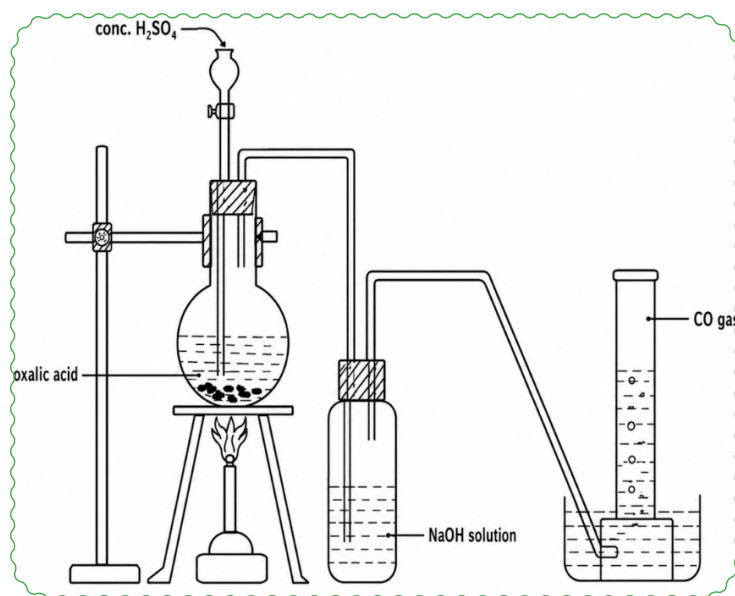
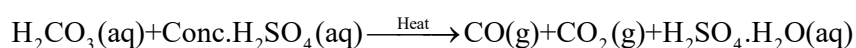
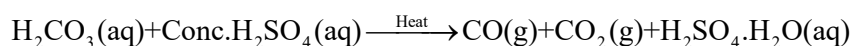
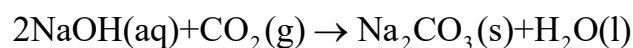


Fig. 1.3.1 (j) Experimental setup for the preparation of carbon monoxide from oxalic acid

Procedure: Small amount of oxalic acid crystal is taken in a round-bottomed flask and all the apparatus are fitted as shown in figure. When concentrated sulfuric acid is poured in the flask through the thistle funnel and heated, then a mixture of carbon monoxide and carbon dioxide gas is formed. The mixture of gas is passed through a bottle containing sodium hydroxide solution where the carbon dioxide gas is absorbed by sodium hydroxide.



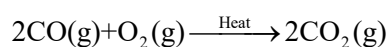
The pure carbon monoxide gas is then collected in a gas jar by downward displacement of water.

B.2. Physical properties of carbon monoxide

1. It is colourless and poisonous gas (with faint smell).
2. It is lighter than air.
3. It is slightly soluble in water.

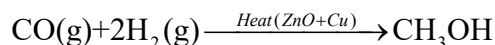
B.3. Chemical properties of carbon monoxide

1. It is neutral to litmus.
2. It is combustible but not a supporter of combustion. It burns in air to form carbon dioxide.

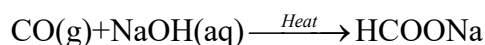


3. Formation of addition products:

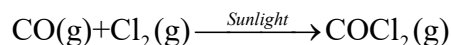
- a) When carbon monoxide gas is heated with hydrogen in presence of Zinc oxide and Copper at then methanol is formed.



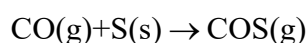
- b) When carbon monoxide is heated with sodium hydroxide then sodium formate is formed.



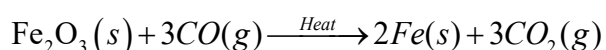
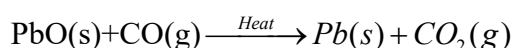
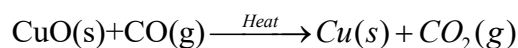
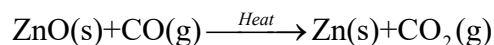
- c) Carbon monoxide reacts with chlorine in presence of sunlight to produce phosgene (carbonyl chloride) which is a poisonous gas.



- d) When a mixture of carbon monoxide and sulphur vapour is heated, then carbonyl sulphide is formed.



- e) Carbon monoxide acts as reducing agent. It is mainly used to reduce metallic oxide to metal in metallurgical process.



B.4. Uses of carbon monoxide

1. Carbon monoxide acts as a reducing agent in blast furnaces. It removes oxygen from iron ore (Fe_2O_3) to produce molten iron, which is essential for making steel.
2. Carbon monoxide is used in modified atmosphere packaging to keep meat looking fresh and red. It stabilizes the red colour in meat by reacting with myoglobin.

3. Carbon monoxide is used to test how efficiently oxygen transfers from the lungs into the bloodstream. Scientists are also studying controlled amounts of CO to reduce inflammation and improve organ transplant outcomes. Carbon monoxide can be used as a fuel in high-temperature fuel cells. These fuel cells produce electricity efficiently with minimal pollution.
4. Carbon monoxide levels in the air are measured to assess air quality and pollution. High carbon monoxide levels indicate incomplete combustion and poor air quality.

Safety Warning: Carbon monoxide is highly toxic. It binds to haemoglobin in blood, preventing oxygen transport. This can cause headaches, dizziness, and even death. Proper ventilation and CO detectors are essential when working with this gas.

1.3.2 Carbonates and hydrogen carbonates

A. Carbonates

A carbonate is a salt formed when all the hydrogen atoms in carbonic acid are replaced by a metal or an ammonium ion. Carbonic acid is formed when carbon dioxide gas dissolves in water. Examples of metallic carbonates include potassium carbonate, sodium carbonate, copper carbonate, zinc carbonate and magnesium carbonate.

On the other hand, a hydrogen carbonate (also called bicarbonate) is formed when the hydrogen atoms in carbonic acids are partially replaced by a metal. Some examples of metallic hydrogen carbonates are sodium hydrogen carbonate and potassium hydrogen carbonate.

i. Preparation of metal carbonates

Learning activity 1.3.2. a



Preparation of carbonates

Materials:

- | | | |
|---------------------|-----------------|----------------------|
| • Calcium chloride | • Filter paper | • Stirring rod |
| • Lead (II) nitrate | • Conical flask | • Spatula |
| • Sodium carbonate | • Beakers | • Measuring cylinder |
| • Filter funnel | | |

Procedure

1. Place 5 g of calcium chloride in a beaker and add to it 20 cm³ of water. Stir until all the solid dissolves.
2. Put 5 g of sodium carbonate in another beaker and add to it 20 cm³ of distilled water, stir the mixture to form a uniform solution.
3. Add the sodium carbonate solution to the calcium chloride solution and stir the mixture. Allow to settle. Note down your observations.
4. Filter off the precipitate obtained and wash it with distilled water.
5. Dry the precipitate between filter papers.

6. Repeat the procedure steps 1-5 using lead (II) nitrate in place of calcium chloride.

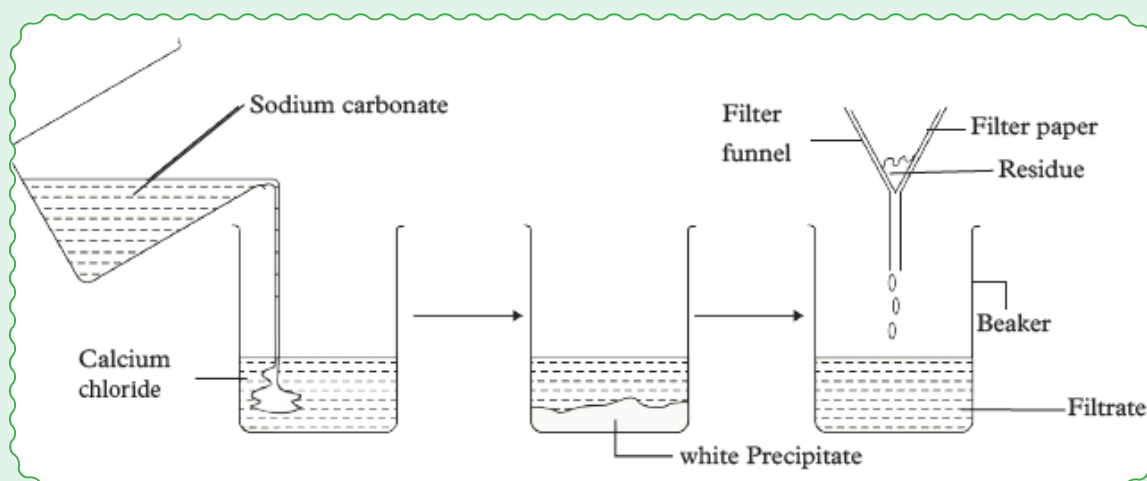
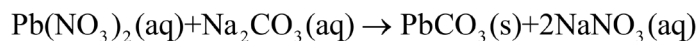
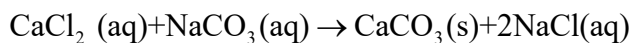


Figure 1.3.2(a): Preparation of calcium carbonate

Guiding questions:

1. What observation did you make when sodium carbonate solution was added to:
 - a) Calcium chloride solution?
 - b) Lead (II) nitrate solution?
2. Write balanced chemical equations for the reactions that occurred during the experiments.
3. Which other solution can be used in place of sodium carbonate solution?

Sodium carbonate reacts with calcium chloride to form sodium chloride solution and insoluble calcium carbonate (white precipitate). Similarly, sodium carbonate reacts with lead (II) nitrate solution to form sodium nitrate solution and an insoluble lead (II) carbonate (white precipitate). The equations for the reactions are as follows:



Filtering is done to separate the solid carbonate from the rest of the solution. The residue is then dried between filter papers or placed outside in the sun. Other carbonates such as magnesium carbonate and zinc carbonate can be prepared using the same method.

B. Hydrogen carbonates

i. Preparation

Learning activity 1.3.2 b



Preparation of hydrogen carbonates

Materials required:

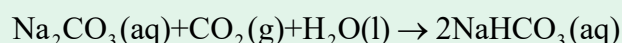
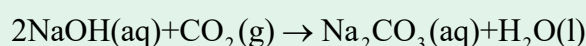
Conical flask, carbon dioxide generator, delivery tubes, cold concentrated solution of sodium hydroxide, filter funnel and filter papers

Procedure

1. Bubble carbon dioxide in excess through a cold concentrated solution of sodium hydroxide for 10 minutes.
2. Filter the resulting mixture.
3. Leave the residue undisturbed for some time.
4. Collect the crystals formed.

Guiding questions:

1. What do you observe when carbon dioxide is bubbled through sodium hydroxide solution for 10 minutes?
2. Write the equation for the reaction that occurs.
3. In the laboratory, sodium hydrogen carbonate is prepared by passing carbon dioxide through cold concentrated solution of either sodium hydroxide or sodium carbonate. The equations for the reactions that take place are as follows:



ii. Properties

(A) Action of heat on carbonates and hydrogen carbonates

Learning activity 1.3.2.c



Experiment to illustrate the action of heat on carbonates and hydrogen carbonates

Materials required:

- | | | |
|------------------------------|------------------------------|------------------------------|
| • Test-tubes | • Red and blue litmus papers | • Potassium carbonate |
| • Spatula | • Copper (II) carbonate | • Sodium hydrogen carbonate |
| • Source of heat | • Magnesium carbonate | • Calcium hydrogen carbonate |
| • Calcium hydroxide solution | • Zinc carbonate | • Ammonium carbonate |
| • Distilled water | • Lead (II) carbonate | |
| | • Sodium carbonate | |

Procedure:

1. Place a spatula full of the carbonate or hydrogen carbonate salts above in different test-tubes then heat.
2. Test any gas formed with calcium hydroxide solution and moist red and blue litmus papers.

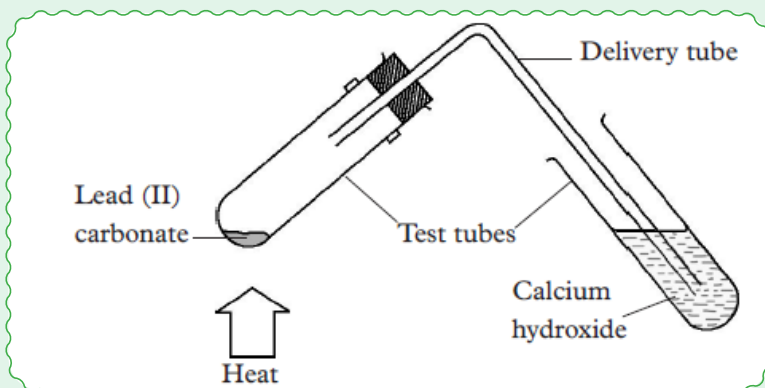


Figure 1.3.2. (c): Heating carbonates and hydrogen carbonates

Note down the colour of the residue when hot and cold for each residue.

1. Record all your observations in a table like the one shown below.

Table 1.3.2(a): Results for action of heat on carbonates and hydrogen carbonates

Name of carbonate/ hydrogen carbonate	Colour before heat- ing	Colour after heat- ing	Colour after cool- ing	Effect of gas evolved on cal- cium hydroxide solution	Effect of gas evolved on moist red and blue litmus papers
Sodium carbonate					
Potassium carbonate					
Magnesium carbonate					
Zinc carbonate					
Lead carbonate					
Copper carbonate					
Ammonium carbonate					
Sodium hydrogen car- bonate					
Calcium hydrogen car- bonate					

Guiding questions

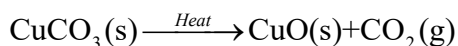
1. Write balanced chemical equations for all the reactions that took place.
2. Identify carbonates that behaved in a similar way.
3. Which carbonate evolved a gas that turned red litmus paper blue?
4. Which carbonate produced a residue that showed colour changes?
5. Which carbonates were not affected by heat?

Sodium carbonate and potassium carbonate were not affected by heat. However, when the carbonates of magnesium, zinc, lead, copper as well as sodium hydrogen carbonate and calcium hydrogen carbonate were heated they produced a gas that formed a white precipitate with calcium hydroxide solution. The gas also turned moist blue litmus paper red. Ammonium carbonate produced two gases, one turned blue litmus paper red while the other turned red litmus paper blue.

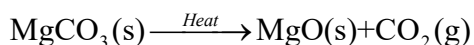
Some carbonates decompose on heating while others do not. Carbonates of magnesium, zinc, lead and copper decompose on heating to produce respective metal oxide and carbon dioxide gas. Carbon dioxide gas produced forms a white precipitate with calcium hydroxide solution. The gas also turns moist blue litmus paper red as it is acidic.

The metal oxides formed are of different colours as shown in the following:

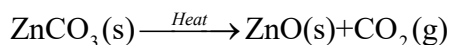
1. Green copper (II) carbonate decomposes on heating to form black copper (II) oxide and carbon dioxide gas.



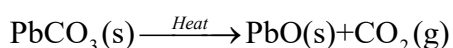
2. White magnesium carbonate decomposes on heating to form white magnesium oxide and carbon dioxide gas.



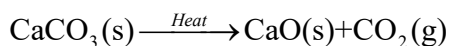
3. White zinc carbonate decomposes on heating to form zinc oxide and carbon dioxide gas. The zinc oxide is yellow when hot but turns white when cooled.



4. White lead carbonate (when hot) decomposes on heating to form lead oxide and carbon dioxide gas. The lead oxide is red-brown when hot but turns yellow when cooled.

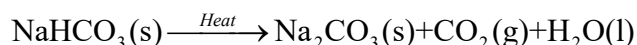


5. White calcium carbonate decomposes on heating to form white calcium oxide and carbon dioxide gas.



6. The carbonates of sodium and potassium are not affected by heat. But sodium hydrogen carbonate and calcium hydrogen carbonate decompose on heating to form a carbonate, water and carbon dioxide.

7. Sodium hydrogen carbonate decomposes on heating to form sodium carbonate, carbon dioxide gas, and water.

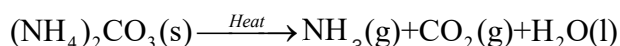


8. Calcium hydrogen carbonate decomposes on heating to form calcium carbonate, carbon dioxide gas, and water.



Note: When calcium hydrogen carbonate is strongly heated, the calcium carbonate formed further decomposes to form calcium oxide and carbon dioxide gas.

Ammonium carbonate decomposes on heating to form ammonia gas, carbon dioxide gas, and water.



Ammonia and carbon dioxide gases are liberated at the same time. If moist red and blue litmus papers are put together at the mouth of the test-tube when ammonium carbonate is heated, ammonia gas is detected first because red litmus paper turns blue.

(B) Action of dilute acids on carbonates and hydrogen carbonates

Learning activity 1.3.2.d



Experiment: Investigating the action of dilute acids on carbonates and hydrogen carbonates.

Materials required

- | | |
|------------------------------|------------------------------|
| • Test tubes in a rack | • Sodium carbonate |
| • Spatula | • Ammonium carbonate |
| • Calcium hydroxide solution | • Sodium hydrogen carbonate |
| • Calcium carbonate | • Calcium hydrogen carbonate |
| • Magnesium carbonate | • Dilute nitric acid |
| • Zinc carbonate | • Dilute sulphuric acid |
| • Lead (II) carbonate | • Dilute hydrochloric acid |
| • Copper carbonate | |

Procedure

1. In different test-tubes, put a spatula ful of the above carbonates and hydrogen carbonates and label them accordingly.

2. Add dilute hydrochloric acid into each test tube and test the gas evolved with calcium hydroxide solution.

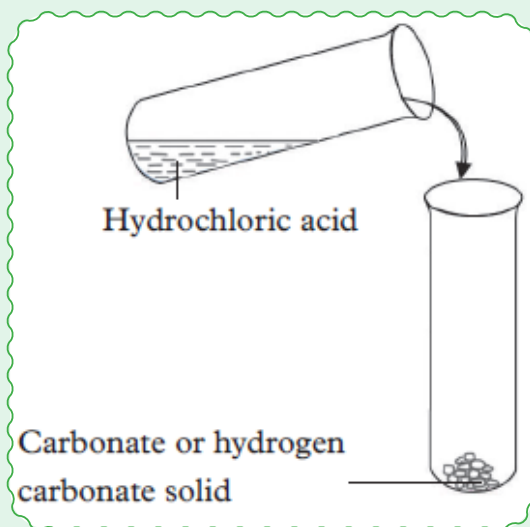


Fig.1.3.2 (d): Reaction of carbonates or hydrogen carbonates with acid

3. Note down the observation made in test tubes A and B.
4. Repeat the experiment using sulphuric acid and then dilute nitric acid on all the carbonates and hydrogen carbonates.
5. Record your observations in both test tubes A (reaction of between the acid and the carbonate/hydrogen carbonate) and B (reaction between the gas produced and the calcium hydroxide solution).
6. Record your observations in a table shown below.

Table 1.3.2 (a) Reaction of carbonate or hydrogen carbonates with acid

Carbonate / hydrogen Carbonate	Acid	Observation in test tube A	Observation in test tube B
Sodium hydrogen carbonate	Hydrochloric acid		
	Sulphuric acid		
	Nitric acid		
Sodium carbonate	Hydrochloric acid		
	Sulphuric acid		
	Nitric acid		
Calcium hydrogen carbonate	Hydrochloric acid		
	Sulphuric acid		
	Nitric acid		

Carbonate / hydrogen Carbonate	Acid	Observation in test tube A	Observation in test tube B
Copper (II) carbonate	Hydrochloric acid		
	Sulphuric acid		
	Nitric acid		
Magnesium carbonate	Hydrochloric acid		
	Sulphuric acid		
	Nitric acid		
Zinc carbonate	Hydrochloric acid		
	Sulphuric acid		
	Nitric acid		
Ammonium carbonate	Hydrochloric acid		
	Sulphuric acid		
	Nitric acid		
Lead (II) carbonate	Hydrochloric acid		
	Sulphuric acid		
	Nitric acid		

Guiding questions:

1. Explain the observations made when dilute hydrochloric acid and sulphuric acid were each added to lead (II) carbonate.
2. What happened when dilute sulphuric acid was added to calcium carbonate solution?
3. Write balanced chemical equations for all the reactions that took place.

Metal carbonates react with dilute mineral acids to produce a salt, water and carbon dioxide gas. The reaction between dilute sulphuric acid and calcium carbonate does not proceed to completion. This is due to formation of an insoluble coat (salt) of calcium sulphate that forms the impervious layer preventing any further reaction. Similarly, the reaction between lead (II) carbonate and dilute sulphuric acid as well as hydrochloric acid does not proceed to completion. This is due to formation of insoluble salts of lead (II) sulphate and lead (II) chloride respectively.

(C) Solubility of carbonates and hydrogen carbonates in water.

Learning activity 1.3.2.e



Investigating the solubility of carbonates and hydrogen carbonates in water

Materials required

- Test tubes
- Distilled water
- Spatula
- Sodium carbonate
- Potassium carbonate
- Ammonium carbonate
- Sodium hydrogen carbonate
- Calcium hydrogen carbonate
- Calcium carbonate
- Zinc carbonate
- Lead (II) carbonate
- Magnesium carbonate

Procedure

1. Place 0.5g of sodium carbonate in a test tube and add 5cm³ of distilled water.
2. Label the test tube and shake its contents thoroughly. Record your observations.
3. Repeat the experiment using the other carbonates and hydrogen carbonates provided and record your results in a table like the one shown below.

Table 1.3.2 (b): Solubility of carbonates and hydrogen carbonates in water

Carbonate/ hydrogen carbonate	Solubility in water (Use ✓ for soluble or × for insoluble salt)
Sodium carbonate	
Potassium carbonate	
Ammonium carbonate	
Sodium carbonate	
Sodium hydrogen Carbonate	
Calcium hydrogen Carbonate	
Calcium carbonate	
Zinc carbonate	
Lead carbonate	
Magnesium carbonate	

Guiding questions

1. Which carbonates are:
 - a) Soluble in water?
 - b) Insoluble in water?
2. Which hydrogen carbonates are:
 - a) Soluble in water?
 - b) Insoluble in water?

- All carbonates are insoluble in water except ammonium carbonate, sodium carbonate and potassium carbonate.
- All hydrogen carbonates are soluble in water.

iii. Uses

Learning activity 1.3.2.f



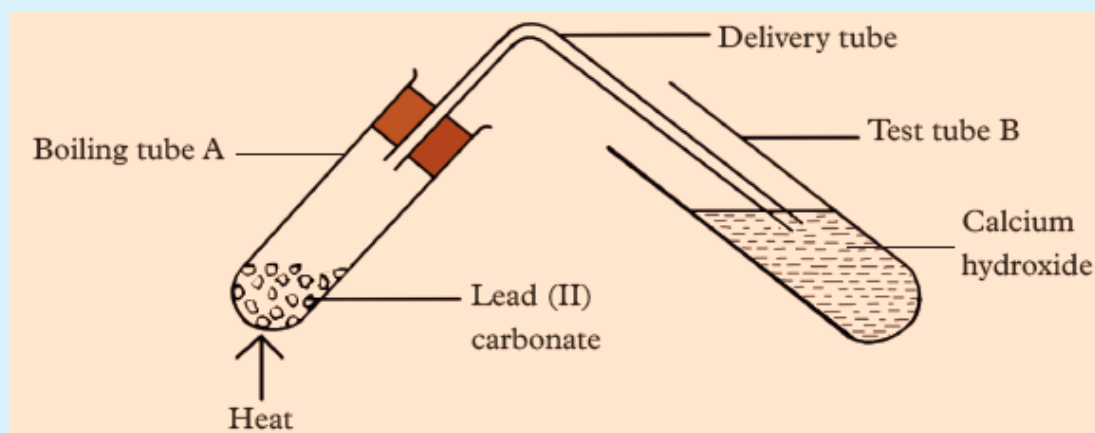
Carry out a research and do a presentation on the uses of carbon compounds. Sodium carbonate has a number of uses but its most common use is in the production of glass.

Uses	Description
Glass making	A mixture of sodium carbonate, calcium carbonate and silicon dioxide is used for making window or bottle glass.
Water softening agent	Carbonate ion from dissolved sodium carbonate can precipitate magnesium and calcium ions from hard water. The reaction leads to the formation of insoluble carbonates, removing them from water hence the water becomes soft. For this reason, sodium carbonate is also known as washing soda.
Baking soda Production	Baking soda or sodium bicarbonate (NaHCO_3) is used in food preparation and in fire extinguishers.
Wool processing	Sodium carbonate removes grease from wool and neutralises acidic solutions.

Application activity 1.3.2



1. Describe a method for the preparation of zinc carbonate solid and write a chemical equation of the reaction that takes place.
2. What happens to solid calcium carbonate when carbon dioxide is passed through water containing it?
3. 2 cm^3 of dilute nitric acid was added to 0.5 g of zinc carbonate in a test tube. The gas produced was then dissolved in calcium hydroxide solution.
 - a) State the observations made.
 - b) Write stoichiometric equations for the reactions that occur.
 - c) Draw a setup of the apparatus that can be used to perform this experiment.
4. The following set up was used to investigate the effect of heat on lead (II) carbonate.



- a) State the observations made in boiling tube A.
 - b) What observation is made in test tube B?
5. Write an equation for the reaction that occurs in test tube:
 - i. A
 - ii. B
 6. Name two carbonates that are:
 - a) Soluble in water
 - b) Insoluble in water

7. A senior 3 student separately heated three solids A, B and C suspected to be carbonates. He tested the gases evolved (if any) with moist blue and red litmus papers. The following are the observations he made.

Carbonate	Observation
A	• Blue litmus paper turned red • No effect on red litmus paper
B	• Red litmus paper turned blue • Blue litmus paper turned red
C	No effect on litmus papers

Suggest the possible identities of carbonates A and C.

- Explain the observations made with carbonate B.
 - A solution reacts with crushed egg shells to give a gas that turns lime water milky.
 - Name the gas that is evolved and give a brief explanation for this observation.
8. Describe a chemical test that you could carry out to confirm the gas evolved in 7(a) above.
9. Name the gas evolved when dilute hydrochloric acid reacts with sodium hydrogen carbonate. How is it recognised?
10. Copy and complete the chemical equations below.
- $\text{H}_2\text{SO}_4(\text{aq}) + \text{Na}_2\text{CO}_3(\text{aq}) \rightarrow$
 - $2\text{HCl}(\text{aq}) + \text{Na}_2\text{CO}_3(\text{aq}) \rightarrow$
 - $2\text{HNO}_3(\text{aq}) + \text{Na}_2\text{CO}_3(\text{aq}) \rightarrow$
 - $\text{H}_2\text{SO}_4(\text{aq}) + \text{CaCO}_3(\text{s}) \rightarrow$
 - $\text{HCl}(\text{aq}) + \text{NaHCO}_3(\text{aq}) \rightarrow$
11. Describe what would happen if the following carbonates were strongly heated: zinc carbonate, copper (II) carbonate and lead (II) carbonate. Write equations for the reactions that would take place.

1.4. Environmental issue of carbon monoxide and carbon dioxide

1.4.1 Environmental issues of carbon dioxide

Application activity 1.4



Illustrating carbon dioxide issues on the environment.

Materials required:

A large glass beaker or glass box, two small plastic beakers full of water, thermometer

Procedure

1. Cover one of the small plastic beakers full of water with the big glass beaker or glass box as shown in Figure 1.4.1. below.

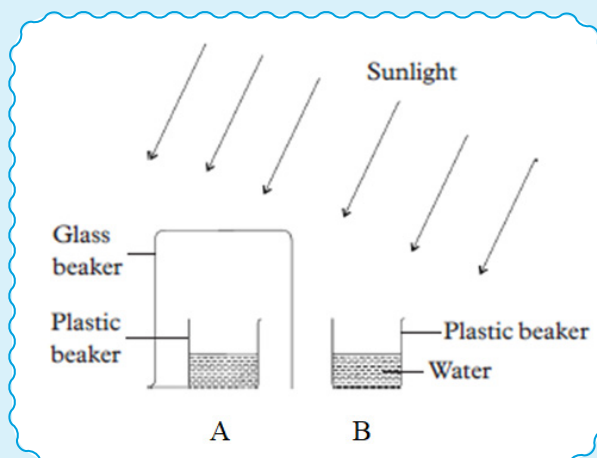


Figure 1.4.1: Simple illustration of the greenhouse effect

2. Leave the other small plastic beaker with water in the open air as shown in figure 1.4.1
3. Put the two beakers under direct sunlight.
4. At the end of the day, measure the temperature of the two beakers.

Guiding question

In which case did you record a higher temperature? Why?

Certain gases in the atmosphere trap heat and keep Earth's surface warm. These gases are called "greenhouse gases." Examples include carbon dioxide, methane, nitrous oxide, and chlorofluorocarbons. Water vapor is also an important greenhouse gas.

This natural warming process is called the "greenhouse effect", and it is essential for life on Earth. Without it, our planet would be too cold to support life. However, when the amount of carbon dioxide and other greenhouse gases in the atmosphere increases, mainly because of human activities, the atmosphere traps more heat. This causes Earth's average temperature to rise, a phenomenon known as global warming.

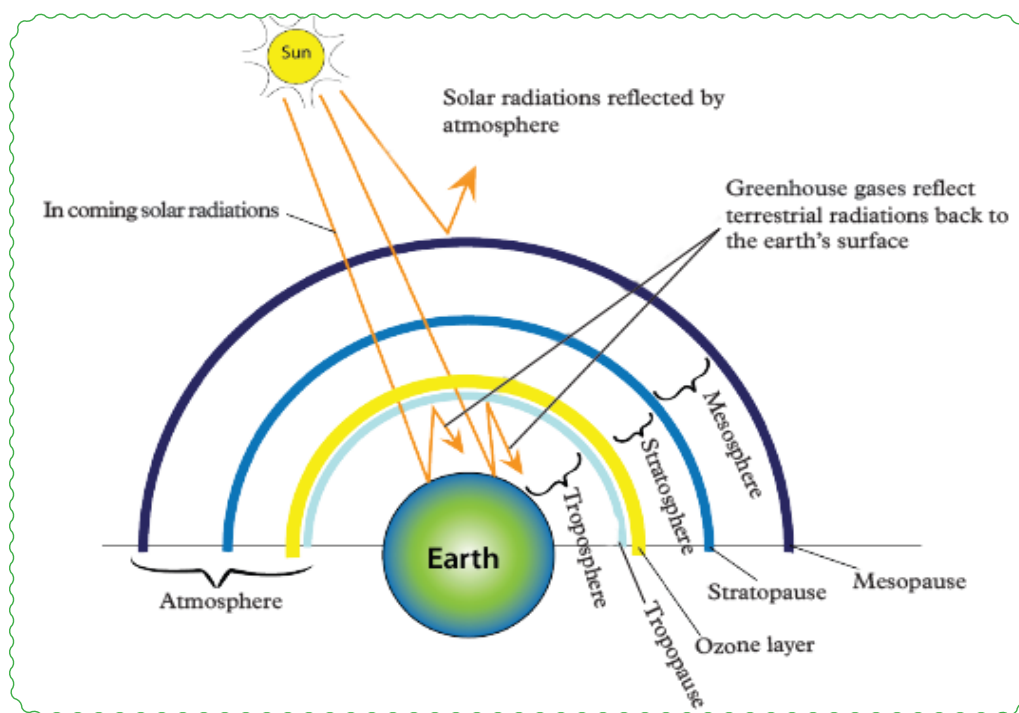


Figure 1.4.1: Greenhouse effect

Human activities such as burning of fossil fuels and clearing forest land for agriculture and settlement have been blamed for the increase in amounts of carbon dioxide in the atmosphere.

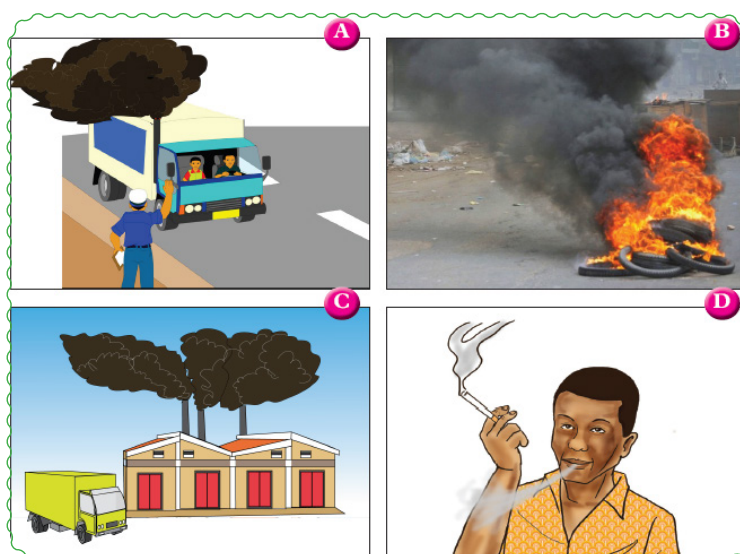


Fig 1.4.2: Human activities contribute significantly to the increase carbon dioxide amount in the atmosphere.

1.4.2 Environmental issues of carbon monoxide

Carbon monoxide forms when fuels such as charcoal, coal, natural gas, or firewood undergo incomplete combustion in poorly ventilated spaces. Carbon monoxide is a colorless, odorless gas that is highly toxic. It binds to hemoglobin (the protein in red blood cells that carries oxygen) more strongly than oxygen does, forming carboxyhemoglobin. This prevents hemoglobin from carrying oxygen to body tissues and cells, leading to oxygen starvation and potentially death. If detected early, carbon

monoxide poisoning can be treated by administering pure oxygen. Severe cases may require a blood transfusion.

Application activity 1.4



1. Carbon monoxide is a poisonous gas. Explain.
2. What is global warming?
3. Name 3 greenhouse gases.
4. Which of the following is not a major contributor to the greenhouse effect?
 - a) Carbon dioxide
 - b) Carbon monoxide
 - c) Chlorofluorocarbons
 - d) Methane gas
 - e) Nitrous oxide
5. Which of the following is not likely a result of global warming?
 - a) Rising sea levels.
 - b) Increased agricultural productivity worldwide.
 - c) Increased storm frequency and intensity.
 - d) Increased coastal flooding.
6. Which gas exists in the highest concentrations in the Earth's atmosphere?

1.5 Hard water and soft water

Learning activity 1.5.1



(a): Investigating the properties of hard water

Materials required:

- Distilled water
- Rain water
- Water from a borehole
- Bar soap
- Three basins
- Three pieces of dirty clothes

Procedure:

1. Put the various kinds of water from each of the three sources in different basins.
2. Using soap, wash the dirty clothes separately in each basin.
3. Note down the water that forms lather easily.

Guiding question

In which water did soap lather easily?

Soap lathers easily in distilled water and rain water than in water from a borehole. Hard water is water that contains salts of calcium and magnesium principally as

bicarbonates, chlorides and sulphates. Hard water does not readily form lather with soap. Soft water on the other hand readily lathers with soap. Distilled water and rain water are examples of soft water while water from borehole is an example of hard water. Water hardness is caused by dissolved calcium and magnesium salts. Hard water therefore contains calcium ions and /or magnesium ions. Soft water on the other hand lacks these ions in solution.

1.5.1 Temporary and permanent water hardness

Learning activity 1.5.1



Investigating the properties of hard water

Using textbooks, journals from the library and from internet, research on the causes, advantages and disadvantages of temporary and permanent water hardness.

Water hardness can either be temporary or permanent. Temporary water hardness is caused by dissolved calcium hydrogen carbonate and magnesium hydrogen carbonate. Permanent water hardness on the other hand is caused by dissolved calcium sulphate and magnesium sulphate.

Table 1.5 Advantages and disadvantages of hard water

Advantages	Disadvantages
The calcium and magnesium ions present in hard water are important for strong development of teeth and bones, and may support cardiovascular health.	Wastes soap as it does not lather easily, requiring more soap for cleaning tasks and increasing household costs.
Helps in formation of an insoluble protective coating inside lead pipes. This prevents lead ions from dissolving in the water, reducing the risk of lead poisoning.	Does not lather easily, leading to formation of scum that leaves clothes with unpleasant marks, stiffness, and grey residue after washing.
Good for formation of animal shells. For example, snails, mussels, and crustaceans obtain calcium from hard water for their shells and skeletal structures.	Forms limescale (boiler scale) inside electric kettles, water heaters, and hot water pipes, requiring significantly more heat energy and increasing electricity costs.
Calcium and magnesium ions are important nutrients for healthy plant growth. Hard water naturally provides these essential minerals to vegetation.	Deposits of calcium and magnesium carbonates reduce the internal capacity of pipes and boilers, decreasing efficiency and shortening the lifespan of appliances.
Many people find hard water tastes better than soft water due to its mineral content, which can enhance the flavor of drinking water, tea, and coffee.	Can cause dry, itchy skin and dull, brittle hair by leaving mineral deposits on the body and preventing proper rinsing of soap and shampoo.

Contains beneficial minerals that can contribute to daily dietary requirements for calcium and magnesium, supporting overall health and nutrition.

Leaves white, chalky spots and film on dishes, glassware, shower doors, sinks, and bathroom fixtures, requiring frequent cleaning and maintenance.

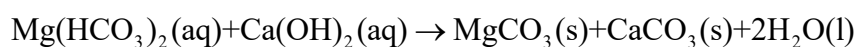
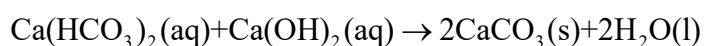
Note:

1. Temporary water hardness can be removed using either of the following ways:

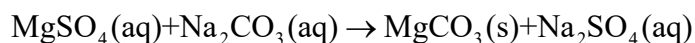
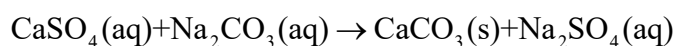
a) Boiling: When water with temporary hardness is boiled, the dissolved calcium hydrogen carbonate and magnesium hydrogen carbonate break down. They form calcium carbonate and magnesium carbonate, which are insoluble and appear as white solid deposits (scale). Carbon dioxide gas and water are also produced.



b) Distillation: Distillation removes all dissolved solids from water. Alternatively, adding calcium hydroxide (lime) to hard water causes calcium carbonate and magnesium carbonate to precipitate out as solids, which can then be filtered out.



2. Permanent water hardness can be removed by addition of washing soda (sodium carbonate). Calcium carbonate and magnesium carbonate are precipitated and therefore removed from the solution.



Application activity 1.5



1. Differentiate between hard water and soft water.
2. Write balanced chemical equations to show how water hardness is removed using the following methods:
(a) Boiling (b) Use of washing soda
3. Hard water is preferred for drinking while soft water is preferred for washing clothes. Explain.
4. The type of water hardness that can be removed by boiling is called
5. Water which forms scum with soap is called

1.6 The carbon cycle

Learning activity 1.6



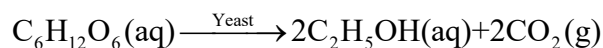
1. Why do plants take in carbon dioxide and release oxygen during the day.
2. Why do animals take in oxygen and release carbon dioxide as a waste product.
3. When we burn wastes, what is produced?
4. What happens when you burn fossil fuels like coal, petroleum and diesel?

Carbon continuously circulates between the atmosphere, living organisms, and the environment through various natural processes. Some processes release carbon dioxide into the atmosphere, while others remove it. This circulation of carbon through nature is called the carbon cycle.

The amount of carbon dioxide in the atmosphere remains relatively constant because of a balance between processes that add carbon dioxide to the atmosphere and those that remove it.

The processes that add carbon dioxide to the atmosphere include the following:

1. **Respiration:** Both plants and animals release carbon dioxide during respiration. This process occurs continuously as organisms break down glucose to produce energy for life processes.
2. **Decomposition:** When plants and animals die, decomposers (bacteria and fungi) break down their remains. This decomposition process releases carbon dioxide back into the atmosphere.
3. **Fossil fuel combustion:** Burning of fossil fuels (coal, oil, and natural gas) in factories, vehicles, and power plants releases large amounts of carbon dioxide into the atmosphere. These fuels formed from ancient decomposed organisms buried underground over millions of years.
4. **Fermentation:** During fermentation, yeast converts sugar into ethanol and carbon dioxide. This process occurs in beer and wine production.



The carbon dioxide formed in this process finally finds its way into the atmosphere.

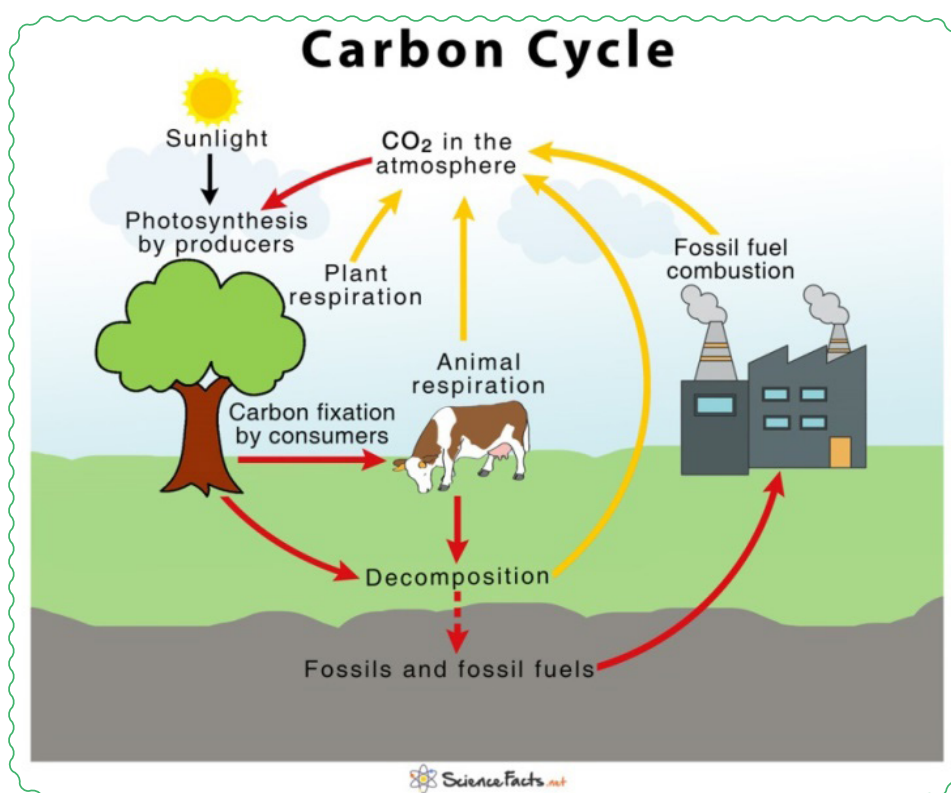
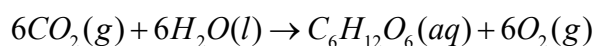


Figure 1.5.1 The carbon cycle illustrating the movement of carbon through different Earth systems.

(Credit: <https://www.sciencefacts.net>)

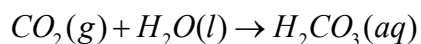
The processes that remove carbon dioxide from the atmosphere include the following:

- a) Photosynthesis:** Green plants (producers) absorb carbon dioxide from the atmosphere and use it along with water and sunlight energy to produce glucose and oxygen. This is the primary process that removes carbon dioxide from the atmosphere.

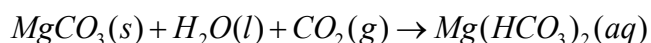
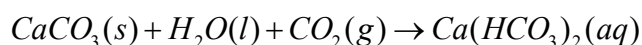


- b)** When animals (consumers) eat plants, the carbon stored in plant tissues is transferred to animal bodies. This temporarily removes carbon from the atmospheric cycle and stores it in living organisms.

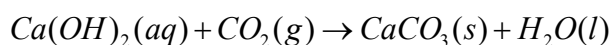
- c) Dissolution in water bodies:** Carbon dioxide dissolves in oceans, rivers, and lakes, forming weak carbonic acid:



This carbonic acid then reacts with calcium carbonate and magnesium carbonate in rocks and sediments to form soluble hydrogen carbonates:



- d) Reaction with calcium hydroxide:** Calcium hydroxide in rocks and soil slowly reacts with atmospheric carbon dioxide to form calcium carbonate:



<https://oceanservice.noaa.gov/facts/carbon-cycle.html#transcript>

Application activity 1.6



1. Carbon dioxide is a necessary evil. Explain.
2. If there is fire in the forest, there will be less in the environment. (oxygen, carbon dioxide)
3. Why is it important to maintain a balance of carbon dioxide in the atmosphere?
4. Describe two processes that remove carbon dioxide from the atmosphere.
5. Carbon cycle is the movement of carbon to the Earth by the process of and

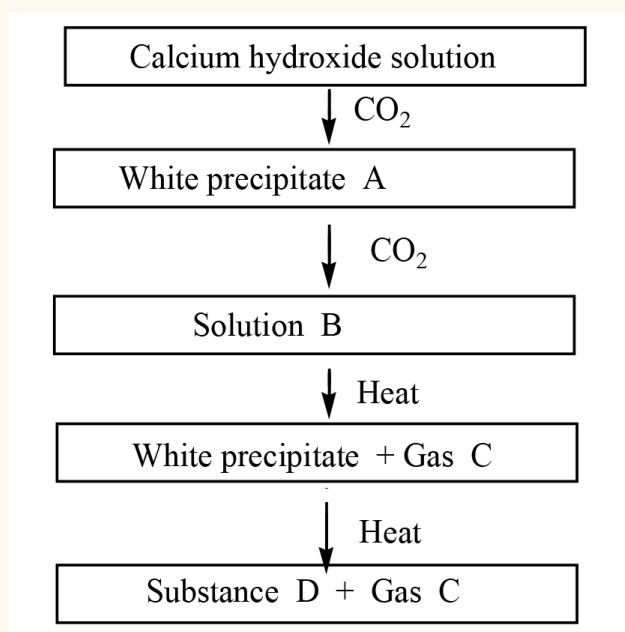
1.7 End of unit assessment

End of unit assessment



A. Knowledge and understanding

1. and are two the crystalline allotropes of carbon while and are the non-crystalline forms of carbon.
2. The flow-chart below shows some processes involving some compounds of carbon. Use it to answer the questions that follow.

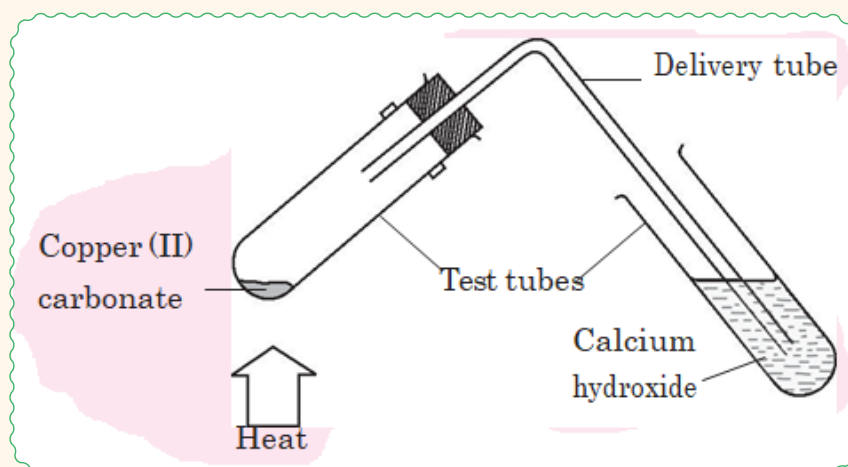


- (a) Name the white precipitate A, solution B and gas C.
 - (b) Describe a test that can be used to identify carbon dioxide in the laboratory.
3. is a gas formed when carbon burns in enough oxygen supply while is formed when carbon burns in inadequate oxygen supply.

4. Zinc oxide is in colour when hot and when cold while lead oxide is in colour when hot and when cold.
5. Natural resources like fossil fuels are important and also harmful if not well utilised. Explain.
6. The following are forms of amorphous carbon. Which one is **not**?
 - A. Lamp black
 - B. Charcoal
 - C. Soot
 - D. Graphite
7. Hard water does not lather easily and hence wastes a lot of soap. Explain using chemical equations how you can convert hard water to soft water and use it in washing clothes. Hard water does not lather easily and hence wastes a lot of soap. Explain using chemical equations how you can convert hard water to soft water and use it in washing clothes.

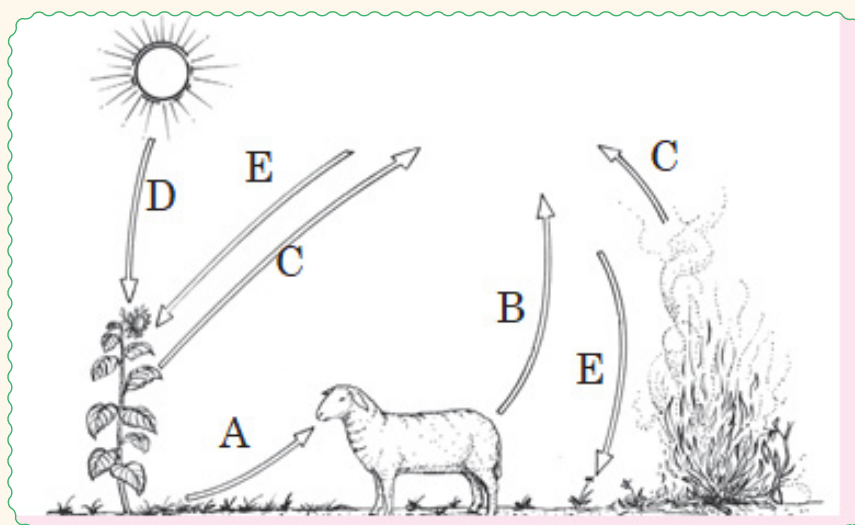
B. Skills

8. Akaliza set up an experiment as shown below.



- a) What observation do you expect her to make in each test tube at the end of the experiment?
- b) Write balanced equations that would lead to the observations expected in both test tubes.
- c) Supposing she used sodium carbonate instead of copper carbonate, what observations would you expect her to make? Explain.
- d) Write a balanced chemical equation for the decomposition of ammonium carbonate by heat.

9. The diagram below represents part of the carbon cycle.



- Name processes A, B, C, D and E
- Explain the relationship between the carbon cycle and the food chain.

C. Attitudes and values

- Explain why it is important to maintain a balance of carbon dioxide in the atmosphere.
- Explain how you can minimise emission of greenhouse gases in your locality.

UNIT 2

NITROGEN AND ITS INORGANIC COMPOUNDS



Key Unit Competence:

Relate the physical and chemical properties of nitrogen and its inorganic compounds to their preparation, uses and effects on the environment.

Learning objectives

By the end of this unit, you should be able to:

1 Knowledge and understanding

- Describe the physical and chemical properties of nitrogen and its uses.
- Describe the impact of nitrogen compounds on the environment.
- Identify methods for preparing nitrogen and its compounds (nitrogen dioxide, ammonia, nitric acid, and nitrates)
- Describe the industrial preparation of nitric acid.
- State the uses of nitric acid, ammonia and some nitrates.

2 Skills

- Prepare nitrogen and its compounds in the laboratory.
- Use experiments to demonstrate the reactions of ammonia and nitric acid, with various substances.
- Interpret the effect of ammonia solution on different cations.
- Prepare ammonia gas and nitric acid.
- Test the presence of ammonia and nitrates.
- Analyze the environmental implications of nitrogen compound usage (e.g. acid rain, eutrophication)

3

Attitudes and values

- Demonstrate awareness of safety precautions when handling nitrogen compounds.
- Develop self- confidence during presentations.
- Develop a culture of working in a team.
- Respect of the procedures in all experiments.
- Appreciate the uses of nitrogen compounds in daily life
- Recognize the environmental hazards associated with nitrogen compounds.
- how concern for environmental pollution caused by nitrogen dioxide and nitrate fertilizers.

2.0. Introduction

Nitrogen is the most abundant component of air, making up approximately 78% of the atmosphere by volume. This unit explores laboratory methods used in manufacturing nitrogen, examines its physical and chemical properties, and discusses the importance of nitrogen and its compounds. Additionally, this unit addresses the environmental issues associated with nitrogen compounds.

Introductory activity



Like carbon, nitrogen is also important in our lives. Study these pictures carefully.

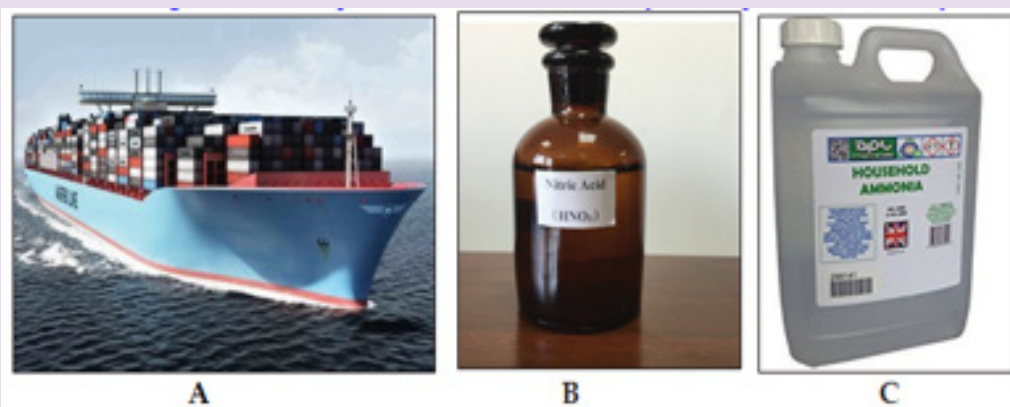


Figure 2. 1 Some nitrogen containing items

Describe preparation, properties and uses of nitrogen and nitrogen compounds.

2.1. Preparation, properties and uses of nitrogen

2.1.1 Laboratory preparation of nitrogen

Learning activity 2.1.1



Experiment illustrating the laboratory preparation of nitrogen

Materials required:

Round-bottomed flask or boiling tube, Bunsen burner/any source of heat, Trough, Gas jars, Sodium nitrite, Ammonium chloride

Procedure

Make a solution of sodium nitrite and ammonium chloride in water.

1. Put the solution in a round-bottomed flask or boiling tube.
2. Arrange the apparatus as shown in figure. 2.1 below.

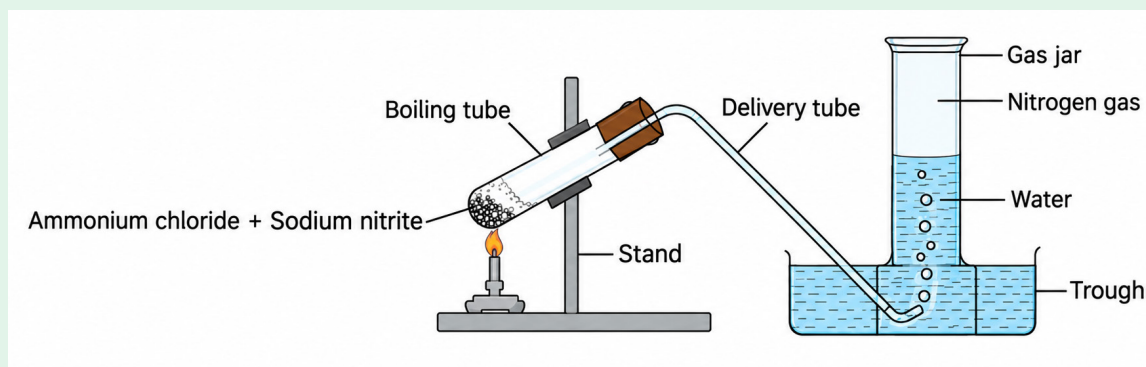


Figure 2.1: Laboratory preparation of nitrogen

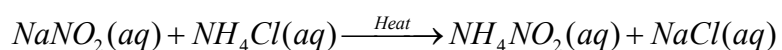
3. Warm the flask slightly and then remove the heat.
4. Record your observations in the flask and gas jar.

Guiding questions

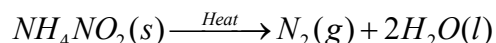
1. Why was warming done only for a short time?
2. Write balanced chemical equations for the reactions that take place.

When the flask is warmed, bubbles are produced and this continues even after warming has been stopped. A colourless gas collects in the upper part of the gas jar.

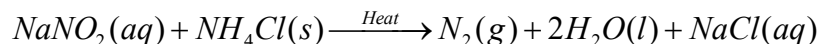
Sodium nitrite reacts with ammonium chloride to form ammonium nitrite and sodium chloride.



Ammonium nitrite decomposes upon heating to produce nitrogen gas.



The overall equation for the reaction:



Note: If required, dry nitrogen gas can be passed through concentrated sulphuric acid. The gas is then collected in a syringe or by upward delivery.

Nitrogen can also be prepared by direct heating of ammonium nitrite. However, when ammonium nitrite is in small quantities, the reaction can be explosive.

2.1.2 Properties of nitrogen

Learning activity 2.1.2



Investigating the properties of nitrogen

Materials required:

Wooden splint, Blue and red litmus papers, Aqueous calcium hydroxide, Magnesium ribbon, Sulphur powder, Six gas jars full of dry nitrogen, A pair of tongs, Deflagrating spoon.

Procedure:

1. Copy Table 2.1.2 in your notebook.
2. Carry out the procedures 1-7 and use your observations and conclusions to fill the table.

Procedure	Observations	Conclusion
1. Observe the colour and smell of nitrogen in the first gas jar.		
2. Put a glowing splint into the first gas jar.		
3. Insert a burning splint into the second gas jar.		
4. Insert wet, blue and red litmus papers into the second gas jar.		
5. Add a few drops of aqueous calcium hydroxide into the third gas jar. Shake gently.		
6. Lower a burning magnesium ribbon into the fourth jar. Add water to the products. Smell the gas produced.		
7. Lower burning sulphur on a deflagrating spoon into the fifth gas jar.		

A. Physical properties of nitrogen

Nitrogen exists as a diatomic gas (N_2) under normal conditions and exhibits the following physical properties:

- **Appearance:** Colorless and odorless gas
- **Solubility:** Almost insoluble in water, allowing collection upward delivery method.
- **Density:** Very slightly lighter than air (density=1.25 g/L at room temperature), with a relative density close to 1.

B. Chemical properties of nitrogen

1. Combustion

Nitrogen neither burns nor supports combustion. That is why a glowing or burning splint goes off when introduced in nitrogen.

2. Effect on litmus papers

Nitrogen is neutral. Litmus papers therefore show no colour change in nitrogen gas.

3. Effect on aqueous calcium hydroxide

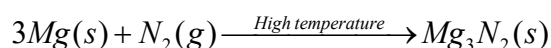
Nitrogen has no effect on calcium hydroxide.

4. Reaction with magnesium and sulphur

Nitrogen is inert at room temperature and does not readily react with other elements. This low reactivity is due to the very strong triple covalent bond ($N \equiv N$) between nitrogen atoms in N_2 molecules, which requires significant energy to break.

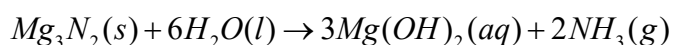
5. Reaction with Magnesium

When magnesium burns, it produces extremely high temperatures that are sufficient to break the triple covalent bonds in nitrogen molecules. This produces single nitrogen atoms that can now react with the burning magnesium to form magnesium nitride, a greenish-yellow solid.



6. Hydrolysis of Magnesium Nitride

Magnesium nitride reacts with water to produce magnesium hydroxide solution and ammonia gas:



2.1.3 Uses of nitrogen

Learning activity 2.1.3



Investigation on the uses of nitrogen in daily life

Use textbooks and electronic means to describe uses of nitrogen in daily life.

Some of the nitrogen uses are as follows:

1. It is used in the Haber process to manufacture ammonia gas.
2. Liquid nitrogen's extremely low boiling point (-196°C) makes it an excellent coolant for freezing and preserving substances. It is widely used for long-term storage of biological materials, including body tissues. Examples include:
 - a) Tissues used in hospitals.
 - b) Bulls' semen for artificial insemination.
 - c) Mending leaking pipes: When liquid nitrogen is poured onto the pipe, it freezes the liquid inside while repair is done.
3. Nitrogen plays an important role in food packaging. When food is packed, nitrogen gas displaces oxygen in the package, preventing oxidation that would otherwise cause spoilage. This oxygen-free environment also inhibits the growth of most bacteria, which cannot survive without oxygen, thereby keeping food fresh for longer periods.
4. Nitrogen's inert nature makes it essential for safety in crude oil transportation. It is pumped into ships' tanks to displace traces of oxygen, thereby preventing dangerous explosions that could occur when oxygen mixes with flammable crude oil vapours.

Application activity 2.1



1. Write a balance equation of the reaction between sodium nitrite and ammonium chloride.
2. Liquid nitrogen boils at 196°C . What is the importance of this?
3. Give three physical properties of nitrogen gas that differentiate it from other gases.
4. Nitrogen gas does not support combustion but a magnesium ribbon continues to burn in the presence of nitrogen. Discuss.
5. In order to attain electronic configuration of noble gas, nitrogen needs to
 - a) Lose one electron
 - b) Gain two electrons
 - c) Loose three electrons
 - d) Gain three electrons

6. Explain how nitrogen can be used:

- a) To Freeze tissues
- b) In food processing

<https://www.vedantu.com/chemistry/preparation-properties-and-uses-of-nitrogen>

2.2. Preparation, properties and uses of nitrogen compounds

2.2.1. Nitrogen dioxide

A. Preparation

Learning activity 2.2.1.a



Illustration of the laboratory preparation of nitrogen dioxide

Materials required:

Thistle funnel or dropping funnel, Flat-bottomed flask, Delivery tube, 3 gas jars, Cardboard cover, Copper turnings, Concentrated nitric acid

Caution: Nitrogen dioxide is poisonous. This experiment should therefore be performed in a fume cupboard or in an open space.

Procedure

1. Put copper turnings into a flat-bottomed flask.
2. Arrange the apparatus as shown in Fig. 2.2.1 (a).

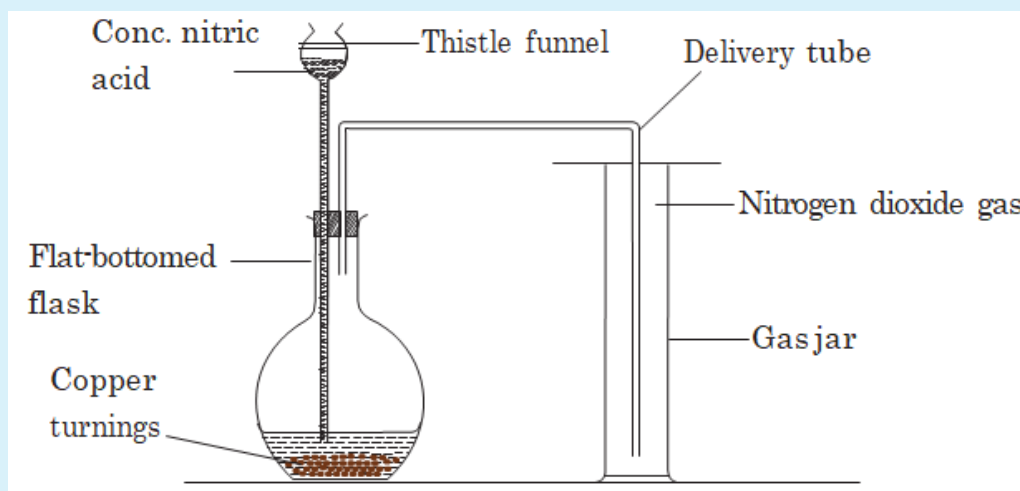


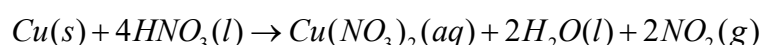
Figure 2.2.1 (a) Laboratory preparation of nitrogen dioxide

3. Add concentrated nitric acid to the copper turnings in the flask.
4. Make your observations in the flask and the gas jar and record in your notebook.

Guiding questions

1. What observations did you make in the flask and the gas jar?
2. Write a balanced chemical equation for the reaction that takes place.

Copper reacts with concentrated nitric acid to form a green solution of copper (II) nitrate, water and a red-brown gas (nitrogen dioxide).



More pure nitrogen dioxide can be prepared by heating a nitrate of a moderately reactive heavy metal. Lead (II) nitrate is preferred because unlike most nitrates, they do not contain water of crystallisation which would otherwise interfere with the preparation process since nitrogen dioxide is highly soluble in water.

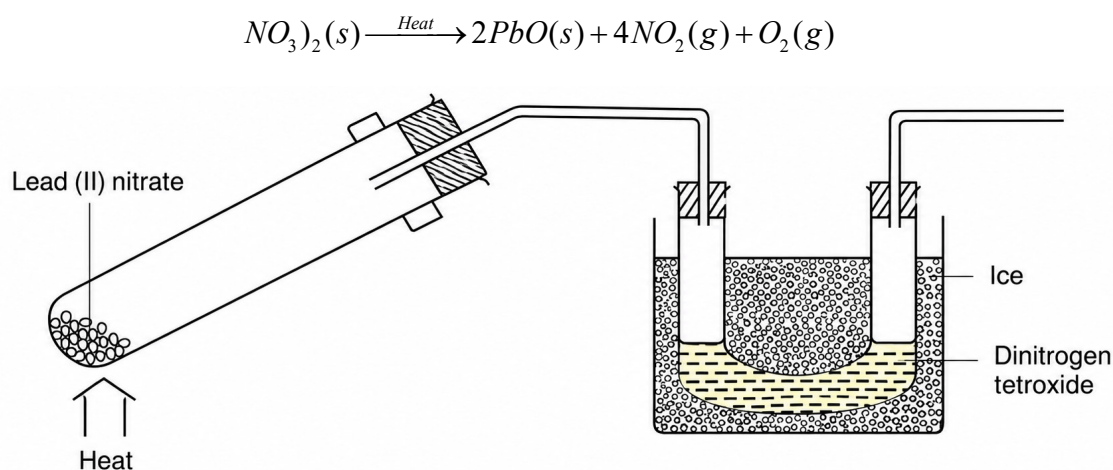
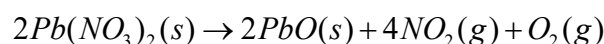
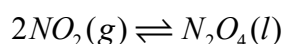


Figure 2.2.1(b): Preparation of nitrogen dioxide by heating lead (II) nitrate

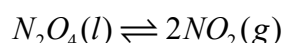
Lead (II) nitrate decomposes on heating to produce nitrogen dioxide and oxygen:



The gaseous mixture is cooled using an ice-salt freezing mixture. Nitrogen dioxide condenses to form pale-yellow liquid dinitrogen tetroxide, while oxygen remains gaseous and is collected over water:



Note: When heated, dinitrogen tetroxide readily dissociates back to nitrogen dioxide:



B. Properties

1) Physical properties of nitrogen dioxide

1. It is a red-brown gas.
2. It has an irritating pungent smell.
3. It is soluble in water forming an acidic solution:
4. $2\text{NO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{HNO}_3(\text{aq}) + \text{HNO}_2(\text{aq})$
5. The density of nitrogen dioxide and that of air are 3.39 kg/m^3 and 1.225 kg/m^3 , respectively. This suggests that nitrogen dioxide is about 2.8 times heavier than air.
6. It is easily liquefied on cooling (boiling point is 21°C).

2) Chemical properties

Learning activity 2.2.1.b



Investigating the chemical properties of nitrogen dioxide

Materials required

Litmus papers (blue and red), magnesium ribbon, phosphorus, sulphur, copper turnings, a pair of tongs, wooden splints

Procedure

1. Copy Table 2. 2.1 in your notebook.
2. Prepare nitrogen gas as shown in previous activity; then carry out the following steps 1-7. Record your observations and conclusions in the blank spaces provided.

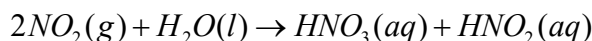
Table 2.2.1: Investigating the chemical properties of nitrogen dioxide

Procedure	Observations	Conclusion
1. Put a glowing splint into the first gas jar.		
2. Put a burning splint into the second gas jar.		
3. Put wet litmus papers (blue and red) into the third gas jar.		
4. Put a burning magnesium ribbon into the fourth gas jar.		
5. Put vigorously burning phosphorus and sulphur into the fifth and sixth jars respectively.		
6. Put red-hot copper into the seventh gas jar.		
7. Heat the gas in the corked test-tube and then place a glowing splint into the tube.		

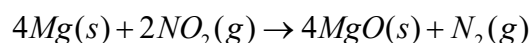
Guiding questions

1. Account for the observations in procedures 4 and 5.
2. Explain the observations made in procedure 6.

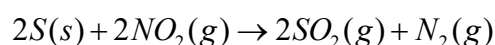
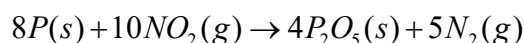
1. Nitrogen dioxide neither burns nor supports combustion. That is why it puts off both the glowing and the burning splints.
2. Nitrogen dioxide reacts with water to form nitric acid and nitrous acid. The gas is therefore acidic and so the solution formed turns wet blue litmus paper red.



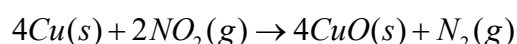
3. Burning magnesium continues to burn in nitrogen dioxide forming magnesium oxide and nitrogen gas.



4. The same type of reaction takes place if vigorously burning phosphorus and sulphur are introduced into gas jar containing nitrogen dioxide.



5. Nitrogen dioxide oxidises red-hot copper to copper (II) oxide while itself is reduced to nitrogen gas.

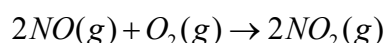


Note: When nitrogen dioxide is heated, it undergoes thermal decomposition in two stages.

First, it breaks down into nitrogen monoxide and oxygen:



The nitrogen monoxide then further decomposes into nitrogen gas and more oxygen:



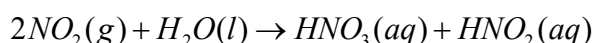
The oxygen released during these reactions readily combines with elements in burning substances to form oxides. This explains why heated nitrogen dioxide easily supports combustion.

6. Nitrogen dioxide reacts with sodium hydroxide solution to form two salts, namely sodium nitrate (NaNO_3) and sodium nitrite (NaNO_2).

This occurs in two stages:

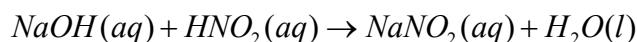
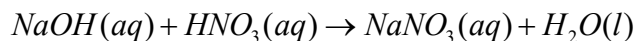
Stage 1:

Nitrogen dioxide dissolves in water to form two acids



Stage 2:

Both acids neutralize sodium hydroxide to form their respective salts:



Overall equation:



C. Uses of nitrogen dioxide

- It is used as an intermediary in the manufacture of nitric acid by Ostwald Process.
- It is an intermediary in the formation of very dilute nitric acid during lightning. This acid is washed into the soil, thereby increasing the amounts of nitrates in the soil.

Application activity 2.2



1. Explain why for the laboratory preparation of nitrogen dioxide Lead (II) nitrate is preferred to most of other nitrates unlike most nitrates.
2. Nitrogen dioxide dissolves in water to form two acids. Write the chemical equation for the reaction that takes place.
3. Nitrogen dioxide neither burns nor supports combustion. However, burning magnesium continues to burn in the presence of the gas. Explain.
4. Give one use of nitrogen dioxide.
5. Write a balanced chemical equation of nitrogen dioxide and sodium hydroxide and give the names of the products.

<https://www.vedantu.com/chemistry/preparation-properties-and-uses-of-nitrogen>

2.2.2 Ammonia

A. Preparation

Learning activity 2.2.2.a



Laboratory preparation of ammonia

Materials required

- | | | |
|------------------------|---------------------|--------------------------------|
| • Round-bottomed flask | • Several gas jars | • Ammonium chloride |
| • Glass tubing | • Mortar and pestle | • Calcium oxide |
| • Spatula | • Cardboard cover | • Litmus papers (blue and red) |
| • Drying tower | • Calcium hydroxide | |

Procedure:

1. Place one spatula full of calcium hydroxide and one spatula full of ammonium chloride in a mortar.
2. Grind the mixture well using the pestle then put it in a round-bottomed flask.
3. Arrange the apparatus as shown in figure 2.2.2(a).

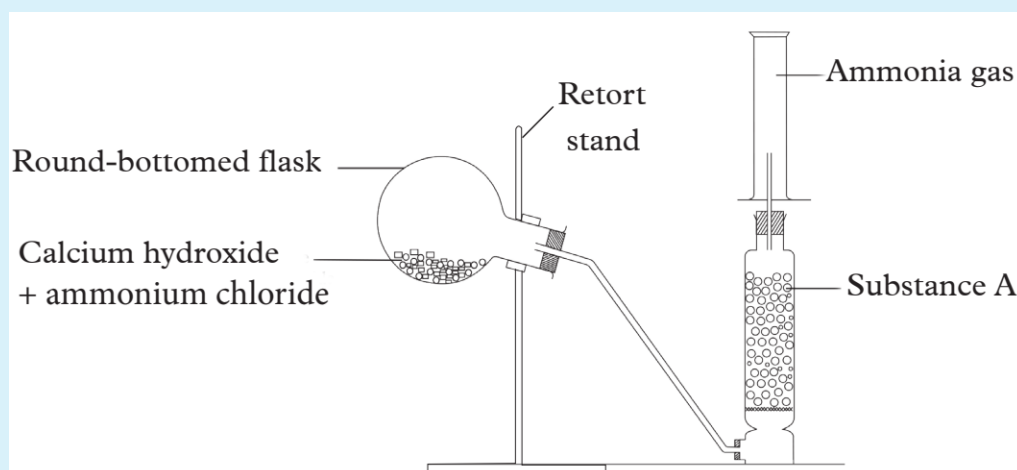


Fig 2.2.2(a): Laboratory set up for the preparation of ammonia

Caution: The flask should be set in a sloping position as shown in the diagram before the experiment starts.

4. Heat the mixture. Note down what you observe on the cooler parts of the flask and the delivery tube.
5. Check whether the gas jar is full, by putting a moist red litmus paper at the mouth of the jar. It should turn blue if the gas jar is full.
6. Collect five jars of this gas for the next activity.

Guiding questions:

1. Why should the flask be in a sloping position?
2. Identify the chemical labeled substance A in the Figure 2.2.2 (a) and state its function.
3. Name the method used to collect ammonia gas and suggest why it is preferred.

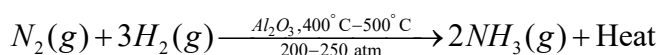
On a large-scale ammonia is manufactured by the **Haber process**. The raw materials required for the manufacture of ammonia by the Haber process are:

- Nitrogen-obtained from air by fractional distillation of liquid air.
- Hydrogen-obtained from natural gas (methane) or by electrolysis of acidified water.

The optimum conditions needed during manufacture of ammonia include:

- A mixture of nitrogen and hydrogen in the volume ratio of 1:3.
- A pressure between 200-250 atmospheres (
- A temperature of between .
- Iron catalyst in finely divided form and impregnated with aluminium oxide as a **promoter of the catalyst**.

The reaction which takes place is represented by the following chemical equation. The reaction is exothermic.



B. Properties

Learning activity 2.2.2.b



Experiment to illustrate properties of ammonia gas:

- Investigating the smell, solubility, effects of burning elements, reaction with water and hydrogen gas

Apparatus and reagents

Four gas jars of ammonia, wooden splints, trough, glass rod and concentrated hydrochloric acid.

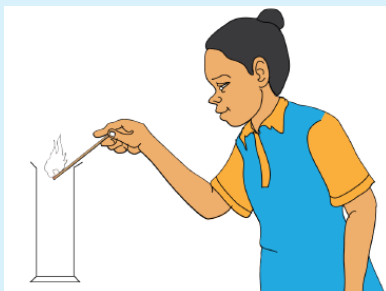


Figure 2.2.2(b): Testing ammonia gas using a burning splint.

Procedure

- Copy table 2.3 in your notebook.
- Carry out procedures 1-5 and use your results to fill the table.

Table 2.2.2(b): Investigating the properties of ammonia gas

Procedure	Observations	Conclusions
1. Watch the gas in the first jar towards your nose. Note the smell and colour of ammonia.		
2. Dip wet red and blue litmus papers into the second gas jar.		
3. Put a glowing splint and then a burning splint into the third gas jar.		
4. Invert the fourth gas jar in a trough of water. Note the level of the water inside the gas jar after 2 to 3 minutes.		
5. Dip a glass rod in concentrated hydrochloric acid and bring the dipped end to the mouth of the fifth gas jar.		

Guiding questions

1. Describe the smell of ammonia gas.
2. Is ammonia an acidic or alkaline gas?

1. Physical properties of ammonia

- Ammonia is a colourless gas.
- It has a pungent choking smell.
- It is very soluble in water. Water rises rapidly in a round-bottomed flask full of ammonia gas and almost fills it immediately if inverted in a trough containing water. This can be demonstrated in a fountain experiment as shown in the set up that follows.

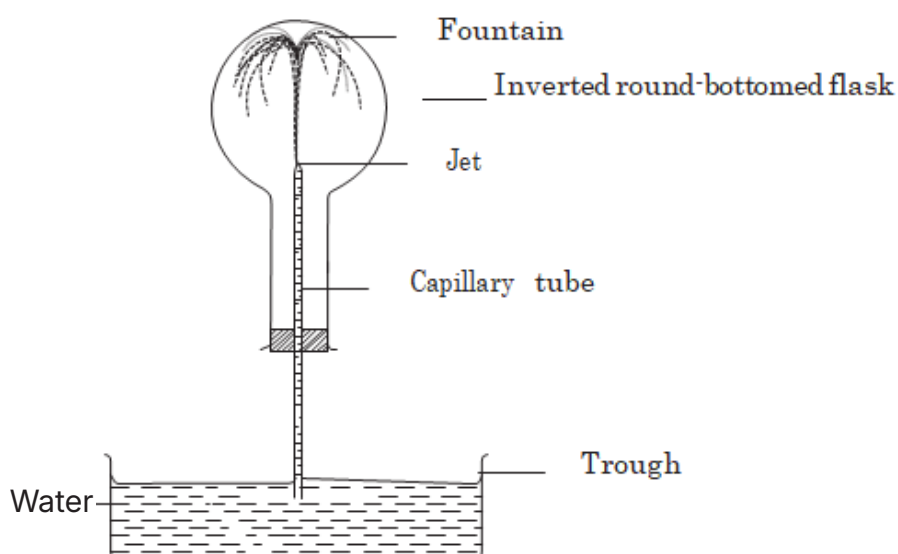


Figure 2.2.2(d): The fountain experiment

In the set up above, when water enters the flask, it dissolves so much ammonia that there is a partial vacuum in the flask. Water is then rapidly forced up the tube and enters the flask as a fountain.

2. Chemical properties

Learning activity 2.2.2.c



Experiment to illustrate properties of ammonia gas:

(i) Investigating the reaction of ammonia gas with oxygen

Ammonia reacts differently with air or oxygen. The reaction is even more different when a catalyst is introduced.

A. In the absence of a catalyst

Materials required:

Open glass tube, delivery tube, glass beads or cotton wool, sources of oxygen and ammonia, wooden splint and cork/rubber bung.

Procedure:

1. Arrange the apparatus as shown in the figure below.

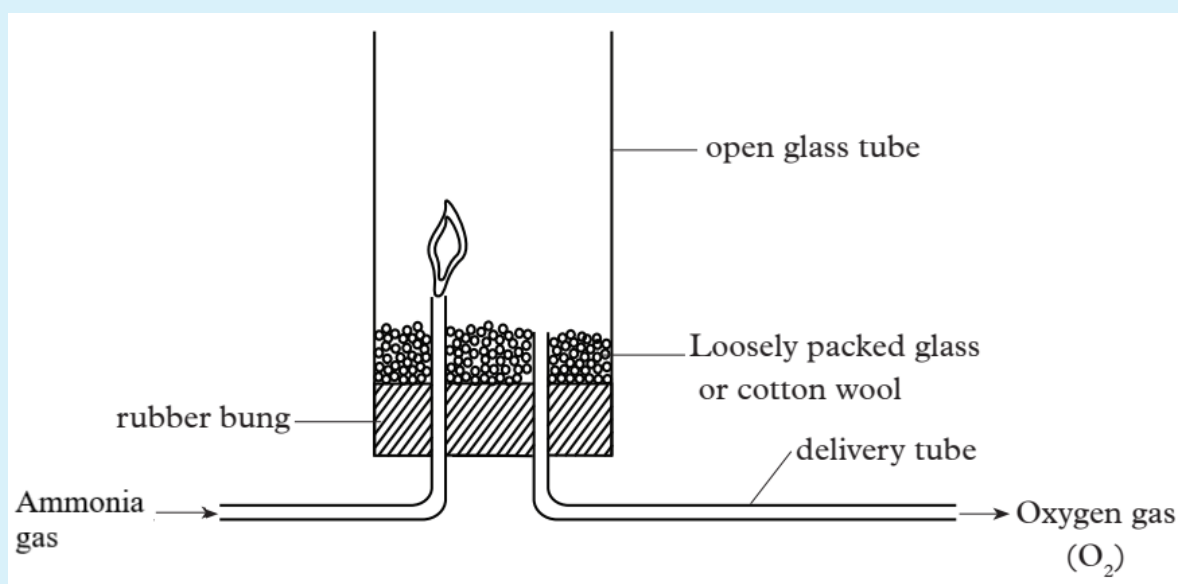


Figure 2.2.2(e): Burning ammonia in oxygen

2. Ignite ammonia gas at the jet delivering ammonia before allowing oxygen in as shown above.
3. Now adjust the rate of flow of both gases until a flame can be lit at the jet of ammonia tube.
4. Record your observations in your notebook.

Guiding questions:

1. What is the use of the glass beads or cotton wool in this experiment?
2. What do you conclude from this experiment?

B. Catalytic oxidation of ammonia

Materials required:

Platinum coil, delivery tube, beaker, concentrated ammonia solution, source of oxygen, litmus papers (blue and red)

Caution: The reaction can sometimes be explosive.

Procedure:

1. Bubble oxygen through the concentrated ammonia solution for a few minutes.
2. Lower a red-hot platinum coil into the mixture of oxygen and ammonia solution.

Note: Concentrated ammonia solution releases ammonia gas.

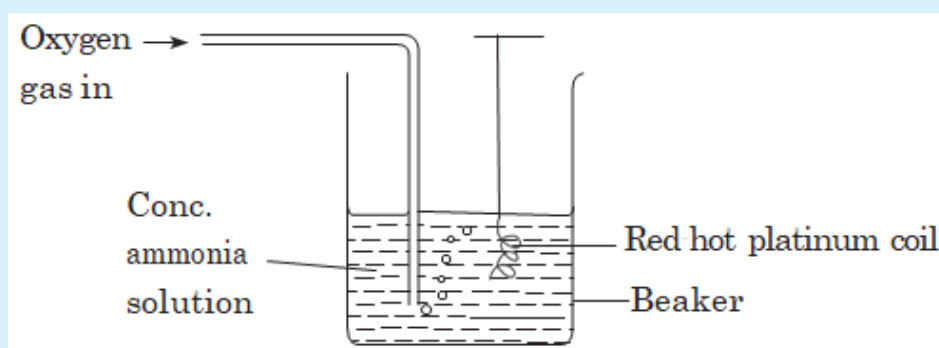


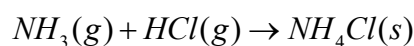
Figure 2.2.2(f): Catalytic oxidation of ammonia

1. Record your observations in your notebook.
2. Test the fumes produced using moist blue and red litmus papers.
3. Dip blue and red litmus papers in the solution after the reaction.

Guiding questions

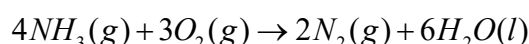
1. What is the colour of the fumes that are seen coming out of the solution?
2. What is the effect of these fumes on moist blue litmus paper?
3. What is the effect of the resulting solution on blue litmus paper?
4. What conclusions can you make from your observations?

1. Ammonia does not burn in air neither does it support combustion.
2. Ammonia turns moist red litmus paper blue. It is the common known alkaline gas.
3. It reacts with hydrogen chloride gas to form dense white fumes of ammonium chloride.

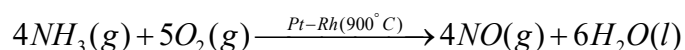


4. Ammonia reacts differently with air or oxygen. The reaction is even more different when a catalyst is introduced.

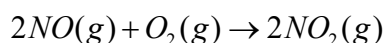
a) Ammonia burns in oxygen to form nitrogen gas and water.



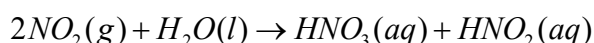
Ammonia reacts with oxygen in the presence of platinum (Pt)-Rhodium (Rh) catalyst to form nitrogen monoxide and water as shown in the equation below.



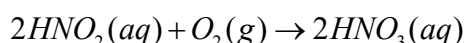
The nitrogen monoxide is further oxidised to form nitrogen dioxide gas characterised by red-brown fumes. This is an acidic gas.



When nitrogen dioxide is dissolved in water, nitrous acid and nitric acid are formed.



Nitrous acid is further oxidised to form nitric acid.



Nitric acid turns blue litmus paper red. It is therefore possible to convert ammonia to nitric acid using platinum as a catalyst.

Reaction of ammonia with hydrochloric, nitric and sulphuric acid

Learning activity 2.2.2.c



Experiments to illustrate the reaction of ammonia with hydrochloric, nitric and sulphuric acid.

Materials required

Test tubes and test-tube rack, 2M nitric acid, 2M hydrochloric acid, 1 M sulphuric acid, methyl orange indicator, aqueous ammonia, labels.

Procedure

1. Place 2cm³ of 1M sulphuric acid, 2M hydrochloric acid and 2M nitric acid in three separate test tubes and label them accordingly.

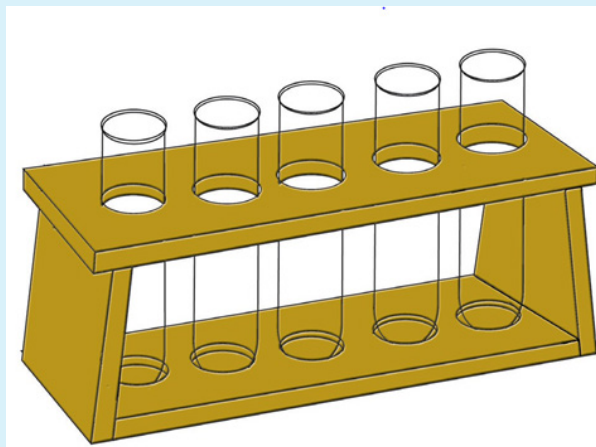


Figure 2.2.2 (g): Experimental set-up for the reaction of ammonia solution with dilute acids

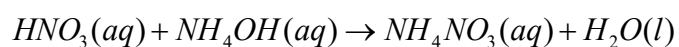
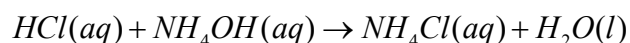
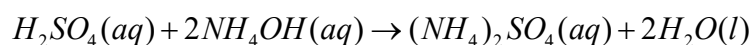
2. Add three drops of methyl orange indicator in each test tube.
3. Add aqueous ammonia drop wise into each test tube until a permanent pink orange colour is formed. Record your observations.

Guiding questions

1. What name is given to the types of reactions that occur between aqueous ammonia and dilute acids?
2. Write the equations for the following reactions between aqueous ammonia and dilute:
 - a) Sulphuric acid
 - b) Hydrochloric acid
 - c) Nitric acid

B. Properties of aqueous ammonia (ammonium hydroxide)

Aqueous ammonia neutralises dilute acids forming the corresponding ammonium salt and water only. The equations for the reactions that occur are as follows:



All ammonium salts are soluble in water and are normally prepared by neutralisation reaction. It is then followed by crystallisation of the salts. In the reactions above ammonia acts as a base by reacting with acids in a neutralisation reaction.

In these reactions, ammonia accepts a hydrogen ion (H^+) to form the ammonium ion (NH_4^+).

These reactions are exothermic.

The ammonium salts formed are important nitrogen containing fertilisers.

Uses of ammonia

- Large quantities of ammonia are used to make fertilisers.
- Ammonia gas is used in the manufacture of nitric acid.
- Liquid ammonia is used as a refrigerant in large-scale refrigerating plants and factories.
- Ammonia solution is used as a solvent during cleaning (in laundries).
- Ammonia is used in the manufacture of ammonium salts such as ammonium.

Application activity 2.2.2



1. What precaution would you take when preparing ammonia in the laboratory?
2. Calcium oxide is preferred in drying ammonia gas over other drying agents. Explain.
3. Uwimana is a senior 3 student. She wanted to confirm whether a gas in a gas jar was ammonia. What chemical test would you advise her to carry out?
4. Draw the apparatus used to prepare aqueous ammonia in the laboratory.
5. Write a balanced chemical equation of the reaction of ammonia
6. In the absence of a catalyst
7. In the presence of a catalyst
8. What is the importance of the reaction (5) b. above?

<https://ammoniagas.com/what-is-ammonia/>

2.2.3. Nitric acid

A. Preparation

Learning activity 2.2.3.a



Laboratory preparation of nitric acid

Materials required

Retort, glass bottle, round or flat-bottomed flask, concentrated sulphuric acid, potassium nitrate solution, glass cork, clamp stand, source of heat, running water from a tap.

Procedure

1. Arrange the apparatus as shown below.

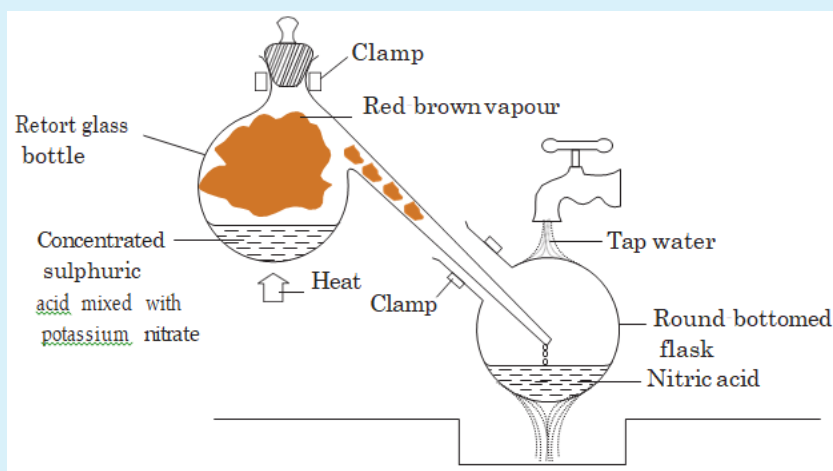


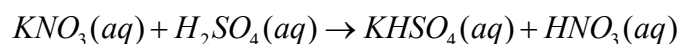
Figure 2.2.3 (a): Laboratory set up for the preparation of nitric acid

2. Put some potassium nitrate in the retort glass bottle. Add concentrated sulphuric acid to cover it.
3. Open the tap water to cool the round or flat-bottomed flask.
4. Heat the retort glass bottle gently.
5. Note down your observations in the retort glass bottle and flask.
6. Keep the acid formed for the next activity.

Guiding questions

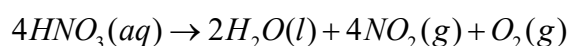
1. Why are all the apparatus made of glass in this experiment?
2. What is the importance of tap water?

When the mixture is heated, effervescence is seen and red-brown fumes are seen in the retort flask. Potassium nitrate reacts with sulphuric acid to form potassium hydrogen sulphate and nitric acid.



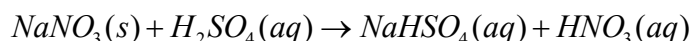
Nitric acid is more volatile than concentrated sulphuric acid and hence vaporises. These vapours condense to form nitric acid (a yellow liquid).

The red-brown vapours seen in the retort flask are as a result of the decomposition of some of the nitric acid by heat, forming red-brown nitrogen dioxide.



The nitric acid collected in this experiment is red-brown because it contains some dissolved nitrogen dioxide, otherwise pure nitric acid is colourless. The red-brown colour can be removed by bubbling air through the acid.

The apparatus used in this experiment is entirely made of glass because hot nitric acid vapours is highly corrosive and would attack cork or rubber stoppers and tubings. Potassium nitrate is preferred in preparing nitric acid in the laboratory because it has no water of crystallisation which can dilute the acid produced. Sodium nitrate can also be used but it is hygroscopic.



Note: Nitric acid is kept in dark bottles. This is because light decomposes the acid into water, nitrogen dioxide and oxygen.

B. Industrial manufacture of nitric acid by Ostwald process

Learning activity 2.2.3.b



Investigation on industrial preparation of nitric acid

Using textbooks and internet carry out research on the industrial preparation of nitric acid.

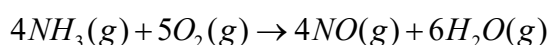
Manufacture of nitric acid by Ostwald Process

This process involves the catalytic oxidation of ammonia. The raw materials required include: ammonia, air and water. The optimum conditions needed are a temperature of 850°C and 900°C and a Platinum -rhodium catalyst.

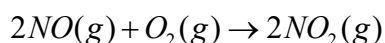
Note: The air and ammonia must be free of dust and other impurities to avoid poisoning of the catalyst.

Reactions that take place in Ostwald Process

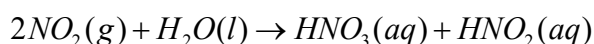
1. **Catalytic oxidation of ammonia to nitrogen monoxide:** When ammonia is oxidized with oxygen in presence of catalyst platinum and rhodium in 9 : 1 ratio at about 850°C and 900°C, nitric oxide is formed.



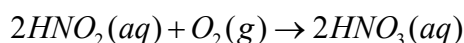
2. **Oxidation of nitric oxide to nitrous oxide:** Nitric oxide is cooled to about 150°C and air oxidized to nitrogen dioxide.



3. In the third stage, nitrogen dioxide is passed through water to form nitric acid and nitrous acid.



4. Excess air is passed to oxidise nitrous acid to nitric acid.



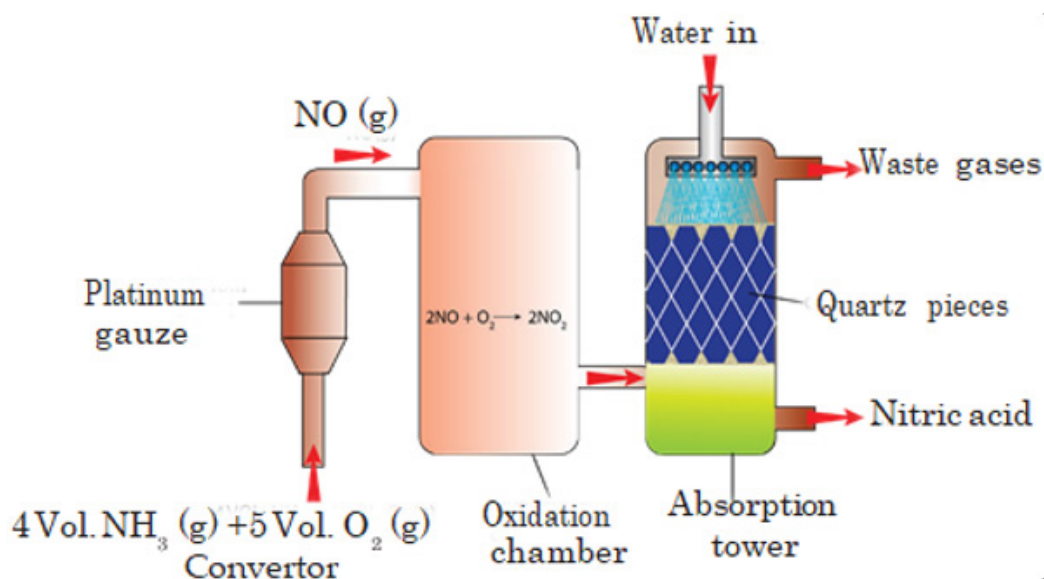


Figure 2.2.3 (b): The Ostwald process for manufacturing nitric acid, showing the catalytic oxidation of ammonia in the converter, further oxidation in the oxidation chamber, and absorption in water to produce HNO_3 .

The nitric acid produced by this process is about 65 % pure. However, the purity of the acid can be increased by distillation. The unreacted air, ammonia and nitrogen dioxide are re-circulated back to the compressor.

C. Properties

(a) Properties of dilute nitric acid

Learning activity 2.2.3.c

Investigating the properties of dilute nitric acid

Materials required

- Test tubes
- Wooden splint
- Sodium carbonate
- Sodium hydroxide
- Copper (II) oxide
- Magnesium ribbon
- Dilute nitric acid
- Phenolphthalein indicator
- Zinc metal

Procedure

1. Copy Table 2.2.3 (c) in your notebook.
2. Carry out activities from (a) to (d) and use your results to fill the table.

Table 2.2.3 (c): Investigating the properties of dilute nitric acid

Activity	Observations	Conclusion
(a) Add 2 or 3 drops of phenolphthalein indicator to a solution of sodium hydroxide in a test tube. Then add dilute nitric acid drop by drop till a change is observed.		
(b) Add dilute nitric acid to copper (II) oxide in a test tube		
(c) Put magnesium ribbon in a test tube. Add dilute warm nitric acid. Repeat the same experiment but now with very dilute cold nitric acid		
(d) Put zinc in a test tube. Add to it dilute nitric acid.		

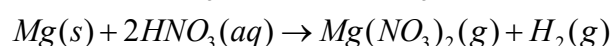
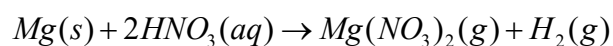
Guiding question

Write chemical equations for the reactions that take place in activities a, b, c and d.

When **very dilute and cold**, nitric acid behaves like other mineral acids, donating H^+ ions without

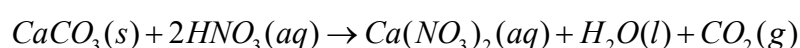
significant oxidizing action. This is observed in reactions with certain reactive metals and with carbonates, hydroxides, and oxides.

1. Reaction with reactive metals: With magnesium ribbon and zinc metal, very dilute nitric acid produces effervescence due to hydrogen gas evolution. The magnesium ribbon gradually disappears, forming a colorless solution.

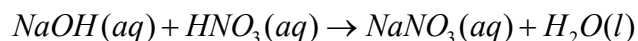


Note: These reactions occur only with very dilute and cold nitric acid.

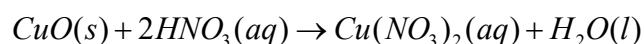
2. Reaction with metal carbonates and hydrogen carbonates: Dilute nitric acid reacts with carbonates to form a salt, water, and carbon dioxide gas observed as effervescence. Effervescence is foaming or fizzing that results from a gas being evolved from a solid or liquid.



3. Neutralization with metal hydroxides: Dilute nitric acid neutralizes metal hydroxides to form a salt and water.

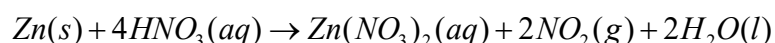


4. Reaction with metal oxides: Dilute nitric acid reacts with metal oxides to form a salt and water. For example, the black copper (II) oxide reacts with and dissolves in nitric acid to give a blue-green solution of copper (II) nitrate and water.



Nitric acid possesses strong oxidizing properties. When moderately dilute or concentrated, it typically does not liberate hydrogen gas with most metals. Instead, the hydrogen is immediately oxidized to water, and various nitrogen oxides are formed depending on the concentration and the metal's reactivity.

For instance, when zinc granules are added to moderately dilute nitric acid, a vigorous exothermic reaction occurs. The grey metal dissolves, producing reddish-brown nitrogen dioxide gas (NO_2) with a pungent odor. A colorless solution of zinc nitrate is formed.



Environmental effects during the manufacture of nitric acid

- Concentrated nitric acid and its vapours are highly corrosive to the eyes, skin and mucous membranes.
- Dilute solutions of the acid on the other hand cause mild skin irritation and hardening of the epidermis.
- Contact with concentrated nitric acid burns the skin and produces deep painful wounds.
- Eye contact can cause severe burns and permanent eye damage. Inhalations of high concentrations can lead to severe respiratory irritations and delayed effects including pulmonary oedema which may be fatal.
- Ingestion of nitric acid may result in burning and corrosion of the mouth, throat and stomach. An oral dose of 10 mL can be fatal in human beings. Always be careful when handling nitric acid.

(b) Properties of concentrated nitric acid

Learning activity 2.2.3.d



Investigating the properties of concentrated nitric acid

Concentrated nitric acid is a very powerful oxidising agent. Its oxidising power depends on the concentration, temperature and the type of reaction.

Material required:

Iron (II) sulphate solution, dilute sulphuric acid, sulphur powder, 50% concentrated nitric acid, distilled water, barium chloride or barium nitrate solution, lead (II) nitrate solution and copper turnings.

Procedure

1. Copy Table 2.2.3 (d) in your notebook.
2. Carry out the procedures outlined in the table below and record your observations and conclusions in the blank spaces.

Table 2.2.3 (d): Properties of concentrated nitric acid

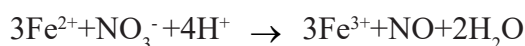
Activity	Observations	Conclusion
Add a few drops of dilute sulphuric acid to about 2 cm ³ of a solution of iron (II) sulphate in a boiling tube.		
(b) Add a few drops of concentrated nitric acid to the solution formed in (a) above.		
(c) Put a few copper turnings in a boiling tube and then add 50% concentrated nitric acid.		
(d) (i) Add about 2 cm ³ of concentrated nitric acid to sulphur powder in a boiling tube and warm it. (ii) Filter the mixture.		
(e) Dilute the filtrate obtained in (d) above with some water and divide it into two portions in two test tubes. To the first portion in test-tube, add a few drops of barium chloride or barium nitrate solution.		
(f) To the second portion in test-tube, add a few drops of lead (II) nitrate solution.		

Guiding question

Which reactions show the oxidation property of concentrated nitric acid?

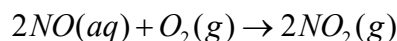
Reaction of concentrated nitric acid with iron (II) sulphate

1. Concentrated nitric acid oxidises iron (II) ions to iron (III) ions. In the process, the acid is reduced to nitrogen monoxide gas.



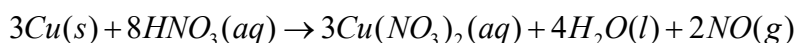
The same type of reaction can take place with iron (II) chloride solution. However, in this case, addition of sulphuric acid is not necessary.

The nitrogen monoxide produced is readily oxidised to red-brown nitrogen dioxide gas by oxygen from the air.

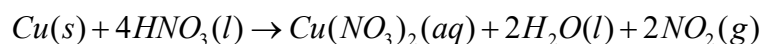


Reaction of concentrated nitric acid with copper metal

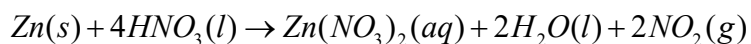
Copper metal reacts with 50% concentrated nitric acid producing copper (II) nitrate, water and nitrogen monoxide gas.



The nitrogen monoxide is oxidised to nitrogen dioxide by oxygen from the air. Concentrated nitric acid reacts with copper to form copper (II) nitrate, water and nitrogen dioxide.



The same type of reaction takes place when concentrated nitric acid reacts with zinc.



Reaction of concentration nitric acid with non-metals

1. Reaction of concentrated nitric acid with sulphur

Concentrated nitric acid reacts with sulphur to form sulphuric acid, water and nitrogen dioxide.



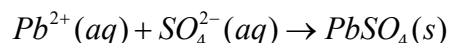
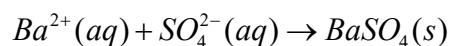
When barium chloride, barium nitrate, or lead (II) nitrate solution is added to the sulfuric acid produced in the reaction, a white precipitate is formed. Barium salts produce barium sulfate, while lead (II) nitrate produces lead (II) sulfate. This confirmatory test identifies the presence of sulfate ions (SO_4^{2-}).

Distinguishing sulfate from sulphite ions

If sulphite ions (SO_3^{2-}) were present instead, any white precipitate formed would dissolve in dilute nitric acid, whereas barium sulfate and lead (II) sulfate precipitates remain insoluble in dilute nitric acid.

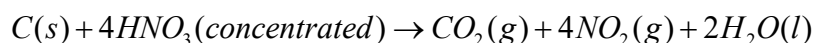
Note: Barium salts (barium chloride or barium nitrate) are preferred over lead (II) nitrate for sulfate testing because barium sulfate is completely insoluble, giving a more definitive and reliable precipitate. Lead (II) chloride cannot be used for this test because it is only slightly soluble in water and therefore cannot provide sufficient Pb^{2+} ions in solution to react with sulfate ions.

Ionic equations for the reactions:



2. Reaction with carbon

When hot, concentrated nitric acid (HNO_3) reacts with carbon (C), carbon dioxide, nitrogen dioxide and water are produced.



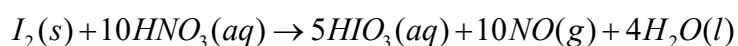
3. Reaction with phosphorus

Nitric acid reacts with phosphorus to produce phosphoric acid, nitrogen dioxide, and water.



4. Reaction with iodine

When nitric acid reacts with phosphorus, iodic acid, nitrogen dioxide and water are formed.



D. Uses

Nitric acid has numerous important industrial and commercial applications, including:

- Manufacture of nitrogenous fertilizers such as sodium nitrate, potassium nitrate, ammonium nitrate, and calcium ammonium nitrate.
- Manufacture of explosives such as trinitrotoluene (TNT) and dynamite.
- Manufacture of dyes.
- Manufacture of plastics.
- Used as an oxidizing agent in textile industries.
- Used in refining gemstones.

Application activity 2.2.3



- A Senior 3 student added concentrated sulphuric acid to potassium nitrate crystals.
- What observation did the student make?
- Write a balanced chemical equation for the reaction that took place.
- Why is potassium nitrate preferred in this reaction over other nitrates?
- Sodium nitrate can also be used in this reaction. Why is it not preferred?

6. Give two raw materials and two conditions required in Ostwald process.
7. Which catalyst is used in large scale preparation of nitric acid?
- A. Vanadium (V) oxide
 - B. Pure platinum
 - C. Iron
 - D. A platinum-Rhodium alloy (Pt - Rh)
8. Write the equation for the action of heat on potassium nitrate
9. Describe what you will observe when concentrated nitric acid is added to copper.
10. Identify the most common fertiliser made from nitric acid.
- A. Nitrogen
 - B. Ammonium sulphate
 - C. Ammonium nitrate
 - D. Manure
11. Write two chemical equations for reactions in which nitric acid acts as an oxidising agent.
12. Nitric acid is not only an important reagent in the Chemistry laboratory but finds application in other fields. Explain.

<https://allen.in/jee/chemistry/nitric-acid>

2.2.4. Nitrates

Preparation

Nitrates are salts formed when the hydrogen ions in nitric acid are replaced by a metal or ammonium ion.

Learning activity 2.2.4.a

Experiment on the preparation of nitrates in the laboratory

Materials required

- | | | |
|-----------------|-------------------------------|---------------------|
| • Spatula | • Beakers | • Magnesium ribbon |
| • Bunsen burner | • Stirring rod | • Calcium carbonate |
| • Filter funnel | • Measuring cylinder | • 1 M nitric acid |
| • Filter papers | • M sodium hydroxide solution | |

Procedure

1. Measure 25 cm³ of 1M nitric acid solution and place in a beaker. To this, add 25 cm³ of 1M sodium hydroxide solution and stir. Heat the resulting solution to dryness. Note down your observations.
2. Add small pieces of magnesium ribbon to 25 cm³ of dilute nitric acid solution in a beaker. Filter the solution to remove excess magnesium and heat the resulting solution to dryness. Record your observations.
3. Add excess calcium carbonate to 25 cm³ of nitric acid in a beaker. When the reaction is complete, filter the mixture and heat the filtrate to dryness. Note down your observations.

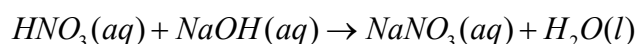
Guiding questions

1. Why is it important to add excess magnesium ribbons and calcium carbonate in procedures 2 and 3 respectively?
2. What is the colour of the salt formed in each procedure above?

The three methods used to prepare nitrates are:

i. Neutralisation of dilute nitric acid by a base/alkali.

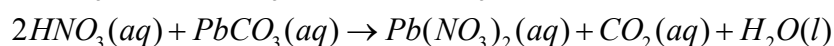
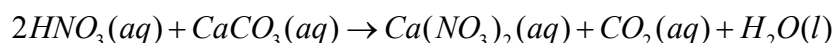
For Example:



When the resulting solution is heated to dryness, a white solid nitrate is obtained. This method is suitable for preparation of nitrates of potassium, sodium and ammonium.

ii. Reaction between dilute nitric acid and a metal carbonate.

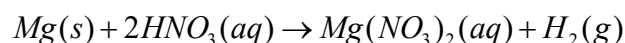
Examples of such reactions include:



The solution formed is then evaporated to concentrate it and allow crystallisation to take place during cooling. The mother liquor is poured off and the crystals dried between filter papers.

(iii) Action of nitric acid on a metal

This method is not suitable for use with very reactive metals like sodium and potassium. It is also not suitable for preparing salts that are readily oxidised. An example of a salt that can be prepared by this method is magnesium nitrate.



B. Properties

Action of heat on nitrates

Learning activity 2.2.4.b



Illustrating the action of heat on nitrates

Materials required

- 5 test tubes
- Labels
- Blue litmus paper
- Wooden splint
- Sodium nitrate
- Potassium nitrate
- Copper (II) nitrate
- Lead (II) nitrate
- Silver nitrate (if available)

Procedure

1. Copy Table 2.2.4 (a) in your notebook.

Table 2.2.4 (a): Action of heat on metal carbonates

Metal nitrate	Observations
Sodium nitrate	
Potassium nitrate	
Copper (II) nitrate	
Lead nitrate	
Silver nitrate	

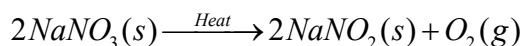
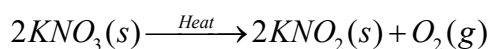
2. Put a spatula full of each of the metal nitrates provided in different test tubes and label them accordingly.
3. Heat each nitrate.
4. Test any gas evolved using a glowing splint and moist blue litmus paper.
5. Observe the residue both when hot and cold.
6. Record all observations in the table above.

Guiding questions

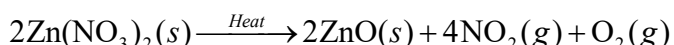
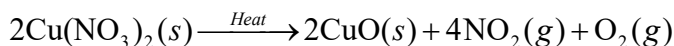
1. What is the colour of the gas or the gas mixtures produced in each case?
2. What is the colour of the hot and cold solid residues in each case?

The ease of decomposition of metal nitrates increases with decrease in reactivity of the metal constituting the nitrate. The nitrates decomposed form a red-brown acidic gas and other reaction produce a gas that relights a glowing splint.

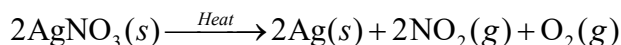
1. All nitrates decompose on heating.
2. Sodium and potassium nitrates decompose on heating to form metal nitrite and oxygen gas.



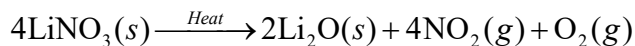
3. Nitrates of moderately reactive metals like copper, lead and zinc decompose to form metal oxides, nitrogen dioxide and oxygen gas.



4. Nitrates of least reactive metals like silver and mercury decompose to form the respective metal, nitrogen dioxide and oxygen gas.



5. Fumes of mercury are poisonous and so the reaction should not be carried out under normal school laboratory conditions.
6. Lithium nitrate behaves differently from other alkali metals as follows.



C. Test for nitrates

Learning activity 2.2.4.c



The Brown ring test for nitrates

Materials required:

- Test tubes
- Freshly prepared iron (II) sulphate solution
- Metal nitrate solution
- Concentrated sulphuric acid

Procedure

1. In a test-tube, mix equal volumes of a metal nitrate solution with freshly prepared iron (II) sulphate solution.
2. With the test-tube tilted at an angle, carefully pour concentrated sulphuric acid down its wall so that two layers of solutions are formed.

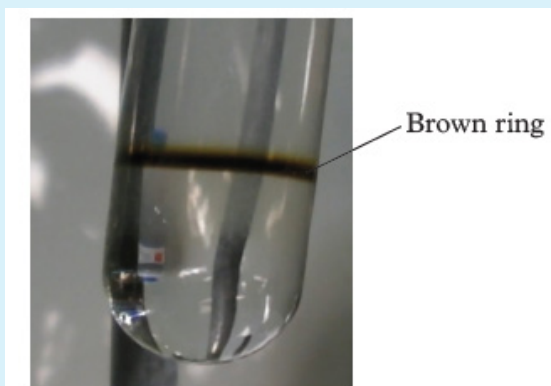


Figure 2.5 (b): Brown-ring test for nitrates

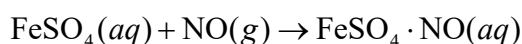
3. Record your observation.

4. Shake the solution and record what happens.

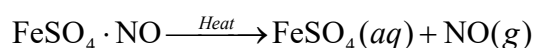
Guiding questions

1. Why should we use freshly prepared iron (II) sulphate solution?
2. Explain what happened when you shook the test tube containing the mixture.

Concentrated sulphuric acid reacts with any nitrate salt to produce nitric acid. The nitric acid is reduced to nitrogen monoxide by iron (II) sulphate. The nitrogen monoxide formed reacts with unreacted iron (II) sulphate to form a brown nitroso-iron (II) sulphate ($\text{FeSO}_4 \cdot \text{NO}$) which appears as a brown ring. This is the test for the presence of nitrate ions in solution.



The ring disappears if the solution is shaken. This is because concentrated sulphuric acid mixes with the aqueous solution producing a lot of heat. This heat decomposes nitroso-iron (II) sulphate as shown in the equation below.



Iron (II) sulphate solution is oxidised by oxygen from the air to iron (III) sulphate within a short time. This is the reason why it should always be freshly prepared.

Note: All nitrate salts are soluble in water. They all form nitrate ions when in solution which gives the "brown ring".

D. Uses

1. They are used as fertilisers for example sodium nitrate, potassium nitrate, ammonium nitrate, calcium ammonium nitrate and NPK fertilizers.



Figure 2.2.4 (a): NPK fertiliser

2. Potassium nitrate is used in the manufacture of gunpowder, which is a mixture of potassium nitrate, charcoal, and sulfur. When ignited, the mixture burns and explodes.
3. Ammonium nitrate is used as a fertilizer and in the manufacture of explosives.
4. Nitrates in form of nitroglycerin are used in medicine for treatment of heart diseases.

Application activity 2.2.4



1. Define a nitrate.
2. Write a balanced chemical equation for the preparation of sodium nitrate from sodium hydroxide and nitric acid.
3. Why is concentrated nitric acid not used with ammonium salts?
 - a) Describe how potassium nitrate can be prepared in the laboratory from potassium carbonate.
 - b) Write a balanced chemical equation for the reaction.
4. Explain what happens when calcium nitrate is heated strongly. Include a balanced equation.
5. Write balanced chemical equation of the action of heat on the following compounds:
 - A. Lead nitrate
 - B. Magnesium nitrate
 - C. Lithium nitrate

<https://www.chemistrylearner.com/nitrate.html>

2.2.5. Environmental issues related to nitrogen dioxide and nitrogen fertilisers

Learning activity 2.2.5



Investigation into the Environmental Impact of Nitrogen Compounds

In groups, research and identify the environmental issues associated with nitrogen oxides and nitrate fertilizers. Based on your findings, propose strategies to prevent or reduce pollution from nitrogenous compounds. Suggest ways that can be used to prevent pollution by nitrogenous compounds. Write a comprehensive report summarizing your research and recommendations, and submit it to your teacher for evaluation

1. Nitrogen dioxide

- Nitrogen dioxide is found in the atmosphere as a result of industrial processes and emissions from automobile exhausts.



Figure 2.3.1 (a): Pollution by industries



Figure 2.3.1(b): Corroded iron sheets due to effects of acidic rain

- Nitrogen dioxide is toxic and can cause respiratory diseases. Additionally, nitrogen dioxide reacts with water vapor in the atmosphere and rainwater to form dilute nitric acid. Rain containing nitric acid or other acids is called "acid rain."
- Acid rain has harmful effects on both terrestrial and aquatic ecosystems. The acid lowers the pH of soil and water bodies, making it difficult for some organisms to survive. Acid rain also corrodes metallic objects such as iron roofs, eventually causing them to leak.

2. Nitrates fertilisers

- The continued use of large quantities of nitrate fertilisers over long period of time pollutes the environment. This is because excess nitrates in the soil are washed down into rivers, lakes and other water bodies.
- Once in water, nitrates cause aquatic plants (e.g. plankton and algae) to grow in large quantities. This process is known as eutrophication. As a result dead plants accumulate faster than can be decomposed by bacteria and they settle to the bottom of the water bodies. Dead plant matter has a high biological oxygen demand (BOD) and some water bodies give up nearly all their dissolved oxygen in order to allow this natural decay process (oxidation) of the dead plant material to take place. This causes a decrease in the concentration of dissolved oxygen in water and this eventually leads to death of aquatic animals such as fish and plants.
- Also, due to large masses of dead plants and animals, the penetration of light in water decreases thereby affecting photosynthesis in aquatic plants.
- If nitrates are consumed in drinking water, they are converted by bacteria in the intestines to more toxic nitrites. After absorption into the blood stream, the nitrites combine with haemoglobin forming methaemoglobin. As a result, the oxygen carrying capacity of blood is reduced.

My environment, my life!

We should advise people in our community to avoid cultivating near river banks. This will prevent the washing away of fertilisers into water bodies hence prevent eutrophication.

<https://www.google.com/search?q=Environmental+issues+related+to+nitrogen+dioxide+and+nitrogen+fertilisers&sc=++issues+related+to+nitrogen+dioxide+and+nitrogen+fertilisers=&gs=>

Application activity 2.2.5



Describe environmental issues related to nitrogen dioxide and nitrogen fertilisers

2.3 End of unit assessment

End of unit assessment



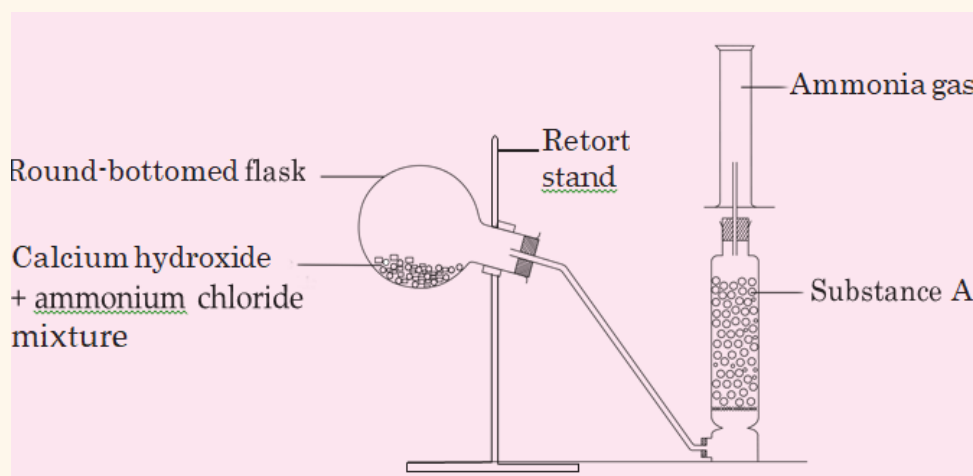
A. Knowledge and understanding

1. Nitrogen dioxide is a gas in colour that has a smell while ammonia is a colourless gas with a smell.
2. Mutesi dissolved nitrogen dioxide gas in water. She later put blue and red litmus papers in the solution formed. The blue litmus paper turned red while there was no effect on the red litmus paper.
 - a) Explain the observations made above.
 - b) Write a chemical equation for the reaction that took place.
3. The following methods are suitable for preparation of nitrates except?
 - A. Reacting dilute nitric acid with sodium hydroxide.
 - B. Reacting sodium metal with nitric acid.
 - C. Reacting nitric acid with lead carbonate.
 - D. Reacting dilute nitric acid with magnesium.
4. Write **True** or **False** for each of the following:
 - A. All nitrates decompose on heating.
 - B. Ammonia is the only common basic gas.
 - C. Nitrogen dioxide dissolves in water to form nitric acid only.....
 - D. Brown-ring test is used to identify nitrates.
5. State three products formed when concentrated nitric acid reacts with sulphur.
6. Write a balanced chemical equation for the reaction between phosphorus and concentrated nitric acid.

7. (a) Describe how potassium nitrate can be prepared in the laboratory from potassium carbonate.
- (b) Write a balanced chemical equation for the reaction.
8. Explain what happens when calcium nitrate is heated strongly. Include a balanced equation.

B. Skills

9. In the preparation of ammonia, a student set up an experiment as shown below.



- A. Identify and correct the mistake in the set up.
- B. Name substance A and state its function.
- C. Name the method used to collect ammonia gas in the set up and explain why it is suitable.
- D. Write a chemical equation for the reaction that took place in the round-bottomed flask.
10. For the laboratory preparation of nitric acid, list all apparatuses that maybe used and describe, specifying the safety precautions that should be taken into consideration to carry out the experiment.

C. Attitudes and values

11. Cultivating near water bodies leads to environmental pollution. Explain this with reference to effects of nitrogen compounds on the environment.

UNIT 3

SULFUR AND ITS INORGANIC COMPOUNDS



Key Unit Competence:

Relate the physical and chemical properties of sulphur and its inorganic compounds to their preparation, uses and effects on the environment.

Learning objectives

At the end of this unit, you should be able to:

Knowledge and understanding

1

- Recall the occurrence, properties and uses of sulphur.
- Explain the impact of sulphur oxides on the environment.
- State the properties and uses of sulphur compounds e.g. sulphuric acid.

Skills

2

- Prepare sulphur dioxide in the laboratory
- Use experiment to demonstrate the reaction of sulphur with water and alkalis.
- Use experiments to show the chemical properties of sulphur and its compounds.
- Test the presence of sulphates, sulphates and sulphur dioxide.

Attitudes and values

3

- Protect natural resources.
- Develop self- confidence in discussions and presentation of research findings.
- Develop a culture of working in a team during research and discussions.
- Respect of procedures during experiments.

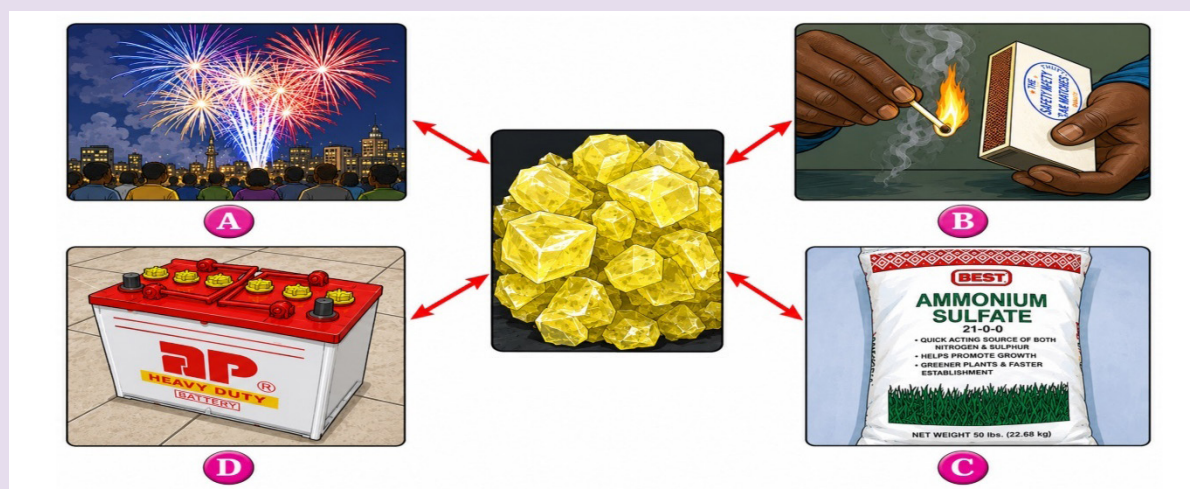
3.0. introduction

Sulfur occurs in nature as a free element or in the form of sulphates in its inorganic derivatives. It is the second member of group 16 elements in the periodic table. This unit explains its physical and chemical behaviour. Preparation methods, common properties, and uses of its compounds as well as environmental issues related are illustrated in the unit.

Introductory activity



Attentively observe the chart below and answer the questions that follow.



Guiding question:

1. What is the chart about ?
2. Investigate what A,B,C,D have in common.

3.1. Occurrence of sulphur

Learning activity 3.1



Carry out, with the help of textbooks and by watching videos, research about the occurrence of sulphur.

In its elemental form, sulphur occurs naturally in volcanic regions or regions where volcanoes were formerly active. Common sources include: Volcanic deposits around the boundaries of extinct and active volcanoes, hot springs where sulphur precipitates out due to geothermal activity.

Sulphur is also found largely as compounds:

- a) Metal sulphides : Galena (PbS), Zinc blende (ZnS), Iron pyrites (FeS_2), Copper pyrites (CuFeS_2)
- b) Metal sulphates: like Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)
- c) Hydrogen sulphide found (H_2S) in petroleum, coal gas and natural gas.

- d) Sulphur compounds like sulphur dioxide (SO_2) occur naturally in the atmosphere due to: volcanic emissions, decomposition and burning of organic matter.

Application activity 3.1



1. State any two natural sources of sulphur.
2. In nature, sulphur occurs as a free element or in combined states. Name any three compounds of sulphur.

3.2. Properties and uses of sulphur

3.2.1. Properties

Learning activity 3.2.1



Carry out, with the help of textbooks and by watching videos, research about the occurrence of sulphur.

Investigating the properties of sulphur

Apparatus and reagents

Test tubes, Beakers, Spatula, Candle, Sulphur powder, Carbon dioxide, Distilled water, Boiling tube, Bunsen burner, Deflagrating spoon, Carbon disulphide, Iron filings, Concentrated sulphuric acid, Concentrated nitric acid

Procedure

1. Put about 3cm^3 of distilled water in a test tube and add a half full spatula of sulphur powder. Shake well and record your observations.
2. Put about 3cm^3 of carbon disulphide in a test tube and add a half full spatula of sulphur powder. Observe and record what you have seen.
3. Place three spatula endful of sulphur in a boiling tube. Heat gently using candle flame. Do not let the molten sulphur darken in colour. Tilt the boiling tube slightly so that you can see how easily the liquid flows. Comment on its colour and viscosity.
4. Now continue heating the dark molten sulphur using a Bunsen burner flame. From time to time tilt the boiling tube to see how easily the molten sulphur flows. Do not let any liquid flow out of the boiling tube. Comment on the colour and viscosity.

5. Place sulphur powder on a deflagrating spoon and heat in a Bunsen burner flame until it starts to burn. Test the gas evolved with wet blue litmus paper and record your observations.

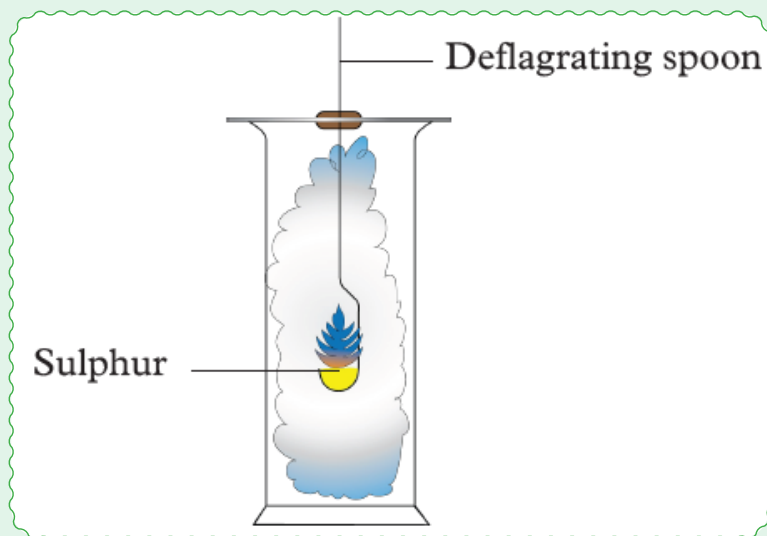


Figure 3.2.1: Burning sulphur in a deflagrating spoon.

6. Mix equal masses of sulphur powder and iron fillings in a boiling tube and heat gently then strongly. Record your observations.
7. To 3 cm³ of concentrated sulphuric acid in a test tube, add a half spatula endful of sulphur powder and warm the mixture gently. Test any gases produced using wet blue litmus paper.

Caution! This activity should be carried out in a fume cupboard since sulphur dioxide is a poisonous gas.

8. Repeat step (7) using concentrated nitric acid and record your observations.

Caution! This activity should also be carried out in a fume cupboard as nitrogen dioxide gas produced is poisonous.

Guiding questions

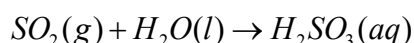
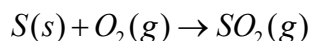
1. What is the colour of flame produced by burning sulphur?
2. Write chemical equations for the reactions that took place in procedures 5-8.

Physical properties of sulphur

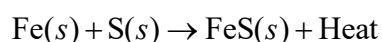
1. It is a bright yellow solid substance at room temperatures.
2. Sulphur is soluble in organic solvents such as carbon disulphide and benzene but it is insoluble in polar solvents like water.
3. It melts at 113°C and boils at 444°C.

Chemical properties of sulphur

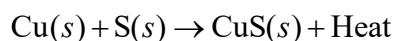
1. Sulphur burns easily with a blue flame in the presence of air to give fumes which turn wet blue litmus paper red. The gas formed is sulphur dioxide. Sulphur dioxide dissolves in water to form sulphurous acid (H_2SO_3).



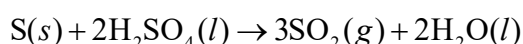
2. Sulphur combines with metals directly and easily to form the corresponding metal sulphides. For example, when a mixture of iron and sulphur is heated, it produces a red-hot glow and a black solid of iron (II) sulphide salt is formed. This reaction is exothermic and once it starts, the glow continues even after heat is withdrawn.



3. Similarly, hot powdered copper combines with sulphur on heating to form copper (II) sulphide.



4. When heated with concentrated sulphuric acid, sulphur is oxidised to sulphur dioxide which turns wet blue litmus paper red.



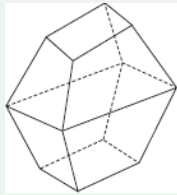
5. Hot concentrated nitric acid oxidises sulphur to sulphuric acid while the acid itself is reduced to brown nitrogen dioxide gas. This is a slow reaction.

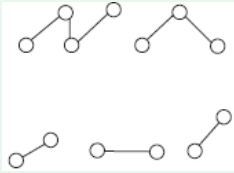


Note: These experiments should be done in the fume cupboard since both SO_2 and NO_2 are poisonous.

6. When heated, sulphur undergoes several changes as shown in the following table 3.1.1.

Table 3.2.1: Changes that take place in sulphur when heated at different temperatures

Temperature /°C	Physical change	Explanation
113	Sulphur melts to give a mobile amber coloured liquid; further increase in temperatures makes the liquid to darken.	Intermolecular forces are broken by heat. 

180	The liquid becomes viscous. Colour changes gradually from red to black.	The S ₈ rings that form sulphur molecules open up forming long chains. These chains entangle making them viscous.
250	Liquid becomes very thick and viscous and cannot flow.	The rings of 8 atoms open up as the molten sulphur is heated more strongly. The long chain entangles and makes liquid sulphur viscous.
Above 250	Colour becomes brighter again and the liquid becomes mobile again.	Short units escape from liquid as gas (sulphur vapour). 
At around 445	Red vapours are observed.	Sulphur boils at around 445°C

Uses of sulphur

1. It is used in the manufacture of sulphuric acid.
2. To vulcanise (harden) rubber; ordinary rubber is soft and elastic. However, when heated with sulphur, rubber becomes hard and tough making it suitable for use in making tyres.
3. Manufacture of matches, fireworks and gun powder. The explosive head of a match stick contains sulphur among other substances.
4. Used as a fungicide. Powdered sulphur is usually mixed with petroleum jelly and the mixture used as a skin ointment to treat fungal diseases in animals and plants.
5. Used in the production of important chemicals such as carbon sulphide (CS₂) and calcium hydrogen sulphite Ca(HSO₃)₂. Calcium hydrogen sulphite is used during the manufacture of paper to bleach wood pulp.
6. Making sulphur dyes which are used for dyeing cotton and artificial hair.
7. Making of sulphur drugs. Some antimalarial drugs contain sulphur.

8. Manufacture of sodium thiosulphate used in photography.

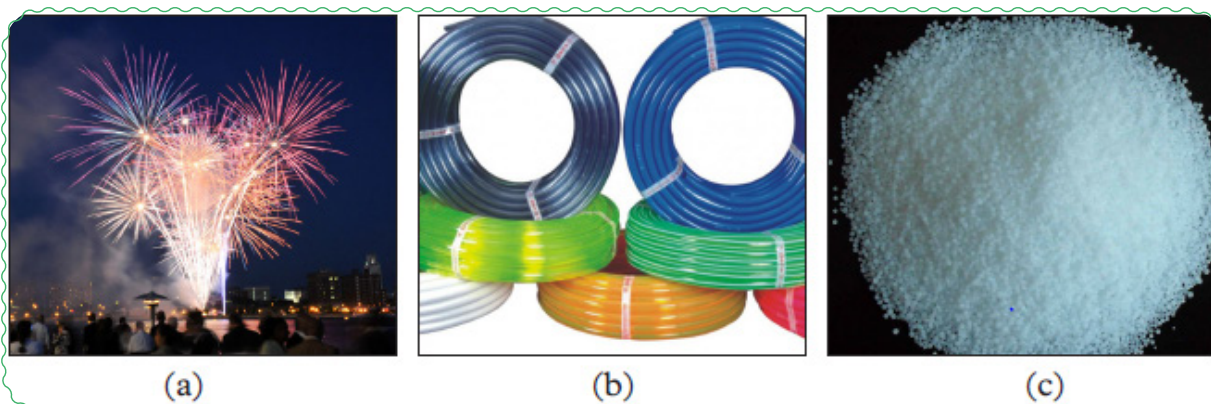


Figure 3.2.2: Products made of sulphur

Application activity 3.2



1. When sulphur is heated, it melts and flows easily. However, on further heating the liquid becomes viscous and does not flow. Explain.
2. State the observations made when:
 - a) Sulphur is heated in air.
 - b) A mixture of sulphur and concentrated nitric acid is heated.
 - c) Write equations for the reactions in (a) and (b) above.
 - d) State four uses of sulphur.

<https://www.samaterials.com/blog/sulfur-element-properties-and-uses.html>

3.3 Inorganic compounds of sulphur and their properties

3.3.1 Laboratory preparation of sulphur dioxide

Learning activity 3.3.1



Experiment illustrating the laboratory preparation of sulphur dioxide

Materials required

Flat bottomed flask, Dropping funnel, Gas jar, Wash bottle, Sodium sulphite, Dilute hydrochloric acid, Bunsen burner, Flower petals, Concentrated sulphuric acid, Sodium hydroxide solution, Blue and red litmus papers, Beakers, Distilled water.

Procedure

Caution: Sulphur dioxide is poisonous. This experiment should be carried out in a fume chamber. Place sodium sulphite in a flat-bottomed flask and set up the apparatus as shown below.

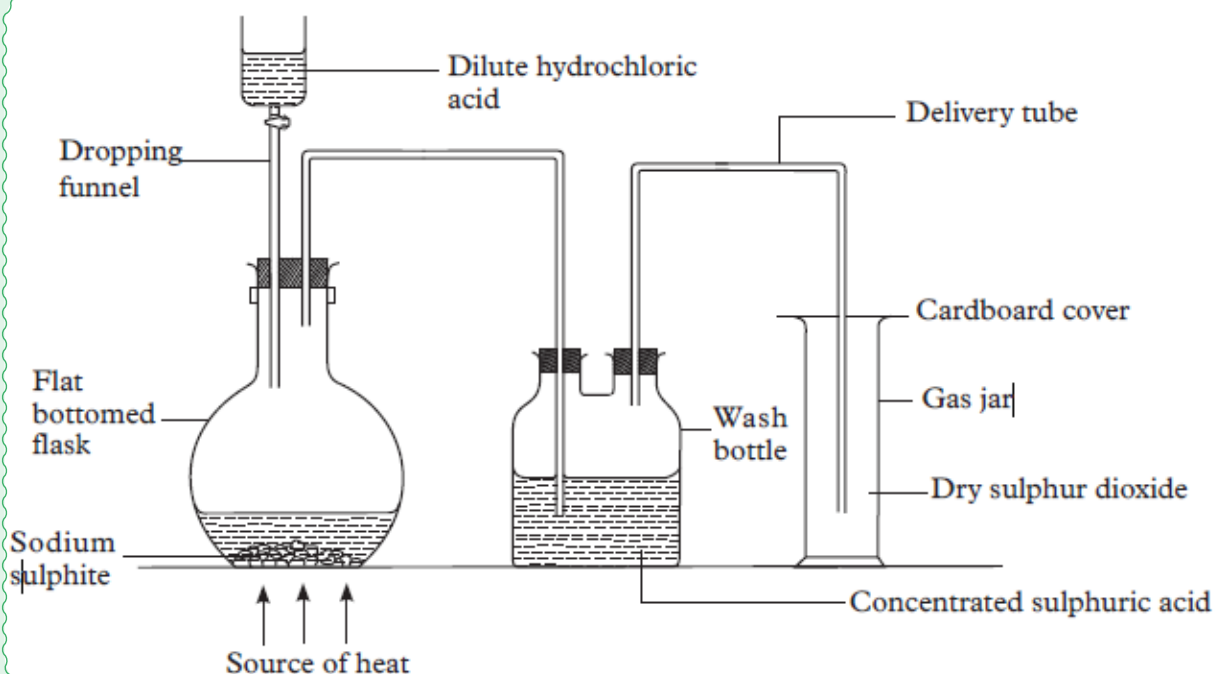


Fig 3.3.1: Preparation of sulphur dioxide using dilute hydrochloric acid and sodium sulphite

1. Run down the hydrochloric acid into flat bottomed flask.
2. Collect six gas jars of sulphur dioxide to be used in the next experiments. Note down the colour of the gas collected.
3. Invert a gas jar full of the gas in water using a funnel as shown below. Drop blue and red litmus papers into the solution.

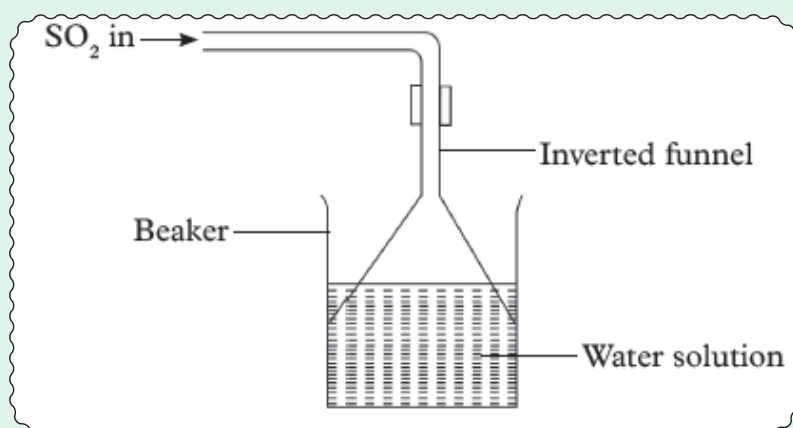


Figure 3.3.2: Reacting sulphur dioxide with water.

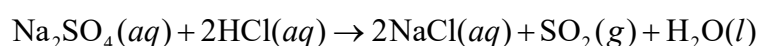
4. Drop pieces of wet blue and red litmus papers into a test tube containing sulphur dioxide. Note down your observations.

- Bubble sulphur dioxide through sodium hydroxide solution. Drop blue and red litmus papers into the resultant solution. Note down your observations.
- Drop red flower petals into a gas jar containing sulphur dioxide. Record your observations after 5 minutes.

Guiding questions

- In the preparation of sulphur dioxide gas, why was the gas passed through concentrated sulphuric acid?
- Name the method used to collect sulphur dioxide gas and explain why it is suitable.

Sulphur dioxide is collected by downward delivery. Wet litmus papers and coloured flower petals when placed in a jar containing sulphur dioxide, lose their colour. Blue litmus paper dropped in a solution of sulphur dioxide turns red. A solution of sulphur dioxide in water turns blue litmus paper red. Sulphur dioxide is prepared in a fume cupboard or in the open air because it is poisonous. When a mixture of sodium sulphite and dilute hydrochloric acid is warmed, effervescence occurs producing a colourless gas. The gas is sulphur dioxide.



The gas is passed through concentrated sulphuric acid to dry it. It is then collected by downward delivery since it is denser than air. Sulphur dioxide gas may also be prepared in the laboratory by heating a mixture of copper turnings with concentrated sulphuric acid using the following set up.

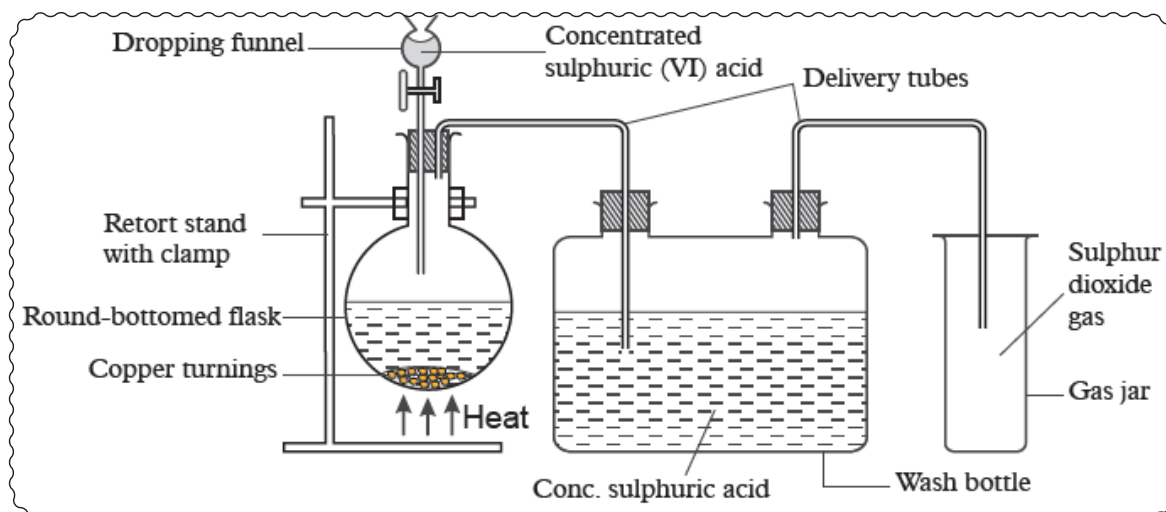


Fig 3.3.3: Preparation of sulphur dioxide gas using concentrated sulphuric acid and copper turnings.

When the reaction mixture in the flask gets hot, effervescence occurs and a colourless gas is collected by downward delivery. The equation for the reaction is as follows.



When the gas jar is full of the gas, the moist potassium dichromate paper placed at the mouth of the gas jar turns from orange to green because it is oxidised.

3.3.2 Physical and chemical properties of sulphur dioxide

Learning activity 3.3.2



Experimental illustration of the properties of sulphur dioxide

Materials required

Gas jars, Flower petals, Sodium hydroxide solution, Blue and red litmus papers, Beakers, Distilled water

Procedure

1. Invert a gas jar full of the gas in water using a funnel as shown below.
2. Drop blue and red litmus papers into the solution.

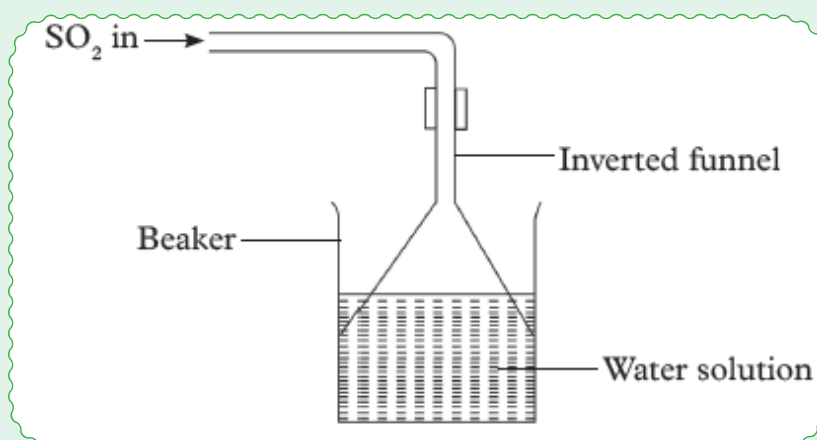


Figure 3.2.2: Reacting sulphur dioxide with water

3. Drop pieces of wet blue and red litmus papers into a test tube containing sulphur dioxide. Note down your observations.
4. Bubble sulphur dioxide through sodium hydroxide solution. Drop blue and red litmus papers into the resultant solution. Note down your observations.
5. Drop red flower petals into a gas jar containing sulphur dioxide. Record your observations after 5 minutes.

Guiding questions

1. What is the colour of sulphur dioxide?
2. Explain the effect of sulphur dioxide on:
 - a) Moist litmus papers
 - b) Flower petals

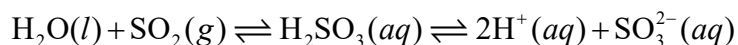
(a) Physical properties of sulphur dioxide

1. Sulphur dioxide is a colourless gas.
2. It has a pungent choking smell.
3. It is very soluble in water. This explains why water rushed into the gas jar in procedure 4 above. It is denser than air. This explains why it is collected by downward delivery.

(b) Chemical properties of sulphur dioxide

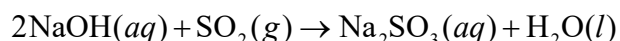
1. Reaction with water

Sulphur dioxide reacts with water to form sulphurous acid. This is a weak acid that partially dissociates to yield few hydrogen and sulphite ions. The presence of hydrogen ions explains why a solution of the gas turns blue litmus paper red.

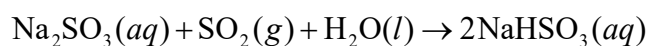
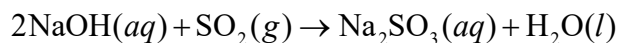


2. Reaction with alkalis

When sulphur dioxide is passed through sodium hydroxide solution, neutralisation reaction occurs. With excess alkali, sodium sulphite is formed as shown in the following equation.



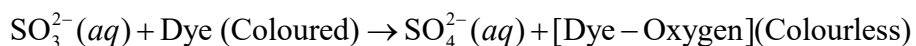
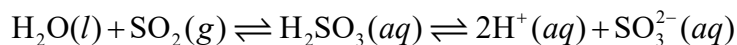
With excess sulphur dioxide, sodium hydrogen sulphite is formed as shown in the following equations:



These solutions have bleaching properties.

3. Bleaching action

Flower petals and wet litmus papers when placed in a gas jar of sulphur dioxide, they become bleached. However, dry litmus papers are not bleached. This is because sulphur dioxide gas must first dissolve in water to form sulphurous acid. It is this acid that thus reduces the dye in petals and wet litmus papers by removing the oxygen atom from them. The dye is thus said to be bleached.



Since this type of bleaching occurs by removal of an oxygen atom from a substance, it is called bleaching by reduction. Sulphur dioxide can only bleach in presence of water. The colour of the dye can be restored if the dye is exposed to environmental oxygen in the presence of sunlight. That is why old newspapers that had been bleached by reduction turn yellow/brown on exposure to air and sunlight after a long time.

Note: Bleaching means removing coloured materials from substances or removing colour.

(c) Uses of sulphur dioxide

1. It is an intermediate in the manufacture of sulphuric acid.
2. Used as a preservative in jam and fruit juices.
3. Used in fumigation to remove sources of infection from a house or room.
4. Used to make calcium hydrogen sulphite, $\text{Ca}(\text{HSO}_3)_2$ that is used to bleach wood pulp in the manufacture of paper.
5. Liquid sulphur dioxide is used in refrigerators as a refrigerant.
6. It is used for bleaching wool, straw, silk and sponges. These cannot be bleached using chlorine as they react with it.

Application activity 3.3



1. The laboratory preparation of sulphur dioxide uses sulphuric acid to dry the produced gas. Why calcium oxide cannot be used for this purpose?
2. Write a balanced chemical equation for the reaction between sodium sulphite and hydrochloric acid.
3. Explain why sulphur dioxide is collected by downward delivery.
4. Write a balanced equation of the reaction between sodium hydroxide with excess:
 - a) Sodium hydroxide
 - b) Sulphur dioxide
5. State any three uses of sulphur dioxide

<https://www.studocu.com/row/document/mount-kenya-university/industrial-chemistry/inorganic-chemistry-sulphur-and-its-compounds-overview-and-properties/127463889?sid=be3d6870-f72f-4107-a532-b737439fa6fc1783165151>

3.4 Properties of concentrated and diluted sulphuric acid and their uses

3.4.1. Properties

A) Concentrated sulphuric acid

Learning activity 3.4.1.a



Investigating the properties of sulphuric acid

Apparatus and reagents

Concentrated sulphuric acid, Beakers, 10 cm³ measuring cylinder, Sugar crystals, Hydrated copper (II) sulphate crystals, Distilled water, Boiling tubes, Pieces of cloth, Piece of paper, Piece of wood, Petri dishes, Stirrer, Test tube holder, Red and blue litmus papers, Felt pen, Copper turnings, Magnesium ribbon.

Procedure

1. Observe the colour and smell of concentrated sulphuric acid. Drop blue and red litmus paper in it. Record your observations.
2. Pour some water in a boiling tube. Holding the tube with a test tube holder in a slanting position, slowly add a little concentrated sulphuric acid by sliding it along the side of the tube and shake gently. Feel the boiling tube with your palm and note down your observations.
3. Place concentrated sulphuric acid into a beaker and mark its level with a felt pen. Place the beaker in an open but secure place and observe after 48 hours and record your observations.
4. Put 3 spatula endful of sugar crystals in 100 mL glass beaker.
5. Add about 5 cm³ of concentrated sulphuric acid into the beaker. Observe the changes in the beaker for about 10 minutes note down your observations.
6. Put 3 spatula endful of hydrated copper (II) sulphate crystals in a 50 mL glass beaker. To it add 5 cm³ of concentrated sulphuric acid note down your observations.
 - a) Place half spatula endful of copper turnings in a test tube and add 3 cm³ of concentrated sulphuric acid. Heat the test tube gently and test for any gas evolved by placing filter paper soaked in acidified potassium dichromate (VI) at the mouth of the test tube note down your observations.
 - b) Repeat procedure 6 (a) using magnesium metal in place of copper turnings. Note down your observations.
 - c) Pour concentrated sulphuric acid onto a piece of cloth, piece of paper and piece of wood. Record your observations.

Guiding question

Write balanced chemical equations for the reactions that took place in procedures 4, 5 and 6 (a) and (b).

Concentrated sulphuric acid converts sugar crystals to yellow, brown and finally to a black mass of substance. Blue copper (II) sulphate crystals in the acid turns into a white powder. On the other hand, brown copper turnings disappear in the acid forming a blue solution. When a piece of paper and a piece of cloth are placed in the acid, they get charred and torn into small pieces.

When concentrated sulphuric acid is placed in an open place for some time its level in the container rises up.

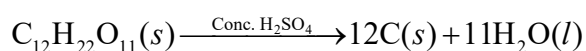
Physical properties of concentrated sulphuric acid

1. It is a colourless, odourless oily liquid.
2. It has a density of 1.84 g/cm³.
3. It dissolves in water with evolution of large amounts of heat, that is, why when added to water in the boiling tube, the boiling tube becomes hot.
4. It freezes at 10°C and boils at 338°C.

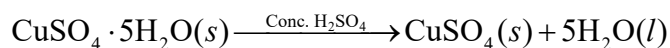
Note: When diluting the acid, always add the acid into water and never water into the acid and stir continuously. If water is added to the acid, a lot of heat is produced causing the solution to boil very violently, splashing the acid out of the container.

Chemical properties of concentrated sulphuric acid

1. **It is hygroscopic:** Concentrated sulphuric acid absorbs water from the atmosphere. That is why the level of water in the beaker in procedure (3) of activity 3.2 rose after 48 hours. This property makes concentrated sulphuric acid a suitable drying agent for gases which do not react with it.
2. **It is a dehydrating agent:** It is capable of removing combined water from substances. This explains the changes that occurred to sugar crystals in procedure (4) of activity 3.2. The black mass formed is carbon. The equation for the reaction that took place is:



Concentrated sulphuric acid as well removes water from the blue crystals of hydrated copper (II) sulphate converting it to a white powder of anhydrous copper (II) sulphate.



Similarly concentrated sulphuric acid removes water from wood converting it into a black mass of carbon.

Note: Dehydration is the removal of chemically combined water or elements of water from a compound. A substance capable of removing chemically combined water or elements of water is called a **dehydrating agent**.

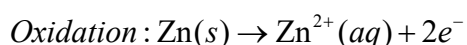
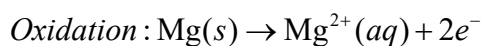
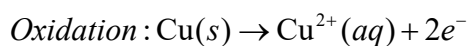
3. **It is corrosive:** Concentrated sulphuric acid should not come into contact with skin or cloth as it will corrode them. The same happens to paper when put into the acid.

4. It is an oxidising agent: Hot concentrated sulphuric acid acts as an oxidising agent. It oxidises almost all metals with the exception of those that are very low in the reactivity series such as gold and platinum. The following equations show the effect of concentrated sulphuric acid with metals.

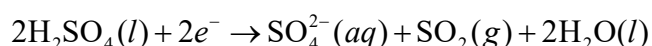


Brown

Blue



As seen from the above equations, the metals lose two electrons to form corresponding metal ions. This is oxidation. Concentrated sulphuric acid accepts the electrons lost and reduced into sulphur dioxide and water. This is reduction.



Note: Oxidation is the loss of electrons while reduction is the gain of electrons.

Mnemonics: **O**xidation **I**s **L**oss of electrons (**OIL**) while **R**eduction **I**s **G**ain of electrons (**RIG**).

B) Dilute sulphuric acid

Learning activity 3.4.1.b



Experiment to illustrate the properties of dilute sulphuric acid

Apparatus and reagents

Dilute sulphuric acid, Test tubes, Magnesium ribbon, Litmus paper (blue and red), Splints, Phenolphthalein indicator, Sodium hydroxide solution, Copper (II) oxide, Calcium hydroxide solution, Sodium carbonate, Sodium hydrogen carbonate.

Procedure

1. Observe the colour of dilute sulphuric acid.
2. Drop blue and red litmus paper in dilute sulphuric acid. Record your observation.
3. Place 3 cm³ of dilute sulphuric acid in a test tube, add magnesium ribbon and test any gas produced using a burning splint. Record your observation.
4. Place 3 cm³ of dilute sulphuric acid in a test tube. Add two drops of phenolphthalein indicator. Add sodium hydroxide solution drop wise until the mixture just changes colour. Record your observations.

5. To 3 cm³ of dilute sulphuric acid in a test tube, add some copper (II) oxide powder little at a time as you stir. Continue until no more copper (II) oxide dissolves. Record your observations.
6. Put 3 cm³ of dilute sulphuric acid into a test tube. Add sodium carbonate and connect it to another test tube containing calcium hydroxide solution as shown below.

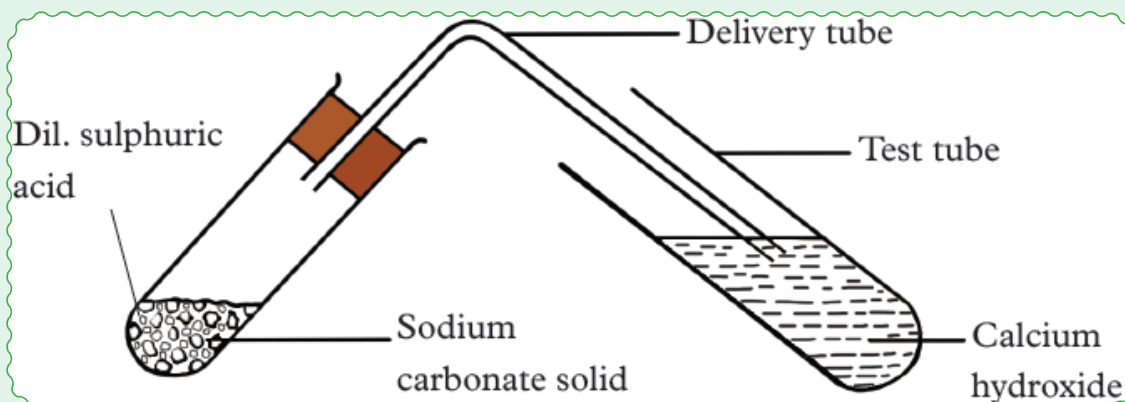


Fig 3.4.2: Reacting dilute sulphuric acid with sodium carbonate

7. Repeat the procedure using sodium hydrogen carbonate. Note down your observations.

Guiding questions

1. Write balanced chemical equations for the reactions taking place in procedures 4, 5 and 6.
2. What was the role of the following:
 - a) Phenolphthalein indicator in procedure 4?
 - b) Burning splint in procedure 3?
 - c) Calcium hydroxide in procedure 6?

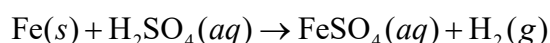
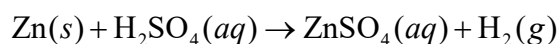
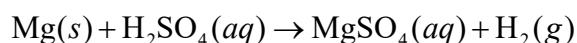
Blue litmus paper in dilute sulphuric acid turns red. Phenolphthalein indicator in a dilute sulphuric acid turns pink. Copper (II) oxide reacts with the acid forming a blue solution. When either sodium carbonate or sodium hydrogen carbonate reacts with dilute sulphuric acid, effervescence is seen due to production of a gas. The gas forms a white precipitate with lime water.

Physical properties of dilute sulphuric acid

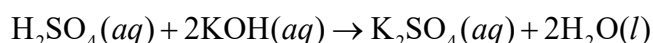
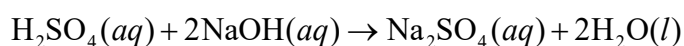
1. It is colourless.
2. It is and odourless.

Chemical properties of dilute sulphuric acid

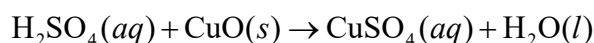
- 1. Reaction with metals:** Dilute sulphuric acid reacts with metals to form salt and hydrogen gas. However, metals like copper, silver and gold do not react with dilute sulphuric acid. They cannot displace hydrogen from the acid. Reaction of dilute sulphuric acid with some metals are shown below.



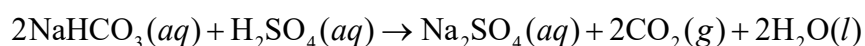
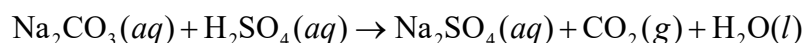
- 2. Reaction with hydroxides and hydroxides:** Dilute sulphuric acid reacts with hydroxides to form salt and water. This is a neutralisation reaction. Phenolphthalein indicator turns pink in a neutral solution. This is shown in the following reactions.



- 3. Dilute sulphuric acid also** neutralises metal oxides to form salt and water. It reacts with copper (II) oxide forming a blue solution of copper (II) sulphate as shown in the following equation.



- 4. Reaction with carbonates and hydrogen carbonates:** Dilute sulphuric acid reacts with metal carbonates and hydrogen carbonates to form salt, water and carbon dioxide. The carbon dioxide produced forms a white precipitate with lime water. The following equations show the reactions that take place when the acid reacts with sodium carbonate and sodium hydrogen carbonate respectively.



Note: When the carbonates of lead, calcium and barium are added to dilute sulphuric acid, the reaction starts soon and then stops. This is because the sulphates formed are insoluble; they coat the rest of the metal carbonate preventing further contact between it and the acid hence the reaction stops.

3.4.2. Uses

Learning activity 3.4.2



What are the uses of sulphuric acid?

A. Uses of concentrated sulphuric acid

1. Concentrated sulphuric acid is one of the most important industrial chemicals. It is used in many processes because it is strongly acidic, dehydrating, and a good oxidizing agent when hot.
2. Concentrated sulphuric acid is used to manufacture fertilisers, detergents, paints and dyes, plastics, explosives and drugs.
3. It is used as an electrolyte.
4. Concentrated sulphuric acid is used for the processing of metal ores during extraction of metals.
5. It is used in petroleum refining.
6. It is used for the production of other chemicals such as hydrochloric acid, hydrofluoric acid and metal sulphates.
7. Used as an electrolyte in car batteries
8. Processing of natural fibres and manufacture of synthetic fibres.
9. Used to dry gases such as carbon dioxide, chlorine, oxygen, and sulphur dioxide.

B. Uses of dilute sulphuric acid

1. Dilute sulphuric acid is used in laboratories as a reagent. It is used to react with metal and with bases and for preparing salts
2. It is used in metals cleaning
3. It removes oxide layers from metals before electroplating or painting
4. It is used as a cleaning agent in toilet cleaners and descaling agents.

Application activity 3.4



1. What happens when concentrated sulphuric acid is added to sugar?
2. State two properties of concentrated sulphuric acid that make it useful as a drying agent.
3. Give any two industrial uses of concentrated sulphuric acid
4. A student reacted dilute sulphuric acid with calcium carbonate. Explain the observation made. Use chemical equation in your explanation.
5. What is observed when dilute sulphuric acid is added to magnesium ribbon?
6. Write the equation of the reaction that takes place?

3.5 Tests for sulphates, sulphites and sulphur dioxide

3.5.1 Test for sulphates and sulphites

Learning activity 3.5.1



Experiment to test sulphate and sulphite ions

Materials

Test tubes, Sodium sulphite, Sodium sulphate solution, Barium nitrate, Dilute nitric acid

Procedure

1. Place 2 cm³ each of sodium sulphate and sodium sulphite in separate test tubes.
2. To each of the solutions in the test tubes, add four drops of barium nitrate solution and record your observations.
3. To each of the test tubes in procedure 2, add 2 cm³ of dilute nitric acid and shake.
4. Record your observations in the following table.

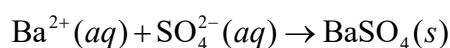
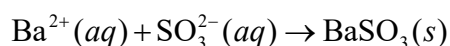
Table 3.5.1 Results for test for sulphite and sulphate ions

Ion in solution	Addition of four drops of barium nitrate	Addition of 2 cm ³ of dilute nitric acid
Sulfite ions SO_3^{2-}		
Sulphate ions SO_4^{2-}		

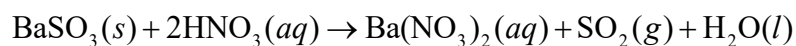
Guiding question

Write ionic equations for the reactions that took place.

Solutions containing sulphite and sulphate ions form white precipitate when barium nitrate or barium chloride solutions are added.



To differentiate between sulphite and sulphate ions, dilute nitric acid or dilute hydrochloric acid is added to the test tube containing the white precipitate. If the salt is a sulphate, the white precipitate persists. However, if the salt is a sulphite the precipitate dissolves giving off sulphur dioxide and a colourless solution is formed.



The identity of the sulphur dioxide gas may be confirmed by testing with a paper dipped in acidified potassium dichromate (VI) solution which changes colour from orange to green.

3.5.2 Test for sulphur dioxide

Learning activity 3.5.2



Laboratory test for sulphur dioxide gas

Materials

Source of sulphur dioxide gas (from a generator), Acidified potassium manganate (VII) solution, Acidified potassium dichromate (VI) solution, Iron (III) chloride solution, Test tubes, Labels.

Procedure

1. Place about 5 cm³ of acidified potassium manganate (VII) solution, acidified potassium dichromate (VI) solution and iron (III) chloride solution into three separate test tubes. Label them as A, B and C.
2. Pass sulphur dioxide gas from a generator through each of the solution in the test tubes.
3. Copy table 3.3.2 in your notebooks and record your observations and conclusions.

Table 3.3.2: Observations and conclusions on test for sulphur (IV) oxide

Solution tested	Observation	Conclusion
Acidified potassium manganate (VII) solution		
Acidified potassium dichromate (VI) solution		
Iron (III) chloride solution		

Sulphur dioxide decolourises an acidified solution of potassium permanganate. The gas also changes the colour of potassium dichromate from orange to green. A solution of iron (III) chloride changes from brown to green.

The three reagents potassium manganate (VII), potassium dichromate (VI) and iron (III) chloride are used to test for sulphur dioxide gas. The above colour changes and the bleaching action of sulphur dioxide are used to identify the gas.

Application activity 3.5



Explain how to test sulfite, sulphate ions and sulphur dioxide.

<https://www.youtube.com/watch?v=U1U1FyWHKM>

3.6. Environmental issues related to sulphur oxides

Learning activity 3.6



Environmental issues related to sulphur oxides

Research on the environmental impact of sulphur oxides. Base your Research on the effect of the oxides on plants and buildings, soil acidity and absorption of minerals by plants, aquatic life and formation of the acid rain.

When sulfur dioxide (SO_2) and sulfur trioxide (SO_3) are released into the atmosphere, they react with water vapor to form sulfurous acid (H_2SO_3) and sulfuric acid (H_2SO_4). These acids have significant environmental consequences.

1. Acid rain formation and its impacts

The acids formed in the atmosphere fall to earth as acid rain, causing soil and water acidification and the corrosion of infrastructures. The raise in acidity of the soil and water leads to the following issues:

- a) The acidification of the lakes, rivers, and soil, disrupting ecosystems
- b) The impairment of fish reproduction and death of acid-sensitive aquatic species
- c) The reduction of soil fertility and damage of forests and crops

The corrosion of infrastructure includes but is not limited to the following:

- a) The deterioration of buildings, monuments, and metal structures.
- b) The damage of limestone and marble through chemical weathering.

2. Air quality degradation

Sulfur dioxide contributes to smog and fog formation, significantly reducing visibility. At high concentrations, SO_2 cause

- a) Respiratory irritation in humans
- b) Aggravation of asthma and bronchitis
- c) Increased respiratory health risks in vulnerable populations

3. Direct damage to vegetation

When plants are directly exposed to sulfur dioxide, the gas damages plant tissues, resulting in:

- a) Chlorosis (yellowing of leaves due to chlorophyll breakdown)
- b) Reduced photosynthesis and stunted growth
- c) Increased susceptibility to diseases and nutrient deficiencies
- d) Premature leaf drops and reduced crop yields

Application activity 3.6



1. Explain ways in which sulphur compounds released to the atmosphere affect plants and aquatic animals.
2. Why is it important to protect our environment from pollution?

<https://www.canada.ca/en/environment-climate-change/services/air-pollution/pollutants/common-contaminants/sulphur-oxides.html>

3.7 END UNIT ASSESSMENT

End of unit assessment



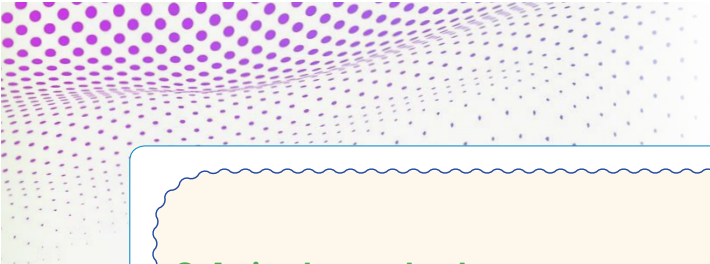
A. Knowledge and understanding

1. Elemental sulphur is commonly found near:
 - A. Deserts
 - B. Volcanoes
 - C. Oceans
 - D. Glaciers
2. Which gas containing sulphur is released during volcanic eruptions?
 - A. Oxygen
 - B. Sulphur dioxide
 - C. Nitrogen
 - D. Carbon dioxide
3. Sulphur occurs naturally in which of the following forms?
 - A. Elemental and combined
 - B. Only elemental
 - C. Only combined
 - D. Only organic

4. Sulphuric acid can be used for all of the following except:
- A. Manufacture of fertiliser
 - B. Making car tyres
 - C. Dehydrating sugar
 - D. Used in car batteries
5. What is the typical colour of elemental sulphur?
6. Write TRUE or FALSE for each of the following statements.
- a) Dilute sulphuric acid is a dehydrating agent.
 - b) Metals react with concentrated sulphuric acid to form salts and hydrogen gas.
 - c) Sulphur dioxide is an alkaline gas.
7. Kampfire and Gasasira wanted to prepare dilute sulphuric acid using concentrated sulphuric acid and distilled water.
- a) What procedure should they follow?
 - b) Why should they follow the procedure you have mentioned in 7 (a) above?
8. Powdered sulphur was placed in deflagrating spoon and heated in a Bunsen burner flame until the sulphur started to burn.
- a) What is observed as the sulphur burns?
 - b) Write an equation for the reaction in 8 (a) above.
9. Briefly explain how sulphur dioxide occurs in the atmosphere.

B. Skills

10. Powdered sulphur was placed in deflagrating spoon and heated in a Bunsen flame until the sulphur started to burn.
- a) What was seen when the sulphur burns.
 - b) Write an equation for the reaction in 10 (a) above.
 - c) When the product formed in 10 (b) is dissolved in water, name the solution that is formed and write the equation for the reaction.
 - d) If a few drops of litmus solution were added to the solution in 10 (c) above, what would be the colour of the litmus solution?
 - e) Write an equation for the reaction that will occur between the solution formed in 10 (c) above and sodium hydroxide solution.



C. Attitudes and values

11.

- a) Suggest ways through which air pollution can be reduced.
- b) To what extent is acid rain a problem in Rwanda?
- c) Acid rain is an international problem. Justify the statement.
- d) Should governments of different countries be doing something about the problem? Discuss.

UNIT 4

CHLORINE AND ITS INORGANIC COMPOUNDS



Key Unit Competence:

Relate the properties of chlorine and its compounds to its use, describe how compounds of chlorine are prepared and discuss the related environmental issues.

Learning objectives

By the end of this unit, you should be able to:

1 Knowledge and understanding

- Explain the preparation of chlorine gas.
- Describe the physical properties of chlorine and its compounds.
- Describe the chemical properties of chlorine and its compounds.
- Describe the toxicity of chlorine and its impact on the environment.

2 Skills

- Prepare chlorine and its compounds.
- Test the presence of chlorine.
- Prepare hydrogen chloride in the laboratory.
- Test for the presence of chlorides and other halides in solutions.
- Use experiments to demonstrate reaction of hydrochloric acid with metals, bases, oxides and carbonates.

3

Attitudes and values

- Protect natural resources.
- Develop self- confidence in the presentation of research, work and group discussions.
- Develop a culture of working in groups
- Appreciate the impacts of CFCs on the environment.
- Respect for the procedures in experiments.

4.0. Introduction

Chlorine in its elemental form does not occur naturally. Unit 4 analyses the ways used for its preparation. It analyses its interactions with others substances and point out its particular importance in the daily life. A particular emphasis is put on its harmful nature when uncombined and the issues related to the use of some its derivatives.

Introductory activity



Look at the picture below and suggest the importance of each represented item in our daily life.



Chlorine belongs to a group of elements in the Periodic Table known as the halogens. It is a highly reactive element. For this reason, chlorine does not occur in its free state in nature. Instead, it is found almost exclusively in combination with other elements. The most common form is sodium chloride, commonly known as rock salt. Rock salt is the source of table salt used in households. Sea water also contains significant amounts of sodium chloride. Other naturally occurring chlorine compounds include potassium chloride, magnesium chloride, and calcium chloride. Both elemental chlorine and its compounds have numerous practical applications.

4.1 Chlorine

4.1.1 Preparation and test for chlorine

Learning activity 4.1.1



Learning activity 4.1.1: Laboratory preparation of chlorine

Materials

Round bottomed flask, Conical flask, Gas jars, Manganese dioxide, Concentrated hydrochloric acid source of heat, Two wash bottles, Delivery tubes, Dropping funnel, Retort stand, Blue and red litmus papers.

Caution: Chlorine gas is poisonous and should be prepared in a fume chamber or in open air.

Procedure

1. Put manganese dioxide in a round bottomed flask and arrange the apparatus as shown in the following set up.

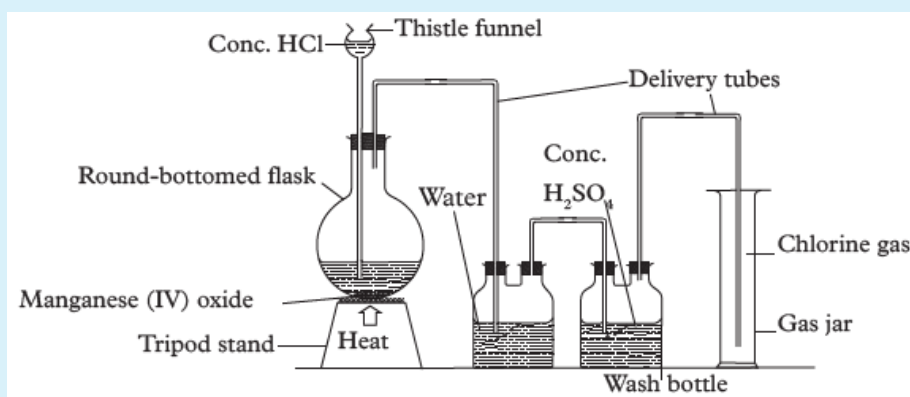


Figure 4.1.1: Preparation of chlorine gas in the laboratory

2. Add concentrated hydrochloric acid into the flask as shown and heat the mixture. Record your observations in the gas jar and in the flask.
3. Test the gas collected with moist blue and red litmus papers. Record your observations.
4. Smell the gas by wafting it towards your nose. Describe the smell of the gas.

Guiding questions

1. What do you observe as soon as concentrated hydrochloric acid is added to manganese dioxide?
2. Write an equation for the reactions that take place.
3. What is the function of water and concentrated sulphuric acid in the two wash bottles?

When dilute hydrochloric acid comes into contact with manganese dioxide, there is effervescence. A green-yellow gas collects in the gas jar. The gas turns wet blue litmus paper red and later bleaches it. Hydrochloric acid reacts with manganese (IV) oxide to form manganese (II) chloride, water and chlorine gas.



Water in the wash bottle absorbs hydrogen chloride fumes. The gas evolved is dried by passing it through concentrated sulphuric acid. Chlorine gas can also be dried by passing it through anhydrous calcium chloride. It is collected by downward delivery since it is denser than air. Chlorine gas can also be prepared using potassium permanganate instead of manganese dioxide as shown in the diagram below.

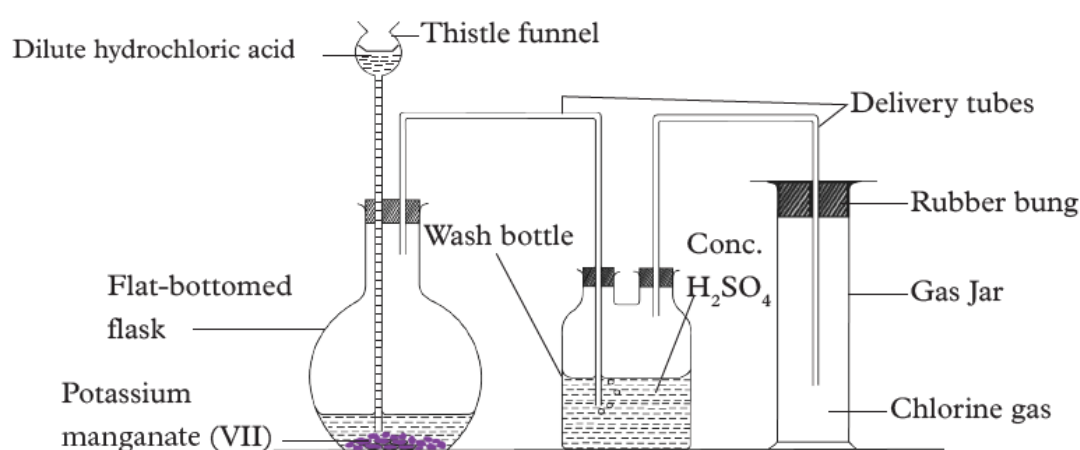
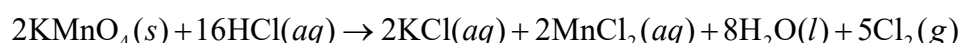
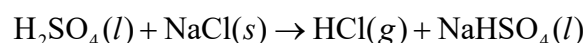


Figure 4.1.1: Preparation of chlorine using potassium permanganate and concentrated hydrochloric acid

In this method, no heating is required. The equation for the reaction that takes place is as follows:



Chlorine also can be prepared by heating a mixture of manganese (IV) oxide and sodium chloride with concentrated sulphuric acid. In this reaction, sulphuric acid first reacts with sodium chloride to form hydrochloric acid. The hydrochloric acid then reacts with manganese (IV) oxide to form chlorine gas.



The chlorine produce is tested using a blue litmus paper. When a moist blue litmus paper is placed in chlorine gas, it immediately turns red and then bleached.

Application activity 4.1.1



1. Chlorine gas is prepared in the laboratory through the reaction between concentrated hydrochloric acid and manganese dioxide. Write a balanced chemical equation for the reaction that takes place.
2. Gatete found a green-yellow gas in a gas jar. He wanted to confirm whether the gas was chlorine. Describe a method you would advise him to use.
3. Which of the following best describes chlorine?
 - A. A solid that readily combines with other elements.
 - B. A halogen.
 - C. A noble gas.
 - D. A mixture of carbon dioxide and ammonia.

<https://www.savemyexams.com/gcse/chemistry/aqa/18/revision-notes/8-chemical-analysis/8-2-identification-of-common-gases/8-2-4-test-for-chlorine/>

4.1.2 Physical properties of chlorine

Learning activity 4.1.2



Using internet, text books and referring to the outcomes of the learning activity 4.1.1, point out the physical properties of chlorine.

The following are the most common physical properties of chlorine:

1. It is a green-yellow gas with a pungent choking smell.
2. Boils at 34.05°C and freezes at -101°C .
3. It is 2.5 times denser than air hence collected by downward delivery.
4. It is fairly soluble in water (in aqueous solution it is called chlorine water).
5. It is easily liquefied under pressure.

Application activity 4.1.2



1. Which of the following statement does not describe physical property of chlorine?
 - A. The density of chlorine gas at standard temperature and pressure is 3.17g/l .
 - B. It combines with sodium to form table salt.

- C. It is a yellow green gas.
D. It freezes at -101°C
2. Chlorine is collected by downward delivery because:
- A. It is insoluble in water.
B. It is heavier than air.
C. It reacts with water.
D. It is lighter than air.
3. State any two physical properties of chlorine.

4.1.3 Chemical properties of chlorine

(I) Reaction of chlorine with metals

Learning activity 4.1.3



Activity 4.1.3 (a): Investigating the reaction chlorine with metals

Materials

Wooden splints, Glass rod, Magnesium ribbon, Iron wire, Hydrogen, Water, Sodium hydroxide, Gas jar, Chlorine, Source of heat, Deflagrating spoon, Conical flask, Combustion tube.

Procedure

1. Wrap cleaned magnesium ribbon on a deflagrating spoon and heat it until it starts burning.
2. Lower the burning magnesium in a gas jar of chlorine. Record your observations.

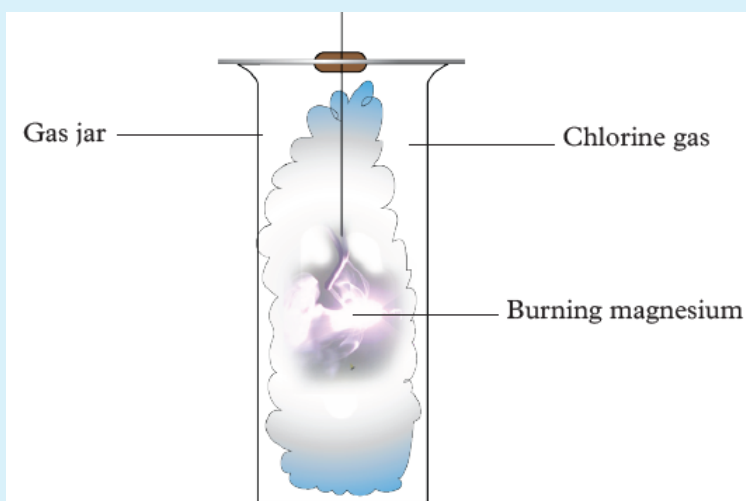


Figure 4.1.3 (a) Reaction between magnesium and chlorine gas

3. Place a piece of iron wire in a combustion tube and arrange the apparatus as follows.

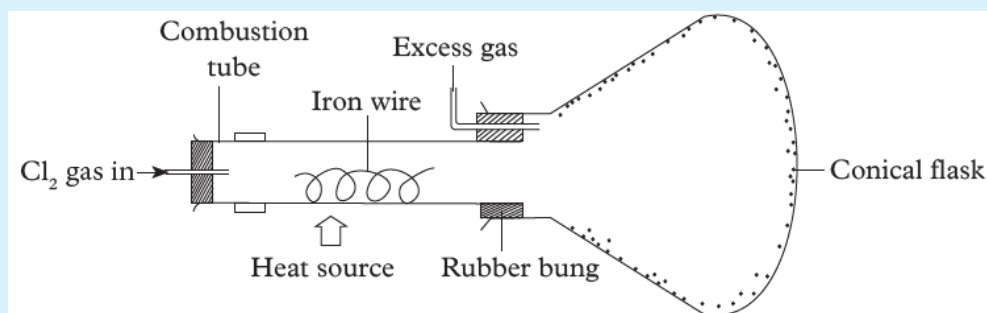


Figure 4.1.3 (b): Reaction of chlorine with iron

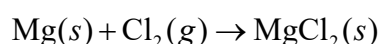
4. Heat the iron wire until it glows red-hot.
5. Pass dry chlorine gas through the combustion tube over the red-hot iron wire.
6. Warm the end of the combustion tube near the conical flask.

Guiding questions

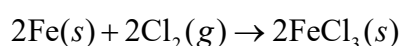
1. Write a chemical equation for the reaction that takes place when chlorine reacts with magnesium.
2. What observation did you make on the combustion tube when chlorine reacted with iron?
3. Write a chemical equation for the reaction between chlorine and iron.
4. Why is it possible to collect the product from the combustion tube into the conical flask as shown in figure 4.1.3(b)?

Magnesium burns vigorously in chlorine, producing a bright white flame and white fumes. When chlorine gas passes over red-hot iron wire, the wire glows continuously and a black substance forms on the combustion tube. Warming the tube causes black crystals to sublime and collect in the receiver. Chlorine combines directly with metals to form chloride salts.

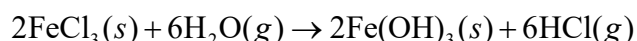
For instance, magnesium produces white magnesium chloride according to the following equation.



Chlorine reacts with iron to form black crystals of iron (III) chloride. When heated, this salt sublimes and condenses on the cooler sides of the combustion tube. This reaction is exothermic and so the glow continues for a long time.



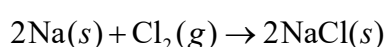
When exposed to moist air, iron (III) chloride slightly fumes. This is because iron (III) chloride reacts with water vapour in air to form white fumes of hydrogen chloride gas.



Therefore, during the preparation of iron (III) chloride, the whole apparatus should be dry.

Note: Where a metal forms two chlorides, the chloride of the higher oxidation number is formed because it is more stable. This explains why iron (II) chloride is not formed in this case.

Similarly, sodium metal burns spontaneously in chlorine to form white solid of sodium chloride.



(II) Reaction of chlorine with hydrogen and water

Learning activity 4.1.3.b

Investigating the reaction of chlorine with hydrogen and water

Materials

Wooden splints, glass rod, concentrated ammonium solution, water, beaker, matchbox, source of hydrogen gas, chlorine gas, gas jar.

Procedure

1. Introduce a burning splint into a gas jar of chlorine. Record your observations.
2. (a) Introduce burning hydrogen from a generator into a gas jar of dry chlorine gas as shown in figure 4.1.3.(c). Record your observations.

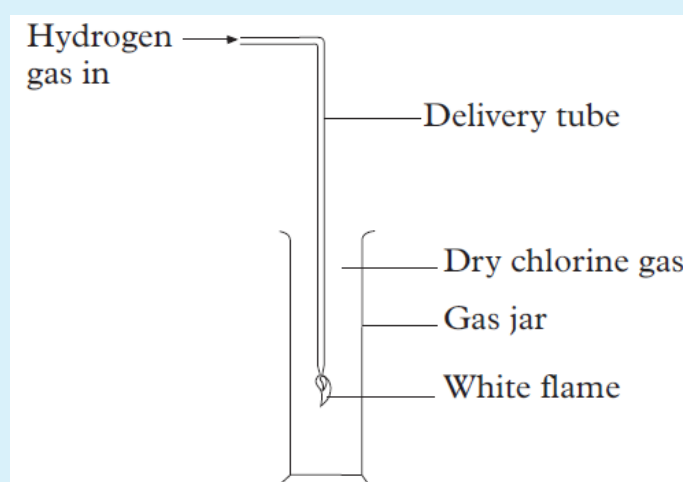


Figure 4.1.3.(c): Experimental set up for the reaction between chlorine and hydrogen

- (b) Dip a glass rod into concentrated ammonium solution and then insert it into the gas jar of chlorine. Record your observations.

3. (a) Pass chlorine gas through distilled water in a gas jar. Record your observations.

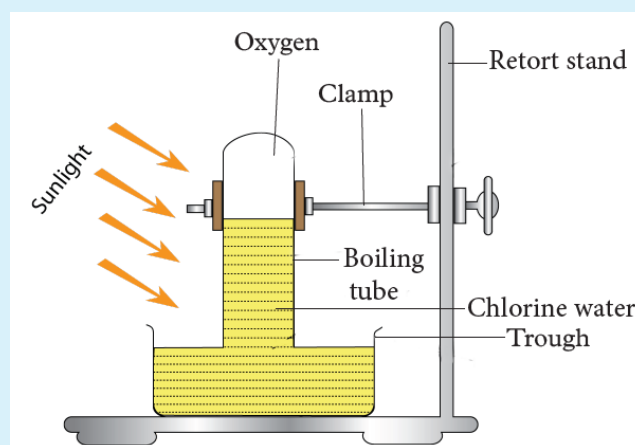


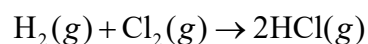
Figure 4.1.3 (d) Chlorine water in sunlight

- (b) Fill a boiling tube with chlorine water. Invert the boiling tube in a beaker containing distilled water as shown in Fig 4.1.3 (d) and leave the set up in sunlight for 2 days. Record your observations.
- (c) Test any gas collected with a glowing splint.

Guiding questions

1. Write chemical equations for the reactions that take place in procedures 2 and 3. Chlorine reacts with hydrogen to form a colourless gas.
2. Hydrogen chloride reacts with ammonia to form ammonium chloride as shown in the equation below. Ammonium chloride appears as dense white fumes.

Industrially, hydrochloric acid is manufactured by the direct reaction of hydrogen and chlorine gases:



Note: This reaction is highly exothermic. In bright sunlight or upon ignition, a mixture of hydrogen and chlorine reacts explosively, producing hydrogen chloride gas with a characteristic sharp odor.

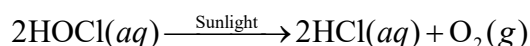
The gaseous HCl is then dissolved in water to produce commercial hydrochloric acid. When chlorine gas is bubbled through water, it undergoes a **disproportionation reaction**:



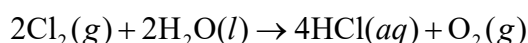
Note: A disproportionation reaction is a redox process in which the same element is simultaneously oxidized and reduced. In this reaction, chlorine (oxidation state 0) is reduced to Cl^- in HCl (oxidation state -1) and oxidized to Cl^+ in HOCl (oxidation state +1).

This produces a yellow-green solution called **chlorine water**, which contains hydrochloric acid (HCl), hypochlorous acid (HOCl) and dissolved chlorine gas (Cl₂). The yellow-green color is due to dissolved chlorine gas.

When chlorine water is exposed to sunlight, the unstable hypochlorous acid decomposes:



Oxygen gas is evolved, which can be observed displacing the solution from a container, as shown in Figure 4.1.3 (d). The solution gradually loses its yellow-green color as chlorine is consumed and oxygen is released. The net effect of chlorine water decomposition in sunlight can be summarized as:



This demonstrates chlorine's bleaching and oxidizing properties, which are exploited in water purification and disinfection.

(III) Reaction of chlorine with strong alkalis

Learning activity 4.1.3.c

Experiment to illustrate the reaction of chlorine with strong alkalis

Materials

Source of chlorine gas, Dilute and concentrated sodium hydroxide solutions, Dilute and concentrated potassium hydroxide solutions.

Procedure

- Bubble chlorine gas into cold dilute sodium hydroxide solution. Record your observations.
- Repeat the procedure using concentrated sodium hydroxide solution. Record your observations.

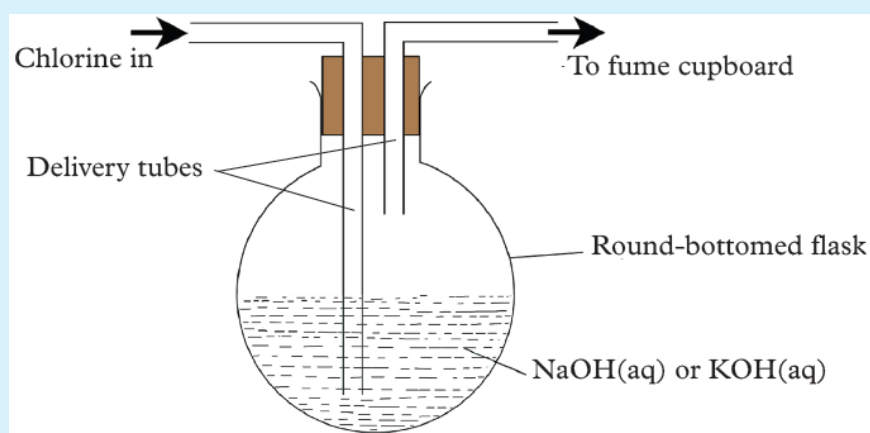


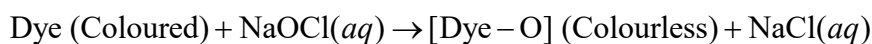
Figure 4.1.3 (c): Reaction of chlorine with concentrated and dilute sodium hydroxide

- Repeat procedures (a) and (b) using potassium hydroxide in place of sodium hydroxide. Record your observations.

1. Chlorine reacts with cold dilute sodium hydroxide solution to form a yellow solution of sodium hypochlorite, and sodium chloride as shown in the following equation.



2. Like chlorine, sodium hypochlorite has bleaching properties. Common household bleaches contain sodium hypochlorite as the active ingredient.



3. Chlorine reacts with hot concentrated sodium hydroxide forming sodium chlorate and sodium chloride according to the following equation.

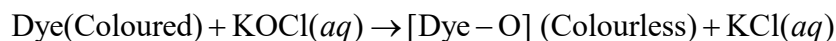


Similar reactions take place when chlorine is bubbled through cold dilute and hot concentrated potassium hydroxide solution. The following are the equations for the reactions that take place:

4. Chlorine reacts with cold dilute potassium hydroxide solution to form a solution of potassium hypochlorite, and potassium chloride as shown in the following equation.



5. Sodium hypochlorite, like chlorine, has bleaching properties. Common household bleaches contain sodium hypochlorite as the active ingredient.



6. Chlorine reacts with hot concentrated potassium hydroxide forming sodium chlorate and sodium chloride according to the following equation.



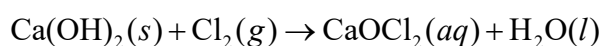
Therefore, the product formed when chlorine reacts with alkalis depends on the temperature and concentration of the alkali.

Note: The chlorates (sodium chlorate and potassium chlorate) are used to make explosives, fireworks, matches, weed killer and medicines such as throat drugs.

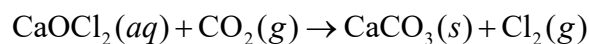
7. When chlorine is bubbled through calcium hydroxide solution, calcium chloride and calcium hypochlorite are formed.



8. When chlorine is passed over dry solid calcium hydroxide (slaked lime) a solid substance called bleaching powder (CaOCl_2) is formed.



The green-yellow colour of chlorine disappears as it reacts with calcium hydroxide. When exposed to the atmosphere, bleaching powder smells strongly of chlorine and deteriorates fast. This is because it reacts with atmospheric carbon dioxide releasing chlorine.



Application activity 4.1.3



1. When sodium reacts with chlorine, the product formed is:
 - A. Sodium oxide
 - B. Sodium hydroxide
 - C. Sodium chloride
 - D. Sodium chlorate
2. The reaction between iron and chlorine produces:
 - A. Iron (II) chloride only
 - B. Iron (III) chloride only
 - C. A mixture of iron (II) and iron (III) chlorides
 - D. Iron oxide
3. Which metal reacts most vigorously with chlorine?
 - A. Copper
 - B. Magnesium
 - C. Sodium
 - D. Iron
4. What type of reaction occurs when metals react with chlorine?
 - A. Neutralisation
 - B. Displacement
 - C. Combination
 - D. Decomposition
5. The general formula of metal chlorides formed is:
 - E. MCl
 - F. MCl_2
 - G. MCl_3
 - H. Depends on the valency of the metal

6. Why during the reaction of chlorine with water the apparatus should be dry?
7. Write a balanced chemical equation of the reaction between chlorine and hydrogen. What is the importance of this reaction?
8. When chlorine is bubbled in water, a mixture of two acids is formed. Illustrate the statement using a balanced chemical equation.
9. Chlorine reacts with cold dilute and hot concentrated potassium hydroxide similarly as it reacts with hot concentrated sodium hydroxide. Write a balanced chemical equation of the reaction.
10. Materials bleached with chlorine do not gain colour with time as opposed to those bleached using sulphur dioxide. Explain.

https://www.brainkart.com/article/Chlorine_38614/

4.1.4 Uses of chlorine and its compounds

Learning activity 4.1.4



What are the uses of chlorine and its compounds.

1. Chlorine has a wide range of uses in our daily lives. Some uses of chlorine are outlined below.
2. It is used in the treatment of water and sewage systems. Low concentration of chlorine is used in killing micro-organisms in domestic water supplies. A higher concentration is used for sterilising swimming pool water. Chlorine kills germs.
3. Chlorine is used in the manufacture of domestic bleaches which consist mainly of sodium hypochlorite used in washing and other bleaching agents used in cotton and paper industries.
4. Manufacture of hydrochloric acid.
5. Manufacture of weed killers such as sodium chlorate. Sodium chlorate is also used to make explosives, fireworks, matches and medicines such as throat lozenges. It is also used in the production of oxygen.
6. Manufacture of germicides, fungicides and pesticides such as DDT.
7. Manufacture of chlorofluorocarbons (CFC's) used to manufacture aerosol propellants, and used in refrigerators as 'freon' and air conditioning units due to their low boiling points.
8. Compounds of chlorine such as vinylchloride are used in the manufacture of plastics (PVC) used in making plastic water pipes, chairs, tables among other.
9. In the manufacture of trichloromethane (chlorofluoros) used as an anaesthetic.
10. Manufacture of cleaning solvents such as carbon tetrachloride.

4.2 Preparation and properties of hydrogen chloride gas and sodium chloride

4.2.1 Preparation and properties of hydrogen chloride gas

Learning activity 4.2.1.a



Preparation and properties of hydrogen chloride gas

Materials required

Round bottomed flask, Gas jars, Splints, Litmus paper, Concentrated ammonia solution, Bunsen burner, Wash bottle, Funnel, Beaker, Water, Glass rod, Magnesium ribbon, Zinc, Iron fillings, Aluminium turnings, Copper turnings, Concentrated sulphuric acid, Sodium chloride, Silver nitrate solution.

Procedure

1. Place sodium chloride in a round-bottomed flask and set up the apparatus as shown in figure 4.2.1 (a).
2. Add concentrated sulphuric acid dropwise until the sodium chloride is immersed.
3. Heat the flask gently and collect several gas jars of the gas that you will use to test for the chemical properties of the gas.

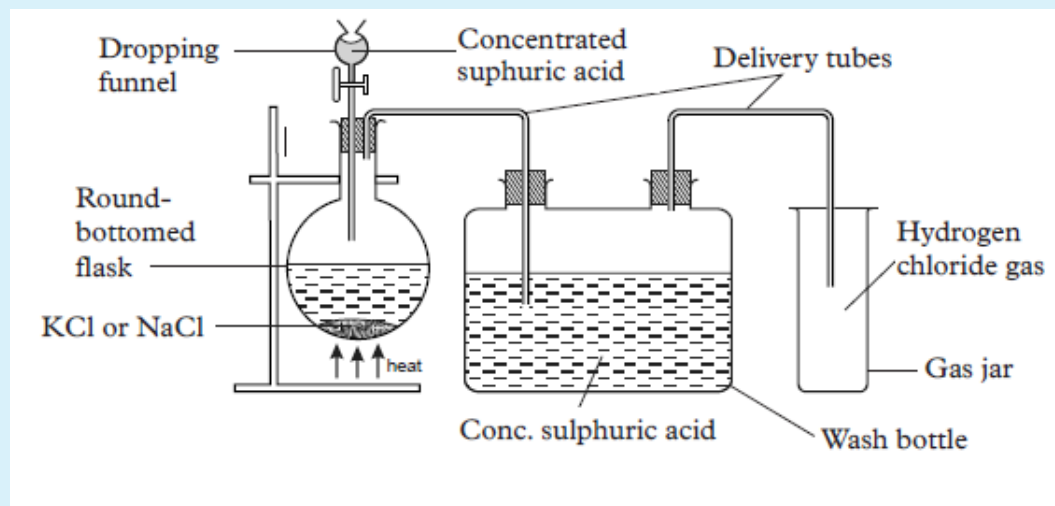


Fig. 4.2.1 (a): Laboratory preparation of hydrogen chloride

- a) Direct dry hydrogen chloride gas through the inverted funnel to a beaker of water as shown in figure 4.2.1 (b). Note down your observations.

- b) Insert moist blue and red litmus papers in a gas jar full of hydrogen chloride gas. Record your observations.

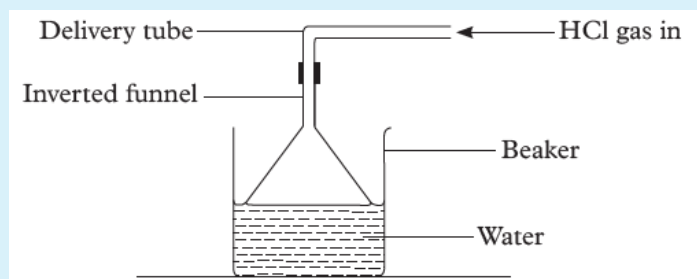


Fig 4.2.1 (b): Dissolving hydrogen chloride gas in water.

- c) Dip a glass rod in concentrated ammonia solution and insert it into a gas jar of hydrogen chloride gas. Record your observations.

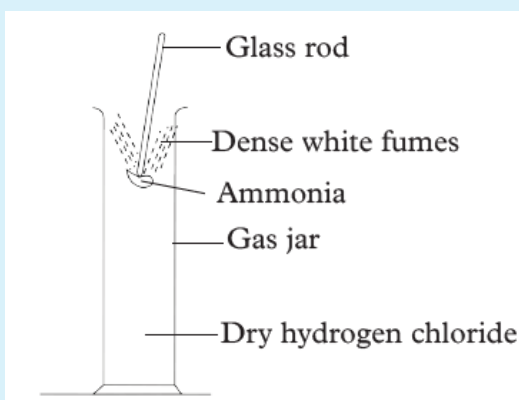


Fig 4.2.1 (c): Testing for hydrogen chloride using ammonia

- d) Lower a burning splint into a gas jar of hydrogen chloride as shown in figure 4.2.1 (c). Record your observation.

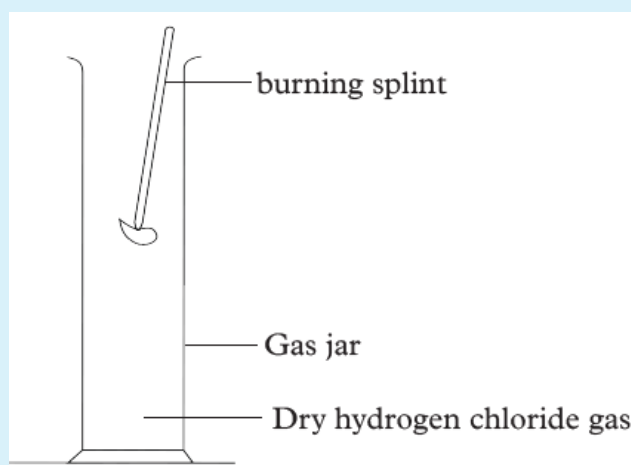


Fig 4.2.1 (d): Testing for hydrogen chloride using silver nitrate solution.

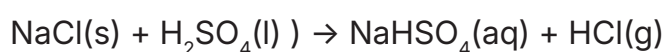
Place a glass rod that has been dipped into silver nitrate solution in a gas jar of hydrogen chloride gas. Record your observations.

Guiding questions

1. Why is the gas passed through concentrated sulphuric acid?
2. Describe two tests for hydrogen chloride gas.
3. Write a balanced chemical equation for the reaction that took place in the flask.

a) Preparation of hydrogen chloride gas

Hydrogen chloride gas is prepared in the laboratory by heating a mixture of sodium chloride and concentrated sulphuric acid. The equation for the reaction is as shown.



The gas is passed through concentrated sulphuric acid to dry it. It is then collected by downward delivery because it is denser than air. The gas can also be dried using anhydrous calcium chloride in a packed U tube. Hydrogen chloride is highly soluble in water forming hydrochloric acid. Thus the 'fountain' experiment can be performed to confirm that the gas is very soluble in water.

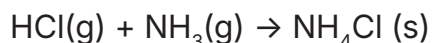
b) Physical properties of hydrogen chloride gas

1. It is a colourless gas.
2. It has a pungent choking smell.
3. Its boiling point is -85°C and its melting point is -114°C .
4. It is very soluble in water.

c) Chemical properties of hydrogen chloride gas

- Reaction with ammonia

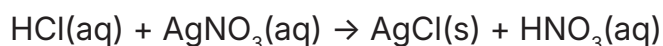
Hydrogen chloride gas reacts with ammonia gas to form dense white fumes of ammonium chloride.



Dense white fumes

- Reaction with acidified silver nitrate solution

When hydrogen chloride reacts with acidified silver nitrate, a white precipitate of silver chloride is formed. In this reaction, hydrogen chloride gas first dissolves in water contained in silver nitrate solution to yield hydrogen ions and chloride ions, (hydrochloric acid). The hydrochloric acid then reacts with silver nitrate to form white insoluble silver chloride.



White precipitate

Combustion

Hydrogen chloride gas extinguishes a burning splint. This means hydrogen chloride neither burns nor supports the burning of substances.

- Reaction with metals

Learning activity 4.2.1.b



Illustrating the reaction of hydrogen chloride with metals

Materials required

Iron fillings, Magnesium ribbon, Aluminium powder, Copper turnings, Glass tube, Hydrogen chloride gas from a generator, Trough, Gas jar, Water, Source of heat, Wooden splint.

Procedure

1. Set up the apparatus as shown in figure 4.2.1 (g).

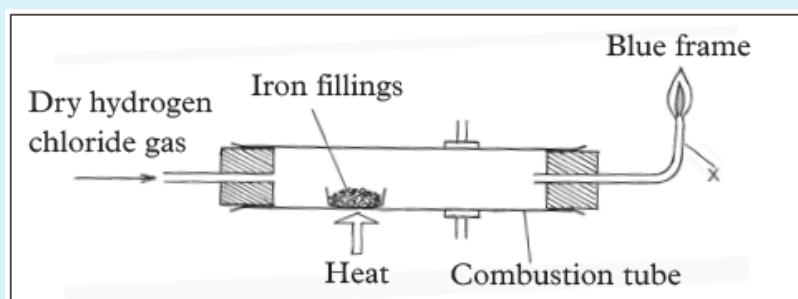


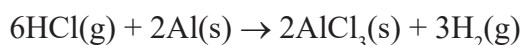
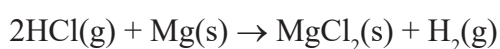
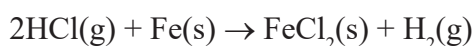
Figure 4.2.1 (g): Reacting metals with hydrogen chloride

2. Heat the iron fillings and record your observations.
3. Test the gas that is produced using a burning splint.
4. Repeat the experiment using cleaned magnesium ribbon, aluminium turnings and copper turnings instead of iron fillings. Record your observations.

Guiding question

Write chemical equations for the reactions that take place.

Hydrogen chloride gas reacts with some heated metals to form the corresponding metal chloride and hydrogen gas. The gas burns with a 'pop' sound. The equations below show the reactions that take place.



Note: Where the metal exhibits more than one valence, the chloride of the lower valence is formed. Hydrogen chloride gas does not react with copper because copper.

4.2.2 Preparation and properties of sodium chloride

Learning activity 4.2.2



Experiment illustrating the preparation and properties of sodium chloride

Materials

Sodium metal, Deflagrating spoon, Gas jars of chlorine gas or chlorine gas generator, Bunsen burner, Knife, Distilled water, Concentrated ammonia solution, Glass rod, Electrical circuit components.

Procedure

1. Place a small piece of sodium metal on a deflagrating spoon and heat it until it starts burning.
2. Insert the burning sodium in a gas jar of chlorine. Record your observations
3. Clean the solid formed and perform the following tests on it.
4. Using a knife cut a small piece of the solid and put it in a test tube. Add 2cm³ of water and shake. Note down your observations.
5. Test the electrical conductivity of the solution formed in 3(i) by connecting it to a electrical circuit.
6. To a small portion of solution in 3 (i) add silver nitrate solution. Note down your observation.
7. To another portion of the solid in a test tube, add concentrated sulphuric acid. Dip a glass rod in a concentrated ammonia solution and introduce it at the mouth of the test tube. Record your observations.

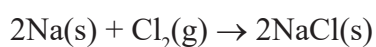
Guiding questions

1. Sodium chloride solution conducts electricity but the solid does not. Explain.
2. Write a chemical equation for the reaction that takes place in procedure 3(iv).

(a) Preparation

Sodium chloride is prepared by direct combination of sodium and chlorine. When burning sodium is introduced in a gas jar of chlorine, a vigorous reaction occurs and a white solid deposit is formed. The solid dissolves in water forming a colourless solution.

Sodium reacts with chlorine to form a white solid of sodium chloride.



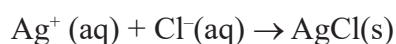
Sodium chloride solids dissolve in water to form sodium chloride solution. This solution is a good conductor of electricity.

b) Physical properties of sodium chloride

1. Sodium chloride is soluble in water.
2. It is a white crystalline solid at room temperature.
3. It has a high melting and boiling points.
4. It does not conduct electricity in solid state, but it conducts electricity in molten or aqueous state.

c) Chemical properties of sodium chloride

1. Sodium chloride reacts with concentrated sulphuric acid to produce hydrogen chloride gas which forms dense white fumes when brought near a bottle of ammonia.
2. A solution of sodium chloride in water forms a white precipitate with silver nitrate showing that it is ionised.



Application activity 4.2



Keza wanted to prepare dry hydrogen chloride gas in the laboratory. Draw a setup of apparatus that she would use.

Uwase found a certain gas in a gas jar. The gas had a pungent irritating smell. She however decided to carry out a chemical test to confirm if the gas was hydrogen chloride. Describe two ways she would do this.

- Write a balanced chemical equation between hydrogen chloride gas and zinc metal.
- Write a chemical equation of the reaction between sodium chloride and concentrated sulphuric acid.

4.3 Reaction of hydrochloric acid with metals, bases, manganese (IV) oxide, and carbonates

Learning activity 4.3



Experiment illustrating the effect of hydrochloric acid on metals, bases, manganese (IV) oxide and carbonates.

Materials required

Hydrochloric acid, Magnesium ribbon, Zinc, Iron, Copper, Sodium hydroxide, Copper (II) oxide, Calcium carbonate, Zinc carbonate, Copper (II) carbonate, Sodium carbonate, Manganese (IV) oxide, Test tubes, Sodium hydroxide, Phenolphthalein indicator, Wooden splint, Calcium hydroxide solution, Flat-bottomed flask, Thistle funnel, Gas jar.

Procedure

1. Put some pieces of zinc metal in a flat-bottomed flask and set up the apparatus as shown in figure 4.3.1 (a) Add 5 cm³ of dilute hydrochloric acid and test for any gas evolved using a burning splint.

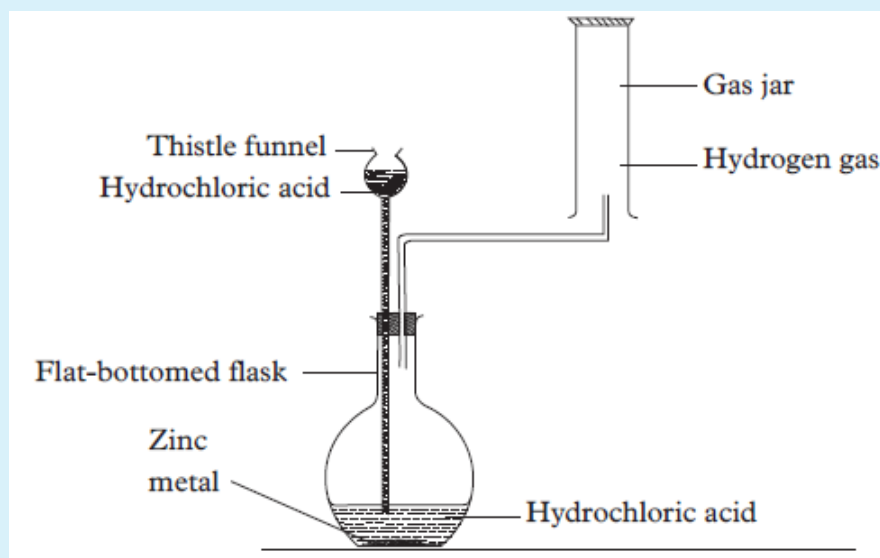


Fig 4.3.1.(a): Reaction between zinc and hydrochloric acid

2. Repeat procedure (1) using magnesium, iron and copper in place of zinc. Record your observations.
3. Place 5cm³ of sodium hydroxide in a test tube and add three drops of phenolphthalein indicator. Add dilute hydrochloric acid to the test tube drop wise until the colour just turns colourless.
 - a) Place some copper (II) oxide in a test tube, add 5cm³ of dilute hydrochloric acid and warm. Record your observations.

- b) Place some calcium carbonate in a test tube and add 5cm³ of dilute hydrochloric acid. Test for any gas produced using calcium hydroxide solution using the set up below.

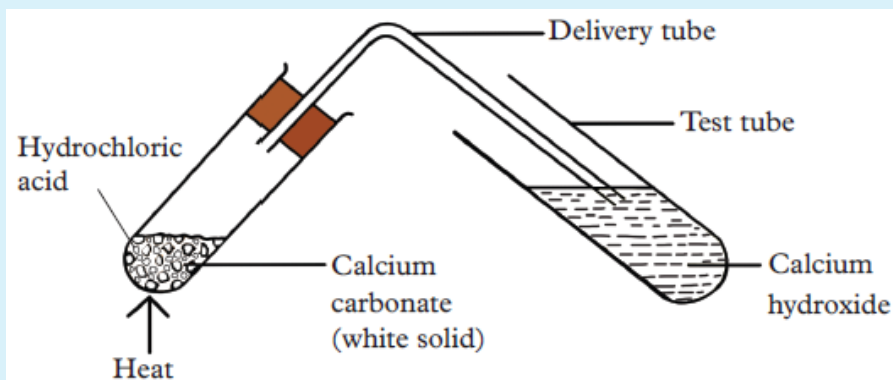


Fig 4.3 (b): Reacting calcium carbonate with hydrochloric acid

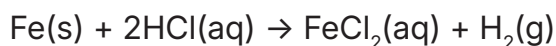
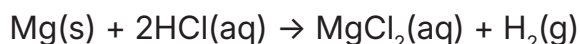
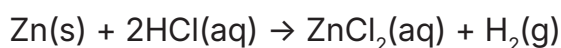
- c) Repeat procedure 3(a) using copper (II) carbonate, sodium carbonate and zinc carbonate. Record your observations. Place a small quantity of manganese (IV) oxide in a test tube and very carefully add some concentrated hydrochloric acid to it. Record your observations.

Guiding questions

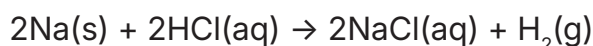
1. Explain the observation made when the gas produced in procedure (1) is tested using a burning splint.
2. Explain the observation made when the gas produced in procedure 3(a) is tested using calcium hydroxide solution.
3. Describe the smell and colour of the gas produced in procedure (4) and give its identity.
4. Write a balanced chemical equation for each of the reaction carried out.

4.3.1 Reaction of hydrochloric acid with Metals

Some reactive metals such as zinc, magnesium and iron react with hydrochloric acid liberating hydrogen gas and forming the corresponding metal chloride. This can be explained using the following equations.



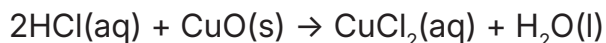
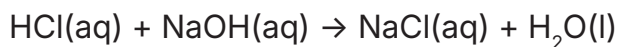
Very reactive metals such as sodium and potassium react violently with dilute hydrochloric acid forming a salt and hydrogen gas.



Copper metal is less reactive therefore, does not react with dilute hydrochloric acid.

4.3.2 Reaction of hydrochloric acid with bases

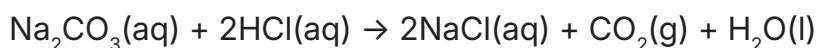
Hydrochloric acid reacts with bases to form salt and water. This is a neutralisation reaction. The equations for the reactions that take place are as follows.



The reaction of hydrochloric acid with oxides is used in the removal of oxides from metal surfaces (pickling) before coating them.

4.3.3 Reaction of hydrochloric acid with carbonates

Hydrochloric acid reacts with metal carbonates to form the corresponding metal chloride, water and carbon dioxide gas as shown in the equations below.



4.3.4 Reaction of hydrochloric acid with manganese (IV) oxide

Hydrochloric acid reacts with manganese (IV) oxide to form manganese chloride, water and chlorine gas.



In this reaction manganese (IV) oxide acts as an oxidising agent. It oxidises hydrochloric acid to chlorine while itself is reduced to manganese chloride and water.

In all these reactions, hydrochloric acid is behaving as a typical acid.

Application activity 4.3



1. Describe the industrial importance of sodium chloride.
2. Munezero reacted dilute hydrochloric acid with the following substances: manganese dioxide, copper granules, zinc granules and sodium carbonate.
3. What observation do you think he made in each case?
4. Explain the observations using chemical equations.
5. Hydrochloric acid is an important reagent. Give four uses of the acid in industries.

<https://allen.in/jee/chemistry/hydrogen-chloride>

4.4 Tests for chlorides and other halides ions in solution

Learning activity 4.4



Testing halides ions

Materials

Dilute nitric acid, silver nitrate, sodium chloride solution, sodium bromide solution, sodium iodide solution, ammonia solution, test tubes.

Procedure

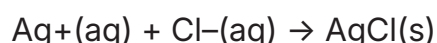
To 2 cm³ of sodium chloride solution in a test tube, add 1cm³ of dilute nitric acid followed by a few drops of silver nitrate. Record your observations.

- (i) To 2 cm³ of sodium bromide solution in a test tube, add 1cm³ of dilute nitric acid followed by a few drops of silver nitrate. Record your observation.
- (ii) To the resulting reaction mixture, add concentrated ammonia solution. Record your observation.
- (iii) To 2 cm³ of sodium iodide solution in a test tube, add 1cm³ of dilute nitric acid followed by a few drops of silver nitrate. Record your observation.
- (iv) To the reaction mixture, add concentrated ammonia solution. Record your observation.

Guiding question

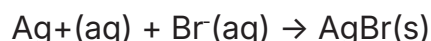
Write balanced ionic equations for each reaction that takes place.

When silver nitrate solution is added to an acidified solution of chloride ions, a white precipitate is formed. The precipitate is silver chloride.



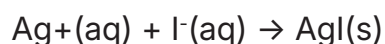
The solution is acidified by adding dilute nitric acid. The nitric acid helps remove other ions that might give confusing precipitates for example, carbonate ions (CO₃²⁻). The white precipitate is soluble in concentrated ammonia solution to form a colourless solution.

When silver nitrate solution is added to an acidified solution containing bromide ions, a pale-yellow precipitate is formed.



The pale-yellow precipitate is soluble in concentrated ammonia solution to form a colourless solution.

When silver nitrate solution is added to an acidified solution containing iodide ions a yellow precipitate is formed.



The yellow precipitate is insoluble in ammonia solution of any concentration.

Note: Silver chloride precipitate is obviously white, but silver bromide and silver iodide precipitates are not very different, hence addition of concentrated ammonia solution is used to differentiate between them as shown in table 4.4 below.

Table 4.4: Differences between the three silver halides

Original precipitate	Observation on addition of concentrated ammonia solution
AgCl	Precipitate dissolves in concentrated ammonia to give a colourless solution.
AgBr	Precipitate dissolves in concentrated ammonia to give a pale-yellow solution.
AgI	Precipitate (yellow in colour) is insoluble in ammonia solution of any concentration.

Application activity 4.4



Describe how you can confirm the presence of the following ions in a given solution.

- a) Cl^-
- b) Br^-
- c) I^-

<https://www.sparkl.me/learn/cambridge-igcse/chemistry-0620-core/test-for-halides-cl-br-i-using-silver-nitrate/revision-notes/3926>

4.5 Uses of hydrochloric acid and sodium chloride

Learning activity 4.5



Identify common uses of hydrochloric acid and sodium chloride

4.5.1 Uses of hydrochloric acid

1. Hydrochloric acid is essential for pickling, the process of removing oxide coatings from iron surfaces before galvanizing, tin plating, or chromium plating.
2. It is used in the production of chlorides such as calcium chloride and ammonium chloride.
3. Hydrochloric acid is used in manufacturing photographic materials.
4. When mixed with concentrated nitric acid in a 3:1 ratio, hydrochloric acid forms aqua regia, the only liquid capable of dissolving gold and platinum. This mixture is used for cleaning and purifying these precious metals.

5. It is used to standardize the pH of beers and wines, in sugar processing, and in gelatin production.
6. Hydrochloric acid is used in the production of photoflash bulbs and fireworks.
7. It serves as a common reagent in school and research laboratories.
8. Hydrochloric acid occurs naturally in the human stomach, where it aids in the digestion process.
9. It is used in the manufacture of aniline dyes.
10. Hydrochloric acid is used in the manufacture of pharmaceutical drugs.

4.5.2 Uses of sodium chloride

1. The following are some of the uses of sodium chloride:
2. It is used in the manufacture of sodium carbonate by the Solvay process.
3. Used in the manufacture of sodium hydroxide, chlorine and hydrogen.
4. Used in the extraction of sodium metal.
5. Used in the manufacture of soap during 'salting out'
6. Used in food as condiment / seasoning food.
7. Medically, sodium chloride solutions are used in catheter flush injections or intravenous injections and cleaning of contact lenses in the form of saline.
8. Sodium chloride is used as a defrosting chemical in cold countries where icing occurs on roads. It is normally added to roads during winter to help melt the ice.
9. Sodium chloride is used as a food preservative where there are no refrigerators for example in preserving fish and other meat foods.

4.6. Environmental and Health Hazards of Chlorine and Its Compounds

Learning activity 4.6



Research on the toxicity and environmental impact of chlorine and its compounds. The research should be guided by the following questions:

1. Name some of the compounds of chlorine that affect the environment.
2. How do the compounds identified in (1) above get into the environment.
3. How does each of the compounds named in (1) affect the environment?
4. How can environmental pollution by chlorine and its compounds be minimised?

Health check!

Chlorine is a poisonous gas. It was used as a chemical weapon during World War 1 where it was dropped from aero airplanes. Due to its high density, it would flow down to the ground mixing with air. Mixing of chlorine bleach (sodium hypochlorite) with ammonia or acidic cleaning agents is a common source of household exposure.

When chlorine is inhaled, it is pulmonary irritating with intermediate water solubility that causes acute damage in the lungs. Respiratory system is also the target of chlorine toxicity in animals. Inhaling of chlorine gas can cause difficulty in breathing, chest pains, cough, eye, irritation. Increases heartbeat, rapid breathing and can result in death.

Chlorine and its compounds are largely used in water treatment, industry, and agriculture, but they present many environmental problems when misused or released into the environment.

Chlorine gas is very toxic and can harm humans, animals, and plants if accidentally released.

In water bodies, chlorine and chlorinated wastewater can kill aquatic organisms even at low concentrations, disturbing ecosystems and food chains.

Some chlorine compounds, such as chlorofluorocarbons (CFCs) are non-biodegradable and hence have a long life span. They contribute to the depletion of the ozone layer. They release chlorine atoms in the upper atmosphere, which destroy ozone molecules, thinning the ozone layer. The destruction of ozone allows more harmful ultraviolet (UV) radiations to reach the Earth, increasing risks of skin cancer and damaging harvests and aquatic life.

Chlorine used in water disinfection can react with organic matter to form harmful by-products like trihalomethanes (THMs), which are persistent and potentially carcinogenic. Some organochlorine compounds such as DDT are persistent pollutants that accumulate in the food chain, causing long-term environmental and health effects. Use of DDT has been banned worldwide including Rwanda and other East Africa countries.

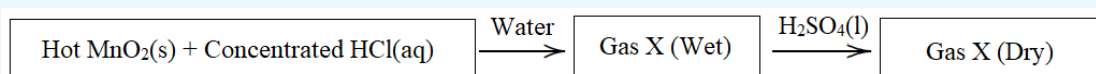
Polychlorinated biphenyl (PCB) is used in transformers in electricity substances, electrical capacitors and in washing machines. PCBs are used in inks, tapes, plastics, paints, glues, waxes and polishes. They are harmful to fish and other aquatic life because they interfere with reproduction. In humans they cause skin lesions, liver diseases and also damage the nervous system.

Inappropriate disposal of chlorine derivatives can lead to soil and water pollution, reducing soil fertility and contaminating drinking water sources.

Application activity 4.6



1. State one harmful effect of chlorine gas on living organisms.
2. How do chlorine derivatives affect aquatic life?
3. Which chlorine derivatives contribute to the depletion of ozone layer?
4. Name gas X.



5. What is the use of water?
6. What is the use of concentrated sulphuric acid?

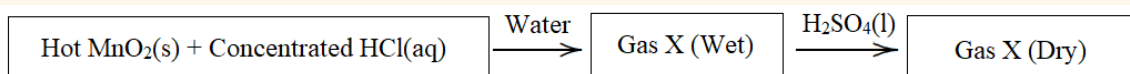
4.7 End of unit assessment

End of unit assessment



A. Knowledge and understanding

1. How does chlorine occur in nature?
2. Study the flow chart below then answer the questions that follow:

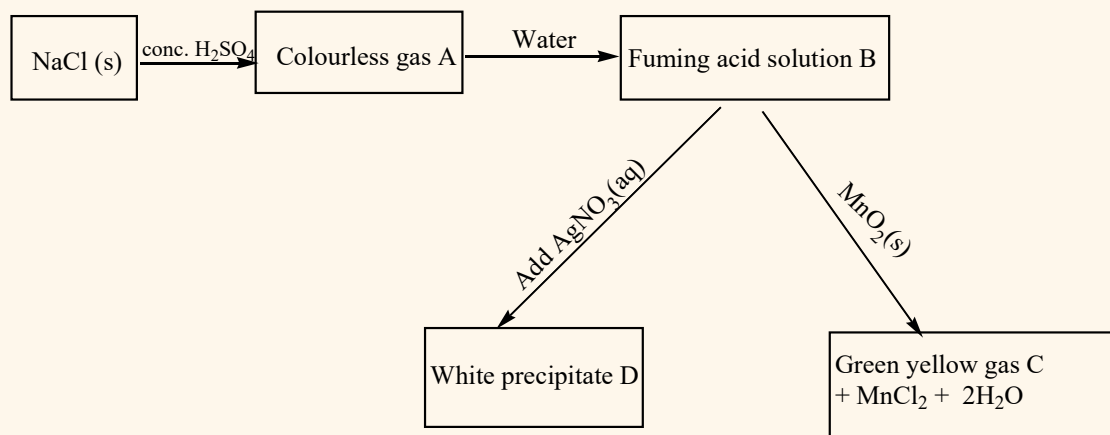


- a) Name gas X.
 - b) What is the use of water?
 - c) What is the use of concentrated sulphuric acid?
3. A small amount of chlorine added to water canit.
 4. In the presence of sunlight, chlorine reacts with hydrogen to yield hydrogen chloride gas.
 - a) Write the chemical equation of the reaction
 - b) Explain why the reaction is explosive.
 - c) What is the role of UV light in this reaction.
 - d) What type of reaction is this?

5. What is observed when the following metals are separately reacted with dry chlorine?
- Burning magnesium
 - Red-hot iron
6. Write an equation for the reaction between chlorine and sodium hydroxide solution.
7. Explain how the resulting solution acts as a bleaching agent.

B. Skills

8. Use the flow chart below to answer the questions that follow.



- Name gas A and C.
 - Name precipitate D.
 - Name the solid which can be heated with solution B to produce gas C.
9. A Senior 3 student prepares chlorine by reacting 12 g of MnO_2 with excess concentrated HCl.
- Write the balanced equation.
 - Calculate the number of moles of MnO_2 used.
 - Calculate the moles of chlorine gas produced.
 - State one assumption made in this calculation.

C. Attitudes and values

10. State any two safety precautions when conducting activities that involve chlorine gas.
11. Explain three effects of chlorine to the environment.
12. Give two measures that can be taken to reduce the effects of chlorine and its compounds to the environment.

UNIT 5

RATE OF REACTIONS



Key Unit Competence:

Describe and explain the effect of different conditions on the speed of reactions.

Learning objectives

Knowledge and understanding

1

- Demonstrate understanding of the rate of chemical reactions.
- Describe the effects of different factors that affect rate of chemical reactions.

Skills

2

- Compare the rates of reactions under different conditions using observable evidence.
- Use experiment to explain that a change in one condition affects the rate of a reaction.
- Evaluate the importance of controlling reaction rates in everyday life and industrial applications.

Attitudes and values

3

- Develop a culture of working in a team during discussions and research.
- Appreciate that reactions in everyday life like burning and rusting happen at different rates
- Respect for the procedures in an experiment.

5.0. Introduction

Processes happening in or outside living systems can proceed very quickly, slowly or even slowly depending on the prevailing conditions. This unit illustrates the common factors that modify the speed at which these processes take place.

Introductory activity



Look at the photograph below. It shows milk processing plant. The raw materials involved are milk, water and empty packets.



In order to increase the amount of packets of milk processed per day, what do you think the management requires to do? In chemical reactions, what does your answer to the question above compare to?

The rate of chemical reactions can be increased or decreased by altering certain factors that affect the reactions. For example, catalysts are used to speed up the rates of chemical reactions and remain unchanged at the end of the reaction. Other factors that affect the rate of chemical reactions are temperature, concentration of reactants and products and pressure. Others are particle size of reactants and light among others.

5.1 Reaction rate

Learning activity 5.1



Illustrating the concept of reaction rate

Materials

Nails, beaker, water in a tin, magnesium ribbon, a pair of tongs and bunsen burner and charcoal.

Procedure

1. Dip nails into a beaker of water.
2. Remove the nail from the beaker and place them in an open place. Note what happens to the nails after some days.
3. Using a pair of tongs place magnesium ribbon on a lit Bunsen burner. Note what happens to the magnesium ribbon.
4. Light the charcoal and record the time it takes to burn completely.

Guiding questions

1. How long does it take the nails to rust?
2. How fast does the magnesium ribbon burn?
3. Give the time taken by the charcoal to burn completely.

The rate of a reaction measures the speed at which a chemical reaction occurs. To determine reaction speed, we measure the rate at which reactants are consumed or products are formed over a given time period.

Therefore, the rate of reaction is defined as the change in amount or concentration of a specific reactant or product per unit time.

Different reactions proceed at different rates. Some are fast, while others are slow or very slow. For example, the combustion of magnesium is very fast, the combustion of charcoal is slow, and rusting of iron is very slow.

Application activity 5.1



1. Suggest two examples of slow reactions and two examples of fast reaction.
2. Name one experiment used to measure the rate of reaction.
3. The rate of a chemical reaction may be defined as:
The change in concentration of any one of the products only.
The change in concentration of any one of the reactants only.
The time it takes for the reaction to be completed.
The change in concentration of any one of the reactants or products.

In an experiment the following carbon dioxide gas volumes were recorded by a student from an acid-limestone reaction.

Time (min)	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0
Volume (cm ³)	0	8	15	21	26	30	33	36	38	39	40	40	40

At what time did the reaction stop?

6.0 min 5.0 min
5.5 min 4.5 min

5.2 Effects of different conditions on the speed of the reaction

5.2.1 Temperature

Learning activity 5.2.1



Investigating the effect of temperature on the reaction rate

Materials

250cm³ conical flask, thermometer, stop clock, piece of white paper, 50cm³ measuring cylinder, sodium thiosulphate solution (40dm³), hydrochloric acid (2mol/dm³), test tubes, clamp and stand, tripod, gauze, graph paper.

It is not necessary for these exact temperatures to be used although the temperature used should **not** exceed 55°C.

Note: Dilute hydrochloric acid will be added to sodium thiosulphate solution at different temperatures in a series of experiments.

Procedure

1. Using a 50 cm³ measuring cylinder, measure out 10 cm³ of the 0.25 moldm⁻³ sodium thiosulfate solution provided and 40 cm³ water and pour this mixture into a conical flask.
2. Using a 10 cm³ measuring cylinder, measure 5 cm³ of the 2.0 moldm⁻³ hydrochloric acid into a test tube.
3. Prepare a water bath by half-filling a beaker with water and place the beaker over a tripod and gauze.
4. Clamp the conical flask in the water bath and also place the test tube in the water bath.

- Place the thermometer in the conical flask and very gently heat the water bath until the contents of the flask reach the required temperature for the experiment.
- Remove the conical flask and the test tube from the water bath.
- Place the conical flask on the paper marked with a cross and immediately add the acid from the test tube as you start the stopwatch. Gently stir the mixture with the thermometer. Record this as the initial temperature.

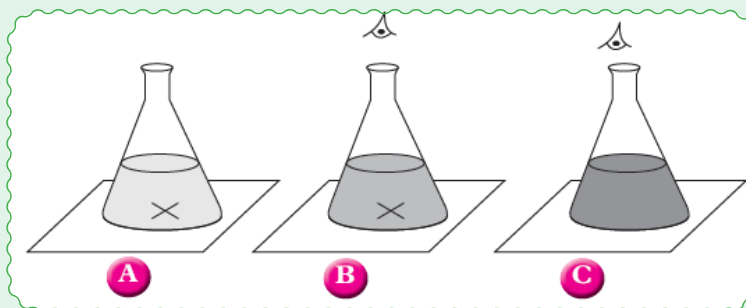


Figure 5.2.1.(a): Experiment on the effects of temperature on rate of reaction

- Observe the cross from above through the solution and stop the stopwatch as soon as the cross disappears from your view.
- Record the time taken for the cross to be obscured and record the final temperature of the mixture.
- Repeat the experiment for each of the five temperatures suggested in the Table 5.2 (a) and record all of your results.

Table 5.2.1: Results of the conditions that affect reaction rate

Experiment number	1	2	3	4	5
Temperature	20	25	30	35	40
Time taken					

- As soon as possible, pour the solution down the sink (in a fume cupboard if possible) and wash away.

Guiding Questions

- What is the name of yellow substance formed?
- Write equation for the reaction that occurs.
- What other observations are made during the reaction?
- For each set of results, calculate the value of $\frac{1}{\text{time}}$. (This value can be taken as a measure of the rate of reaction for this experiment).

- Plot a graph of on the vertical (y) axis and average temperature on the horizontal (x) axis.
- Comment on the shape of the graph.
- What is the effect of increasing temperature on the reaction time?
- What is meant by saying that two quantities are directly proportional?

The rate of a reaction measures the speed at which a chemical reaction occurs. To determine reaction speed, we measure the rate at which reactants are consumed or products are formed over a given time period. Therefore, the rate of reaction is defined as the change in amount or concentration of a specific reactant or product per unit time. The rate of a reaction can be determined by measuring either the decrease in reactant concentration or the increase in product concentration per unit time.

Where $\Delta[\text{Reactant}]$ and $\Delta[\text{Product}]$ represent the changes in reactant and product concentrations, respectively, and Δt represents the time interval.

Example: The Sodium Thiosulphate-Acid Reaction

When sodium thiosulphate reacts with an acid, a yellow precipitate of sulphur forms:



The rate of this reaction can be measured by observing the time required for a black cross marked on white paper beneath the reaction vessel to become obscured by the sulphur precipitate. Since the amount of sulphur needed to obscure the cross is constant, the reaction rate is inversely proportional to time: Rate $\propto 1/\text{time}$.

Several factors influence the rate at which chemical reactions proceed, including temperature, concentration of reactants, surface area of solid reactants, presence of catalysts, and the nature of the reactants themselves.

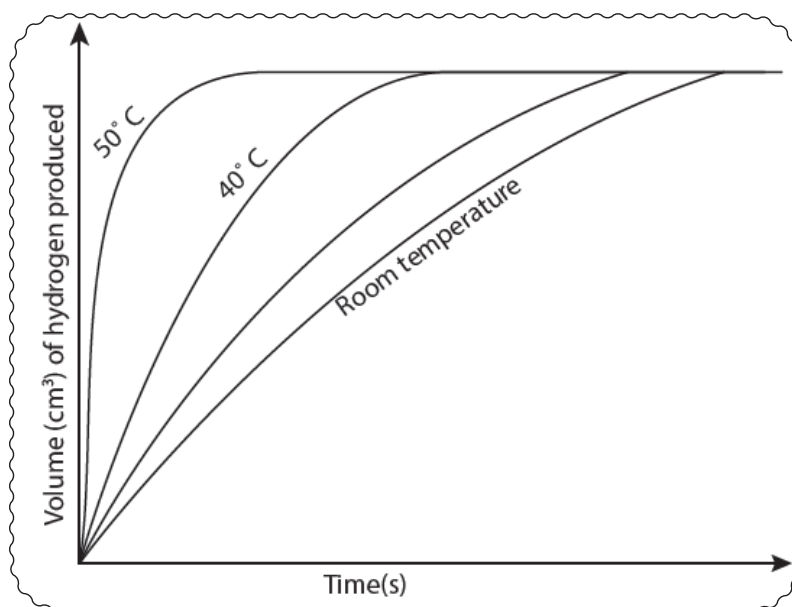


Figure 5.2.1 (b): Graph of volume versus time

When the temperature of the reactants is increased a higher volume of hydrogen is produced per unit time. The volume of hydrogen produced per unit time is a measure of the rate of reaction because the velocity of all the reacting particles increases as the temperature increases.

Application activity 5.2.1



Calcium carbonate in the form of limestone dissolves in hydrochloric acid giving off carbon dioxide gas. Which of the following conditions would produce the fastest reaction?

- Temperature of 40°C and 10cm³ of dilute acid
- Temperature of 20°C and 1010cm³ of dilute acid
- Temperature of 40°C and 510cm³ of dilute acid and 5cm³ of water
- Temperature of 20°C and 510cm³ of dilute acid and 5cm³ of water

Potatoes cook more quickly in boiling vegetable oil than in boiling water. This is because oil

- Boils at a higher temperature than water.
- Contains water.
- Replaces the water in the potatoes.
- Catalyzes the cooking process.

5.2.2 Concentration

Learning activity 5.2.2



Illustrating the effect of the concentration on the reaction rate

Materials

- 250 cm³ conical flask
- Piece of paper marked X
- 100 cm³ measuring cylinder
- Sodium thiosulphate solution of different concentrations (0.32M, 0.25M, 0.18M, 0.13M and 0.06M)
- 2M hydrochloric acid
- Stop watch

Procedure

1. Put 50 cm³ of 0.32M sodium thiosulphate solution into a conical flask.
2. Measure 5 cm³ of 2M hydrochloric acid into the measuring cylinder.

3. Add the acid to the conical flask and immediately start the stopwatch. Swirl the flask to mix the solutions and place it on a piece of paper marked with a cross.

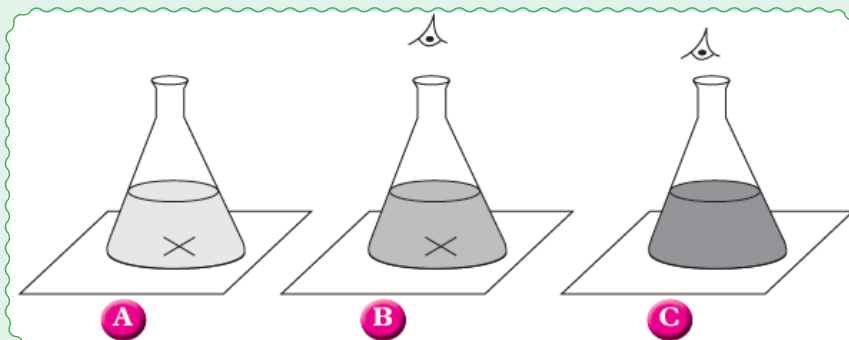


Fig 5.2.2.(a): Experiment on effects of concentration on rate of reaction

4. Look down at the cross from above. When the cross disappears stop the watch and note the time taken. Record this in a table like the one below.

Table 5.2.2(b) Results of the experiment

Experiment N ^o	1	2	3	4	5
Concentration of Na ₂ S ₂ O ₃ [mol/dm ³]	0.32	0.25	0.18	0.13	0.06
Time taken by the cross to disappear (s)					
1/t (1/s)					

5. Repeat the experiment using the different concentrations of sodium thiosulphate solution. As soon as possible, pour the solution at the end of every reaction down the sink (in a fume cupboard if possible) and wash away.

Guiding questions

1. Write the equation for the reaction that occurs.
2. For each set of results, calculate the value of 1/t and fill the table. (This value can be taken as measure of the rate of reaction).
3. Plot a graph of 1/t on the vertical (y) axis and concentration of the thiosulphate on the horizontal (x) axis.
4. Comment on the shape of the graph.
5. What is the effect of raising the concentration on the reaction?

When sodium thiosulphate and dilute hydrochloric acid are mixed, a yellow precipitate of sulphur is formed. The higher the concentration of sodium thiosulphate solution the less the time taken for the cross to disappear (get obscured).

The higher the concentration of the reactants (see Figure 5.2.e), the faster the rate of reaction. This is because increasing the concentration of reactants increases the frequency of fruitful collisions between reacting particles. The greater the number of collisions, the faster the rate of reaction.

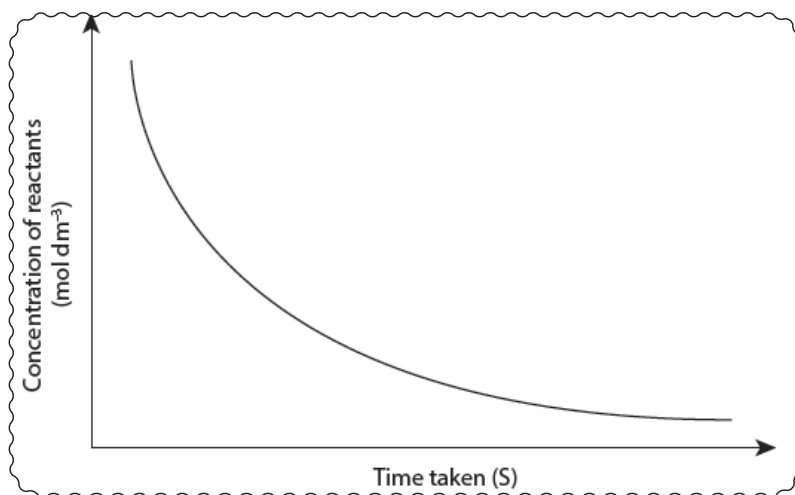


Fig 5.2.2.(b): Graph of concentration of reactants versus time

The frequency of collisions between reacting particles is influenced by the number of particles closer to each other in given volume unit.

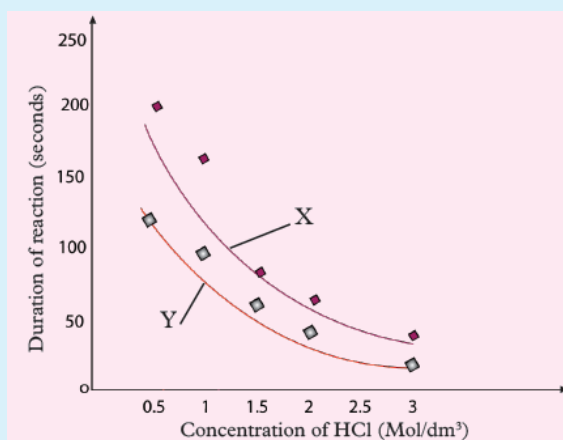
Application activity 5.2.2



Which one of the following reactions will start with the highest rate?

1. 5 g of limestone powder in 100 mL of 0.1 M HCl at 45°C.
2. 5 g of limestone chips in 100 mL of 0.1 M HCl at 25°C.
3. 5 g of limestone powder in 100 mL of 0.1 M HCl at 25°C.
4. 5 g of limestone chips in 100 mL of 0.1 M HCl at 45°C.

The following graphs were obtained in an experiment to determine the rate of the reaction between nickel carbonate and hydrochloric acid.



Write a word equation for the reaction that is taking place.

Which of the graphs shows the reaction that had the most concentrated hydrochloric acid? Explain your answer.

5.2.3 Pressure

Learning activity 5.2.3



Investigating the effect of the pressure on the reaction rate

Look at the syringes below. If the end is sealed, how can you increase the pressure of the gases inside the syringe?

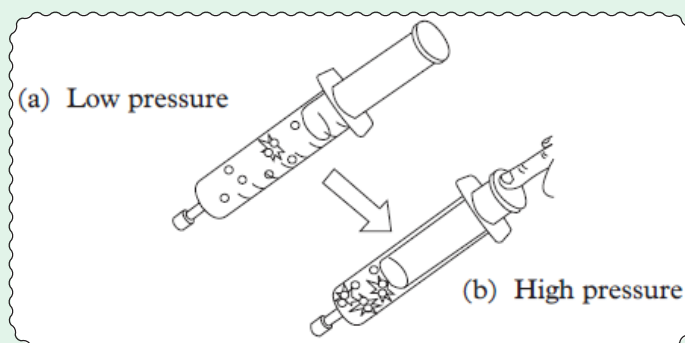


Fig 5.2.3: Syringes showing how pressure can be increased

To increase pressure, you push the plunger downwards (inward as shown). By pressing the plunger in, you will have the same number of gas particles in a smaller volume. This reduces the space in which collisions take place thereby increasing the chances of fruitful collisions in a given time. As a result, the rate of reaction is increased.

Materials

Two identical plastic syringes (without needles), Two effervescent tablets, Water, Stopwatch, Clamp or rubber band, Measuring cylinder

Procedure

1. Add 10 cm^3 of water to each syringe.
2. Drop an effervescent tablet into each syringe.
3. Immediately:
4. Syringe A: Leave the plunger free (low pressure).
5. Syringe B: Push the plunger down to reduce volume and hold it fixed with a clamp or rubber band (high pressure).
6. Start the stopwatch for both syringes at the same time.
7. Observe and record:
8. Speed of gas production
9. Time taken for the plunger to move noticeably or for fizzing to stop.

Guiding question

In which syringe the gas is produced more quickly? Explain.

In syringe A, the reaction is slower than in syringe B because a higher pressure is applied in syringe B. For reactions involving gases, higher pressure means gas particles are closer together. This leads to more frequent collisions and therefore the speed of the reaction increases.

Application activity 5.2.3



1. Which apparatus is most suitable for measuring the rate of a gas-producing reaction?
 - A. Measuring cylinder
 - B. Burette
 - C. Gas syringe
 - D. Thermometer
2. When pressure is increased at constant temperature, gas particles:
 - A. Become larger
 - B. Move further apart
 - C. Are forced closer together
 - D. Stop moving
3. Pressure has little effect on reactions involving only solids because:
 - A. Solids do not react
 - B. Solid particles are already closely packed
 - C. Solids have no kinetic energy
 - D. Solids have low mass

5.2.4 Catalyst

Learning activity 5.2.4



Effect of a catalyst on the reaction rate

Materials

3 conical flasks, Measuring cylinders (10 cm³), Dropping pipette, Zinc granules, Copper turnings or powder, Dilute sulphuric acid, 0.5M copper (II) sulphate solution

Procedure

1. Put a few pieces of zinc granules into each of the three conical flasks. Name them A, B and C. (Weigh approximately 5 grams of zinc into each conical flask).
2. Add 5 cm³ of dilute sulphuric acid to conical flask A. Measure the amount of gas produced as shown below..

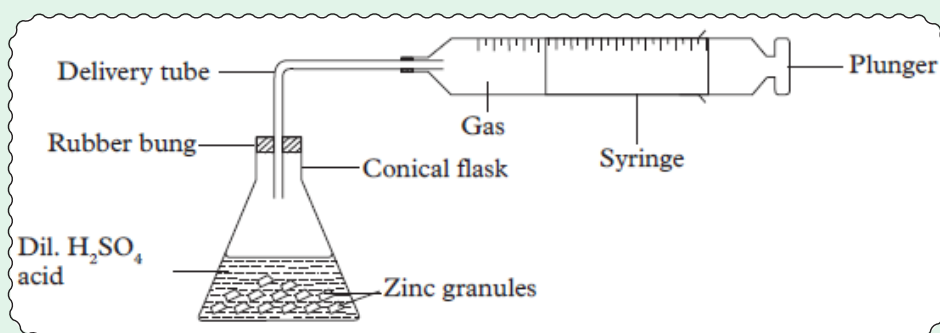


Figure 5.2.4: Experimental set-up for the reaction between HCl and magnesium

3. In conical flask B, add a few copper turnings. Make sure they are in contact with the zinc granules. Add 5 cm³ of dilute sulphuric acid and insert the rubber bung quickly. Measure the amount of gas produced.
4. Add 5 cm³ of dilute sulphuric acid to conical flask C. Add about 1 cm³ of copper (II) sulphate solution using a dropping pipette. Insert the rubber bung quickly. Measure the amount of gas produced as in procedure 2 above.
5. Record the results in a table like the one shown below.

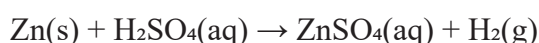
Table 5.2.4 Results of the experiment

Experiment	A	B	C
Amount of gas produced (cm ³)			

Guiding questions

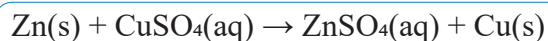
1. Write equation for the reaction between zinc and dilute sulphuric acid.
2. What was the purpose of adding copper (II) sulphate solution in experiment C?
3. Write the equation for the reaction between copper (II) sulphate and zinc metal.
4. Explain how you would test for the gas evolved.

Flask 1 (Pure zinc and sulfuric acid): Very pure, granulated zinc reacts slowly with dilute sulfuric acid. The shiny surface shows the zinc is pure with very few impurities. Surprisingly, this purity actually slows down the reaction. Impure zinc reacts faster because the impurities help the reaction occur. Bubbles of hydrogen gas form on the zinc surface:



Flask 2 (Zinc, sulfuric acid, and copper): Adding copper turnings makes the reaction faster than in Flask 1. The copper acts as a catalyst, reducing the amount of energy needed to start the reaction. However, the reaction is still slower than in flask 3.

Flask 3 (Zinc, sulfuric acid, and copper sulfate solution): This flask shows the fastest reaction. First, the zinc displaces copper from the copper sulfate solution:



The copper forms a black coating on the zinc surface. This copper then acts as a catalyst, making the hydrogen gas production much faster. The reaction is faster than flask 2 because the copper deposits directly onto the zinc, creating better contact between them and making the catalyst work more effectively.

Application activity 5.2.4



1. Hydrogen peroxide decomposes at room temperature to give oxygen and water according to the equation:
 - A. Which piece of apparatus could be used to measure the rate of this reaction?
 - B. What effect would be adding manganese dioxide have on the rate of this reaction?
 - C. What name do we give to chemicals like manganese dioxide?
2. A catalyst can increase the rate of a chemical reaction. Which of the following is also TRUE about catalysts?
 - A. They have to be a solid.
 - B. They are chemically the same at the end of the reaction.
 - C. Their appearance is unchanged at the end of the reaction.
 - D. They have to be a metal.
3. Calcium carbonate in the form of limestone dissolves in hydrochloric acid giving off carbon dioxide gas. Which of the following conditions would produce the fastest reaction?
 - A. Temperature of 40°C and 10cm³ of dilute acid
 - B. Temperature of 20°C and 10 cm³ of dilute acid
 - C. Temperature of 40°C and 5 cm³ of dilute acid and 5 cm³ of water
 - D. Temperature of 20°C and 5 cm³ of dilute acid and 5 cm³ of water
4. Platinum catalyses the reaction between sulphur dioxide (SO₂) and oxygen (O₂) to form sulphur trioxide (SO₃). Which statement(s) is not correct?

Less sulphur trioxide is formed when more catalyst is used.

 - A. The catalyst decreases the rate of reaction between sulphur dioxide and oxygen.
 - B. When the reaction is complete the platinum catalyst has gained mass.
 - C. A, B and C
 - D. Only A and C
 - E. Only A
 - F. Only B and C

5.2.5 Particle size

Learning activity 5.2.5



Effect of the particle size on the rate of the reaction

Materials

Magnesium ribbon, 1 M hydrochloric acid, Distilled water, 2 conical flasks (100 cm³), Measuring cylinder (100 cm³), Clamp stand, Glass trough, Stopwatch, Sand paper and weighing balance

Procedure

1. Clean the magnesium ribbon using a sand paper to remove any oxides that may be coating its surface. This will reduce reaction errors related to impurities.
2. Cut three equal sizes (10 cm each) of magnesium from the freshly cleaned magnesium ribbon, weigh each of them using a digital weighing balance and record their weights. Are they the same?
3. Wrap the magnesium pieces immediately in an aluminium foil to prevent them from being re-oxidised.
4. Fold the first magnesium ribbon and keep it for the next procedure.
5. Take the second magnesium ribbon cut it into smaller pieces, repeat this until you obtain very fine pieces of magnesium.
6. Measure 20 cm³ of 1M hydrochloric acid using a clean dry measuring cylinder and pour into a clean 100 cm³ conical flask. Repeat this one more time in the second conical flask. Label them A and B.

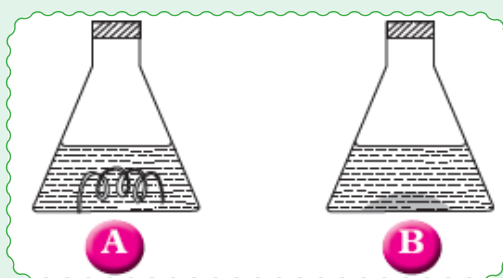


Figure 5.2.5.(a): Experimental set-up for measuring reaction rates between HCl and magnesium of different surface areas

7. Simultaneously put the folded magnesium ribbon and the fine pieces magnesium powder into conical flasks A and B respectively as shown above. Immediately start the stopwatch.
8. Monitor the reaction progress closely and stop your stopwatch when the magnesium completely dissolves in the acid or reaction comes to stop. In each case, record the reaction duration in seconds in the table like the one shown below.

9. Repeat steps 7-8 using 0.5M, 1.0M, 1.5M, 2.0M, 2.5M and 3.0M HCl.

Table 5.2.5: Table of results showing HCl-Mg ribbon reaction

Concentration of HCl (M)	3.0	2.5	2.5	1.5	1.0	0.5
Reaction duration for ribbon (in seconds)						
Reaction duration for fine pieces (in seconds)						

Guiding questions

1. What observations did you make during the reaction?
2. Write a chemical equation for the reaction that occurs.
3. What conclusion can you make from this experiment? Why?

The reaction involving magnesium pieces came to an end much faster than magnesium ribbon. Also, effervescence was more rapid in this experiment than in the reaction involving magnesium ribbon. If solid reactants are broken down into smaller pieces, the rate of reaction increases. The increase in rate of reaction happens because many smaller pieces of the same mass of solid provide a greater surface area compared to larger pieces. This generally increases the contact surface of the reactants colliding with each other thereby increasing reaction rate. At higher acid concentrations, the rate of reaction is even more rapid than in lower acid concentrations as shown in the graph below.

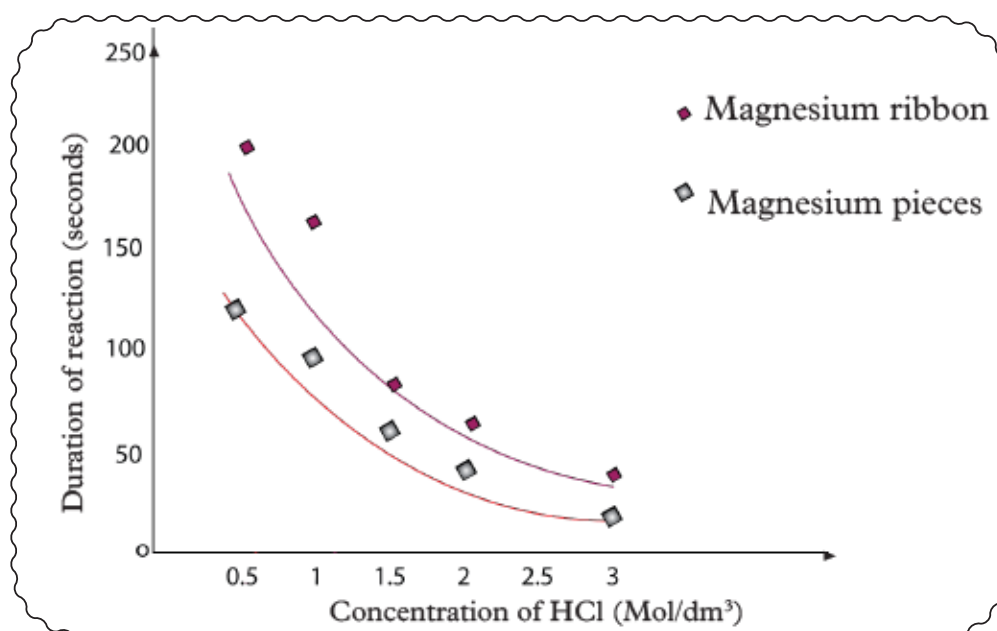


Figure 5.2.5.(b) Graph of reaction rate of magnesium of different particle sizes and hydrochloric acid at different concentrations

Application activity 5.2.5



1. Why does the rate of reaction increase when powdered calcium carbonate is used instead of marble chips?
2. Which chemical species below will react the fastest with hydrochloric acid?
 - A. Powdered marble at 40°C
 - B. Powdered marble at 25°C
 - C. Marble chips at 25°C
 - D. Marble chips at 40°C

5.2.6 Light

Learning activity 5.2.6



Investigating the effect of light on the rate of the reaction

Materials

Pair of scissors, Cardboard, A4 paper, Silver bromide solution, Art brush, Dark room.

Procedure

1. Cut a piece of paper into a square shape using a pair of scissors then cut a piece of cardboard to the same size as paper.
2. In a dark room smear the paper with silver bromide solution.
3. Cover the lower part of the paper with card board as shown in the fig 5.2.6
4. Cover the upper part with a thin piece of paper.

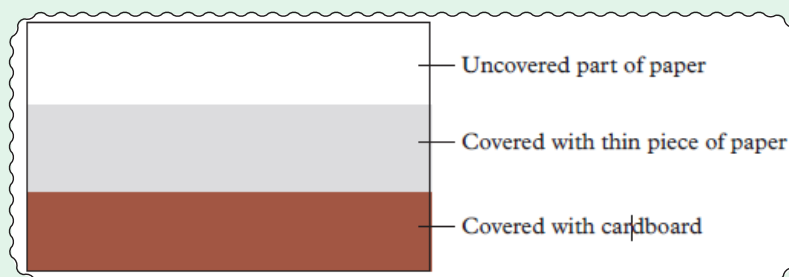


Figure 5.2.6: Experimental set up for determining the effect of light on silver bromide

5. Expose the improvised film to light, leave it for about 5 minutes.
6. Remove all the coverings then make your observations.

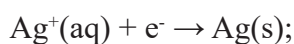
Guiding questions

1. Why was the paper smeared in dark place?

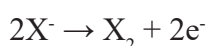
2. Why was the cardboard used in the experiment?
3. What observations did you make after exposure to light?

Silver bromide decomposes to silver metal which is seen as black spots. On the cardboard area no black spots seen because light cannot pass through it.

Silver chloride (AgCl), silver bromide (AgBr) and silver iodide (AgI) are all sensitive to light (i.e. are 'photosensitive') and all the three are used in the production of various types of photographic film to detect visible light, beta and gamma radiations. Each silver halide salt exhibits different sensitivity towards light. When the radiation hit the film the silver ions in the salt are reduced by electron gained to silver metal according to the following equation.



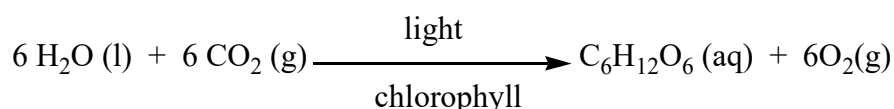
The halide ion is oxidised to the halogen molecule i.e.



Note: The above reaction does not occur in absence of light.

AgI is used in X-ray radiography, AgCl is used in 'fast' film for cameras.

Photosynthesis in green plants is also sensitive to light. Light energy is used to combine water and carbon dioxide to form glucose during photosynthesis. The equation of the reaction is:



Application activity 5.2.6



State any 3 examples of reactions that proceed only in the presence of light.

<https://courses.lumenlearning.com/suny-albany-chemistry/chapter/factors-affecting-reaction-rates-2/>

5.3 END UNIT ASSESSMENT

End of unit assessment



A. Knowledge and understanding

1.
 - a) Explain why increasing the (i) concentration (ii) temperature of reacting particles increases the rate of reaction.
 - b) Explain why decreasing the particle size of the reactants increases the rate of reaction.
 - c) Explain why the usage of catalyst increases the rate of reaction.
 - d) State whether addition of catalyst increases the amount of energy released during a chemical reaction.
2. Describe briefly how the rates of reactions below could be monitored:
 - a) Calcium carbonate with hydrochloric acid.
 - b) Sodium thiosulphate with hydrochloric acid producing sodium chloride, water, sulphur dioxide and sulphur solid.
 - c) Decomposition of hydrogen peroxide producing water and oxygen gas.
3. Fill the blank spaces below with an appropriate word.
 - a) Chemical reactions can only take place when particles _____ with each other.
 - b) There are several ways to change the _____ of a reaction.
 - c) Increasing the _____ increases the _____ energy of the particles and they _____ more _____ and with greater energy.
 - d) Increasing the _____ of a reactant in solution increases the _____ of a 'fruitful' collisions between particles.
 - e) A _____ reactant will react faster if it is broken into _____ pieces. This increases the _____, _____ and allows more _____ between the reactants.
 - f) Adding a catalyst can also speed up the rate of a reaction because it helps in providing binding surfaces for reactants bringing them closer to each other for successful collisions as a result of lowered activation energy. However, although it helps the reaction to take place much faster, a _____ is not _____ at the end of the reaction and could be _____ again.

B. Skills

4. A student performed an experiment to establish how effective manganese dioxide was as catalyst in the decomposition of hydrogen peroxide. The results below were obtained using two different quantities of manganese dioxide. The volume of the gas produced was recorded against time.

Time(s)	0	30	60	90	120	150	180	210
Volume (cm ³) for 0.3g	0	29	55	79	98	115	133	146
Volume (cm ³) for 0.5g	0	45	84	118	145	162	174	182

Draw a diagram of the apparatus you could use to carry out the experiments.

Plot a graph of these results.

Is the manganese dioxide acting as a catalyst in this reaction?

- At what stage does the reaction proceed most quickly?
- In terms of particles, explain why the reaction is quickest at the point you have chosen in d (i) above.

Why does the shape of the graph become less steep as the reaction proceeds?

What volume of gas was produced when using 0.3 g of manganese dioxide after 50s?

How long did it take for 60 cm³ of the gas to be produced when the experiment was carried out using 0.5g of manganese dioxide?

Write a balanced chemical equation for the decomposition of hydrogen peroxide.

- A flask containing dilute hydrochloric acid was placed on a digital balance. Excess limestone chips were added to this acid, a plug of cotton wool was placed in the neck of the flask and the initial mass of the apparatus was calculated. The table below shows the results obtained.

Time (min)	0	2	4	6	8	10	12	14	16
Loss in mass(g)	0	2.1	3.0	3.1	3.6	3.8	4.0	4.0	4.0

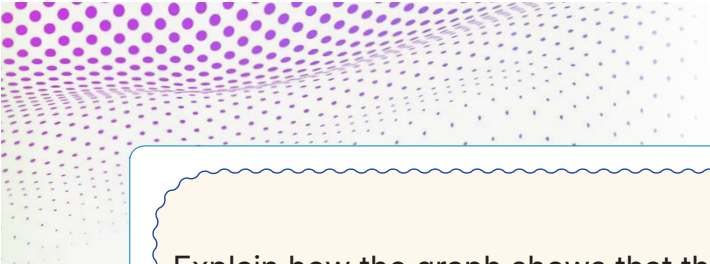
- Plot the results of this experiment in your notebook.
- Which of the results would appear to be incorrect? Explain your answer.
- Write a balanced equation to represent the reaction that takes place.

- The table below shows the volume of gas produced during a reaction.

Time (s)	0	20	40	60	80
Volume (cm ³)	0	30	55	70	70

Plot a graph of volume of gas vs time.

At what time does the reaction stop?



Explain how the graph shows that the reaction has finished.

State whether the reaction is fastest at the beginning or end. Explain your answer.

C. Attitudes and values

7. Explain how temperature do affect the rate of reaction.
8. Point out two examples in the daily life that show the of the temperature on the rate of reactions

UNIT 6

CONCENTRATION OF SOLUTION



Key Unit Competence:

Determine the concentrations of solutions from data obtained by simple acid-base titration

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Demonstrate understanding concept of concentration
- Explain the difference between end point and equivalence point.
- Describe the process of titration.

Skills

2

- Prepare solutions of given concentrations.
- Use experiments to explain a simple acid-base titrations.
- Use data obtained from the titration of an acid and base to calculate the concentration of either acid or base.

Attitudes and values

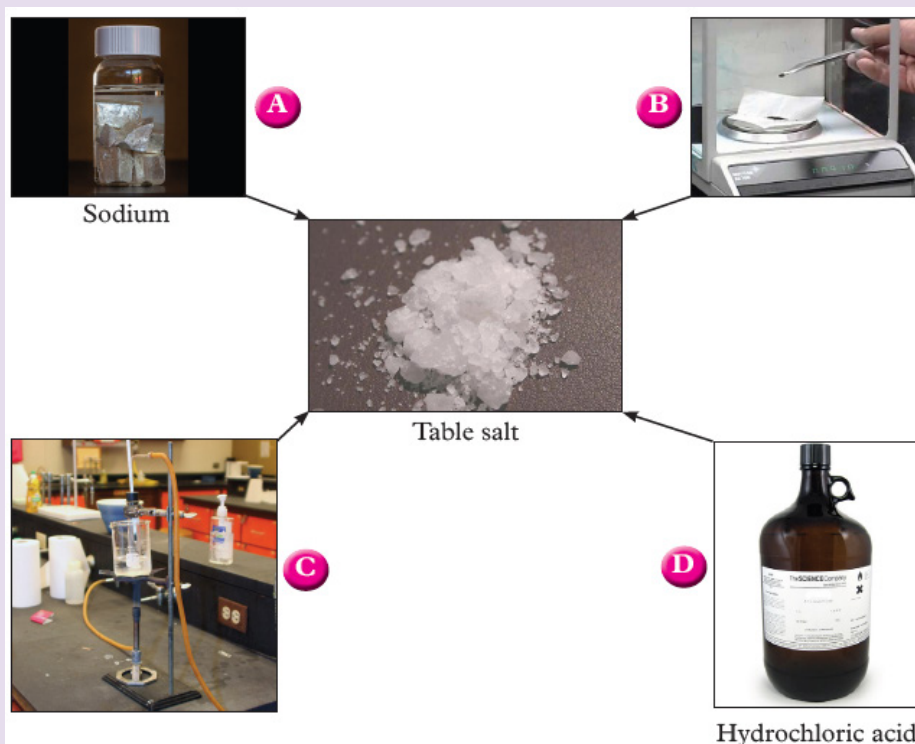
3

- Develop a culture of working in a team.
- Develop self- confidence during presentations and discussion in groups.
- Appreciate the importance of accuracy in measurement.
- Appreciate the importance of repeating procedures to obtain more accurate results

6.0 Introduction

Solutions are homogeneous mixture obtained by dissolving one or many substances. These solutions have precise concentration. Ways of expressing the concept of concentration are developed in this units as well methods used to prepare liquid solutions. Simple experimental methods like, titrations, used to determine unknown concentrations are illustrated.

Introductory activity



What is the relationship between the items in the picture above? Put the activities in the circled letters into two categories that is raw materials and processes. Identify each

The importance of solutions of known concentration (standard solutions) is not debatable. Indeed they are essential in chemistry for **titrations**, determining unknown concentrations, calibrating instruments, and in industry for medicines production, and certifying product quality, enabling precise analysis, reaction control, and safe product creation. They act as a reference to find moles, mass, or reaction endpoints accurately.

6.1 Concept of concentration

Learning activity 6.1



Determining the concentration of solutions

Materials

Sugar, teaspoons, 3 cups, water

Procedure

- Put equal amounts of drinking or potable water in each of the three cups and label them A, B and C.

Add sugar as follows:

- In cup A, add half teaspoonful of sugar.
- In cup B, add one spoonful of sugar.
- In cup C, add two spoonful of sugar.
- Stir and taste each sugar solutions prepared in step 2 above.

Guiding questions

1. In which cup was the solution sweetest?
2. How can we tell the cup with a lot of sugar?
3. How can we decrease the taste of sugar in each cup?
4. What would happen when the water used was warmed first before step 2 in the procedure above?
5. How can concentrations of solutions be measured?

Cup C had the sweetest taste because it contained most sugar particles. By adding more water that is diluting the solution, we reduce the taste of sugar.

Many chemical reactions are performed with reagents in solution. It is necessary to know the amount of solute dissolved in a given amount of solution. If a large volume of water is added to a small amount of solute, the solution is said to be dilute. Similarly if the solution contains a larger amount of solute, the solution is said to be concentrated.

Concentration of solutions

The term concentration refers to the amount of solute dissolved in a specific volume of solvent. In a given amount of water, the more solute dissolved, the more concentrated the solution. If the solution contains a little solute, it is described as a dilute solution.

Similarly, if the solution contains large amount of solute, then the solution becomes more concentrated.

How concentrations are expressed

In consumer and industrial world, the most common method of expressing the concentration is based on the quantity of solute in a fixed quantity of solution.

The quantities referred to here can be expressed in weight (w), volume (v) or both (that is the weight of a solute in a given volume of solution). In order to distinguish among these possibilities the abbreviations (w/w, v/v and w/v) are used. There are a number of ways to express the relative amounts of solute and solvent in a solution.

Units of concentration

Once you have identified the solute and solvent in a solution, you are ready to determine its concentration. Concentration may be expressed in several ways, using percent composition by mass, volume percent, mole fraction, molarity, molality or normality.

6.1.1 Molarity

Molarity (M) tells us the amount (expressed in moles (n)) of a solute in exactly one litre (one cubic decimetre) of a solution. Note: Molarity is represented by a capital M.

We need two pieces of information to calculate the molarity of a solute in a solution:

The moles of solute present in the solution.

The volume of solution (in litres or cubic decimetre) containing the solute.

To calculate molarity we use the equation:

Molarity = (moles of solute (n))/(volume of solution in litres or cubic decimetre (mol. dm³))

The units of molarity are mol/l or mol/dm³ (mol dm⁻³)

Molarity or **molar concentration** is probably the most commonly used unit of concentration. It is the number of moles of solute per litre (or cubic decimetre) of solution (not necessarily the same as the volume of solvent).

Example 6.1.1

What is the molarity of a solution made when water is added to 11g calcium chloride (CaCl₂) to make 100 mL of solution? (Ca = 20, Cl = 35)

Answer:

Mm of CaCl₂ = 40 + (35 × 2) = 110

$$\text{Moles of CaCl}_2 = \frac{\text{mass of CaCl}_2}{\text{molar mass of CaCl}_2} = \frac{11\text{g}}{110\text{g/mol}} = 0.1\text{mol} \quad \text{or} \quad 110\text{g} \rightarrow 1\text{mole}$$

$$11\text{g} \rightarrow x \text{ mole} \quad x = \frac{1 \times 11}{110} = 0.1\text{mol}$$

$$\text{Molarity} = \frac{\text{moles of solute (n)}}{\text{volume of solution in litres (v)}} = \frac{0.1\text{mol}}{100 \times 10^{-3}\text{l}} = 1\text{mol/l or } 1\text{mol l}^{-1} \text{ or } 1\text{mol dm}^{-3} \text{ or } 1\text{M}$$

6.1.2 Mass concentration

Mass concentration is the amount (expressed in grams) or mass of solute dissolved in 1 litre or 1 cubic decimetre (litre 1 or 1 dm³) of solution. It shows how many grams are present in one litre (one cubic decimetre) of the solution.

To calculate mass concentration we use the equation:

$$\text{Mass concentration} = \frac{\text{mass of solute in grams (} m \text{)}}{\text{volume of solution in litres or cubic decimetre (} v \text{)}}$$

The units for mass concentration are g/l or g/dm³.

Example

Calculate the mass concentration of a solution obtained by dissolving 10g of sugar to get 500mL of solution.

Answer:

Mass of sugar: 10g

Volume of solution 500mL or 0.5l or 0.5dm³

$$\text{Mass concentration} = \frac{\text{grams of solute}}{\text{litres or cubic decimetres (} v \text{)}} = \frac{10\text{g}}{0.5\text{l}} = 20\text{g/l or } 20\text{g/dm}^3 \text{ (} 20\text{g dm}^{-3} \text{)}$$

Relationship between molarity and mass concentration

The molar concentration is the number of moles of the solution in 1 litre (1 cubic decimetre) of the solution.

The mass concentration is the number of grams of the solution in 1 litre (1 cubic decimetre) of the solution.

The two concentrations are related through the molar mass.

Mass concentration = molar concentration x molar mass or

$$\text{Molar concentration} = \frac{\text{mass concentration}}{\text{molar mass}}$$

Example: Calculate the mass concentration of a solution of sodium chloride which is 0.5M.

There are 0.5 moles of sodium chloride in 1 litre

Molar concentration = 0.5 mol. l⁻¹

Molar mass of sodium chloride = (23 + 35.5) g mol⁻¹ = 58.5 g mol⁻¹

Mass concentration = 0.5 mol l⁻¹ x 58.5 g mol⁻¹ = 29.25g l⁻¹

6.1.3 Volume percentage and mass percentage

(a) Volume percentage

Volume percentage or volume percent or percent by volume is most often used when preparing solutions of liquids. Volume percent is given by the formula:

$$\text{Volume percent} = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100\%$$

It shows how many mL of solute are present in **100 mL of solution**.

It is written as **% v/v**

Note: Volume percent is relative to volume of solution, not volume of solvent. For example, if wine is 12% ethanol, it means that there are 12 mL of ethanol for every 100 mL of wine solution. In addition it is important to realise that liquid and gas volumes are not necessarily additive. For example, if you mix 12 mL of ethanol with 100 mL of wine, you will get a solution which is less than 112 mL in volume.

Example

Determine the volume percent of a solution made by combining 25 mL of ethanol with enough water to produce 200 mL of the solution.

Answer:

- Volume of solute is 25 mL
- Volume of the solution is 200 mL
- Volume percent = $\frac{25}{200} \times 100 = 12.5\%$

(b) Mass percentage

We can consider mass percentage or mass percent (or percent by mass) in two ways:

- The parts of solute per 100 parts of solution.
- The fraction of a solute in a solution multiplied by 100.

You need two pieces of information to calculate the percent by mass of a solute in a solution:

- The mass of the solute in the solution.
- The mass of the solution.

Percent by mass (or mass percent) is the mass of the solute divided by the mass of the solution (mass of solute plus mass of solvent), multiplied by 100. It shows how many grams of solute are present in **100 g of solution**. It is written as **% m/m**

Use the following equation to calculate percent by mass:

$$\text{Mass percent} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 100\%$$

Example

1. Determine the mass percent of a 100 g salt solution which contains 20 g salt.

Answer:

$$\text{Mass percent} = \frac{20\text{g}}{100\text{g}} \times 100\% = 20\%$$

Application activity 6.1



1. Calculate the molarity of a sodium hydroxide obtained by dissolving 20g to obtain 500mL of solution.
2. Calculate the mass concentration of sodium carbonate obtained by dissolving 5g of sodium carbonate to get 250mL of solution.
3. A solution of sodium hydroxide (NaOH) has a mass concentration of 12.0 g dm⁻³. Calculate its molar concentration.

6.2 Preparation of solutions

Preparing solutions is one of the most fundamental tasks performed in a laboratory. Many solutions are obtained by dissolving solid substances in distilled water or by diluting concentrated stock solutions.

6.2.1 Preparation of molar solutions from solid substances

Learning activity 6.2.1



Preparation of a 1 mol dm⁻³ sodium chloride solution (1M NaCl solution)

Materials

Weighing balance, 1 dm³ volumetric flask, Stirrer/glass rod, Wash bottle, Distilled water, Beaker, Funnel, Sodium chloride.

Procedure

- Weigh 58.5 g (Molar mass of NaCl) of sodium chloride crystals.
- Place 400 cm³ of distilled water in a beaker.
- Add at a time small amount of the salt into the water, stirring continuously until all the salt dissolves.
- Using a filter funnel transfer the salt solution into a 1 dm³ volumetric flask Figure 6.2.1
- Rinse the beaker with distilled water carefully and add the washings into the flask.

- Add more distilled water and shake the flask well. Add more water until the solution level is just below the calibration mark.
- Using a wash bottle add more distilled water drop by drop until the bottom of the meniscus is at the same level with calibration mark Fig. 6.2.(c).
- Use a stopper to cover the flask and invert it several times to make sure the solution mixes thoroughly. Transfer the solution into a reagent bottle and label it 1M NaCl, see figure below.

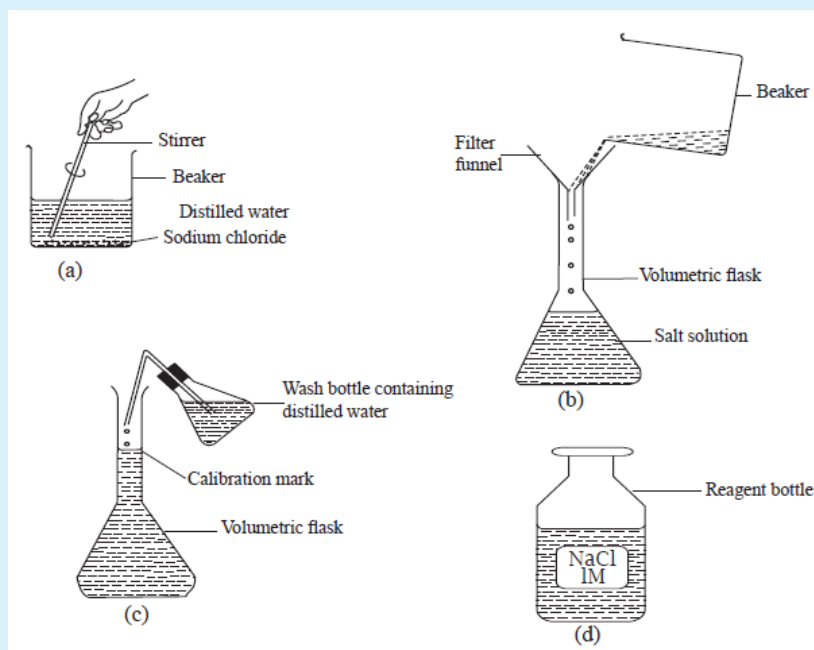


Figure 6.2.1: Preparation of 1M sodium chloride solution

Guiding questions

1. Did all the salt dissolve in the water?
2. Write the formula of sodium chloride.
3. Using the relative atomic masses, find the molar mass of sodium chloride.
4. What conclusion can we make from this experiment?

To prepare 1M solution of sodium chloride, 58.5g of NaCl is added to water and the volume made up to 1 litre.

Finding the mass of the solute (NaCl) needed:

NaCl 1M means one mole of sodium chloride (58.5 g) in 1dm^3 of the solution. Therefore when we dissolve this amount of salt in distilled water and the volume is made up to 1000 mL (or 1L) of solution, the molarity or molar concentration of the solution is 1mol/L or mol dm^{-3} or 1M.

Suppose we did not wish to prepare so much solution that is 1000mL (1L) but rather only 250 mL, of a less concentrated solution, for example 0.2M sodium chloride, what

would be the mass of sodium chloride required for this preparation. We need first to know the mass of the solute which is required.

Calculations involved in this preparation.

Solution needed: 250 mL of 0.2M sodium chloride

Number of moles of NaCl in 250 mL

$$M = n/v, \quad n = M \times V = \frac{0.2\text{mol}}{l} \times 0.250 = 0.05\text{mol}$$

Mass of NaCl needed

Molar mass of NaCl = (23 + 35.5) = 58.5g mol⁻¹

$$n = \frac{m}{M_m} \text{ or } m = n \times M_m = 0.05\text{mol} \times 58.5\text{g mol}^{-1} = 2.925 \text{ of sodium chloride.}$$

The mass of sodium chloride required for this preparation is 2.925g.

The mass to be weighed is 2.925g. A 250 mL (instead of a 1000mL) volumetric flask is required. Then, the procedure described above is followed.

Note:

- Use clean and dry apparatus.
- Read the meniscus at eye level.
- Ensure the solute is fully dissolved before making up to volume.

To prepare an aqueous solution of a specific molarity using a solid solute, follow these general steps:

- Calculate the required mass:** Determine the number of moles (n) needed for your desired volume (V) using the equation $n = M/V$. Convert the moles of the solute into mass (grams) using its molar mass ($m = M_m \times n$).
- Weigh the solute:** Accurately measure the calculated mass (m) of the solid solute using an analytical balance.
- Dissolve the solute:** Place the solid into a beaker and add a portion of the solvent (usually distilled water), which should be less than the final desired volume, and stir until the solute is completely dissolved.
- Transfer the solution:** Quantitatively transfer the dissolved solution into a volumetric flask of the appropriate volume, rinsing the beaker multiple times with the solvent to ensure all solute is transferred.
- Dilute to volume:** Add the remaining solvent to the volumetric flask until the bottom of the meniscus reaches the calibration mark on the flask's neck.
- Mix thoroughly:** Stopper the flask and invert it multiple times to ensure the solution is homogeneous.

Exercise:

1. Calculate the mass of NaOH needed to prepare NaOH 0.5 M when having 50 ml volumetric flask.

Solution: M_m of NaOH = $23 + 16 + 1 = 40\text{g/mol}$

$$V = 50\text{ ml} = 0.05\text{ l}$$

Number of moles of NaOH to be dissolved to make 0.05 ml of solution

$$n = M \cdot V = 0.5 \times 0.05 = 0.025\text{ mol. of NaOH}$$

$$\text{Mass of NaOH to be dissolved: } m = n \times M_m = 0.025 \times 40 = 1\text{g}$$

Then dissolve 1g of NaOH to prepare 50 ml of NaOH 0.5 M following above proposed steps

6.2.2 Preparation of solutions by dilution

During a dilution process, adding more solvent to a given solution changes (decreases) the concentration without changing the number of moles of solute. That is,

Number of moles before dilution = Number of moles after dilution

i.e

$M_i V_i = M_f V_f$, Where

- M_i and M_f are the initial and the final concentrations, respectively.
- V_i and V_f are the initial and the final volumes, respectively.

Learning activity 6.2.2



Preparation of 250mL of a 0.5M from a stock solution of HCl with the following specifications:

- Molar weight: 36.5g/mol
- Percentage of HCl: 36% by mass
- Density: 1.18g/mL

Materials required

250mL volumetric flask, 25mL pipette or a burette, Wash bottle, Funnel, Dropper pipette

Procedure

1. Using a graduated pipette or a burette, measure 10.74mL of the 36% hydrochloric acid.
2. Using a funnel, carefully transfer the concentrated acid into a 250mL volumetric flask containing about 150mL of distilled water.
3. Put a stopper and gently shake the mixture.

- Using a wash bottle, add more distilled water until the level is 1cm below the calibration mark.
- Using a dropper pipette, make up to 250mL.
- Stopper the volumetric flask and gently shake.

The solution prepared is 0.5M HCl.

Note:

- Use clean and dry apparatus.
- Read the meniscus at eye level.
- Always add acid to water, never water to acid.

Calculations involved

Mass of HCl contained in one litre: $1.18 \text{ g/mL} \times 1000 \text{ mL} = 1180 \text{ g}$

Mass of 36% HCl in one litre: $1180 \text{ g} \times \frac{36}{100} = 424.8 \text{ g}$

Number of moles of HCl in one litre: $\frac{424.8 \text{ g}}{36.5 \text{ g/mol}} = 11.64 \text{ mol}$

The molar concentration of 36% HCl is 11.64 mol/L also written as 11.64 M

Number of moles of 0.5M needed in 500mL(0.500L): $0.5 \text{ mol/L} \times 0.500 \text{ L} = 0.25 \text{ mol}$

Volume of 36% HCl needed to prepare 500mL of 0.3M HCl: $\frac{0.25 \text{ mol}}{11.64 \text{ mol/L}} = 0.02148 \text{ L}$

Initial molarity M_i : 11.64M; initial volume V_i = ?

Final molarity M_f : 0.5M; final volume: 250mL (or 0.250L)

From the expression $M_i V_i = M_f V_f$ we get

$$V_i = \frac{M_f V_f}{M_i} = \frac{0.5 \text{ mol/L} \times 0.250 \text{ L}}{11.64 \text{ mol/L}} = 0.01074 \text{ L (10.74 mL)}$$

The volume of 36% HCl needed to prepare 250 mL of 0.5 M HCl is 10.74 mL.

Application activity 6.2



- (a) How many grams of potassium chloride are needed to make 250 cm³ of a 0.050 M solution?
(b) Describe the required procedure to prepare that solution.
- Describe, with calculations, the procedure for preparing 250 mL of 0.05 M potassium bromate solution from a 2.5 M KBrO₃ stock solution.

6.3 Simple acid-base titration

Learning activity 6.3



Research activity on acid base titration

Using the internet and textbooks from the library, research about the following:

- Acid-base indicators
- Apparatus used during titration
- Handling titration apparatus
- Conducting simple titration
- Calculations involving acid-base titration.

Titration is a volumetric analysis technique used to determine the unknown concentration of a solution (the analyte) by gradually adding a solution of known concentration (the titrant) until the reaction is complete. The titrant is added from a burette, and an indicator signals when the reaction is complete by changing color at a specific pH. For example, phenolphthalein changes from colorless to pink in basic solutions.

The point where the indicator changes color is called the end point, while the equivalence point is where reactants have reacted in exactly the mole ratio given in the balanced equation. Ideally, these two points should coincide, but they may differ slightly depending on the indicator chosen.

Table 6.3: Apparatus commonly used in volumetric analysis and their uses

Balance	To weigh solids to be dissolved.
Measuring cylinder	Measure volume of solution to be diluted.
Volumetric flask	To make solution of known concentration.
Pipette	To deliver fixed volumes of liquid or solution into conical flasks.
Burette	To release small amounts of solutions into the conical flasks.
Beaker	To collect and pour solutions into the burette.
Funnel	To facilitate pouring solutions into the burette.
White tile or paper	To help in the observation of colour change in the flask more clearly.
Dropper	To add 2-3 drops of an indicator into the conical flask.

6.3.1. Titration involving sodium hydroxide and hydrochloric acid

Learning activity 6.3.1

Performing a titration of NaOH with HCl solution

Materials required

Retort stand, Clamp, Burette, Distilled water, Phenolphthalein or methyl orange, Pipette, Conical flask, Standard solution of HCl, NaOH with unknown concentration, Beaker, Wash bottle, Glass rod/ stirrer

Procedure

1. 25.0 cm^3 of sodium hydroxide is pipetted into a conical flask to which a few drops of phenolphthalein indicator are added.

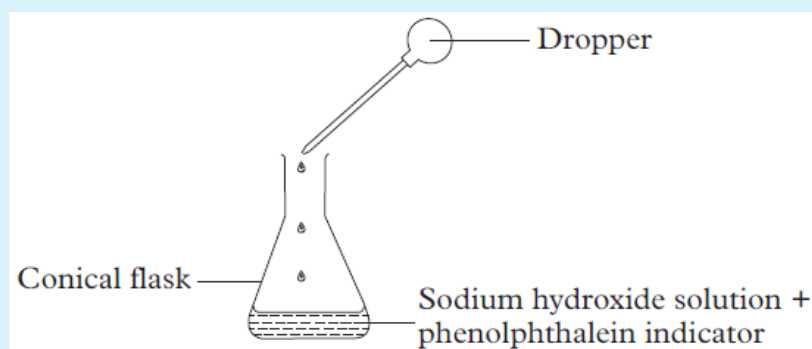


Figure 6.3.1(a): Dropping phenolphthalein indicator in a conical flask

2. A 0.1M hydrochloric acid solution is then filled in the burette using a filter funnel until it is up exactly to zero mark. The filter funnel is then removed.

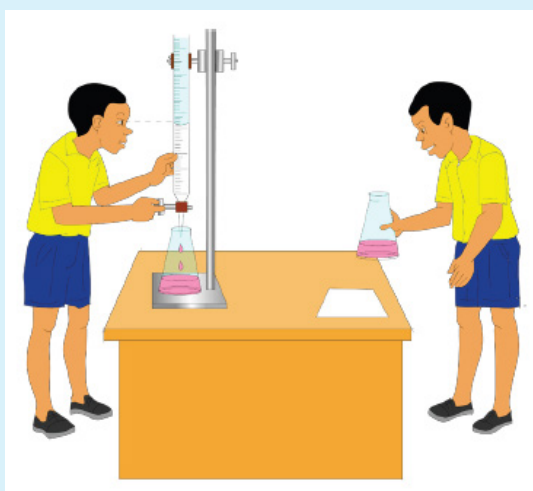


Figure 6.3.1(b): Titration process

Place a white tile or paper under the conical flask to clearly see the colour changes. The hydrochloric acid is added to the sodium hydroxide solution dropwise usually no more than 0.5 cm^3 at time.

Note: The contents of the flask is swirled after each addition of acid for thorough mixing while at the same time keeping an eye on the solution in the flask to notice any colour change.

The acid is added until the alkali has been completely neutralised. This is shown by the disappearance of the pink colour of the indicator.

Note: When you swirl the flask and the new colour developing persists for a while, the end point is near. Add one drop of the acid at a time until you get a permanent colour. This is the end-point of the titration.

Read and record the volume of the acid to the nearest 0.1 cm³. Record your results in a table like the one shown below.

Table 6.3.1: An example of recording of volume of titration results

Burette reading	1	2	3
Final burette reading (cm ³)	21.50	43.20	21.10
Initial burette reading (cm ³)	0.00	22.00	0.00
Volume of acid used (cm ³)	21.50	21.20	21.10

Repeat the titration two more times using a different clean conical flask. Add the same number of drops of the indicator as used in the first titration. However, if you use the same flask, you must wash it thoroughly with distilled water and then rinse with a little of the solution you intend to suck into the pipette.

Note: The first volume obtained should guide you to get the second volume fairly quickly as follows: "If your first titre volume was 25.0 cm³, you may add 21 cm³ quickly while swirling the flask. Then add the acid dropwise while swirling the conical flask until you get the same permanent colour change as in previous titration."

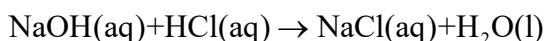
Repeat the titration using the second volume as a guide to the volume required. You can add the acid until you are about 1 cm³ from the end point, then you add the acid drop by drop.

Get the average of any two values for the volume of the acid added, which differ by not more than ± 0.1 cm³. In other words, the three titrations should be fairly consistent and accurate.

Guiding questions

1. Write the chemical equation of the reaction that takes place during the titration.
2. Calculate the average volume of acid used (use concordant results).
3. Calculate the number of moles of acid used.
4. Calculate the number of moles of base neutralized (number of moles in 25mL).
5. Calculate the molarity of the base.

In most volumetric calculations of this type, you first calculate the known moles of one reactant from a volume and its molarity. Then, from the equation, you relate this to the number of moles of the other reactant, and then with the volume of the unknown concentration, you work out its molarity as well. The neutralisation reaction which taking place represented by the following chemical equation.



Considering the example of results from Table 6.2, we notice that the average volume of used acid

$$\text{is } \frac{21.2 + 21.1}{2} = 21.15 \text{ cm}^3$$

Note: The value 22.15 in the table is omitted because it is off the limit of ± 0.2 from the other readings.

Therefore, the number of moles of the acid use is:

$$n_a = M_a V_a = 0.1 \text{ mol L}^{-1} \times 21.15 \times 10^{-3} \text{ L} = 2.115 \times 10^{-3} \text{ mol}$$

For the reaction between sodium hydroxide and hydrochloric acid, 1 mole of NaOH (aq) reacts with 1 mole of HCl (aq), giving a 1:1 mole ratio. Therefore,

$$n_b(\text{NaOH}) = n_b(\text{HCl}) = 2.115 \times 10^{-3} \text{ mol}$$

The molar concentration of sodium hydroxide is:

$$M_b = \frac{2.115 \times 10^{-3} \text{ mol}}{25 \times 10^{-3} \text{ L}} = 0.084 \text{ mol L}^{-1}$$

Generally, in a neutralization reaction between a mono base and a monoacid, 1 mole of base reacts with 1 mole of acid, giving a 1:1 mole ratio. Therefore, which can be expressed as $M_a V_a = M_b V_b$ where M represents molarity and V represents volume.

Where

n_a = Number of moles of the monoacid

n_b = Number of moles of the monobase

M_a = Molarity of the acid

M_b = Molarity of the base

V_a = Volume of the acid

V_b = Volume of the base

In the example above,

$M_a = 0.1 \text{ mol/dm}^3 = 0.1 \text{ M}$, $V_a = 21.15 \text{ cm}^3 = 0.02115 \text{ dm}^3$, $M_b = \text{Unknown}$, $V_b = 25 \text{ cm}^3 = 0.025 \text{ dm}^3$

Rearrange the formula to solve for Mb:

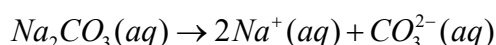
$$M_b = (M_a V_a) / V_b$$

$$M_b = (0.1 \times 0.02115) / 0.025$$

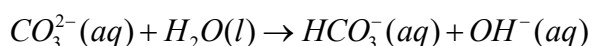
$$M_b = 0.084 \text{ M}$$

6.3.2 Titrations involving sodium carbonate and hydrochloric acid

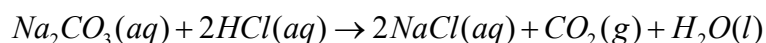
Sodium carbonate is classified as a **base** because of how it behaves **in water** and **with acids**. When sodium carbonate dissolves in water, it dissociates into its component ions as follows:



The carbonate ion (CO_3^{2-}) reacts with water:



This reaction produces hydroxide ions (OH^-), making the solution alkaline. Since sodium carbonate forms a basic solution in water, its reaction with an acid is a neutralization reaction. Sodium carbonate reacts with hydrochloric acid to produce sodium chloride, carbon dioxide, and water:



Carbonate ions can be qualitatively detected by adding acid to release CO_2 gas, which turns lime water milky, while quantitative analysis is done by titrating sodium carbonate against a standard hydrochloric acid solution.

Learning activity 6.3.2



Determining the molarity of sodium carbonate by titration

Materials required

Burette (50 cm^3), Retort stand and clamp, Pipette (25.0 cm^3), Pipette filler, Conical flask, White tile, Funnel, Beaker, Wash bottle, **0.1 M Hydrochloric acid (HCl)**, **Sodium carbonate (Na_2CO_3)** of *unknown concentration*, **Methyl orange indicator**

Procedure

1. Rinse the **burette** with hydrochloric acid and fill it with the acid. Record the **initial burette reading**.
2. Rinse the pipette with sodium carbonate solution.
3. Use the pipette to transfer 25.0 cm^3 of sodium carbonate solution into a clean conical flask.
4. Add 2–3 drops of methyl orange indicator. What is the colour of the solution?

5. Place the conical flask on a white tile under the burette.

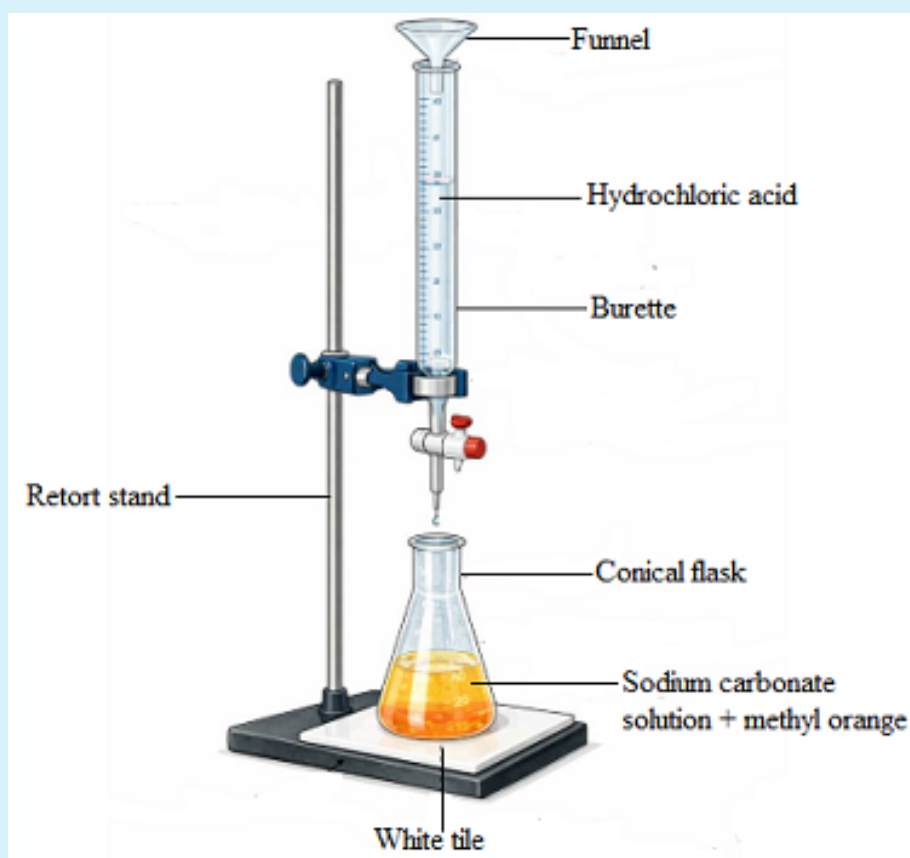


Figure 6.3.2: Titration of sodium carbonate against hydrochloric acid

6. Add hydrochloric acid from the burette slowly, swirling the flask continuously.
7. Near the end point, add acid dropwise until the colour changes. Record your observations.
8. Record the final burette reading.
9. Repeat the titration until two concordant results are obtained (within 0.10 cm^3).
10. Record your results in a table like the one shown below.

6.3.2: An example of recording of volume of titration results

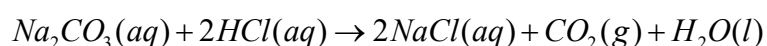
Burette reading	rough	1	2	3
Final burette reading (cm^3)	24.50	47.30	23.00	45.30
Initial burette reading (cm^3)	0.00	25.00	0.00	23.00
Volume of acid used (cm^3)	24.50	22.30	23.00	22.30

Guiding questions

1. Write a balanced chemical equation for the reaction between sodium carbonate and hydrochloric acid.
2. Calculate the average volume of acid used (use concordant results).

3. Calculate the number of moles of acid used.
4. Calculate the number of moles of sodium carbonate reacted (no. of moles in 25cm^3).
5. Calculate the molarity of sodium carbonate.

Sodium carbonate is soluble in water. The concentration of its solution is determined using titration against a hydrochloric acid of known concentration and methyl orange indicator. The equation of the reaction that occurs is:



1 : 2 2 : 1 : 1

When the sodium carbonate has completely reacted, the colour of the solution changed from yellow to orange/pink.

Considering the results from Table 6.3.2 as an example, we notice that the average volume of used acid

is $(23.00 + 22.30) / 2 = 22.65 \text{ cm}^3$.

The number of moles of hydrochloric acid used is:

$$n_{\text{HCl}} = 2.265 \times 10^{-3} \text{ dm}^3 \times 0.1 \text{ mol dm}^{-3} = 2.265 \times 10^{-4} \text{ mol}$$

The balanced chemical equation of the reaction between sodium carbonate and hydrochloric acid shows that the mole ratio of Na_2CO_3 of 2HCl is 1:2.

$$n_{\text{Na}_2\text{CO}_3} = \frac{n_{\text{HCl}}}{2}$$

$$n_{\text{Na}_2\text{CO}_3} = \frac{2.265 \times 10^{-4}}{2}$$

$$n_{\text{Na}_2\text{CO}_3} = 1.1325 \times 10^{-4} \text{ mol}$$

$$\text{Molarity of } \text{Na}_2\text{CO}_3 = \frac{1.1315 \times 10^{-4} \text{ mol}}{25 \times 10^{-3} \text{ dm}^3} = 0.045 \text{ mol dm}^{-3}$$

The sodium carbonate solution titrated is **0.045M**

6.3.3 End points and equivalent point

Learning activity 6.3.3

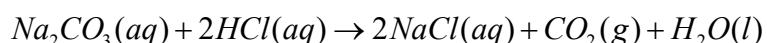


Using textbooks and internet investigate on the meaning of end and equivalent points

The end point in a titration is the point at which the indicator changes colour, showing that the reaction between reactants (in this context, the acid and the base) is just complete. The end point is very important because, it tells the experimenter when to stop adding the titrant, helps to choose the right indicator and allows accurate calculation of the unknown concentration.

For example, in the titration of sodium hydroxide using hydrochloric acid, as HCL is added the solution becomes acidic, phenolphthalein changes from pink to colourless. At this point neutralisation is complete.

The equivalence point is the stage in a titration where the reacting substances have reacted in exact stoichiometric proportions as shown by the balanced equation. At the equivalent point, the reactants are completely used up. There is no reactant in excess. The number of moles added follows the mole ratio in the equation. Sodium carbonate reacts with hydrochloric acid to the following chemical equation.



The equivalence point occurs when 2 moles of HCl react with 1 mole of Na_2CO_3 . At this point, sodium carbonate is just completely neutralized. The equivalence point is the point in a titration at which the acid and base have reacted in exactly the required mole ratio according to the balanced chemical equation.

Note: In a correct titration, the end point is very close to the equivalence point.

The table below summarises the differences between end point and equivalence point.

Table 6.3.3: Differences between end point and equivalence point

Feature	End point	Equivalence point
Meaning	The point at which the indicator changes colour.	The point at which the exact stoichiometric amounts of acid and base have reacted.
Nature	Observed/ experimental point	Theoretical / chemical point
Detection method	Seen by colour change of indicator	Calculated from chemical equation
Indicator Dependence	Yes	No
Position in titration	Slightly before or after equivalence point	Exact completion of reaction

Application activity 6.3



1. 24.2 cm^3 of a solution containing 0.2 mol/dm^3 of hydrochloric acid neutralised 25 cm^3 of potassium hydroxide solution. What is the molar concentration of potassium hydroxide solution?
2. 25 cm^3 of sodium hydroxide solution was pipetted into a conical flask and titrated with 0.2 M hydrochloric acid. Using a suitable indicator, it was found that 15 cm^3 of acid was needed to neutralise the alkali. Calculate the molarity of the sodium hydroxide.
3. A solution of sodium carbonate has an unknown concentration. It is titrated with a standard solution of hydrochloric acid of concentration 0.10 mol dm^{-3} using methyl orange as an indicator.
 - (i) State the purpose of this titration
 - (ii) Name the type of titration involved.
 - (iii) Explain why hydrochloric acid is described as a standard solution in this experiment
 - (iv) Name any five pieces of apparatus used in this titration.
 - (v) State the apparatus used to:
 - a) measure an accurate fixed volume of sodium carbonate solution
 - b) deliver hydrochloric acid during the titration
4. Write the balanced chemical equation for the reaction between sodium carbonate and hydrochloric acid. During the titration, 25.0 cm^3 of sodium carbonate solution required 20.0 cm^3 of 0.10 mol dm^{-3} hydrochloric acid for complete neutralisation.

Calculate the number of moles of HCl used.

Calculate the number of moles of Na_2CO_3 that reacted.

Calculate the concentration of the sodium carbonate solution in mol dm^{-3} .

Differentiate between end point and equivalence point.

6.4 End of unit assessment

End of unit assessment



A. Knowledge and understanding

- Which one of the following pieces of apparatus is not used during titration?
a) Liebig condenser; b) Burette; c) Pipette filler; d) Conical flask.
- Which one of the following equations is correct for the reaction between hydrochloric acid and sodium carbonate?
a) $\text{Na}_2\text{CO}_3(\text{s}) + \text{HCl}(\text{s}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$
b) $\text{Na}_2\text{CO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$
c) $\text{Na}_2\text{CO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
d) $\text{Na}_2\text{CO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{CO}_2(\text{g})$
- Before starting titration, the pipette, burette and conical flask are rinsed with deionised water and not tap water. Why is this?**
a) Ions in the tap water may 'clog up' the tap of the burette.
b) The acid, when placed in the burette will become more dilute if tap water rather than deionised water is used to rinse the burette.
c) Tap water contains ions that may affect the titration result.
d) The colour change at the end-point of the titration is more noticeable if deionised water is used.
- When placing the acid in the burette, which one of the following procedures is incorrect?**
a) The top of the meniscus is read when doing the titration.
b) Any air bubble below the tap is removed before the titration.
c) The funnel is removed from the top of the burette before the titration.
d) The burette is filled to the zero mark when filling it up.
- When pipetting sodium carbonate solution into the conical flask, which one of the following procedures is always carried out?**
a) The sodium carbonate solution must be pipetted directly from the volumetric flask into the conical flask.
b) When emptying the contents of the pipette into the conical flask, the last few drops are blown out.

- c) A pipette filler is used instead of using one's mouth to suck up the solution.
- d) The pipette is always rinsed with sodium carbonate solution first and then with deionised water.

6. Why is the sodium carbonate mixed with the acid in a conical flask rather than a beaker?

- a) Its easier to see the colour change in a conical flask.
- b) The conical flask will be less likely to break during the titration.
- c) To prevent any of the solutions splashing out when swirling the flask.
- d) The conical flask is more accurate for measuring volume.

7. When adding the indicator to the conical flask, only two or three drops are added because

- (i) too much indicator may make it difficult to see the colour change at the end-point clearly,
- (ii) the indicator is corrosive, and adding too much poses a greater risk,
- (iii) the indicator may be a weak acid or base and may interfere with the titration result.

Which of the above statements is/are correct?

- (i) and (ii) only
- (ii) and (iii) only
- (iii) only
- All of them

8. When adding the acid from the burette into the conical flask

- (i) the acid is initially added drop by drop,
- (ii) the conical flask is swirled while adding the acid,
- (iii) the conical flask is placed over a white tile or white sheet of paper,
- (iv) the inside of the conical flask is washed down from time to time with deionised water.

Which of the above procedures is/are not carried out during the titration?

- (i) and (iii) only
- (i) only
- (iv) only
- (ii) and (iii) only

9. If the colour of the indicator at the end-point changes from yellow to peach/pink, then which one of the following indicators was most likely to have been used?

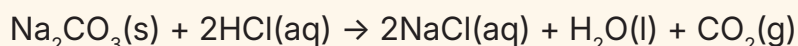
- a) Litmus indicator, b) Universal indicator, c) Phenolphthalein, d) Methyl orange

10. How many titrations should be carried out in order to give a valid result?

- a) One rough titration and two accurate titrations that agree to within 0.1 cm^3 of each other.
b) Three titrations which are then averaged to get a result.
c) One rough titration and one accurate titration.
d) Two rough titrations and one accurate titration that agrees to within 0.5 cm^3 of the rough titrations.

B. Skills

11. The equation for the reaction taking place in this titration is



If A represents the acid, and B represents the base, then what are the values of n_A and n_B in calculating the concentration of the hydrochloric acid solution?

- a) $n_A = 1$ and $n_B = 1$, b) $n_A = 1$ and $n_B = 2$,
c) $n_A = 2$ and $n_B = 2$, d) $n_A = 2$ and $n_B = 1$

12. Using the correct values of n_A and n_B from the previous question, and given that 25 cm^3 of 0.1 M sodium carbonate solution was used in neutralising 21 cm^3 of hydrochloric acid solution, what is the concentration of the acid in moles per litre?

- a) 0.024 mol/litre , b) 0.06 mol/litre , c) 0.12 mol/litre , d) 0.24 mol/litre

13. Given that the concentration of a hydrochloric acid solution is 0.24 mol/litre , what is this concentration in grams per litre? The molar mass of hydrochloric acid is 36.5 grams .

- a) 8.52 grams/litre , b) $0.0066 \text{ grams/litre}$,
c) 8.76 grams/litre , d) $36.74 \text{ grams/litre}$

14. 4.9 g of sulphuric acid is dissolved in water and the final solution has a volume of 220 cm^3 . Using titration, it was found that 20 cm^3 of this solution was able to completely neutralise 10 cm^3 of a sodium hydroxide solution. Calculate the concentration of the sodium hydroxide in mol.dm^{-3} .

15. In a titration involving 24 cm^3 KOH solution against a solution containing 1 mol/dm^3 of H_2SO_4 , 28 cm^3 of the acid was found to neutralise the alkali completely.

Write a word and chemical balanced equation for the reaction.

Name a suitable indicator for the titration and state the colour change you would observe at the end point.

Calculate the concentration of the alkali in mol/dm^3 .

Describe a chemical test which you would use to identify the type of salt produced during the reaction.

16. You are provided with the following solutions:

Solution **A**: Hydrochloric acid of concentration $0.150 \text{ mol dm}^{-3}$

Solution **B**: 25cm^3 Sodium carbonate solution of unknown concentration You are requested to determine the concentration of solution **B** by titration. List the apparatus and reagents you would use to carry out this titration. Draw a labelled diagram of the experimental set-up for this titration. Describe clearly how you would carry out the titration to obtain accurate results.

Draw a suitable results table for recording your titration readings.

During the titration 25.0 cm^3 of sodium carbonate required on average 25.10cm^3 of hydrochloric acid, calculate the molarity of sodium carbonate.

The following results were obtained:

- (i) Name a suitable indicator for this titration.
- (ii) State the colour change observed at the end point.

17. **Attitudes and values**

17. Give one example illustrating the importance in daily life of preparing solutions and one importance of titration in the daily life;

UNIT 7

ELECTROLYSIS AND ITS APPLICATIONS



Key Unit Competence:

Analyse the process of electrolysis and its applications in industry and daily life.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Explain the concept of electrolysis
- Describe products of electrolysis for different solutions (e.g., concentrated sodium chloride solution, acidified water).
- Explain the applications of electrolysis.

Skills

2

- Use experiments to show the products of the electrolysis of: concentrated sodium chloride (brine), diluted sodium chloride, Copper (II) sulphate solution, molten sodium chloride and electrolysis of water (dilute sulphuric acid).
- Write balanced equations to represent reactions that take place at each electrode during electrolysis.

Attitudes and values

3

- Develop a culture of working in a team.
- Develop orderliness especially during experiments and presentations.
- Respect of the procedures to perform experiments.
- Appreciate the importance of electrolysis in daily life.
- Develop self-confidence in handling electrical apparatus and chemicals.

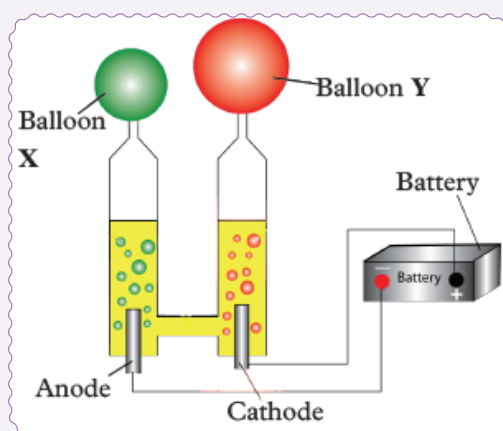
7.0. Introduction

There are many types of reactions as you have learned in senior 2. Among them, we can cite the decomposition reaction. Electrolysis is a decomposition reaction performed using electric current. This book describes and explains the electrolysis of some substances like water, sodium chloride, among others. It also points out the most common applications of the electrolysis processes.

Introductory activity



A Senior 3 student set up the apparatus as shown below. To his surprise, the balloons started expanding after sometime. Balloon Y at the end of the experiment was twice the size of balloon X.



What could be the possible explanation of the observations in this experiment? Can you make a conclusion regarding the behaviour of ionic solutions such as sodium chloride (salt) when an electric current is passed through them?

Electrolysis is the decomposition of an electrolyte by an electric current that is passed through it. The products of electrolysis vary depending on certain factors. Among such factors include the nature of the electrodes used and the concentration of the electrolyte. Electrolysis has found application in various fields, such as electroplating and in purification and extraction of metals.

7.1. Concept of electrolysis

Learning activity 7.1



Research on the concept of electrolysis

The research will focus on the following items:

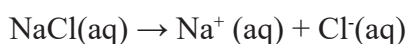
- Definition of electrolysis, electrode, anode and anode
- Role of water in electrolysis
- Factors affecting preferential discharge

An electrolyte, when dissolved in water, breaks up into two types of charged particles, one carrying a positive charge (cations) and the other a negative charge (anions).

Therefore, an electrolyte is a substance that conducts electricity when it is dissolved in water or melted, because it contains free-moving ions (charged particles).

Electrolytes are usually ionic compounds (like sodium chloride, or acids, bases, and salts dissolved in water).

Example



An **electrolysis** is the decomposition of an electrolyte by passing an electric current through it. In other words, electricity is used to cause a chemical reaction that would not happen on its own.

The electrolyte is a compound which when in solution or molten state, conducts an electric current and is decomposed by it.

During electrolysis, electrodes are used. **Electrodes** are electric conductors usually metals or carbon used as the two terminals during electrolysis.

There are two types of electrodes:

- **Anode (+):** The positive electrode. It allows electric current to enter the electrolyte.
- **Cathode (-):** The negative electrode. It allows electric current to leave the electrolyte.

As a result, positive and negative ions move to the correct electrodes, where chemical reactions occur and new substances are produced.

That is, during electrolysis, an electrical energy is used to cause a non-spontaneous chemical reaction.

The electrolyte must contain **mobile ions**, which carry electric charges through the liquid.

For electrolysis to occur, the following are required:

- An electrolyte: dissolved or molten ionic compound
- Two electrodes: a positive electrode (Anode) and a negative electrode (Cathode) and
- A direct current power supply

A direct current (D(C) is used because it flows in one direction: from the positive terminal to the negative terminal. This keeps the anode always positive and the cathode always negative. That is, ions move to fixed electrodes. This allows substances to be formed and collected at the electrodes.

Application activity 7.1



1. What is electrolysis?
2. Name the liquid that allows electricity to pass during electrolysis.
3. Which type of current is used in electrolysis?
4. What is the name of the positive electrode?
5. What is the name of the negative electrode?
6. State two things needed to carry out electrolysis.
7. Which ions move to the cathode during electrolysis?
8. Which ions move to the anode during electrolysis?
9. Explain why the electrolyte must be a liquid.

7.2 Electrolysis

7.2.1 Electrolysis of acidified water or (dilute sulphuric acid)

Learning activity 7.2.1



Experiment on the electrolysis of water

Material required

6-volt or 9-volt battery, Two alligator clip leads or insulated wire, Beaker or glass, Piece of thin cardboard or cardstock, Two sharpened pencils.

Procedure

- Fill the beaker or glass with warm water.
- Carefully remove the erasers and metal sleeves so you can sharpen both ends of each pencil. These pencils are your electrodes. The graphite in them will conduct electricity, but will not dissolve into the water.

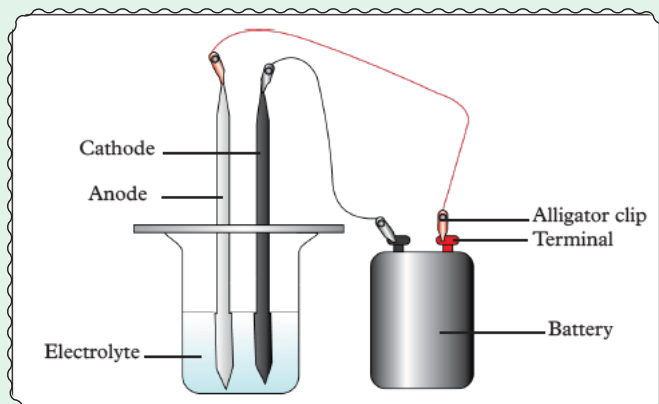


Figure 7.2.1: Electrolysis of water

- Cut a piece of the cardboard to fit over the beaker, then punch two holes in the centre of the cardboard about one inch apart. Push the pencils through the holes and set them in the glass. They should extend into the water,

but not touch the bottom of the glass. The cardboard will hold them in place.

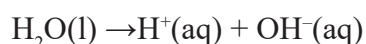
- Connect each pencil to the battery with a crocodile clip lead attached to the exposed graphite (pencil lead). If you do not have crocodile clip leads, use two lengths of wire and strip an inch of insulation off each end.
- Wrap the wire around the graphite of each pencil and connect the wires to the battery. You may need to use tape to hold the wires in place.

Guiding questions

1. Does one pencil collect more bubbles than the other? Which one? Why do you think this is?
2. Why is electrolysis of dilute sulphuric acid referred as electrolysis of acidified water?
 - Write equations taking place at each electrode.
 - Identify which electrode produces more gas bubbles.

Electrolysis of water is the process of using **direct current (DC) electricity** to decompose water into **hydrogen gas** and **oxygen gas**. Because pure water conducts electricity very poorly, a small amount of an electrolyte, such as dilute sulfuric acid, is usually added.

Water is a weak electrolyte. It is only slightly dissociated. Its molecules dissociate into hydrogen ions (H^+) and hydroxide ions (OH^-) i.e.



Hydrogen gas (H_2) is produced at the **cathode (negative electrode)** and **Oxygen gas (O_2)** is produced at the **anode (positive electrode)**.

During electrolysis, the hydrogen ions(H^+) are attracted to the cathode and hydroxide ions (OH^-) to the anode.

- Reaction at cathode (negative electrode)
 $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$; Hydrogen gas liberated at cathode.
- Reaction at anode (positive electrode)
 $4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$, Oxygen gas liberated at anode

Overall reaction:



The cathode produces more gas bubbles because **twice** as much hydrogen gas is formed as oxygen gas. The volume ratio of hydrogen to oxygen is **2 : 1**.

Application activity 7.2.1



1. Electrolysis of water is the process of:
A. Freezing water, B. Dissolving water in acid, C. Burning water, D. Splitting water using electricity
2. During the electrolysis of water, hydrogen gas is produced at the:
A. Anode, B. Power supply, C. Cathode, D. Both electrodes
3. The gas produced at the anode during electrolysis of water is:
A. Hydrogen, B. Nitrogen, C. Oxygen, D. Carbon dioxide
4. Which type of electricity is used in electrolysis?
A. Alternating current (A(C), B. Direct current (D(C), C. Static electricity, D. Solar energy
5. Why is an electrolyte added to water during electrolysis?
6. Name the two gases produced during the electrolysis of water and state one test for each of them.

7.2.2 Electrolysis of diluted sodium chloride

Learning activity 7.2.2



Investigating the products of electrolysis of dilute sodium chloride

Materials required

Beaker, Carbon (graphite) rods, Wooden splint, Connecting wires fitted with crocodile clips, Dry cells (1.5V) with cell holder/ battery, Stirrer, Bulb/ ammeter, Test tube rack with test tubes, Distilled water, Sodium chloride

Procedure

- Set up the apparatus as shown in figure 7.2.2

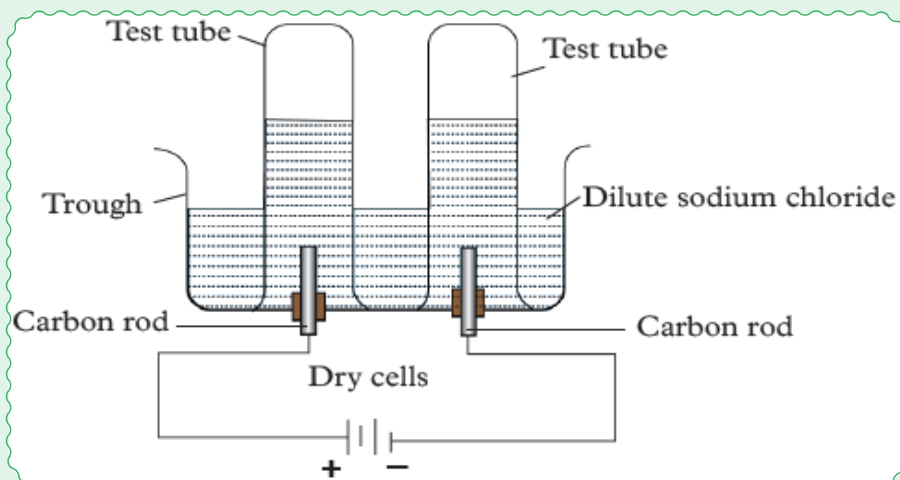


Figure 7.2.2: Electrolysis of dilute sodium chloride solution

- Half-fill the trough with a dilute solution of sodium chloride.
- Switch on the current. What do you observe on the electrodes?

Guiding questions:

1. What is the colour of the gas collected at:
 - a) the anode?
 - b) the cathode?
2. Test the gases collected with a glowing and burning splints.
3. Copy Table 7.2.2 in your notebook and record your observations.

Table 7.2.2: Electrolysis of dilute sodium chloride solution

Gas collected	Colour	Test with glowing / burning splint
Anode		
Cathode		

A colourless gas, which re-lights a glowing splint, is collected at the anode. The gas is therefore oxygen. At the cathode, a colourless gas is collected which burns with a 'pop' sound. This gas is therefore hydrogen.

Now, let us see how these gases are formed at the respective electrodes. We start by finding out which ions are present in the electrolyte, then see how they migrate and finally find out which ones are selectively discharged.

Ions present are $\text{Na}^+(\text{aq})$, $\text{Cl}^-(\text{aq})$ from sodium chloride and $\text{H}^+(\text{aq})$, $\text{OH}^-(\text{aq})$ from water.

The positive ions (cations), Na^+ and H^+ migrate to the cathode. The negative ions, (anions) Cl^- and OH^- , migrate to the anode. This is because of the attraction of opposite charges.

At the cathode	At the anode
Ions present, $\text{Na}^+(\text{aq})$, $\text{H}^+(\text{aq})$	Ions present, $\text{Cl}^-(\text{aq})$, $\text{OH}^-(\text{aq})$
Both ions move to the cathode. H^+ ion is discharged in preference to Na^+ ion because it is lower than Na^+ ion in the electrochemical series. It gains an electron to form H atom. $\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{H}(\text{g})$ $\text{H}(\text{g}) + \text{H}(\text{g}) \rightarrow \text{H}_2(\text{g})$ $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	Both ions move to the anode. OH^- ion is discharged because it is lower in electrochemical series OH^- is discharged by losing an electron. $\text{OH}^-(\text{aq}) \rightarrow \text{OH}(\text{g}) + \text{e}^-$ 4OH^- ions interact to form one mole of oxygen and two moles of water as shown below, releasing 4 electrons which move to the cathode. Overall reaction $4\text{OH}^-(\text{aq}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$
But the 4e^- produced at the anode must react hence overall equation is $4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2(\text{g})$	

Note: The 4e^- released at the anode must be taken up by H^+ ions. So, we must multiply the last equation under cathode by 2 to get the overall equation.

Finally, Na^+ and Cl^- ions remain in solution, resulting in sodium chloride solution of higher concentration.

7.2.3 Electrolysis of concentrated sodium chloride

Learning activity 7.2.3



Investigating the products of the electrolysis of brine

Materials required

- The apparatus is the same as those in experiment 7.2.2. You can also use glass cell fitted with inert electrode of platinum.
- 1 M sodium chloride solution

Procedure

- Set up the apparatus as shown in the figure 7.2.3

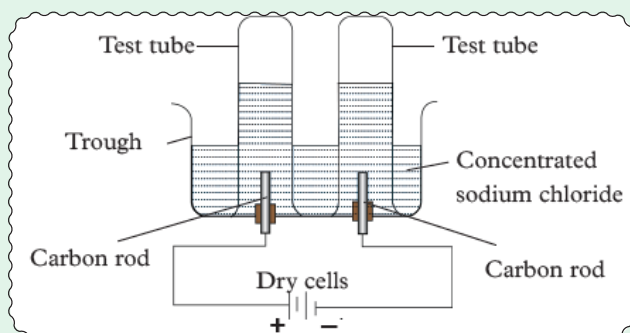


Figure 7.2.3: Electrolysis of brine

- Half-fill the beaker or glass cell with 1M sodium chloride solution.
- Switch on the current.
- Note the colour of the gas collected at the anode and the cathode.

Guiding questions

1. At which electrode is a gas with a smell produced?
2. Test the gas produced at each electrode with a burning splint.
3. At which electrode does the gas burn with a "pop" sound?
4. Write equation of the reactions which take place at electrodes.

When the circuit is completed, bubbles are observed at the electrodes. A green-yellow poisonous gas which has a choking irritating smell is collected at the anode. This gas bleaches moist litmus papers and therefore the gas is chlorine. At the cathode, a colourless gas which burns with a 'pop' sound is collected. This gas is therefore hydrogen.

Ions present are $\text{Na}^+(\text{aq})$, $\text{Cl}^-(\text{aq})$ from sodium chloride and $\text{H}^+(\text{aq})$, OH^- from water

At the cathode	At the anode
$\text{Na}^+(\text{aq})$, H^+	$\text{Cl}^-(\text{aq})$, $\text{OH}^-(\text{aq})$
H^+ ions are discharged in preference to Na^+ ions because H^+ ion is lower in electrochemical series than Na^+ ion.	Although OH^- ion is lower in the electrochemical series than Cl^- , the concentration of Cl^- ions at the cathode is greater than the OH^- ions.
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	Therefore, Cl^- ions are discharged in preference to OH^- ions
	$2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$

Note

- The concentration has determined the product at the cathode.
- H^+ and Cl^- ions are discharged leaving Na^+ and OH^- ions. Finally, sodium chloride solution is converted to sodium hydroxide solution.

The chlorine produced at the anode reacts with sodium hydroxide solution to form sodium hypochlorite which is a bleaching agent, and sodium chloride.

Ionic equation for the reaction is $\text{Cl}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{Cl}^-(\text{aq}) + \text{OCl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$

Application activity 7.2.3



- During the electrolysis of brine, explain why the sodium chloride solution must be concentrated.
- For the process above, name the products formed at electrodes and state a test to identify each of them.

7.2.4 Electrolysis of molten sodium chloride

Learning activity 7.2.4



Research using books and internet about the electrolysis of molten sodium chloride. Answer the following questions:

- What are the products formed at the anode and the cathode during the electrolysis of molten sodium chloride?
- Why must the sodium chloride be in a molten state rather than an aqueous solution for this specific process?
- What is the half-equation for the reaction occurring at the anode?
- What is the half-equation for the reaction occurring at the cathode?

Electrolysis of molten sodium chloride is a process in which electricity is used to decompose molten sodium chloride into sodium metal and chlorine gas.

Sodium chloride is an ionic compound made of sodium ions (Na^+) and chloride ions (Cl^-) held together by strong electrostatic forces.

In the solid state, these ions are fixed in position and cannot move. Therefore, solid sodium chloride does not conduct electricity.

When sodium chloride is heated until molten, the ions become free to move, allowing the molten salt to conduct electricity and undergo electrolysis.

The requirements are:

Electrolyte: Molten sodium chloride

Electrodes: Graphite (carbon) electrodes, because they are inert and can withstand high temperatures

Power supply: Direct current (D(C)

– Reactions at the electrodes

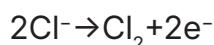
At the cathode (negative electrode):

Sodium ions gain electrons (reduction) to form sodium metal.

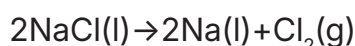


At the anode (positive electrode):

Chloride ions lose electrons (oxidation) to form chlorine gas.



Overall reaction



- Observations during the experiment

Silvery molten sodium metal is formed at the cathode. Greenish-yellow chlorine gas is released at the anode. Chlorine gas has a choking smell and bleaches damp litmus paper.

Safety precautions

Molten sodium chloride is at a very high temperature and must be handled carefully.

Chlorine gas is toxic, so the experiment should be done in a well-ventilated area.

Protective equipment such as gloves and goggles should be worn.

While the lab setup uses a simple crucible, on a large scale, the industrial **Downs Cell** uses a more complex design to ensure the products do not touch each other, as they would react explosively.

Application activity 7.2.4



1. Explain the differences in the electrolysis products of:
 - a) Molten sodium chloride
 - b) Dilute aqueous solution of sodium chloride.
2. Predict the products formed at the anode and cathode when the following molten compounds are electrolysed.
 - a) Sodium fluoride
 - b) Copper (II) iodide
 - c) Zinc bromide

7.2.5 Electrolysis of copper (II) sulphate

Learning activity 7.2.4



(a): Electrolysis of copper(II) sulphate using carbon electrodes

Materials required

Battery/dry cells, Carbon rods, Connecting wires fitted with crocodile clips, Beaker, Bulb (6V) or ammeter, Copper (II) sulphate solution

Procedure

1. Half-fill a beaker with copper (II) sulphate solution
2. Set up the apparatus as shown in Figure 7.2.5.(a).

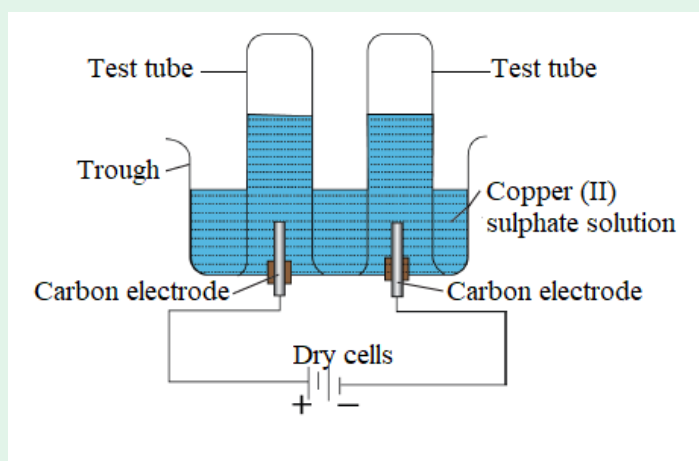


Figure 7.2.5.(a): Apparatus showing electrolysis of copper (II) sulphate using graphite electrodes.

3. Switch on the current and after a short while, observe what happens on the carbon electrodes at the:
 - (i) anode
 - (ii) cathode
4. Record your observations.
5. What do you conclude?

Guiding questions

1. What is the purpose of the bulb?
2. What is the purpose of the switch?
3. What did you observe at each of the electrodes?
4. What happens to the colour of the solution at the end of the experiment?

Ions present are $\text{Cu}^{2+}(\text{aq})$, SO_4^{2-} from copper (II) sulphate and $\text{H}^+(\text{aq})$, $\text{OH}^-(\text{aq})$ from water

At the cathode	At the anode
- $\text{Cu}^{2+}(\text{aq})$, $\text{H}^+(\text{aq})$ both move to the cathode. - Cu^{2+} ions gain electrons, to form copper which appears as a brown deposit, at the cathode. That is, Cu^{2+} ions are preferentially discharged. Ionic equation: - $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ Account for the 4e^- from the anode	- $\text{SO}_4^{2-}(\text{aq})$, $\text{OH}^-(\text{aq})$ both move to the anode. - OH^- ions preferentially lose electrons being lower in the electrochemical series. They combine to form water and oxygen Ionic equation: $4\text{OH}^-(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g}) + 4\text{e}^-$ Bubbles of oxygen gas are observed at the anode.

Copper (II), Cu^{2+} , ions, give copper (II) sulphate solution its blue colour. As they are being discharged and deposited at the cathode as copper metal, the colour of the solution fades and if we continue with electrolysis the solution finally turns colourless and acidic.

If we continue passing electricity in the colourless solution of copper sulphate what would be the product at the cathode is hydrogen gas and at the anode is oxygen gas.

Note Further electrolysis of the colourless solution of copper sulphate would give the same products as electrolysis of dilute sulphuric acid.

Learning activity 7.2.4



7.2.4 (b) Electrolysis of copper (II) sulphate using copper electrodes

Materials required

Copper electrodes, Electronic balance, Distilled water, Propanone

Procedure

- Put copper (II) sulphate solution in a beaker.
- Clean the copper electrodes until they shine.

- Weigh the electrodes and record the mass in your notebook as in Table 7.2.5.(b) below.
- Set up the apparatus as shown in Fig. 7.2.5 (b) and switch on electric current.

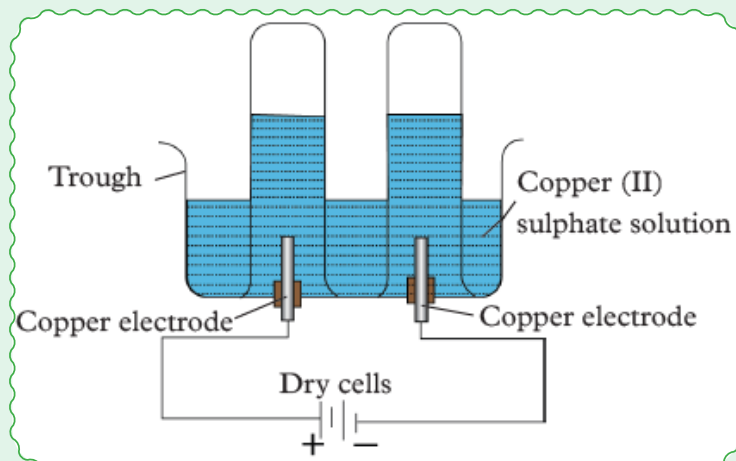


Figure 7.2.5.(b): Electrolysis of copper (II) sulphate using copper electrodes

After 10 minutes, note the colour of the electrolyte and rinse the electrodes with distilled water.

- Dip the electrodes in propanone and then let them dry off.
- When dry, re-weigh the electrodes.
- Record the mass obtained after the experiment.
 - Why must electrodes be cleaned before electrolysis?
 - Why must the electrodes be washed with distilled water and propanone after electrolysis?

Table 7.2.5.(b): Results of electrolysis of copper (II) sulphate solution using copper electrodes

	Before passing electricity		After passing electricity	
	Anode	Cathode	Anode	Cathode
Colour of copper (II) sulphate solution				
Mass of electrode				
Change in mass of electrode				

Guiding questions

1. Observe and describe what happens at each electrode?
2. What happens to the colour of the solution at the end of the experiment?

The electrodes must be cleaned to remove any oxide that might have formed on the surface. The electrodes are washed with water and propanone to ensure they are completely dry.

The masses of both electrodes change during electrolysis. The anode loses mass while the cathode gains mass.

Ions presents are $\text{Cu}^{2+}(\text{aq})$, $\text{SO}_4^{2-}(\text{aq})$ from copper (II) sulphate and $\text{H}^+(\text{aq})$, $\text{OH}^-(\text{aq})$ from water:

At the cathode	At the anode
$-\text{Cu}^{2+}(\text{aq}), \text{H}^+(\text{aq})$ These ions move to the cathode. Cu^{2+} ions are lower in electrochemical series and are preferentially discharged by gaining electrons to form a brown solid which is copper. $-\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ Copper is deposited as shown in figure 7.2.5.(c) and it increases the mass of cathode.	$-\text{SO}_4^{2-}(\text{aq}), \text{OH}^-(\text{aq})$ These ions move to the anode. We would expect OH^- ions to be oxidised. But because of the nature (type) of electrodes used i.e copper; none of the ions is discharged. Instead, copper electrode dissolves i.e. $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$ Copper is an active electrode and it therefore gets dissolved. This process is called electrode ionisation and it causes a decrease in anode mass.

Note: Copper (II) ions, Cu^{2+} , make copper (II) sulphate solution blue. For each ion removed at the cathode, it is replaced when the anode copper dissolves at the anode. Therefore, the solution remains blue and the concentration of the solution remains constant.

When we electrolyse copper (II) sulphate solution using copper electrodes, the mass of both electrodes changes. The anode loses mass while the cathode gains mass. The loss in mass at the anode is equal to the mass gained at the cathode. Remember that metals are always formed at the cathode. This is the reason why even if we use carbon (graphite) as the cathode, the metal will still be deposited on it.

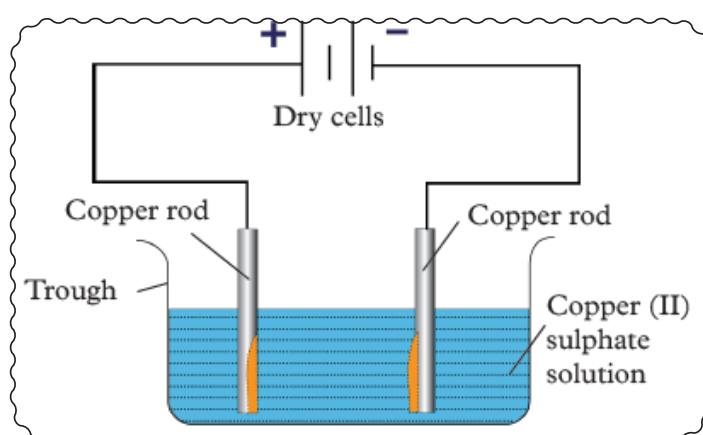


Figure 7.2.5.(c): Loss in mass at the anode is equal to gain in mass at the cathode

Application activity 7.2.5



1. Draw a table to summarise the similarities and differences when copper (II) sulphate solution is electrolysed using:
 - a) Platinum electrodes
 - b) Copper electrodes
2. Complete the following passage using the words below:

(Cathode, copper, electrolysed, impure, solution)

When a solution of copper (II) sulphate is _____ using _____ electrodes the copper atoms at the _____ anode go into _____ as copper ions. At the _____ the copper ions turn into copper atoms.

7.3 Applications of electrolysis

Research activity

1. Using the internet and textbooks from the library, research on the applications of electrolysis. Use the following subheadings:
 - (i) Extraction of metal
 - (ii) Refining metals
 - (iii) Electroplating metal

7.3.1 Electroplating

Electroplating is a process whereby a thin layer of desired material is applied on a required metal of interest. Electroplating is mostly done on stainless steel to prevent rusting. It is also done on some objects to make them look shiny and attractive.

Examples

- (i) Most metal objects are made of iron or steel. To prevent rusting, they are nickel-chromium plated e.g. bicycle handlebars, car rims, among others.
- (ii) Silver or gold plating is done e.g. on spoons, plates and jewelry items to improve their appearance.
- (iii) Tin cans – Cans made of steel are electroplated using a thin layer of tin to prevent them from rusting.

Learning activity 7.3.1



Electroplating a spoon with copper

Materials required

1.5-volt D battery with battery holder, Two crocodile clip leads or insulated wire, Beaker or glass, Copper (II) sulphate, Copper electrode (or coil of copper wire), Spoon, Safety equipment.

Procedure

- Prepare the spoon for copper-plating by cleaning it with toothpaste or soap and water. Dry it off on a paper towel.
- Stir copper sulphate into some hot water in a beaker until no more will dissolve. Your solution should be dark blue. Let it cool.
- Use one alligator clip to attach the copper electrode to the positive terminal of the battery (this is now the anode) and the other to attach the spoon to the negative terminal (now called the cathode).

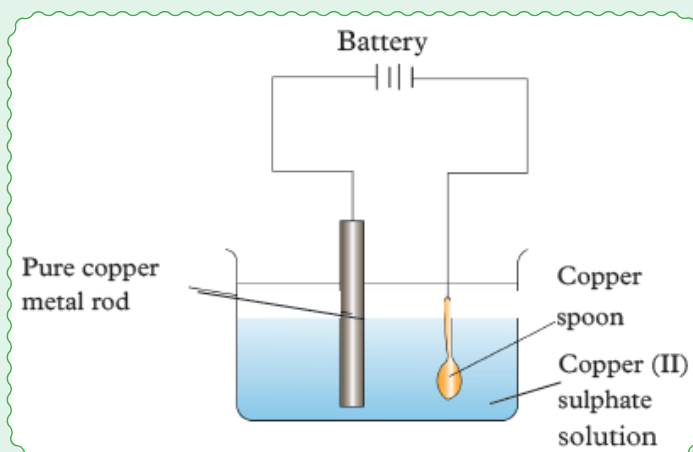


Figure 7.4.1: Electroplating a spoon using copper

- Partially suspend the spoon in the solution by wrapping the wire loosely around it and placing it across the mouth of the beaker. The wrapping wire should not touch the solution.
- Place the copper electrode into the solution, making sure it does not touch the spoon and the solution level is below the alligator clip. An electrical circuit has now formed and current is flowing.
- Leave the circuit running for 20-30 minutes.

Guiding questions

1. Explain what happens at each electrode.
2. Write equations for the reaction that take place at each electrode.
3. What is the colour of the spoon at the end of the experiment?
4. Explain why the electrolyte process was continuous.

Copper sulphate solution is an electrolyte. When electric current is flowing, oxidation (loss of electrons) takes place at the copper anode, adding copper ions to the solution. Those ions move to the cathode, where reduction (gain of electrons) takes place, plating or depositing the copper metal onto the spoon. There were already copper ions present in the copper sulphate solution before the reaction, but the oxidation reaction at the anode kept replacing them in the solution as they were plated onto the key, keeping the reaction going.

The coating of a metal object with another metal object is called **electroplating**. It is carried out in a cell called plating bath. It contains an electrolyte. For silver plating electrolyte is a solution of silver salt. The article to be plated is made the cathode in the cell so that metal ions move to it when inserted in electrolyte.

Reaction at cathode (negative electrode)

Copper ions + 2electrons $\rightarrow$ Copper atoms

$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$ Copper metal deposit at cathode.

Reaction at anode (positive electrode)

- Copper atoms dissolve from anode.
- Copper metal of anode will dissolve and added copper ions the solution.

Copper atoms $\rightarrow$ copper ions + 2 electrons

$\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-}$

7.3.2 Extraction of metals

Extraction of metals by the process of electrolysis is known as **electro-metallurgy**. This process is used to extract highly reactive metals such as sodium, magnesium and aluminium from their molten ionic compounds, e.g.

- Sodium is extracted from molten sodium chloride.
- Aluminium is extracted from molten bauxite (hydrated aluminium oxide, $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$).

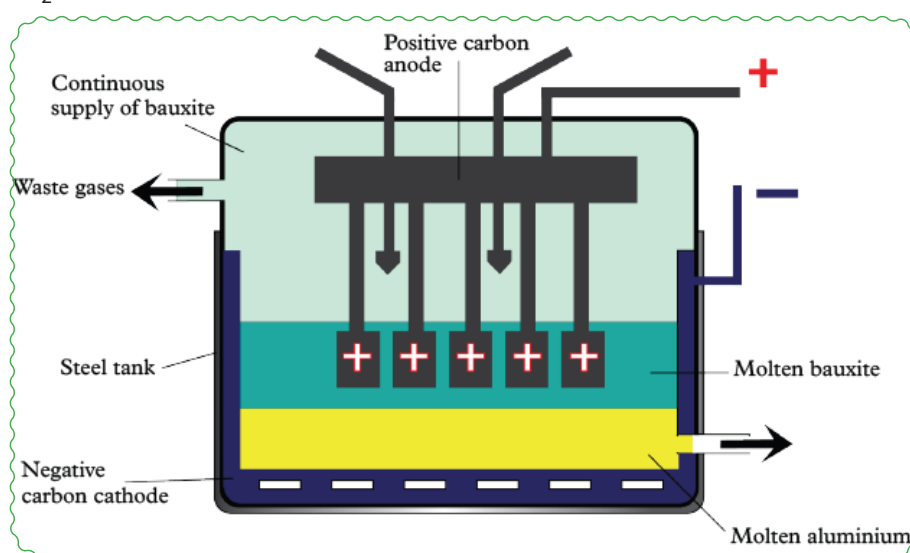


Figure 7.3.2): Hall's cell for electrolysis of aluminium oxide

- Reaction at cathode (negative electrode)

Aluminium ions + 3 electrons → Molten aluminium



Molten aluminium will be collected at the bottom near cathode.

- Reaction at anode (positive electrode)

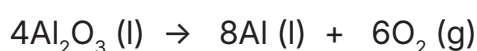
Oxide ions oxygen gas + 12 electrons



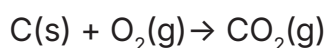
Oxygen gas will be discharged at anode.

Overall reaction:

Aluminium oxide → Molten aluminium + oxygen gas



Oxygen is given off at the positive anode. Carbon dioxide is also given off at the carbon anode because hot oxygen reacts with carbon anode to form carbon dioxide gas.



The carbon anodes slowly disappear because each molecule of carbon dioxide which is given off takes a little piece of carbon away with it.

The carbon anodes need to be replaced when they become too small. Continual replacement of the anodes is a major expense.

7.3.3 Manufacture of some gases such as oxygen, hydrogen, chlorine and manufacture of sodium hydroxide

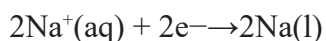
Electrolysis is an important industrial process used to manufacture useful gases and chemicals such as **chlorine (Cl₂)**, **hydrogen (H₂)**, **oxygen (O₂)** and **sodium hydroxide (NaOH)**.

Raw material: Concentrated sodium chloride solution (brine, NaCl(aq))

Electrolysis of Brine: In industry, brine is electrolysed using Mercury cathode (negative electrode) and Graphite anode (positive electrode)

- **At the cathode (Mercury)**

Sodium ions gain electrons to form sodium metal.



The sodium dissolves in mercury to form sodium-mercury amalgam (**NaHg**).

The sodium-mercury amalgam (NaHg) reacts with water to produce sodium hydroxide and hydrogen gas.

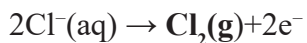


The mercury produced is recycled and used again.

Products formed: Sodium hydroxide (NaOH) and Hydrogen (H₂)

– At the anode (Graphite)

Graphite is made the anode because it is an inert electrode hence resists corrosion by chlorine which is discharged and collected at the anode i.e. Chloride ions lose electrons to form chlorine gas.



Product: Chlorine gas (Cl_2)

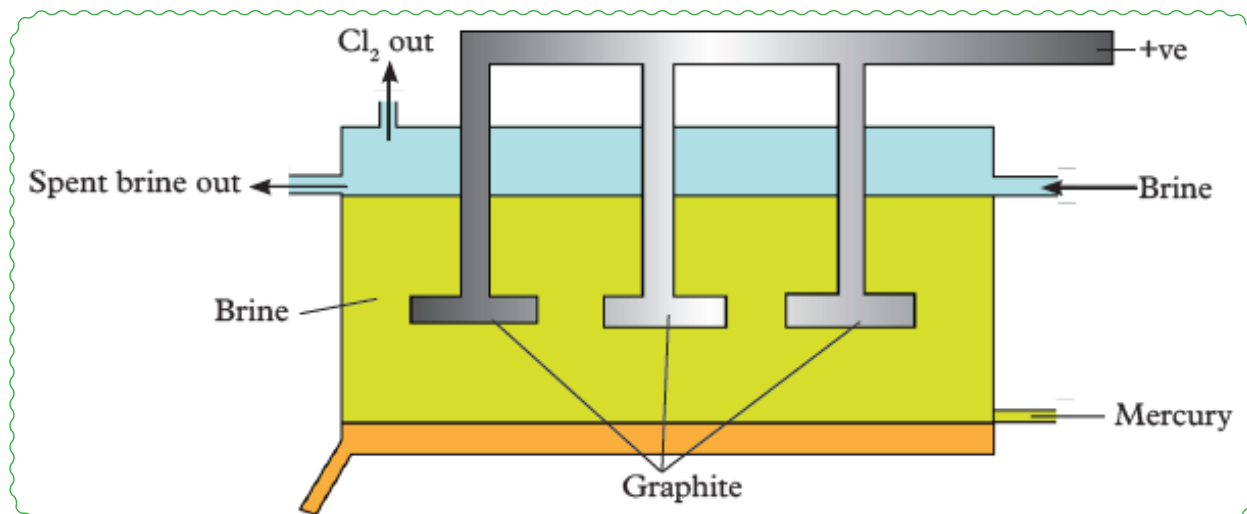


Figure 7.3.3: The flowing mercury cell for the manufacture of sodium hydroxide, chlorine and hydrogen

The mercury produced is recycled. Chlorine, sodium hydroxide and hydrogen are packaged in readiness for use.

Note: Mercury is a heavy toxic metal. It can cause heavy metal poisoning to animals and humans. We should therefore ensure that all measures are taken to avoid any traces of mercury finds its way to the environment through the effluents from industries or any other means.

By electrolysis, practically pure oxygen can be obtained on a large scale by electrolysis of water. Hydrogen which is obtained is used as fuel and oxygen is a by-product that can be delivered to some industries or hospitals.

Application activity 7.3



1. Complete the passage using the words below:

(Anode, cathode, electrolyte, plate, solution)

To electroplate an object you make the object to be electrolyzed the _____. The _____ is the metal that will _____ the surface.

The _____ is a _____ of the salt compound of the plating metal.

2. Using half equations explain how silver can be used to electroplate an object.

You want to electroplate a piece of copper with chromium. Draw a labelled diagram of the apparatus you could use.

Complete the passage using appropriate words.

Aluminium is extracted by electrolysis of molten aluminium oxide _____ in _____. This mixture melts at much _____ temperature than _____ aluminium oxide. Therefore, a lot of _____ is saved and the electrical _____ of the electrolyte is improved.

7.4 End of unit assessment

End of unit assessment



7.4.1. Project activity: Construction of a simple apparatus to demonstrate electrolysis of solutions.

Context

Electrolysis is an important chemical process in which an electric current causes a chemical change in a solution or molten substance. It is widely used in industries for electroplating, purification of metals, and the production of chemicals such as hydrogen and chlorine. Understanding electrolysis helps students connect concepts of electricity, ions, and chemical reactions with real-life applications.

In many schools, laboratory equipment may be limited. Therefore, students can apply scientific knowledge and creativity to construct a simple electrolysis apparatus using locally available and affordable materials.

Challenge

How can you design and construct a simple, safe, and functional apparatus using readily available materials to demonstrate the process of electrolysis and observe the products formed at the electrodes?

Task

Working in groups, you will:

1. Research the principles of electrolysis.
2. Identify suitable materials that can be obtained locally.
3. Design and construct a simple electrolysis apparatus.
4. Use the apparatus to perform electrolysis of a selected solution (e.g., copper sulfate solution, sodium chloride solution, or acidified water).

5. Observe and record changes occurring at the electrodes.
6. Explain the observations using scientific concepts of ions, oxidation, and reduction.
7. Present the apparatus, procedure, observations, and conclusions to the class.

Guiding Questions

1. What is electrolysis, and what conditions are necessary for it to occur?
2. What materials can be used to construct a simple electrolysis apparatus?
3. What role does each component of the apparatus play?
4. Why is a source of direct current required?
5. What happens at the anode and cathode during electrolysis?
6. How can the products formed during electrolysis be identified?
7. What safety precautions should be followed during the experiment?
8. How can the design be improved to make the apparatus more effective and reliable?
9. What are some industrial and everyday applications of electrolysis?

Expected Outcome

By the end of the project, you should be able to:

- Construct a working electrolysis apparatus from locally available materials.
- Demonstrate the electrolysis of a solution successfully.
- Identify and explain reactions occurring at the electrodes.
- Record and interpret experimental observations accurately.
- Apply scientific reasoning to explain the results obtained.
- Communicate findings through a report, presentation, or demonstration.
- Appreciate the practical applications of electrolysis in everyday life and industry.

7.4 End of unit assessment answers

A. Knowledge and understanding

- Which of the following statements concerning electrolysis is incorrect?
 - During electrolysis, chemical energy is converted into electrical energy.
 - Electrolysis involves decomposition of an electrolyte.
 - An energy supply is required for electrolysis.
 - Graphite is commonly used as the inert electrodes in electrolysis.
- Which of the following is not an essential component of the set-up for electrolysis?
 - Electrolyte
 - Electrodes
 - Ammeter
 - Energy supply
- Which of the following substances would not be decomposed by electricity?
 - Molten silver
 - Dilute sulphuric acid
 - Concentrated sodium chloride solution
 - Aqueous silver nitrate
- Which of the following ions are reduced during electrolysis of molten silver chloride?
 - Silver ions
 - Chloride ions
 - Hydrogen ions
 - Hydroxide ions
- Which of the following sets of changes would occur during the electrolysis of dilute sulphuric acid using platinum electrodes?

Product at cathode	Product at anode	Concentration of sulphuric acid
Oxygen	Hydrogen	Increases
Hydrogen	Oxygen	Increases
Oxygen	Hydrogen	Decreases
Oxygen	Oxygen	Decreases

6. Which of the following combinations is correct?

Gas X	Gas Y
Hydrogen	Oxygen
Oxygen	Hydrogen
Hydrogen	Chlorine
Chlorine	Hydrogen

7. The products of the electrolysis of dilute magnesium iodide solution using carbon electrodes are

- A. Hydrogen and iodine
- B. Hydrogen and oxygen
- C. Magnesium and iodine
- D. Magnesium and oxygen

8. Which of the following substances, when electrolyzed using graphite electrodes, will liberate gaseous products at both electrodes?

- A. Molten sodium chloride
- B. Concentrated sodium bromide solution
- C. Dilute zinc sulphate solution
- D. Molten silver nitrate

9.

- a) The (higher/lower) the concentration of an ion in a solution, the greater is the probability of its being discharged at its appropriate electrode.
- b) Cations are formed by (Loss/gain) of electrons and anions are formed by (Loss/gain) of electrons.
- c) Electrolysis is the passage of (electricity/ electrons) through a liquid or a solution accompanied by a (physical/chemical) change

10. A student carries out electrolysis of acidified water using graphite electrodes.

Explain why the volume of hydrogen collected is twice the volume of oxygen.

- a) Write:
 - (i) The word equation for the electrolysis of water
 - (ii) The balanced chemical equation
- b) Why are graphite electrodes used?
- c) State two safety precautions during the experiment.
- d) Name one use of hydrogen gas produced.

B. Skills

11. You are provided with the following apparatus and materials:

Beaker, DC power supply (battery), Two carbon (graphite) electrodes, Copper (II) sulphate solution, Connecting wires, Switch, Balance

- Draw a well-labelled set-up used to electrolyse copper (II) sulphate solution using carbon electrodes.
- Describe the procedure you can use to carry out the electrolysis of copper (II) sulphate.
- State the observations made during the experiment at the cathode and at the anode

In the colour of the solution

- Name the product formed at: i. anode ii. cathode
- Write the ionic equation for the reaction at each electrode cathode
- State three precautions that should be taken during the electrolysis of copper (II) sulphate.

A thin layer of metal can be deposited on an object by the process of electroplating.

- List two uses of electroplating.
- Which of the following objects cannot be electroplated: Clay bowl, Iron tray, Porcelain plate, Steel fork, Wooden spoon? Why ?
- (i) Draw a labelled diagram of a set – up that can be used to coat a layer of nickel on the surface of a steel coin.
(ii) Write balanced equations, including state symbols, for the reactions at the cathode and the anode.

C. Attitudes and values

12. During the electrolysis of concentrated sodium chloride,

- Name the product formed at:
 - the anode
 - the cathode
- Name the substance remaining in solution after electrolysis.
- State one industrial use of:
 - chlorine
 - sodium hydroxide
- State one safety or environmental precaution associated with this electrolysis.

UNIT 8

INTRODUCTION TO ORGANIC CHEMISTRY



Key Unit Competence:

Analyse the nature, properties, compositions, and applications of organic compounds in relation to their occurrence and differences from inorganic substances.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Demonstrate understanding of organic chemistry.
- Describe the concept of a homologous series and its characteristics.

Skills

2

- Use simple experiments to classify compounds into organic and inorganic.
- Compare the organic and inorganic compounds.
- Relate the applications of organic compounds to everyday life and industrial processes.

Attitudes and values

3

- Develop a team approach and sense of responsibility in group discussions and experiments.
- Appreciate the economic importance of organic chemistry in daily life such as fuels, drugs, plastics, etc.

8.0. Introduction

Organic compounds are carbon-based compounds found in all living organisms, including plants and animals. Certain carbon compounds like oxides, carbonates, hydrogen carbonates, and carbides are not considered as organic. The term "organic" originates from "organism," reflecting the historical belief that these compounds could only be produced by living beings. This view changed in 1828 when Friedrich Wöhler synthesized urea in the laboratory, proving organic compounds could be artificially made. Organic compounds are present in daily life in foods, clothing, fuels, paper, plastics, and leather, and they are essential for life processes (e.g., carbohydrates, proteins, vitamins, fats, and enzymes). Organic chemistry studies the structure, properties, reactions, and applications of carbon-containing molecules, both naturally occurring (e.g., cotton, wool, petroleum) and man-made (e.g., plastics, synthetic fabrics, pharmaceuticals).

Introductory activity



1. Make a list of things which you
 - A. eat or drink daily
 - B. wear daily
 - C. Use daily
2. Can you distinguish the objects you have listed as organic or inorganic? Explain reason.
3. Can you name some materials around you that are made of carbon compounds?

8.1. Definition of organic chemistry and occurrence of organic compounds.

Learning activity 8.1



Using textbooks, internet resources, and other reference materials, research and answer the following questions:

1. Where are organic compounds commonly found in nature and everyday life? Provide at least two specific examples with brief descriptions.
2. In your own words, explain what organic compounds are and what makes them different from inorganic compounds.
3. Why do you think organic compounds are so abundant and important in living organisms and modern industry?
4. Present your findings clearly, using complete sentences and scientific terminology where appropriate.

8.1.1. Definition of organic chemistry

Organic chemistry is the branch of chemistry that studies compounds containing carbon chemically bonded to hydrogen. These compounds may also contain oxygen, nitrogen, sulfur, phosphorus, and halogens.

Not all carbon-containing compounds are organic. Exceptions include carbonates (CaCO_3), cyanides (KCN), carbides (CaC_2), carbon oxides (CO , CO_2), carbonic acid (H_2CO_3), and carbon disulfide (CS_2), which are classified as inorganic despite containing carbon.

Since organic compounds contain carbon bonded to hydrogen, they are considered derivatives of hydrocarbons, the simplest organic compounds consisting only of carbon and hydrogen.

Organic chemistry examines the structure, properties, reactions, and applications of carbon-hydrogen containing molecules. These molecules form essential biological compounds including proteins, enzymes, carbohydrates, lipids, nucleic acids, and vitamins.

Organic compounds exist naturally (cotton, wool, petroleum) and synthetically. Using organic chemistry principles, industries have developed plastics, synthetic fibers, pharmaceutical drugs, dyes, and pesticides that are essential to modern life.

8.1.2. Occurrence of organic compounds

The simplest organic compounds are naturally present in crude oil and natural gas. These two resources are often found together and are formed from the gradual decomposition of marine plants and animals over millions of years. Crude oil is a complex mixture of various hydrocarbons, predominantly alkanes. These alkanes are extensively used as fuels and are extracted from crude oil through a process called fractional distillation, which separates the crude oil into different fractions based on the boiling points of its hydrocarbon components.

Methane (CH_4), the simplest alkane, is obtained from several natural sources:

(i) Natural gas: Methane is the main component of natural gas, comprising approximately 90% of its composition. Natural gas also contains smaller amounts of ethane (C_2H_6), propane (C_3H_8), and butane (C_4H_{10}). It is typically found in deposits above petroleum layers beneath the Earth's surface.

(ii) Coal gas: Methane constitutes 32–34% of coal gas, which is produced during the destructive distillation of coal.

(iii) Biogas and related gases: Methane is the major constituent of biogas, gobar gas (produced from cattle dung), and sewage gas, which are generated through anaerobic digestion processes.

(iv) Marshlands: Methane occurs naturally in marshy lands and wetlands, where it is formed through microbial decomposition of organic matter such as plant wastes, animal dung, and other biological materials. This naturally occurring methane is sometimes called "marsh gas."

Application activity 8.1



1. What is the definition of organic chemistry?
2. Why is carbon considered the backbone of organic chemistry?
3. How does organic chemistry relate to everyday life?

8.2. Differences between organic and inorganic chemistry

Learning activity 8.2



Experiment: Observing the behaviour of petroleum, water, and their mixture

Objective: To investigate the flammability and behaviours of petroleum, water, and a petroleum-water mixture.

Materials: Petroleum, Water, Containers with a small hole, Pieces of cloth (cotton), Matchstick

Procedure:

1. Pour petroleum into a container with a small hole and insert a piece of cloth so it absorbs the petroleum.
2. Pour water into a separate container with a small hole and insert a piece of cloth so it absorbs the water.
3. Carefully light a match and bring it near the petroleum-soaked cloth. Observe and record the behaviour.
4. Repeat the process with the water-soaked cloth and record the observation.
5. Mix petroleum and water in a container.
6. Observe and record what happens.

Observation:

1. Note the behaviour of each cloth (petroleum-soaked, water-soaked, and mixture-water and petroleum).
2. Record differences in flammability, burning rate, and any other noticeable reactions.

Organic compounds are chemical compounds that mainly contain carbon and hydrogen, and often oxygen, nitrogen, or other elements. They are usually found in living things or substances derived from living organisms, such as sugars, oils, and proteins. **Inorganic compounds**, on the other hand, generally do not contain carbon-hydrogen bonds and are found in non-living things, such as salts, metals, and minerals.

Table 8.2: Differences between organic and inorganic compounds

No.	Property	Organic Compounds	Inorganic Compounds
1	Composition	Usually contain carbon, hydrogen, and oxygen. They always contain carbon-hydrogen (C-H) bonds.	Contain all the known elements. They may contain carbon (e.g., CO_2 , CaCO_3) but do not contain carbon-hydrogen (C-H) bonds.
2	Solubility	Generally insoluble in water but soluble in organic solvents (e.g., alcohol, ether).	Generally soluble in water and non-soluble in organic solvents.
3	Flammability	They are highly flammable (burn easily).	They are non-inflammable (do not burn easily).
4	Electrical Conductivity	In most aqueous solutions, they are poor conductors of heat and electricity.	In aqueous solutions, they are known to be good conductors of heat and electricity (as they form ions).
5	Melting and Boiling Points	They have low melting and boiling points (due to weaker intermolecular forces).	They have high melting and boiling points (due to strong ionic or metallic bonds).

Application activity 8.2



List three differences between organic and inorganic compounds.

How do the structures of organic compounds differ from inorganic compounds?

Give two examples of organic compounds and two of inorganic compounds.

8.3. Homologous series and application of organic chemistry

Learning activity 8.3



Observe the list of compounds below

Organic compounds: methane (CH_4), ethane ($\text{CH}_3\text{-CH}_3$), propane ($\text{CH}_3\text{-CH}_2\text{-CH}_3$), butane ($\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$)

Inorganic compounds: NaCl , CaCO_3 , H_2O , KBr

Can you identify any pattern in the formulas of the organic compounds?

How do the formulas of inorganic compounds differ from those of organic compounds?

Do the organic compounds share a common structure? What is it?

Identify the repeating unit ($\text{-CH}_2\text{-}$) in the homologous series of organic compounds.

Identify two organic compounds used as medicines or drugs and describe their purpose

8.3.1. Homologous series

Organic compounds with related structures and properties are usually separated into "families" known as homologous series. Each member of a series differs from the next member by a single carbon atom and two hydrogen atoms. With each successive member, the properties of the compounds, such as melting and boiling points, change in a regular way.

- All members of a homologous series can be represented by same general formula.
- All the compounds of a homologous series show similar chemical properties.
- The members of a homologous series show a gradual change in their physical properties with increase in molecular mass.
- Any two adjacent homologous differ by a -CH_2 group.

Example of homologous series.

1. $\text{CH}_3\text{-CH}_3$; $\text{CH}_3\text{-CH}_2\text{-CH}_3$; $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$; $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_3$.
2. $\text{CH}_3\text{-CH}_2\text{-OH}$; $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-OH}$; $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$; $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$
3. $\text{CH}_3\text{-COOH}$; $\text{CH}_3\text{-CH}_2\text{-COOH}$; $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-COOH}$; $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-COOH}$

The list contains three homologous series:

1. Alkanes – Ethane, Propane, Butane, Pentane
2. Alcohols – Ethanol, Propanol, Butanol, Pentanol
3. Carboxylic acids – Ethanoic acid, Propanoic acid, Butanoic acid, Pentanoic acid

Note: In each homologous series, successive members differ by one $\text{-CH}_2\text{-}$ group, have similar chemical properties, and show a gradual change in physical properties such as boiling point.

8.3.2. Applications of organic chemistry

1) Application in daily life.

In our day-to-day life, we find many substances or materials that are commonly used and the later are made of organic compounds.

- Food: starch, fats, proteins, vegetables, ...
- Clothes: cotton, wool, nylon, Dacron,
- Fuels: petrol, diesel oil, and kerosene
- Dyes of all kinds
- Cosmetics (body lotion, ...)
- Soaps and detergents

- Medicine: cortisone, sulphonamide, penicillin, ...
- Drugs: morphine, cocaine, ...
- Stationery: pencils, paper, writing ink, ...
- Insecticides, rodenticides, ovicides ...

2) Applications in industry

The knowledge of organic chemistry is required in many industries such as manufacture of food, pharmacy, manufacture of dyes and explosives, alcohol industry, soil fertilisers, petroleum industry, etc.

Application activity 8.3



Define the term homologous series.

Outline three characteristics of a homologous series

Give three day-to-day life applications of organic chemistry.

Some organic compounds are used as solvents in laboratories. Give two examples and explain why they are preferred over inorganic solvents.

8.4 End of unit assessment

8.4 End of unit assessment



A. knowledge and understanding

- 1) Which of the following compounds is organic?
(a) NaCl, (b) CH₄, (c) H₂O, d) CaCO₃
- 2) Members of a homologous series differ from one another by:
One carbon atom only
One hydrogen atom
A -CH₂- group
Two carbon atoms
- 3) Give two examples of organic compounds and state where they occur.
- 4) How does organic chemistry differ from inorganic chemistry?
- 5) Define a homologous series and explain its main characteristics.
- 6) List three applications of organic chemistry in everyday life.

B. Skills

Classify the following compounds as organic or inorganic: CO_2 , CH_4 , NaCl , C_2H_6 .

Identify the next member of the homologous series for the following compound: C_3H_8 .

Using a table, give examples of organic compounds that are obtained from living and non-living sources.

Compare the physical properties (melting and boiling points) of members of a homologous series.

C. Attitudes and values

7. Discuss how the application of organic chemistry can improve daily life or industry.

UNIT 9

ALKANES



Key Unit Competence:

Analyse the structure, properties in relation to their uses of alkanes.

Learning objectives

By the end of this unit, you should be able to:

1 Knowledge and understanding

- Describe the physical and chemical properties of alkanes
- State the uses of methane and some other alkanes.
- Explain structural isomerism.

2 Skills

- Prepare methane gas in the laboratory.
- Apply IUPAC rules to name alkanes (C1-C10).
- Write balanced chemical equations of combustion, halogenation and cracking reactions.

3 Attitudes and values

- Develop a team approach and sense of responsibility in group discussions and in performing experiments.
- Appreciate the economic importance of alkanes in daily life such as fuels, lubricating, and preservation of fruits.

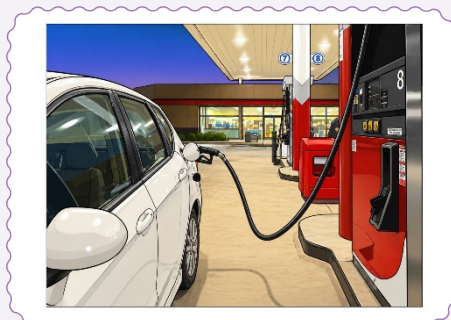
9.0 INTRODUCTION

Alkanes are the simplest and most fundamental class of organic compounds, forming the basis of many other organic substances. They are hydrocarbons, meaning they contain only carbon and hydrogen atoms, and are commonly found in natural sources such as natural gas, petroleum, and biogas. Because of their simple structure and relatively low reactivity, alkanes are often referred to as **saturated hydrocarbons**.

Introductory activity



Look at the picture below and answer the questions that follow.



1. What type of fuel is commonly used to run cars like the one shown in the picture?
2. From which natural resources do fuels such as petrol and diesel come?
3. What kind of chemical compounds make up petrol and diesel?
4. What elements are mainly present in fuels used for cooking gas and vehicle fuel?

9.1. Alkanes

Learning activity 9.1



1. Identify materials used in everyday life that are composed of carbon and hydrogen.
2. Which gas is extracted from Lake Kivu?

In alkanes, each carbon atom is bonded directly to four other atoms. Since in these compounds, all the carbon atoms utilise their full combining capacity by forming single covalent bonds with other atoms, these compounds are called **saturated compounds**.

9.1.1. Definition

Alkanes are hydrocarbons made up of only carbon and hydrogen atoms, with all carbon-carbon bonds being single bonds.

Examples: Ethane, Propane, Butane, Pentane

9.1.2. General formulae and nomenclature of alkanes (from C₁- C₁₀)

General formula of homologous series of alkanes is C_nH_{2n+2} , where n is a positive integer. Thus, alkanes may be defined as the saturated hydrocarbons having general formula C_nH_{2n+2} .

The hydrocarbon compounds were named using the International Union of Pure and Applied Chemistry IUPAC Rules for naming straight-chain alkanes.

The name of a straight chain alkane, may be divided into two parts: **Prefix** and **suffix**. Prefix designates the number of carbon atoms in the chain. For chains containing one to four carbon atoms, special prefixes are used and for chains of five or more carbon atoms, Greek number prefixes are used. IUPAC prefixes for a few carbon chains are given in Table 9.1.

Table 9.1.2 (a): Prefixes for Carbon Chain Lengths

Chain length	Prefix
C ₁	Meth-
C ₂	Eth-
C ₃	Prop-
C ₄	But-
C ₅	Pent-
C ₆	Hex-
C ₇	Hept-
C ₈	Oct-
C ₉	Non-
C ₁₀	Dec-

In order to derive the IUPAC name, a suffix is added to the prefix. The primary suffix used for alkanes is "ane". The IUPAC names of straight chain alkanes containing carbons 1–10 are given in Table 9.2.

Table 9.1.2 (b): Names of Alkanes

Condensed Structural Formula	IUPAC Name	Molecular Formula
CH ₄	Methane	CH ₄
CH ₃ CH ₃	Ethane	C ₂ H ₆
CH ₃ CH ₂ CH ₃	Propane	C ₃ H ₈
CH ₃ CH ₂ CH ₂ CH ₃	Butane	C ₄ H ₁₀
CH ₃ CH ₂ CH ₂ CH ₂ CH ₃	Pentane	C ₅ H ₁₂
CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₃	Hexane	C ₆ H ₁₄
CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ CH ₃	Heptane	C ₇ H ₁₆
CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ CH ₃	Octane	C ₈ H ₁₈

Condensed Structural Formula	IUPAC Name	Molecular Formula
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	Nonane	C_9H_{20}
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	Decane	$\text{C}_{10}\text{H}_{22}$

Application activity 9.1.2



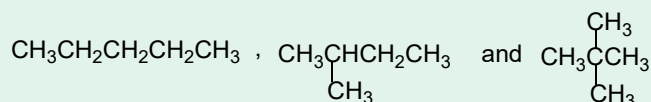
- Given the name, write the chemical formula of Propane, Butane, Pentane, Hexane, Octane, Nonane, Decane.
- Given the chemical formula, write the name of CH_4 , C_2H_6 , C_6H_{14} , C_7H_{16} .
- Which of the following is not the correct formula for alkanes?
 C_2H_2 , C_2H_4 , C_3H_6
- Complete: The systematic nomenclature of organic compounds follows rules established by the
- Name the alkane with the molecular formula C_8H_{18} .

9.1.3. Structural isomerism up to C_5

Learning activity 9.1.3



Look at the structure of the alkanes below



(a)

(b)

(c)

- Determine the molecular formula of the alkane given above.
- Identify whether there are any differences among a, b, and c if yes, explain.

Compounds (a), (b), and (c) all have the molecular formula C_5H_{12} , but their atoms are arranged differently. Compound (a) has a straight chain with no branches, whereas compounds (b) and (c) have branched structures with side chains.

Isomers are compounds that have the same molecular formula but different arrangements of atoms. Therefore, **isomerism** is the existence of two or more compounds with the same molecular formula but different arrangement of their component atoms.

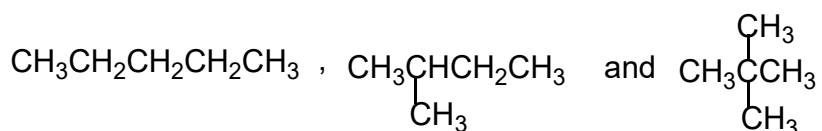
Structural isomerism is the existence of compounds with the same molecular formula but different structural formulae. These compounds are called **structural isomers**.

Each of the first three alkanes, i.e methane (CH_4), ethane (C_2H_6), and propane (C_3H_8) has only one possible structure. Therefore, **structural isomerism is not possible** for alkanes with fewer than four carbon atoms.

Butane has 2 possible structural isomers:



Pentane has 3 possible structural isomers:



An alkane with unbranched chain of carbon atoms is called **normal** alkane or *n*-alkane. The number of isomers increases with increasing number of carbon atoms.

Application activity 9.1.3



1. Using examples explain the concept of "structural isomerism".
2. Explain the term "normal alkane" and give an example.

9.1.4. Physical properties of alkanes

Learning activity 9.1.4



Physical properties of alkanes

Using internet resources and textbooks, research and identify the main physical properties of alkanes.

- **Physical state:** Alkanes contain only C-C and C-H bonds. They have very small difference of electronegativity between carbon and hydrogen atoms. The first four members (C_1 to C_4) are colourless gases; the next thirteen members, (C_5 to C_{17}) are colourless liquids while the higher members are waxy solids.
- **Solubility:** alkanes are soluble in organic solvents such as ether, ethanol, carbon tetrachloride, etc. and insoluble in polar solvents such as water.
- **Boiling Points:** Alkanes generally have low boiling points. The boiling points of *n*-alkanes increase regularly with the increase in the number of carbon atoms.
- **Melting Points:** The melting points of alkanes generally increase with the increase in molecular mass. For example, let us look at the melting points of propane to *n*-octane in Table 9.1.4 The physical properties of the first ten straight chain alkanes are given in Table 9.1.4
- **Density:** The densities of alkanes increase with increasing molecular masses but become constant at about 0.8 g cm^{-3} . This means that liquid alkanes are lighter than water.

Table 9.1.4: Physical Properties of the First 10 n-Alkanes with the physical state at 20°C.

Alkane	Formula	Boiling Point (K)	Melting Point (K)	Density (gcm ⁻³)
Methane	CH ₄ (g)	109	90	0.55
Ethane	C ₂ H ₆ (g)	184	90	0.51
Propane	C ₃ H ₈ (g)	231	84	0.50
Butane	C ₄ H ₁₀ (g)	273	135	0.58
Pentane	C ₅ H ₁₂ (l)	309	143	0.63
Hexane	C ₆ H ₁₄ (l)	342	178	0.66
Heptane	C ₇ H ₁₆ (l)	371	182	0.68
Octane	C ₈ H ₁₈ (l)	399	216	0.70
Nonane	C ₉ H ₂₀ (l)	424	222	0.72
Decane	C ₁₀ H ₂₂ (l)	447	243	0.73

Application activity 9.1.4



- Alkanes are insoluble in water because they are:
A. Ionic, B. Polar, C. Non-polar, D. Acidic
- Which property increases as the chain length of alkanes increases?
A. Volatility, B. Flammability, C. Boiling point, E. Solubility in water

9.1.5. Chemical properties of alkanes

Learning activity 9.1.5



(a): Chemical properties of Alkanes

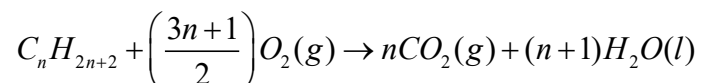
Using internet resources and textbooks, research and identify the main chemical properties of alkanes. Focus on how alkanes react with different substances and under different conditions.

Alkanes are **saturated hydrocarbons** containing only strong carbon-carbon (C-C) and carbon-hydrogen (C-H) single bonds. Due to this molecular structure, alkanes exhibit relatively low chemical reactivity and are referred to as *paraffins*, derived from the Latin *parum* (little) and *affinis* (affinity or reactivity). Alkanes are generally **chemically inert** toward acids, bases, oxidizing agents, and reducing agents under normal conditions. As saturated compounds, alkanes **cannot undergo addition**

reactions because they lack double or triple bonds. Instead, alkanes primarily undergo **combustion reactions** and **substitution reactions**. The most important and characteristic reactions of alkanes are **substitution reactions**.

a) Reaction with oxygen: Combustion

In the presence of air or oxygen, alkanes burn with a clean flame to give carbon dioxide and water with the evolution of large amount of heat. The general chemical equation for combustion of alkanes may be written as:



This hot carbon dioxide, contributes, if released in the atmosphere to global warming and the formation of the acid rain.

Example



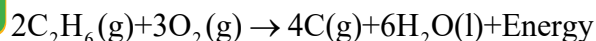
The reaction provides energy that can be used for various purposes.

In the presence of limited supply of air, alkanes burn to give carbon monoxide, which is a highly poisonous gas.



Incomplete combustion may even give soot, which is nothing other than carbon deposit.

Example



Carbon monoxide and soot are air pollutants.

Application activity 9.1.5



- Complete combustion of methane produces:
 - C and H_2O
 - CO_2 and H_2
 - CO and H_2O
 - CO_2 and H_2O
- Alkanes are also known as
- On burning in air/oxygen, alkanes get completely oxidised into and
- Write balanced chemical equation of the combustion of propane (C_3H_8).

9.1.6 Laboratory preparation of alkanes

Learning activity 9.1.6



Laboratory preparation of methane

Materials required

Retort stand, Bunsen burner, Boiling tube, Test tube/ gas jar, Delivery tube, rubber stoppers, Sodium ethanoate, Soda lime

Procedure

1. Mix sodium ethanoate with soda lime in the ratio 2: 1 and transfer in a dry boiling tube.
2. Clamp the boiling tube containing the reacting mixture horizontally on a retort stand.
3. Fit the boiling tube with a stopper connected to a delivery tube.
4. Place the delivery tube under an inverted test tube filled with water in a trough as shown on figure 9.5.

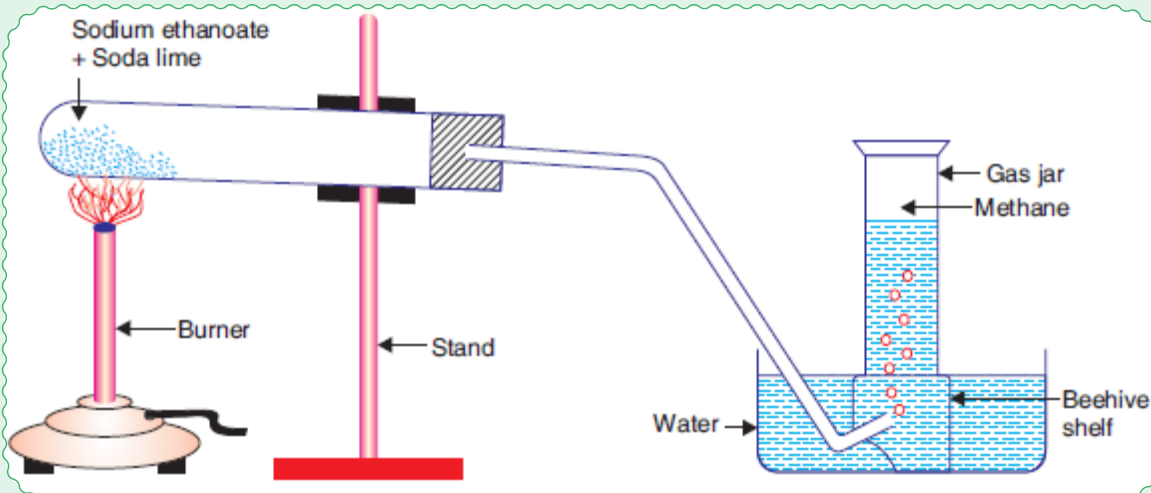


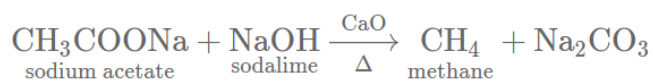
Figure 9.5: Laboratory preparation of methane.

5. Heat the mixture first gently, then strongly using a Bunsen burner.
6. Methane gas is produced and collected over water.
7. When enough gas is collected, remove the gas jar and cover it.

Guiding questions

1. Why the boiling tube must be dry?
2. Write balanced chemical equation of the reaction that takes place in the boiling tube
3. Explain why methane gas is collected by downward displacement of water.

When the sodium salt of a monocarboxylic acid is heated with soda lime (a mixture of NaOH and CaO), an alkane is produced with the elimination of a carbon dioxide molecule. This process is known as a decarboxylation reaction.

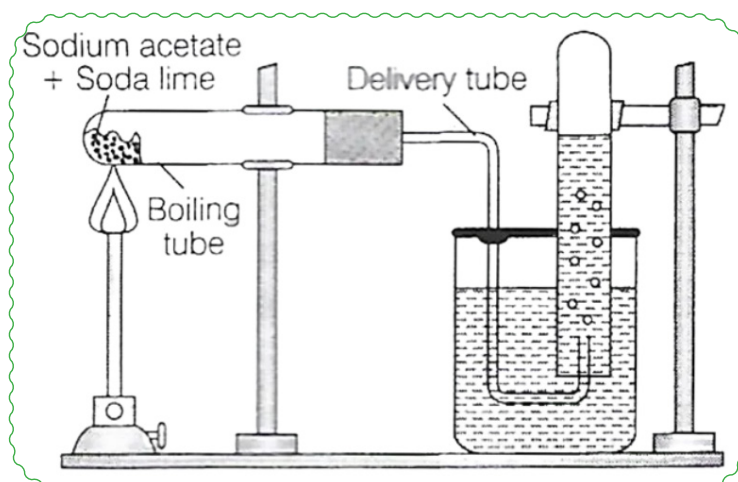


The gas is collected by downward displacement of water.

The boiling tube containing the reaction mixture must be completely dry for several important reasons. Soda lime contains sodium, which is highly hygroscopic and reacts readily with moisture. If the tube is wet, the NaOH absorbs water and may partially dissolve, significantly reducing its effectiveness in the decarboxylation reaction. The presence of water also dilutes the reactants, consequently slowing down the reaction rate and leading to a poor yield of the desired alkane product. Additionally, if water is present in the tube, it will vaporize upon heating, producing steam pressure that can cause sudden violent bubbling or even cracking of the boiling tube, posing a safety hazard. In summary, the boiling tube must be completely dry to ensure the effective action of soda lime, maximum yield of the alkane product, and safe, controlled heating of the reaction mixture.

Note: Avoid any contact of the methane produced with naked flame because it is highly flammable.

In Rwanda, methane gas is extracted in Lake Kivu. Methane is used as source of heat.



Application activity 9.1.6



1. Methane gas can be prepared in the laboratory by heating sodium ethanoate with soda lime.
2. State any four pieces of apparatus required for the experiment.
3. State two safety precautions to be taken into account during this preparation.
4. Write a balanced chemical equation for the reaction taking place.

9.1.7 Uses of alkanes

Learning activity 9.1.7



Investigating the uses and applications of alkanes

Using internet resources and textbooks, research and identify the main uses and applications of alkanes in everyday life and industry. Present your findings clearly with specific examples.

- Petroleum is largely a mixture of different alkanes. Upon refining, petroleum yields various products including Liquefied Petroleum Gas (LPG), gasoline, kerosene, diesel, furnace oil, and wax, all of which serve as fuels. The specific applications of alkanes depend primarily on their chain length and physical properties.
- **Short-chain alkanes (C_1 – C_4)** are primarily used as fuels for cooking, heating, and electricity generation. Examples include methane (natural gas) and the components of LPG (propane and butane). These small alkanes also serve as starting materials in the industrial production of hydrogen, methanol, plastics, and fertilizers.
- **Medium-chain alkanes (C_5 – C_{10})** find application as transportation fuels, particularly petrol (gasoline) for cars and motorcycles. Hexane, a six-carbon alkane, is widely used as a solvent in the extraction of vegetable oils and in dry-cleaning processes due to its ability to dissolve non-polar substances.
- **Long-chain alkanes (C_{11} – C_{20})** are utilized in various capacities depending on their exact chain length. Lighter fractions serve as jet fuel for aircraft and kerosene for lamps, while heavier fractions are used as diesel fuel for trucks, buses, generators, and tractors. The heaviest members of this range function as lubricating oils for engines and machinery.
- **Very long-chain alkanes (C_{20} and above)** are solid at room temperature and are processed into products such as paraffin wax for candles, polishes for furniture and floors, and waterproofing materials for various applications.

Application activity 9.1.7



1. Which alkane is used mainly as cooking gas?
A. Pentane, B. Methane, C. Hexane, D. Decane
2. Paraffin wax is mainly used for making:
A. Petrol, B. Lubricating oil, C. Candles, D. Diesel
3. Explain the uses of alkanes in daily life.

9.2 END OF UNIT ASSESSMENT

End of unit assessment



A. Knowledge and understanding

1. Which alkane fraction is used as fuel for jet aircraft?
 - A. Petrol
 - B. Diesel
 - C. Kerosene
 - D. Lubricating oil
2. (a) What are alkanes?
(b) Write the general formula of alkanes.
3. (a) Explain why alkanes are good fuels.
(b) State three disadvantages of using alkanes as fuels.
4. (a) Explain the concept of homologous series.
(b) Show all structural formulae of all compound of molecular formula C_5H_{12} .

B. Skills

5. A student uses hexane to remove oil stains from clothes.
6. Identify the class of hydrocarbons hexane belongs to.
7. State one property that makes hexane suitable for this use.
8. Describe the procedure used to prepare and collect methane gas in the laboratory.

C. Attitudes and values

9. Explain **four reasons** why alkanes are widely used as fuels.
10. State three safety precautions when using alkanes in the laboratory or at home.
11. State two environmental problems caused by alkane combustion.

UNIT 10

ALKENES AND ALKYNES



Key Unit Competence:

Analyse the structure, properties, and uses of alkenes and alkynes.

Learning objectives

By the end of this unit, you should be able to:

1 Knowledge and understanding

- Describe the structures of alkenes, alkynes and their physical properties.
- Name alkenes and alkynes using the IUPAC system of naming - up to C₁₀.
- State the chemical properties of alkenes and alkynes.
- Describe the uses of alkenes and alkynes in daily life.

2 Skills

- Test alkenes with bromine water.
- Draw the structure of alkenes and alkynes.
- Differentiate alkenes, and alkynes based on their degree of unsaturation and chemical properties.

3 Attitudes and values

- Develop a culture of working in a team during discussions and experiments.
- Develop orderliness in work to present results.
- Respect for the procedures in an experiment.
- Develop self- confidence in the presentation of research work and discussions.

10.0 Introduction

Alkenes and alkynes are unsaturated hydrocarbons distinguished by the presence of double and triple carbon-carbon bonds, respectively. Alkenes and alkynes can undergo addition reactions, polymerization, and serve as precursors for alcohols, halides, and other functional groups. Alkenes and alkynes are widely used in industry and everyday life, serving as raw materials for plastics, fuels, medicines, and more. Hydrocarbons are organic compounds composed exclusively of carbon (C) and hydrogen (H) atoms. They serve as fundamental building blocks in chemistry, forming the basis of crude oil, natural gas, and coal. Structurally, they consist of a carbon framework bonded to hydrogen atoms, classified into aliphatic (alkanes, alkenes, alkynes) and aromatic compounds.

Introductory activity



Observe the following picture and answer the question that follow



Which compounds are used in the manufacture of plastics?

10.1 Alkenes

10.1.1 Structure and nomenclature of alkenes.

Learning activity 10.1.1



Draw the structural formulae of the following compounds and suggest their IUPAC names:

1. C_2H_6
2. C_4H_8

Alkenes are hydrocarbons containing a carbon-carbon double bond ($C=C$) as well as single bonds between carbon atoms ($C-C$) and between carbon and hydrogen atoms ($C-H$). The $C=C$ double bond is called the **functional group** of alkenes. A **functional**

group is a specific atom or group of atoms within a molecule that is responsible for the characteristic chemical reactions and properties of that compound.

Alkenes form a **homologous series**, a family of organic compounds that share the same functional group, have similar chemical properties, show gradual changes in physical properties, and have successive members differing by a $-\text{CH}_2-$ unit.

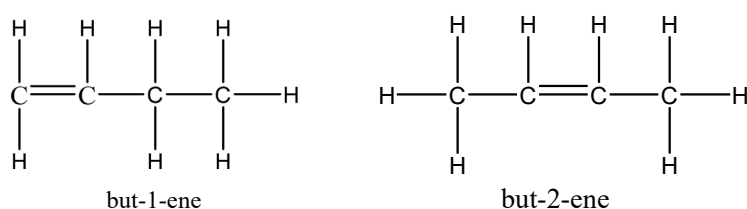
Alkenes have the general formula C_nH_{2n} , where $n = 2, 3, 4$, etc. For example: C_2H_4 (ethene), C_3H_6 (propene), and C_4H_8 (butene). In this formula, n represents the number of carbon atoms, from which the number of hydrogen atoms can be deduced. This general formula applies only to alkenes with one $\text{C}=\text{C}$ double bond in the molecule.

Note: The smallest possible alkene is **ethene (C_2H_4)** because a double bond requires at least two carbon atoms.

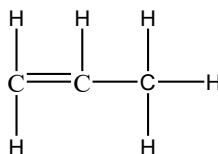
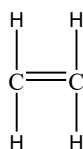
Table 10.1: The first five members of alkenes

Number of carbon atoms	Molecular formula	Displayed structural formula	Name
2	C_2H_4	<pre> H H C = C H H </pre>	Ethene
3	C_3H_6	<pre> H H H C = C - C - H H H </pre>	Propene
4	C_4H_8	<pre> H H H H C = C - C - C - H H H H </pre>	But-1-ene
5	C_5H_{10}	<pre> H H H H H C = C - C - C - C - H H H H H </pre>	Pent-1-ene
6	C_6H_{12}	<pre> H H H H H H C = C - C - C - C - C - H H H H H H </pre>	Hex-1-ene

The IUPAC rules used in naming alkanes can also be used in systematic naming for straight chain alkenes. The rules demand that the position of the double bond must be included in the name. This is why number 1 is included in some alkenes in Table 10.1. It tells us where the double bond is placed in that particular alkene as shown in the examples below:



The number shows the position of the double bond, and we write the lower number of the two carbon atoms bearing the double bond. In case of an alkene like ethene or propene the number is not necessary because if you number the carbon chain from left or right hand side the double bond will be on the first carbon.



Note: In alkenes, you must count the number of bonds around each carbon atom properly to ensure each carbon atom is surrounded by maximum of 4 bonds.

Application activity 10.1.1



Name the following organic compounds.

- (a) $\text{CH}_3\text{CH}_2\text{CH} = \text{CH}_2$
- (b) $\text{CH}_2 = \text{CHCH}_2\text{CH}_2\text{CH}_3$
- (c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH} = \text{CHCH}_3$

What is the main feature in the bonding of an alkene?

10.1.2. Structural isomerism of alkenes

Learning activity 10.1.2

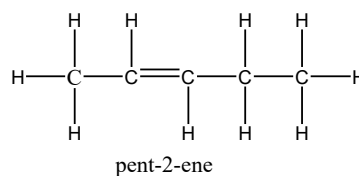
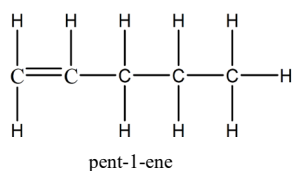


1. Review the concept of isomerism in alkanes.
2. Use the working knowledge in (1) above to try writing isomerism in alkenes.
3. Compare your findings with the content below.

In previous classes you learned about chain isomerism in alkanes. In this section, you will learn about two types of structural isomerism that occur in alkenes, namely **position isomerism** and **chain isomerism**.

(i) Position isomerism

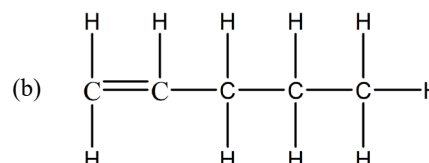
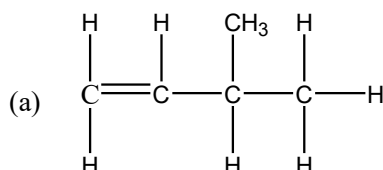
Sometimes the position of a double bond in an alkene can be shifted to a different position within the molecule. This type of isomerism is known as **position isomerism**. This is shown in the following examples.



Note that in the above isomers, the position of the double bond shifts while the position of the carbon atoms remains the same. Such isomers are known as **position isomers**.

(ii) Chain isomerism

It is possible to draw other different structural formulae of pent-1-ene. The atoms of carbon are arranged in a different way in each structure as shown below.



The above compounds (a) and (b) have the same molecular formula (i.e. C₅H₁₀), but they have different structural formulae. The general name of such structures is isomers. In (a) and (b), the position of the double bond remains the same but the length of the chain is different. Such isomers are known as chain isomers.

Application activity 10.1.2



1. Write the structural formulae for the following alkenes.

(a) Hex-2-ene

(b) But-2-ene

2. Draw and name all position isomers of hexene.

10.1.3. Physical and Chemical properties of alkenes

a) Physical properties of alkenes

Learning activity 10.1.3.a



Trends in physical properties of alkenes

Observe the following table and state how the physical properties (melting and boiling points) of alkenes change as the number of carbon atoms increases.

Table 10.2: Trends in physical properties of alkenes at room temperature

Alkene	Formula	Melting point	Boiling point	Density	Physical state	Solubility In water	Solubility in organic solvent
Ethene	C ₂ H ₄	-169	-104	-	Gas	Insoluble	Soluble
Propene	C ₃ H ₆	-185	-47.7	-	Gas	Insoluble	Soluble
Butene	C ₄ H ₈	-185	-6.2	-	Gas	Insoluble	Soluble
Pentene	C ₅ H ₁₀	-138	30.0	0.640	Liquid	Insoluble	Soluble
Hexene	C ₆ H ₁₂	-98	63.9	0.674	Liquid	Insoluble	Soluble

1. The melting and boiling points of alkenes increase as the number of carbon atoms increase. This is because with increasing carbon atoms the molecular mass increases. This also causes an increase in the intermolecular forces of attraction. To break these intermolecular forces, more energy is required hence the increase in melting and boiling points.
2. The first three alkenes, i.e ethene, propene and but-1-ene are gases at room temperature while pent-1-ene and hex-1-ene are liquids. This also has to do with increase in intermolecular forces of attractions. Alkenes are organic compounds hence they are insoluble in water but soluble in organic solvents.

Physical properties of ethene

Ethene, the simplest alkene, exhibits the following physical properties:

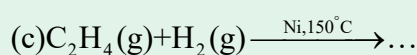
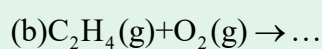
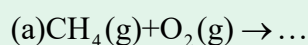
1. Ethene is a colourless gas.
2. It has a pleasant, slightly sweet odour.
3. It is completely insoluble in water but dissolves readily in organic solvents.
4. Ethene has a boiling point of -104°C and a melting point of -169°C.
5. It exhibits anaesthetic effects when inhaled.

b) Chemical properties of alkenes

Learning activity 10.1.3.b



1. Differentiate the reactivity of alkenes and alkanes.
2. Complete the following reactions:



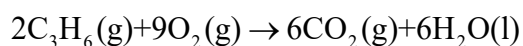
Alkenes are far more reactive than alkanes due to the carbon-carbon double bond.

Alkenes are unsaturated and they can easily undergo addition reactions to yield saturated products.

(i) Combustion of alkenes

Alkenes readily burn, just like alkanes, to give carbon dioxide and water if combustion is complete. However, they are not used as fuels for two reasons.

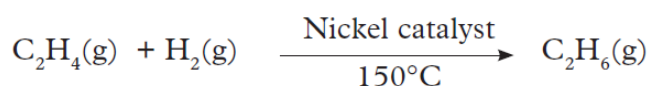
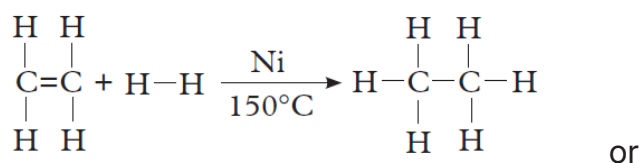
1. They are far too valuable for use to make plastics and numerous other useful compounds.
2. They burn with a smokier flame than alkanes.



(ii) Addition reactions

- Addition of hydrogen to alkenes (Hydrogenation of alkenes)

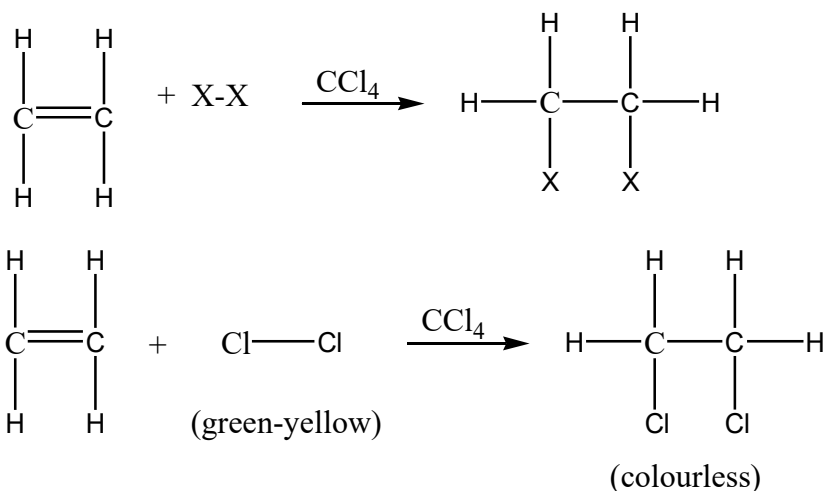
Alkenes react with hydrogen in the presence of nickel catalyst at 150°C to form alkanes.

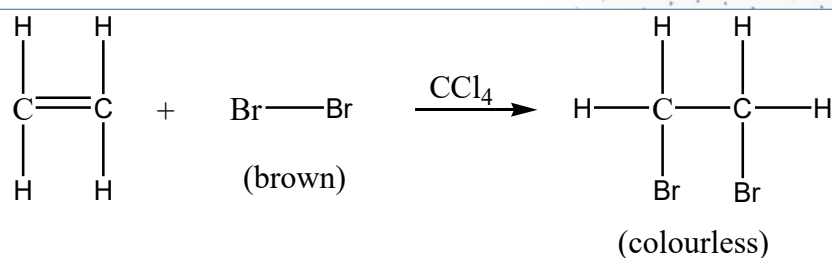


Note: Reduction with platinum and palladium as catalyst occurs at room temperature, while using nickel requires a temperature of about 150°C.

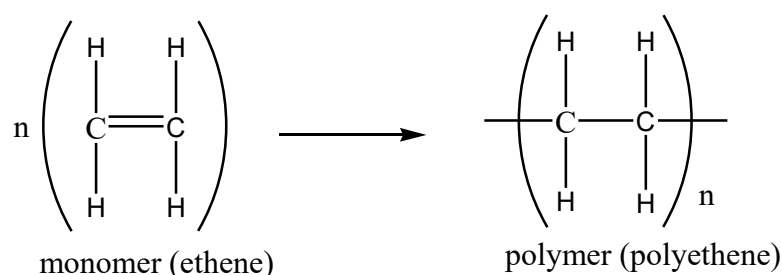
- Addition of halogens to alkenes (halogenation of alkenes)

Halogens readily add across a double bond of alkenes to form compounds called **dihalides**. This process is called **halogenation**. We may represent a halogen molecule with X_2 or $\text{X}-\text{X}$, for example Cl_2 or Br_2 .

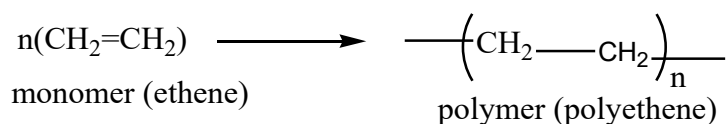




- **Polymerisation reactions (self-addition reactions):** Many monomers can join together to form a long chain consisting of thousands of carbon atoms. The process by which many small molecules called monomers join to form a large molecule is called polymerisation. The large molecule formed is called a polymer. The process can be represented by the general equation shown below.



It can also be represented as:



As shown above ethene molecule undergoes addition reaction. The process is therefore called **addition polymerisation**.

(iii) Reaction of alkenes with bromine water (test for unsaturation)

Learning activity 10.1.3.c



Test for unsaturation

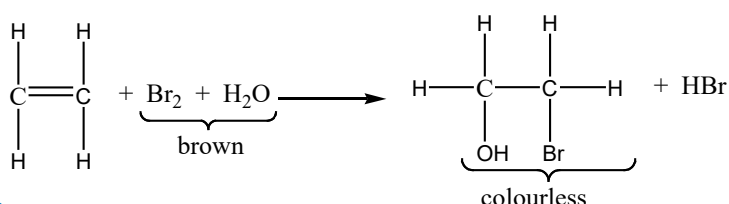
Apparatus and chemicals

Test tubes, teat pipette, bromine water, ethene

Procedure

Bubble ethene through bromine water and record your observations and conclusions.

When ethene reacts with bromine water, brown color of bromine water disappears (ethene decolourises bromine water). This decoloration of bromine is often used as a test for a carbon-carbon double bond.



Application activity 10.1.3



(a)

1. Alkenes are generally insoluble in water. What is the main reason?
2. What is the typical physical state of alkenes with 2–4 carbon atoms at room temperature?
3. An alkene has a melting point of -140°C and a boiling point of 64°C . Would you expect this alkene to be a solid, liquid or gas at room temperature (25°C)? Explain.
4. Complete the following chemical equations of reactions:
 - (a) $\text{C}_6\text{H}_{12} + \text{Br}_2 \longrightarrow$
 - (b) $\text{C}_4\text{H}_8 + \text{O}_2(\text{g}) \longrightarrow$
5. Write an equation to show how propene reacts with hydrogen to produce propane.
6. Predict the product formed when but-1-ene reacts with hydrogen.

10.1.4. Uses of alkenes

Learning activity 10.1.4



Describe the uses of alkenes

Alkenes have numerous important applications in industry, agriculture, and medicine:

- a) Alkenes are extremely important in the manufacture of plastics which have many applications such as packaging, wrapping, making clothes, artificial flowers, pipes, cups, and windows.
- b) Ethene is a plant hormone involved in the ripening of fruits, seed germination, and bud opening.
- c) Ethene is used in medicine as a precursor for anaesthetics.
- d) Alkenes are used as raw materials in industry for the manufacture of alcohols through hydration.

Application activity 10.1.4



State the uses of alkenes

10.2. Alkynes

10.2.1 Structure of alkynes and nomenclature

Learning activity 10.2.1



1. Define alkynes and give the general formula.
2. How does an alkyne differ from an alkene?

Alkynes are unsaturated hydrocarbons with at least one carbon–carbon triple bond $\text{—C}\equiv\text{C—}$. Their general formula is $\text{C}_n\text{H}_{2n-2}$, where $n = 2, 3, 4$ etc. For example: C_2H_2 , C_3H_4 , C_4H_6 , C_5H_8 , etc. The simplest alkyne is C_2H_2 ethyne.

Alkynes are named using the suffix “**-yne**” in IUPAC nomenclature. The longest chain containing the triple bond is chosen, numbered from the end nearest the triple bond, and the position of the triple bond is indicated by the lowest possible index.

Table 10.2 The first five members of alkynes.

Number of carbons	Molecular formula	Structural Formula	Name
2	C_2H_2	$\text{H—C}\equiv\text{C—H}$	Ethyne
3	C_3H_4	$\text{H—C}\equiv\text{C—CH}_3$	Propyne
4	C_4H_6	$\text{H—C}\equiv\text{C—CH}_2\text{CH}_3$	But-1-yne
5	C_5H_8	$\text{H—C}\equiv\text{C—CH}_2\text{CH}_2\text{CH}_3$	Pent-1-yne
6	C_6H_{10}	$\text{CH}_3\text{—C}\equiv\text{C—CH}_2\text{CH}_2\text{CH}_3$	Hex-2-yne

Application activity 10.2.1



Name the following compounds:

- (a) $\text{CH}_3\text{—C}\equiv\text{C—CH}_2\text{CH}_3$
- (b) $\text{CH}_3\text{—C}\equiv\text{C—CH}_3$
- (c) $\text{HC}\equiv\text{C—CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

10.2.2. Structural isomerism of alkynes

Learning activity 10.2.2

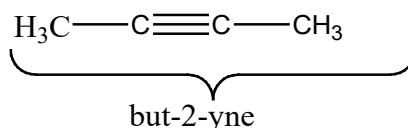
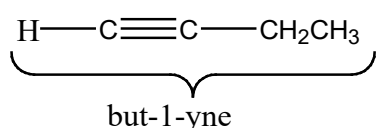


Draw and name the structural formulae of two isomers of C_4H_6 .

Structural isomers are compounds with the same molecular formula but different spatial arrangement of component atoms. Alkynes exhibit various structural isomers, primary through position isomerism and chain isomerism, depending on the arrangement of carbon atoms and the location of the triple bond.

a) Position Isomerism

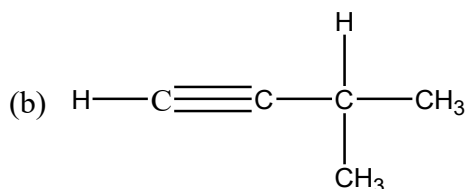
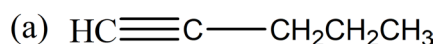
In position isomers, the triple bond is located at different positions in the same carbon chain. For example, butyne has two position isomers.



But-1-yne has a triple bond at carbon 1 whereas **but-2-yne** has a triple bond at carbon 2.

b) Chain Isomerism

Chain Isomerism occurs when compounds with the same molecular formula differ in the arrangement of their carbon skeleton (straight chain versus branched chain). For example,



The above compounds (a) and (b) have the same molecular formula (i.e. C_5H_8), but they have different structural formulae.

Application activity 10.2.2



Draw and name the three possible position isomers of C_6H_{10} .

10.2.3. Laboratory preparation of ethyne (acetylene) from CaC_2

Learning activity 10.2.3



Apparatus and chemicals

Dropping funnel, Conical flask, Delivery tube, Test tubes, Trough, Calcium carbide, Water

Procedure:

1. Set up the apparatus as shown in the diagram below.

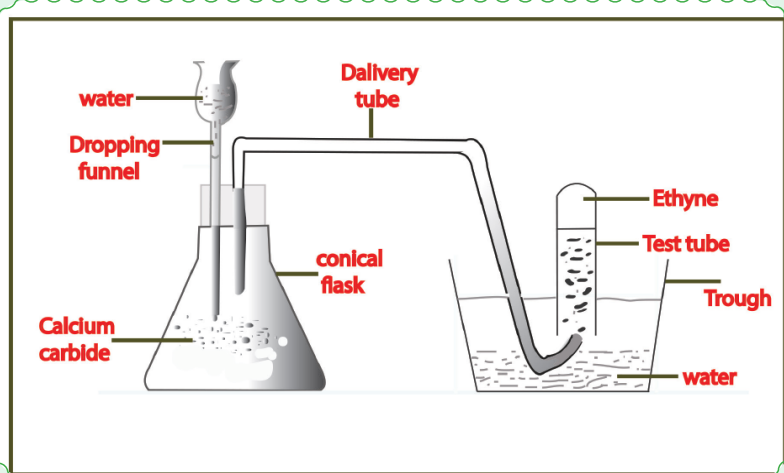
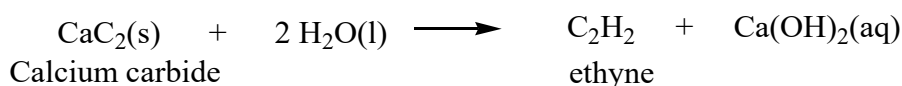
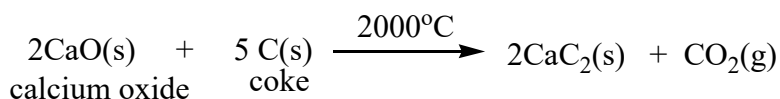


Figure 10.2.3. Laboratory preparation of ethyne

2. Place 2g of calcium carbide in a conical flask
3. Using the dropping funnel, add water drop by drop.
4. Collect the gas produced in the test tube.
5. To the collected gas add two drops of bromine water. Record your observations.

Ethyne (acetylene) can be prepared on a large scale (industrial preparation) from calcium carbide which is obtained by reduction of calcium oxide (quick lime) by coke (a form of carbon) at high temperature (between 2000°C and 2500°C giving calcium carbide. Then, calcium carbide is hydrolysed to give ethyne and slaked lime.



Application activity 10.2.3



Describe the industrial method of preparing acetylene from calcium oxide.

10.2.4 Physical and chemical properties of alkynes

(a) Physical properties of alkynes

Learning activity 10.2.1 a



Alkynes have the general formula C_nH_{2n-2} . They have two fewer hydrogen atoms than alkenes, and four fewer H than alkanes. Do you expect alkynes to be more or less volatile than alkenes?

The physical properties of alkynes, such as boiling point, melting point, and solubility are largely determined by their molecular structure, particularly the presence of the carbon-carbon triple bond, and the resulting intermolecular forces.

- a) Alkynes are less dense than water, similar to other hydrocarbons.
- b) Alkynes are **nonpolar** and insoluble in water but soluble in organic solvents.
- c) Short-chain alkynes (C_2 – C_4) are gases;
- d) Medium-chain (C_5 – C_{15}) are liquids;
- e) Longer chains are solids.
- f) Alkynes have boiling points slightly higher than alkenes with the same number of carbon atoms. The linear structure of alkynes gives them greater intermolecular forces than alkenes.

Example

Compound	Melting point/ $^{\circ}C$	Boiling point/ $^{\circ}C$
Propene	-185	-47
Propyne	-101	-23
Pent-1-ene	-138	30
Pent-1-yne	-90	40

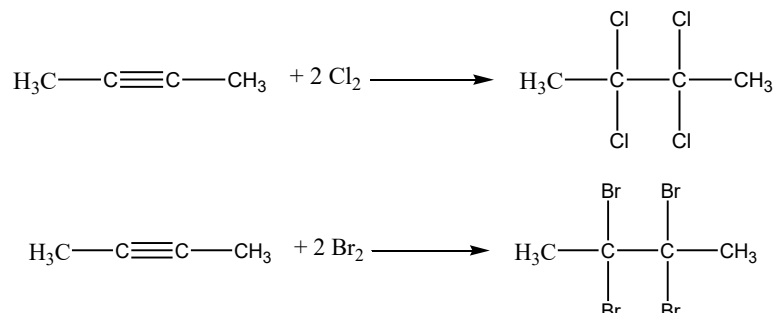
Application activity 10.2.4 (a)



1. Which of the following statements about the boiling points of alkynes is correct?
 - A. They are lower than those of alkanes with the same carbon number.
 - B. They are higher than those of alkanes and alkenes with the same carbon number.
 - C. They are identical to those of alkanes.
 - D. They do not depend on carbon number.

2. Addition of halogens (Halogenation)

Similar to the case of alkenes, the brown colour of bromine disappears when alkyne reacts with bromine suggesting the presence of multiple bonds. Hence, this reaction is used as a **test for unsaturation**. The addition of halogens (chlorine and bromine) to alkynes is as follows:



Application activity 10.2.4 (b)



1. Write a balanced chemical equation to show the complete hydrogenation of propyne. State the conditions required and name the product formed.
2. Write a balanced chemical equation for the reaction of but-1-yne with hydrogen gas in the presence of Lindlar's catalyst. Name the product and explain why Lindlar's catalyst is used in this reaction.
3. Write a balanced chemical equation for the reaction of propyne with excess chlorine gas (Cl_2). Name the organic product formed and explain why two moles of Cl_2 are required.
4. Write a balanced chemical equation for the reaction of but-2-yne with hydrogen chloride (HCl). Show that the equation is balanced and name the product formed.

10.2.5 Uses of alkynes

Learning activity 10.2.5



Look at the following pictures and suggest the uses of alkynes.



Alkynes have several important industrial and medical applications:

- Ethyne (acetylene) is used as a fuel in welding and cutting metals because it produces an extremely hot flame when burned with oxygen.
- Propyne serves as a substitute for acetylene as a fuel for welding operations.
- Propyne and acetylene are used as starting materials for producing vinyl chloride (used to make PVC) and other polymers.
- Certain alkyne derivatives are used in pharmaceutical synthesis, including the production of contraceptives and cancer treatment drugs.

Application activity 10.2.5



- Which alkyne is commonly used in welding due to its high flame temperature?
A. Ethyne (acetylene), B. Butyne, C. Hexyne, D. Propyne
- Alkynes are used in the production of which type of synthetic material?
A. Glass, B. Ceramics, C. Plastics, D. Wood
- Which industry benefits from alkyne derivatives in drug development?
A. Pharmaceutical
B. Automotive
C. Textile
D. Agriculture
- Which polymer is produced using alkynes as a starting material?
A. Polystyrene
B. Polyvinyl chloride (PVC)
C. Polyethylene
D. Polypropylene

10.3 END OF UNIT ASSESSMENT

End of unit assessment



A. Knowledge and understanding

- Which of the following is given off during ripening of fruits and vegetables?
A. Ethane, B. Ethene, C. Ethyne, D. Methane
- The molecular formula of----- fit the general formula (C_nH_{2n-2}).
A. Alkanes, B. Alkynes, C. Alkenes, D. Hydrocarbons

3. Which property of alkenes makes them useful as starting materials in polymer production?
- Ability to undergo addition reactions
 - High solubility in water
 - High boiling points
 - Strong hydrogen bonding
4. Which of the following is an alkene?
- C_2H_{2l}
 - C_2H_{4l}
 - C_2H_{6l}
 - CH_4
5. What is the name of the alkyne which has the formula C_5H_8 ?
- Ethane
 - Pentyne
 - Pentane
 - Pentene
6. How do you test for an alkene?
- They turn universal indicator purple.
 - They turn bromine water colourless.
 - They turn limewater cloudy.
 - They do not react.

II. Filling in and matching questions

1. Copy and complete the passage using the list of words in the brackets provided.

(bromine, decolorises, double, ethane, ethene and reddish-brown)

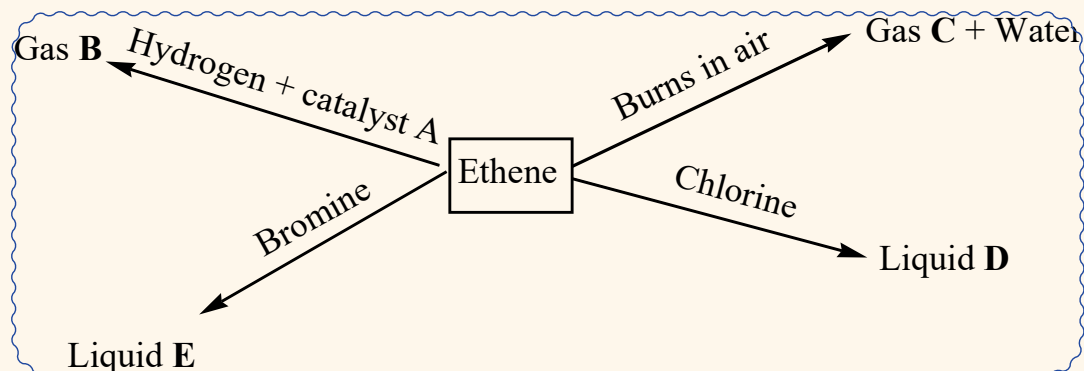
Ethane can be distinguished from ethene by adding water to each compound. Bromine water has a colour. Ethene bromine water but does not. The bromine adds across the bond in and no other compound is formed.

2. Match each word on the left with its description on the right.

Saturated	A reaction in which two or more compounds combine to form only one compound.
Unsaturated	Organic compounds containing only single bonds.
Addition	Organic compounds containing $C=C$ double bond.

B. Skills

1. Draw two position isomers of pentene and three chain isomers of pentene.
2. How does the presence of a double bond in alkenes influence their chemical reactivity compared to alkanes?
3. The following question is about some of the reaction of ethene.



- A. Give the formulae for substances A, B, C, D and E.
- B. Name the homologous series that gas B belongs to.

C. Attitudes and values

1. Explain why ethane is described as a plant hormone.

REFERENCE

1. <https://flexbooks.ck12.org/cbook/cbse-chemistry-class-10/section/4.2/primary/lesson/allotropic-forms-of-carbon-10378583/>
2. <https://unacademy.com/content/jee/study-material/chemistry/uses-of-allotropes-and-oxides-of-carbon/>
3. <https://study.com/academy/lesson/inorganic-carbon-compounds-definition-examples.html>
4. <https://kunakair.com/carbon-monoxide/>
5. <https://byjus.com/chemistry/hard-water-and-soft-water/>
6. <https://oceanservice.noaa.gov/facts/carbon-cycle.html#transcript>
7. <https://www.vedantu.com/chemistry/preparation-properties-and-uses-of-nitrogen>
8. <https://ammoniagas.com/what-is-ammonia/>
9. <https://allen.in/jee/chemistry/nitric-acid>
10. <https://www.chemistrylearner.com/nitrate.html>
11. <https://www.samaterials.com/blog/sulfur-element-properties-and-uses.html>
12. <https://www.studocu.com/row/document/mount-kenya-university/industrial-chemistry/inorganic-chemistry-sulphur-and-its-compounds-overview-and-properties/127463889?sid=be3d6870-f72f-4107-a532-b737439fa6fc1783165151>
13. <https://allen.in/jee/chemistry/sulphuric-acid>
14. <https://www.youtube.com/watch?v=UI1U1FyWHKM>
15. "New Certificate Chemistry" (or "Certificate Chemistry") by G.C. Ashworth and B.P. Bass.